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(54) ORGANOMETALLIC COMPOUND, LIGHT-EMITTING DEVICE INCLUDING THE SAME, AND ELECTRONIC APPARATUS INCLUDING THE LIGHT-EMITTING DEVICE

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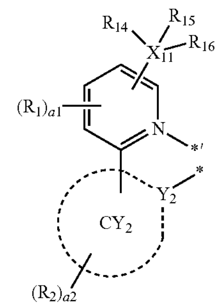
(57) **ABSTRACT**

An organometallic compound represented by Formula 1:

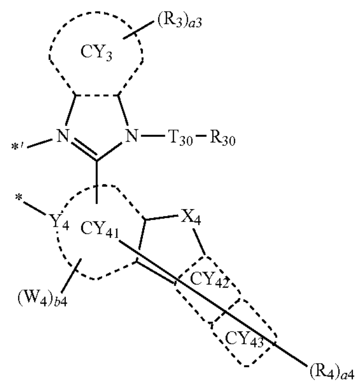
$M(L_1)_{n1}(L_2)_{n2}$ Formula 1

wherein M is a transition metal, L_1 is a ligand represented by Formula 2-1, L_2 is a ligand represented by Formula 2-2, n1 and n2 are each independently 1 or 2, and L_1 and L_2 are different from each other:

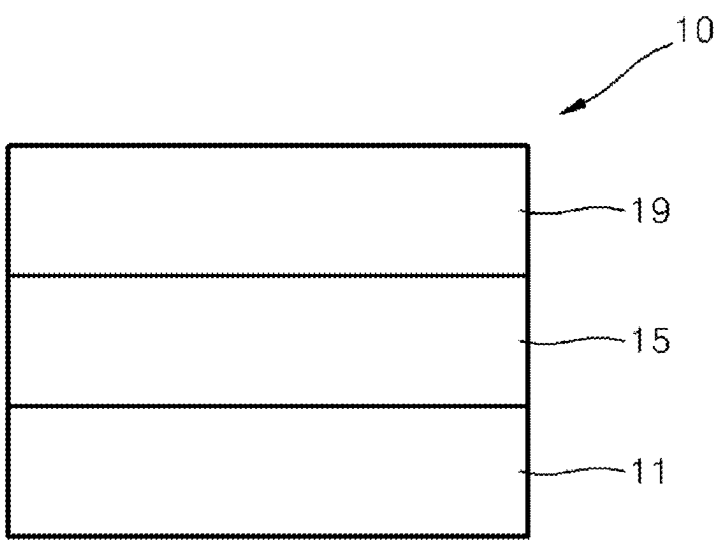
Formula 2-1



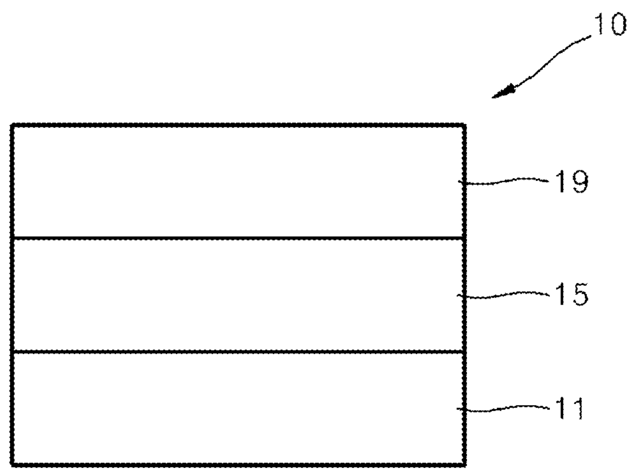
Formula 2-2



wherein Y_2 and Y_4 are each independently C or N; ring CY_2, ring CY_3, and ring CY_{41} to ring CY_{43} are each independently a C_5-C_{30} carbocyclic group or a C_1-C_{30} heterocyclic group; X_{11} is C, Si, or Ge; X_4 is O, S, Se, $N(R_{48})$, $C(R_{48})(R_{49})$, or $Si(R_{48})(R_{49})$; b4 is an integer from 1 to 10; * and *' each indicate a binding site to M in Formula 1; and the remaining substituents are as described herein.



FIGURE



ORGANOMETALLIC COMPOUND, LIGHT-EMITTING DEVICE INCLUDING THE SAME, AND ELECTRONIC APPARATUS INCLUDING THE LIGHT-EMITTING DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0032837, filed on Mar. 7, 2024, in the Korean Intellectual Property Office, and all the benefits accruing therefrom under 35 U.S.C. § 119, the entire content of which is incorporated by reference herein.

BACKGROUND

1. Field

[0002] The disclosure relates to an organometallic compound, a light-emitting device including the same, and an electronic apparatus including the light-emitting device.

2. Description of the Related Art

[0003] From among light-emitting devices (OLEDs), organic light-emitting devices are self-emission devices, which have improved characteristics in terms of viewing angles, response time, luminance, driving voltage, and response speed. In addition, OLEDs can produce full-color images.

[0004] In an example, an organic light-emitting device includes an anode, a cathode, and an interlayer that is arranged between the anode and the cathode and includes an emission layer. A hole transport region may be arranged between the anode and the emission layer, and an electron transport region may be arranged between the emission layer and the cathode. Holes provided from the anode may move toward the emission layer through the hole transport region, and electrons provided from the cathode may move toward the emission layer through the electron transport region. The holes and the electrons recombine in the emission layer to produce excitons. When the excitons transition from an excited state to a ground state, light is emitted.

SUMMARY

[0005] Provided are an organometallic compound, a light-emitting device using the same, and an electronic apparatus including the light-emitting device.

[0006] Additional aspects will be set forth in part in the detailed description that follows and, in part, will be apparent from the detailed description, or may be learned by practice of the presented exemplary embodiments described herein.

[0007] According to an aspect, an organometallic compound represented by Formula 1 is provided:

$$M(L_1)_{n1}(L_2)_{n2} \quad \text{Formula 1}$$

wherein, in Formula 1,

[0008] M is a transition metal,

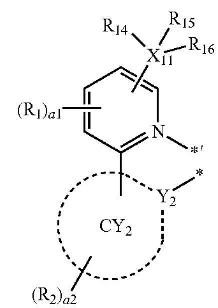
[0009] L_1 is a ligand represented by Formula 2-1,

[0010] L_2 is a ligand represented by Formula 2-2,

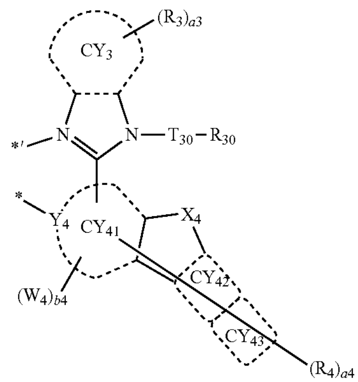
[0011] n1 and n2 are each independently 1 or 2, wherein, when n1 is 2, two of L_1 are identical to or different from each other, and when n2 is 2, two of L_2 are identical to or different from each other,

[0012] L_1 and L_2 are different from each other,

Formula 2-1



Formula 2-2



wherein, in Formulae 2-1 and 2-2,

[0013] Y_2 and Y_4 are each independently C or N,

[0014] ring CY_2, ring CY_3, and ring CY_{41} to ring CY_{43} are each independently a C_5-C_{30} carbocyclic group or a C_1-C_{30} heterocyclic group,

[0015] X_{11} is C, Si, or Ge,

[0016] T_{30} is a single bond, a C_1-C_{20} alkylene group unsubstituted or substituted with at least one R_{10a}, a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a},

[0017] X_4 is O, S, Se, $N(R_{48})$, $C(R_{48})(R_{49})$, or $Si(R_{48})(R_{49})$,

[0018] R_1 to R_4, R_{14} to R_{16}, R_{30}, R_{48}, and R_{49} are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted C_1-C_{60} alkoxy group, a substituted or unsubstituted C_1-C_{60} alkylthio group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_7-C_{60} alkyl aryl group, a substituted or unsubstituted C_7-C_{60} aryl alkyl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted C_2-C_{60} alkyl heteroaryl group, a substituted or unsubstituted C_2-C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1-C_{60} heteroaryloxy group, a substituted or unsubstituted C_1-C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q_1)(Q_2)(Q_3), —Ge(Q_1)(Q_2)(Q_3), —N(Q_4)(Q_5), —B(Q_6)(Q_7), —P(═O)(Q_8)(Q_9), or —P(Q_8)(Q_9),

[0019] a1 is an integer from 0 to 3,

[0020] a2 to a4 are each independently an integer from 0 to 20,

[0021] W_4 is a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted C_1-C_{60} alkoxy group, a substituted or unsubstituted C_1-C_{60} alkylthio group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_7-C_{60} alkyl aryl group, a substituted or unsubstituted C_7-C_{60} aryl alkyl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted C_2-C_{60} alkyl heteroaryl group, a substituted or unsubstituted C_2-C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1-C_{60} heteroaryloxy group, a substituted or unsubstituted C_1-C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group,

[0022] b4 is an integer from 1 to 10,

[0023] two or more of a plurality of R_1 are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a},

[0024] two or more of a plurality of R_2 are optionally be linked to each other to form a C_5-C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

[0025] two or more of a plurality of R_3 are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a},

[0026] two or more of a plurality of R_4 are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a},

[0027] two or more of R_1 to R_4 and R_{30} are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a},

[0028] R_{10a} is as described in connection with R_2,

[0029] * and *' each indicate a binding site to M in Formula 1,

[0030] a substituent of the substituted C_1-C_{60} alkyl group, the substituted C_2-C_{60} alkenyl group, the substituted C_2-C_{60} alkynyl group, the substituted C_1-C_{60} alkoxy group, the substituted C_1-C_6o alkylthio group, the substituted C_3-C_{10} cycloalkyl group, the substituted C_1-C_{10} heterocycloalkyl group, the substituted C_3-C_{10} cycloalkenyl group, the substituted C_1-C_{10} heterocycloalkenyl group, the substituted C_6-C_{60} aryl group, the substituted C_7-C_{60} alkyl aryl group, the substituted C_7-C_{60} aryl alkyl group, the substituted C_6-C_{60} aryloxy group, the substituted C_6-C_{60} arylthio group, the substituted C_1-C_{60} heteroaryl group, the substituted C_2-C_{60} alkyl heteroaryl group, the substituted C_2-C_{60} heteroaryl alkyl group, the substituted C_1-C_{60} heteroaryloxy group, the substituted C_1-C_{60} heteroarylthio group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is:

[0031] deuterium, —F, —Cl, —Br, —I, —SF_5, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy group, or a C_1-C_{60} alkylthio group,

[0032] a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy group, or a C_1-C_{60} alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF_5, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_3-C_{10} cycloalkyl group, a C_1-C_1 heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_1 heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a C_2-C_{60} alkyl heteroaryl group, a C_1-C_{60} heteroaryloxy group, a C_1-C_{60} heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —Si(Q_{11})(Q_{12})(Q_{13}), —Ge(Q_{11})(Q_{12})(Q_{13}), —N(Q_{14})(Q_{15}), —B(Q_{16})(Q_{17}), —P(═O)(Q_{18})(Q_{19}), —P(Q_{18})(Q_{19}), or a combination thereof,

[0033] a C_3-C_{10} cycloalkyl group, a C_1-C_1 heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_1 heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a C_2-C_{60} alkyl heteroaryl group, a C_1-C_{60} heteroaryloxy group, a C_1-C_{60} heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each unsubstituted or substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF_5, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy group, a C_1-C_{60} alkylthio group, a C_3-C_{10} cycloalkyl group, a C_1-C_1 heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_1 heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a C_2-C_{60} alkyl heteroaryl group, a C_1-C_{60} heteroaryloxy group, a C_1-C_{60} heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —$Si(Q_{21})(Q_{22})(Q_{23})$, —$Ge(Q_{21})(Q_{22})(Q_{23})$, —$N(Q_{24})(Q_{25})$, —$B(Q_{26})(Q_{27})$, —$P(=O)(Q_{28})(Q_{29})$, $P(Q_{28})(Q_{29})$, or a combination thereof,

[0034] —$Si(Q_{31})(Q_{32})(Q_{33})$, —$Ge(Q_{31})(Q_{32})(Q_{33})$, —$N(Q_{34})(Q_{35})$, —$B(Q_{36})(Q_{37})$, —$P(=O)(Q_{38})(Q_{39})$, or —$P(Q_{38})(Q_{39})$, or

[0035] a combination thereof, and

[0036] Q_1 to Q_9, Q_{11} to Q_{19}, Q_{21} to Q_{29} and Q_{31} to Q_{39} are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_7-C_{60} alkyl aryl group, a substituted or unsubstituted C_7-C_{60} aryl alkyl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted C_2-C_{60} alkyl heteroaryl group, a substituted or unsubstituted C_2-C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1-C_{60} heteroaryloxy group, a substituted or unsubstituted C_1-C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

[0037] According to another aspect, a light-emitting device includes a first electrode, a second electrode, and an interlayer disposed between the first electrode and the second electrode, wherein the interlayer includes an emission layer, and wherein the interlayer further includes at least one of the organometallic compounds represented by Formula 1.

[0038] According to another aspect, an electronic apparatus includes the light-emitting device.

BRIEF DESCRIPTION OF THE DRAWING

[0039] The above and other aspects, features, and advantages of certain exemplary embodiments will be more apparent from the following detailed description taken in conjunction with the FIGURE, which shows a schematic cross-sectional view of a light-emitting device according to one or more embodiments.

DETAILED DESCRIPTION

[0040] Reference will now be made in detail to exemplary embodiments, examples of which are illustrated in the accompanying drawing, wherein like reference numerals refer to like elements throughout the specification. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the FIGURE, to explain certain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0041] The terminology used herein is for the purpose of describing one or more exemplary embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The term “or” means “and/or.” It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof.

[0042] It will be understood that, although the terms first, second, third etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of the present embodiments.

[0043] Exemplary embodiments are described herein with reference to cross section illustrations that are schematic illustrations of idealized embodiments. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments described herein should not be construed as limited to the particular shapes of regions as illustrated herein but are to include deviations in shapes that result, for example, from manufacturing. For example, a region illustrated or described as flat may, typically, have rough and/or nonlinear features. Moreover, sharp angles that are illustrated may be rounded. Thus, the regions illustrated in the figures are schematic in nature and their shapes are not intended to illustrate the precise shape of a region and are not intended to limit the scope of the present claims.

[0044] It will be understood that when an element is referred to as being "on" another element, it can be directly in contact with the other element or intervening elements may be present therebetween. In contrast, when an element is referred to as being "directly on" another element, there are no intervening elements present.

[0045] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this general inventive concept belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0046] "About" or "approximately" as used herein is inclusive of the stated value and means within an acceptable range of deviation for the particular value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the particular quantity (i.e., the limitations of the measurement system). For example, "about" can mean within one or more standard deviations, or within ±30%, 20%, 10%, 5% of the stated value.

[0047] As used herein, an "energy level" (e.g., a triplet (Ti) energy level) is expressed as an absolute value from a vacuum level. In addition, when the energy level is referred to as being "deep," "high," or "large," the energy level has a large absolute value based on "0 electron Volts (eV)" of the vacuum level, and when the energy level is referred to as being "shallow," "low," or "small," the energy level has a small absolute value based on "0 eV" of the vacuum level.

[0048] An organometallic compound according to an aspect is represented by Formula 1:

$$M(L_1)_{n1}(L_2)_{n2} \qquad \text{Formula 1}$$

wherein M in Formula 1 is a transition metal.

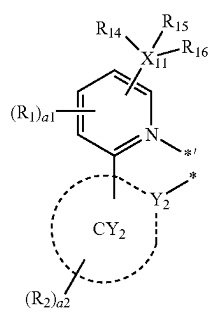
[0049] For example, M may be a first row transition metal of the Periodic Table of Elements, a second row transition metal of the Periodic Table of Elements, or a third row transition metal of the Periodic Table of Elements.

[0050] In one or more embodiments, M may be iridium (Ir), platinum (Pt), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), thulium (Tm), or rhodium (Rh).

[0051] In one or more embodiments, M may be iridium (Ir), platinum (Pt), osmium (Os), or rhodium (Rh).

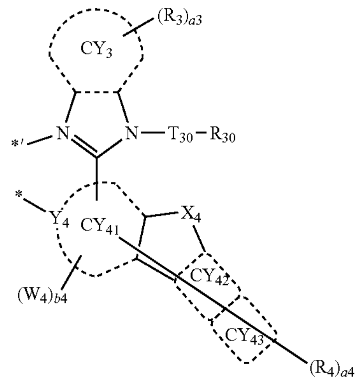
[0052] In Formula 1, L_1 is a ligand represented by Formula 2-1, and L_2 is a ligand represented by Formula 2-2:

Formula 2-1



-continued

Formula 2-2



[0053] Formula 2-1 and Formula 2-2 may each be the same as described herein.

[0054] n1 and n2 in Formula 1 respectively represent the number of ligand(s) L_1 and the number of ligand(s) L_2, and are each independently 1 or 2. When n1 is 2, two Li are the same as or different from each other, and when n2 is 2, two L_2 are the same as or different from each other.

[0055] For example, in Formula 1, i) n1 may be 2 and n2 may be 1; or ii) n1 may be 1 and n2 may be 2.

[0056] In one or more embodiments, in Formula 1, i) M may be iridium (Ir) or osmium (Os) and a sum of n1 and n2 may be 3 or 4; or ii) M may be platinum (Pt) and the sum of n1 and n2 may be 2. In Formula 1, M may be iridium (Ir), and a sum of n1 and n2 may be 3.

[0057] L_1 and L_2 in Formula 1 are different from each other.

[0058] Y_2 and Y_4 in Formula 2-1 and Formula 2-2 are each independently C or N.

[0059] For example, Y_2 and Y_4 may each be C.

[0060] Ring CY_2, ring CY_3, and ring CY_{41} to ring CY_{43} in Formula 2-1 and Formula 2-2 are each independently a C_5-C_{30} carbocyclic group or a C_1-C_{30} heterocyclic group.

[0061] For example, ring CY_2, ring CY_3, and ring CY_{41} to ring CY_{43} may each independently be i) a first ring, ii) a second ring, iii) a condensed ring group in which two or more first rings are condensed with each other, iv) a condensed ring group in which two or more second rings are condensed with each other, or v) a condensed ring group in which one or more first rings and one or more second rings are condensed with each other,

[0062] wherein the first ring may be a cyclopentane group, a cyclopentadiene group, a furan group, a thiophene group, a pyrrole group, a silole group, a germole group, a borole group, a selenophene group, a phosphole group, an oxazole group, an oxadiazole group, an oxatriazole group, a thiazole group, a thiadiazole group, a thiatriazole group, a pyrazole group, an imidazole group, a triazole group, a tetrazole group, an azasilole group, an azagermole group, an azaborole group, an azaselenophene group, or an azaphosphole group, and

[0063] wherein the second ring may be an adamantane group, a norbornane group (bicyclo[2.2.1]heptane group), a norbornene group, a bicyclo[1.1.1]pentane group, a bicyclo[2.1.1]hexane group, a bicyclo[2.2.2]octane group, a cyclohexane group, a cyclohexene group, a benzene group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, or a triazine group.

[0064] According to one or more embodiments, ring CY_2, ring CY_3, and ring CY_{41} to ring CY_{43} may each independently be a cyclopentane group, a cyclohexane group, a cyclohexene group, a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a 1,2,3,4-tetrahydronaphthalene group, a thiophene group, a furan group, an indole group, a benzoborole group, a benzophosphole group, an indene group, a benzosilole group, a benzogermole group, a benzothiophene group, a benzoselenophene group, a benzofuran group, a carbazole group, a dibenzoborole group, a dibenzophosphole group, a fluorene group, a dibenzosilole group, a dibenzogermole group, a dibenzothiophene group, a dibenzoselenophene group, a dibenzofuran group, a dibenzothiophene 5-oxide group, a 9H-fluoren-9-one group, a dibenzothiophene 5,5-dioxide group, an azaindole group, an azabenzoborole group, an azabenzophosphole group, an azaindene group, an azabenzosilole group, an azabenzogermole group, an azabenzothiophene group, an azabenzoselenophene group, an azabenzofuran group, an azacarbazole group, an azadibenzoborole group, an azadibenzophosphole group, an azafluorene group, an azadibenzosilole group, an azadibenzogermole group, an azadibenzothiophene group, an azadibenzoselenophene group, an azadibenzofuran group, an azadibenzothiophene 5-oxide group, an aza-9H-fluoren-9-one group, an azadibenzothiophene 5,5-dioxide group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a 5,6,7,8-tetrahydroisoquinoline group, a 5,6,7,8-tetrahydroquinoline group, an adamantane group, a norbornane group, or a norbornene group.

[0065] According to one or more embodiments, ring CY_2 may be a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a 1,2,3,4-tetrahydronaphthalene group, a carbazole group, a fluorene group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, a dibenzoselenophene group, a pyridine group, a benzoxazole group, a benzothiazole group, or a benzene group that is condensed with a norbornane group.

[0066] According to one or more embodiments, in Formula 2-2, ring CY_3 and ring CY_{41} to ring CY_{43} may each independently be i) a Group A, ii) a polycyclic group in which two or more Group A are condensed with each other, or iii) a polycyclic group in which one or more Group A and one or more Group B are condensed with each other,

[0067] Group A may be a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, or a triazine group, and

[0068] Group B may be a cyclohexane group, a norbornane group, a furan group, a thiophene group, a selenophene group, a pyrrole group, a cyclopentadiene group, or a silole group.

[0069] According to one or more embodiments, ring CY_3 and ring CY_{41} to ring CY_{43} may each independently be:

[0070] a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, a triazine group, a pyrrole group, a cyclopentadiene group, a silole group, a thiophene group, or a furan group; or

[0071] a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, a triazine group, a pyrrole group, a cyclopentadiene group, a silole group, a thiophene group, or a furan group, each condensed with at least one of a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, a triazine group, a cyclohexane group, a norbornane group, an adamantane group, a pyrrole group, a cyclopentadiene group, a silole group, a thiophene group, a furan group, or a combination thereof.

[0072] According to one or more embodiments, ring CY_3 and ring CY_{41} to ring CY_{43} may each independently be:

[0073] a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, a triazine group, a naphthalene group, a quinoline group, an isoquinoline group, a quinazoline group, a quinoxaline group, a phenanthrene group, a benzoquinoline group, a benzoisoquinoline group, an anthracene group, or a chrysene group; or

[0074] a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, a triazine group, a naphthalene group, a quinoline group, an isoquinoline group, a quinazoline group, a quinoxaline group, a phenanthrene group, a benzoquinoline group, a benzoisoquinoline group, an anthracene group, or a chrysene group, each condensed with at least one of a cyclohexane group, a norbornane group, an adamantane group, or a combination thereof.

[0075] According to one or more embodiments, ring CY_{41} and ring CY_{42} in Formula 2-2 may each independently be a benzene group, a naphthalene group, a benzene group condensed with a cyclohexane group, or a benzene group condensed with a norbornane group.

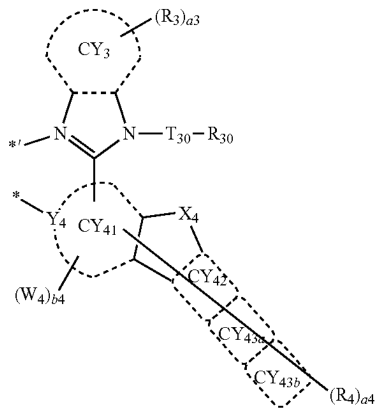
[0076] According to one or more embodiments, ring CY_{43} in Formula 2-2 may be:

[0077] a benzene group, a naphthalene group, a phenanthrene group, an anthracene group, or a chrysene group; or

[0078] a benzene group, a naphthalene group, a phenanthrene group, an anthracene group, or a chrysene group, each condensed with at least one of a cyclohexane group, a norbornane group, or a combination thereof.

[0079] According to one or more embodiments, L_2 may be a ligand represented by Formula 2-2A:

Formula 2-2A



wherein, in Formula 2-2A,

[0080] each of Y_4, ring CY_3, ring CY_{41}, ring CY_{42}, T_{30}, X_4, R_3, R_4, R_{30}, a3, a4, W_4, and b4 is as described herein,

[0081] ring CY_{43a} and ring CY_{43b} may each independently be a C_5-C_{15} carbocyclic group or a C_1-C_{15} heterocyclic group, and

[0082] * and *' each indicate a binding site to M in Formula 1.

[0083] For example, ring CY_{43a} and ring CY_{43b} may each independently be:

[0084] a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, a triazine group, a naphthalene group, a quinoline group, an isoquinoline group, a quinazoline group, a quinoxaline group, a phenanthrene group, a benzoquinoline group, a benzoisoquinoline group, an anthracene group, or a chrysene group; or

[0085] a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, a triazine group, a naphthalene group, a quinoline group, an isoquinoline group, a quinazoline group, a quinoxaline group, a phenanthrene group, a benzoquinoline group, a benzoisoquinoline group, an anthracene group, or a chrysene group, each condensed with a cyclohexane group, a norbornane group, an adamantane group, or a combination thereof.

[0086] X_{11} in Formula 2-1 is C, Si, or Ge.

[0087] T_{30} in Formula 2-2 is a single bond, a C_1-C_{20} alkylene group unsubstituted or substituted with at least one R_{10a}, a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a}.

[0088] For example, T_{30} in Formula 2-2 may be:

[0089] a single bond; or

[0090] a C_1-C_{20} alkylene group, a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a furan group, a thiophene group, a silole group, an indene group, a fluorene group, an indole group, a carbazole group, a benzofuran group, a dibenzofuran group, a benzothiophene group, a dibenzothiophene group, a benzosilole group, a dibenzosilole group, an azafluorene group, an azacarbazole group, an azadibenzofuran group, an azadibenzothiophene group, an azadibenzosilole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isooxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, or a benzothiadiazole group, each unsubstituted or substituted with at least one R_{10a}.

[0091] According to one or more embodiments, T_{30} in Formula 2-2 may be:

[0092] a single bond; or

[0093] a benzene group unsubstituted or substituted with at least one R_{10a}.

[0094] According to one or more embodiments, T_{30} may be:

[0095] a single bond; or

[0096] a C_1-C_{20} alkylene group, a benzene group, a naphthalene group, a dibenzofuran group, or a dibenzothiophene group, each unsubstituted or substituted with at least one of deuterium, —F, a cyano group, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, or a combination thereof.

[0097] According to one or more embodiments, T_{30} may be a benzene group having substituents on 2nd position and 6th position and optionally 4th position. The substituents on the 2th position, 6th position and 4th position may each independently deuterium, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, or a combination thereof. The C_1-C_{20} alkyl group, the deuterated C_1-C_{20} alkyl group, and the fluorinated C_1-C_{20} alkyl group may be linear or branched. For example, the C_1-C_{20} alkyl group, the deuterated C_1-C_{20} alkyl group, and the fluorinated C_1-C_{20} alkyl group may be a C_1-C_{20} linear alkyl group or C_3-C_{20} branched alkyl group, a deuterated C_1-C_{20} linear alkyl group or C_3-C_{20} branched alkyl group, and a fluorinated C_1-C_{20} linear alkyl group or C_3-C_{20} branched alkyl group.

[0098] X_4 in Formula 2-2 is O, S, Se, $N(R_{48})$, $C(R_{48})(R_{49})$, or $Si(R_{48})(R_{49})$. Each of R_{48} and R_{49} is as described herein.

[0099] For example, X_4 may be O or S.

[0100] R_1 to R_4, R_{14} to R_{16}, R_{30}, R_{48}, and R_{49} in Formula 2-1 and Formula 2-2 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted C_1-C_{60} alkoxy group, a substituted or unsubstituted C_1-C_{60} alkylthio group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_1a heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_7-C_{60} alkyl aryl group, a substituted or unsubstituted C_7-C_{60} aryl alkyl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted C_2-C_{60} alkyl heteroaryl group, a substituted or unsubstituted C_2-C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1-C_{60} heteroaryloxy group, a substituted or unsubstituted C_1-C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —$Si(Q_1)(Q_2)(Q_3)$, —$Ge(Q_1)(Q_2)(Q_3)$, —$N(Q_4)(Q_5)$, —$B(Q_6)(Q_7)$, —$P(=O)(Q_8)(Q_9)$, or —$P(Q_8)(Q_9)$. Q_1 to Q_9 are each as described herein.

[0101] According to one or more embodiments, R_1 to R_4, R_{14} to R_{16}, R_{30}, R_{48}, and R_{49} in Formula 2-1 and Formula 2-2 may each independently be:

[0102] hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{20} alkyl group, a C_1-C_{20} alkoxy group, or a C_1-C_{20} alkylthio group;

[0103] a C_1-C_{20} alkyl group, a C_1-C_{20} alkoxy group, or a C_1-C_{20} alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF_5, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, a adamantanyl group, a norbornanyl(bicyclo[2.2.1]heptyl group), a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, a (C_1-C_{20} alkyl)cyclopentyl group, a (C_1-C_{20} alkyl)cyclohexyl group, a (C_1-C_{20} alkyl)cycloheptyl group, a (C_1-C_{20} alkyl)cyclooctyl group, a (C_1-C_{20} alkyl)adamantanyl group, a (C_1-C_{20} alkyl)norbornanyl group, a (C_1-C_{20} alkyl)norbornenyl group, a (C_1-C_{20} alkyl)cyclopentenyl group, a (C_1-C_{20} alkyl)cyclohexenyl group, a (C_1-C_{20} alkyl)cycloheptenyl group, a (C_1-C_{20} alkyl)bicyclo[1.1.1]pentyl group, a (C_1-C_{20} alkyl)bicyclo[2.1.1]hexyl group, a (C_1-C_{20} alkyl)bicyclo[2.2.2]octyl group, a phenyl group, a (C_1-C_{20} alkyl)phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a 1,2,3,4-tetrahydronaphthyl group, a pyridinyl group, a pyrimidinyl group, or a combination thereof;

[0104] a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, cycloheptenyl group, a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, a phenyl group, a (C_1-C_{20} alkyl)phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a 1,2,3,4-tetrahydronaphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, or an azadibenzothiophenyl group, each unsubstituted or substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF_5, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a C_1-C_{20} alkoxy group, a C_1-C_{20} alkylthio group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, a adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, a (C_1-C_{20} alkyl)cyclopentyl group, a (C_1-C_{20} alkyl)cyclohexyl group, a (C_1-C_{20} alkyl)cycloheptyl group, a (C_1-C_{20} alkyl)cyclooctyl group, a (C_1-C_{20} alkyl)adamantanyl group, a (C_1-C_{20} alkyl)norbornanyl group, a (C_1-C_{20} alkyl)norbornenyl group, a (C_1-C_{20} alkyl)cyclopentenyl group, a (C_1-C_{20} alkyl)cyclohexenyl group, a (C_1-C_{20} alkyl)cycloheptenyl group, a (C_1-C_{20} alkyl)bicyclo[1.1.1]pentyl group, a (C_1-C_{20} alkyl)bicyclo[2.1.1]hexyl group, a (C_1-C_{20} alkyl)bicyclo[2.2.2]octyl group, a phenyl group, a (C_1-C_{20} alkyl)phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a 1,2,3,4-tetrahydronaphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazole group, an isothiazole group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzoisothiazolyl group, a benzoxazolyl group, a benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, an azadibenzofuranyl group, an azadibenzothiophenyl group, —$Si(Q_{31})(Q_{32})(Q_{33})$, —$Ge(Q_{31})(Q_{32})(Q_{33})$, or a combination thereof; or

[0105] —$Si(Q_1)(Q_2)(Q_3)$, —$Ge(Q_1)(Q_2)(Q_3)$, —$N(Q_4)(Q_5)$, —$B(Q_6)(Q_7)$, —$P(=O)(Q_8)(Q_9)$, or —$P(Q_8)(Q_9)$,

[0106] wherein Q_1 to Q_9 and Q_{31} to Q_{33} are each independently:

[0107] —CH_3, —CD_3, —CD_2H, —CDH_2, —CH_2CH_3, —CH_2CD_3, —CH_2CD_2H, —CH_2CDH_2, —$CHDCH_3$, —$CHDCD_2H$, —$CHDCDH_2$, —$CHDCD_3$, —CD_2CD_3, —CD_2CD_2H, or —CD_2CDH_2; or

[0108] an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, a phenyl group, a biphenyl group, or a naphthyl group, each unsubstituted or substituted with at least one of deuterium, a C_1-C_{20} alkyl group, a phenyl group, or a combination thereof.

[0109] According to one or more embodiments, R_1 to R_4, R_{30}, R_{48}, and R_{49} may each independently be:

[0110] hydrogen, deuterium, —F, or a cyano group; or a C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, or a dibenzothiophenyl group, each unsubstituted or substituted with at least one of deuterium, —F, a cyano group, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, —$Si(Q_{31})(Q_{32})(Q_{33})$,

[0111] —$Ge(Q_{31})(Q_{32})(Q_{33})$, or a combination thereof.

[0112] According to one or more embodiments, R_{14} to R_{16} may each independently be a C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, or a dibenzothiophenyl group, each unsubstituted or substituted with at least one of deuterium, —F, a cyano group, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, or a combination thereof.

[0113] According to one or more embodiments, R_{14} to R_{16} in Formula 2-1 may each independently be —CH_3, —CH_2CH_3, —CD_3, —CD_2H, —CDH_2, —CH_2CD_3 or -CD_2CH_3.

[0114] According to one or more embodiments, R_{14} to R_{16} in Formula 2-1 may be identical to or different from each other.

[0115] a1 to a4 in Formula 2-1 and Formula 2-2 each represent the number of R_1 to the number of R_4, where a1 is an integer of 0 to 3, a2 to a4 are each independently an integer of 0 to 20 (for example, an integer of 0 to 10, an integer of 0 to 6, an integer of 0 to 9, or an integer of 0 to 4). When a1 is 2 or more, two or more of R_1 may be identical to or different from each other, when a2 is 2 or more, two or more of R_2 may be identical to or different from each other, when a3 is 2 or more, two or more of R_3 may be identical to or different from each other, and when a4 is 2 or more, two or more of R_4 may be identical to or different from each other.

[0116] W_4 in Formula 2-2 is a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted C_1-C_{60} alkoxy group, a substituted or unsubstituted C_1-C_{60} alkylthio group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_1 heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_1 heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_7-C_{60} alkyl aryl group, a substituted or unsubstituted C_7-C_{60} aryl alkyl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted C_2-C_{60} alkyl heteroaryl group, a substituted or unsubstituted C_2-C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1-C_{60} heteroaryloxy group, a substituted or unsubstituted C_1-C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

[0117] According to one or more embodiments, W_4 may be —CH_3 (a methyl group).

[0118] According to one or more embodiments, W_4 may be —CD_3 (a fully deuterated methyl group).

[0119] According to one or more embodiments, W_4 may not be —CH_3 (a methyl group).

[0120] According to one or more embodiments, W_4 may not be —CD_3 (a fully deuterated methyl group).

[0121] According to one or more embodiments, the number of carbon atoms included in W_4 may be 2 or greater.

[0122] According to one or more embodiments, W_4 may be —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a substituted or unsubstituted C_2-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted

C_1-C_{60} alkoxy group, a substituted or unsubstituted C_1-C_{60} alkylthio group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_7-C_{60} alkyl aryl group, a substituted or unsubstituted C_7-C_{60} aryl alkyl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted C_2-C_{60} alkyl heteroaryl group, a substituted or unsubstituted C_2-C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1-C_{60} heteroaryloxy group, a substituted or unsubstituted C_1-C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

[0123] According to one or more embodiments, W_4 may be a C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, or a dibenzothiophenyl group, each unsubstituted or substituted with at least one of deuterium, —F, a cyano group, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, —Si(Q_{31})(Q_{32})(Q_{33}), —Ge(Q_{31})(Q_{32})(Q_{33}), or a combination thereof.

[0124] According to one or more embodiments, W_4 may be a C_2-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, or a dibenzothiophenyl group, each unsubstituted or substituted with at least one of deuterium, —F, a cyano group, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, —Si(Q_{31})(Q_{32})(Q_{33}), —Ge(Q_{31})(Q_{32})(Q_{33}), or a combination thereof.

[0125] b4 in Formula 2-2 represents the number of W_4 and is an integer from 1 to 10. When b4 is 2 or more, two or more of W_4 may be identical to or different from each other.

[0126] In one or more embodiments, b4 may be 1, 2, or 3.

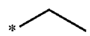
[0127] According to one or more embodiments, b4 may be 1 or 2.

[0128] According to one or more embodiments, R_1 to R_4, R_{14} to R_{16}, R_{30}, R_{48}, and R_{49} in Formula 2-1 and Formula 2-2 may each independently be hydrogen, deuterium, —F, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, —SF_5, —CH_3, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, —OCH_3, —$OCDH_2$, —OCD_2H, —OCD_3, —SCH_3, —$SCDH_2$, —SCD_2H, —SCD_3, a group represented by one of Formulae 9-1 to 9-39, a group in which at least one hydrogen of one of Formulae 9-1 to 9-39 is substituted with deuterium, a group in which at least one hydrogen of one of Formulae 9-1 to 9-39 is substituted with —F, a group represented by one of Formulae 9-201 to 9-230, a group in which at least one hydrogen of one of Formulae 9-201 to 9-230 is substituted with deuterium, a group in which at least one hydrogen of one of Formulae 9-201 to 9-230 is substituted with —F, a group represented by one of Formulae 10-1 to 10-145, a group in which at least one hydrogen of one of Formulae 10-1 to 10-145 is substituted with deuterium, a group in which at least one hydrogen of one of Formulae 10-1 to 10-145 is substituted with —F, a group represented by one of Formulae 10-201 to 10-354, a group in which at least one hydrogen of one of Formulae 10-201 to 10-354 is substituted with deuterium, and a group in which at least one hydrogen of one of Formulae 10-201 to 10-354 is substituted with —F, —Si(Q_1)(Q_2)(Q_3), or —Ge(Q_1)(Q_2)(Q_3), wherein each of Q_1 to Q_3 is as described herein.

[0129] According to one or more embodiments, at least one of a1 R_1 of Formula 2-1 (for example, R_{11} in Formula CY1-1) may be a group represented by one of Formulae 9-1 to 9-39, a group in which at least one hydrogen of one of Formulae 9-1 to 9-39 is substituted with deuterium, a group in which at least one hydrogen of one of Formulae 9-1 to 9-39 is substituted with —F, a group represented by one of Formulae 9-201 to 9-230, a group in which at least one hydrogen of one of Formulae 9-201 to 9-230 is substituted with deuterium, a group in which at least one hydrogen of one of Formulae 9-201 to 9-230 is substituted with —F, a group represented by one of Formulae 10-1 to 10-145, a group in which at least one hydrogen of one of Formulae 10-1 to 10-145 is substituted with deuterium, a group in which at least one hydrogen of one of Formulae 10-1 to 10-145 is substituted with —F, a group represented by one of Formulae 10-201 to 10-354, a group in which at least one hydrogen of one of Formulae 10-201 to 10-354 is substituted with deuterium, or a group in which at least one hydrogen of one of Formulae 201 to 10-354 is substituted with —F.

[0130] According to one or more embodiments, R_{30} in Formula 2-2 may be a group represented by one of Formulae 10-12 to 10-145, a group in which at least one hydrogen of one of Formulae 10-12 to 10-145 is substituted with deuterium, a group in which at least one hydrogen of one of Formulae 10-12 to 10-145 is substituted with —F, a group represented by one of Formulae 10-201 to 10-354, a group in which at least one hydrogen of one of Formulae 10-201 to 10-354 is substituted with deuterium, or a group in which at least one hydrogen of one of Formulae 10-201 to 10-354 is substituted with —F.

[0131] According to one or more embodiments, W_4 in Formula 2-2 may be —CH_3, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a group represented by one of Formulae 9-1 to 9-39, a group in which at least one hydrogen of one of Formulae 9-1 to 9-39 is substituted with deuterium, a group in which at least one hydrogen of one of Formulae 9-1 to 9-39 is substituted with —F, a group represented by one of Formulae 9-201 to 9-230, a group in which at least one hydrogen of one of Formulae 9-201 to 9-230 is substituted with deuterium, a group in which at least one hydrogen of one of Formulae 9-201 to 9-230 is substituted with —F, a group represented by one of Formulae 10-1 to 10-145, a group in which at least one hydrogen of one of Formulae 10-1 to 10-145 is substituted with deuterium, a group in which at least one hydrogen of one of Formulae 10-1 to 10-145 is substituted with —F, a group represented by one of Formulae 10-201 to 10-354, a group in which at least one hydrogen of one of Formulae 10-201 to 10-354 is substituted with deuterium, or a group in which at least one hydrogen of one of Formulae 201 to 10-354 is substituted with —F: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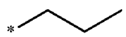
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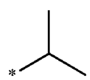
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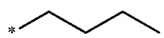
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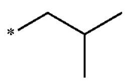
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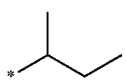
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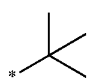
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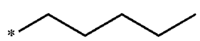
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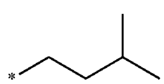
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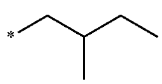
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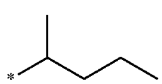
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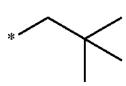
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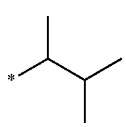
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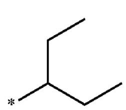
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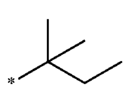
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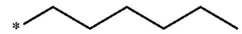
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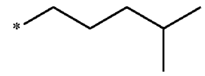
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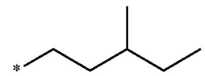
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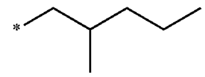
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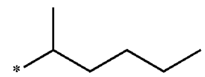
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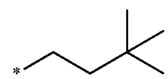
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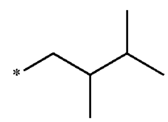
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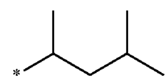
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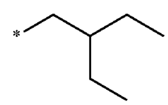
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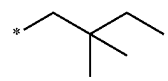
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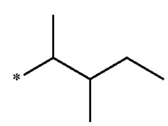
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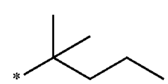
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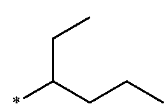
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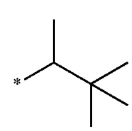
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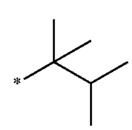
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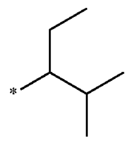

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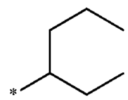

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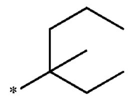

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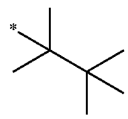

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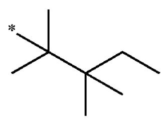

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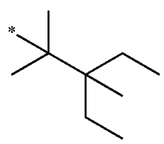

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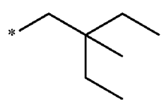

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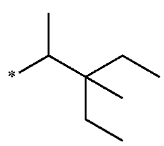

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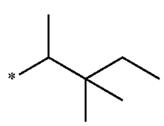

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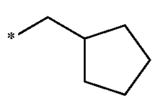

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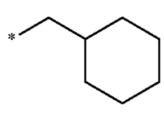

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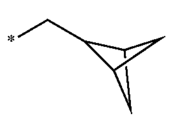

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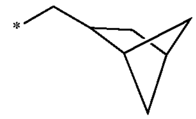

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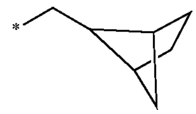

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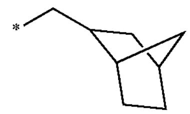

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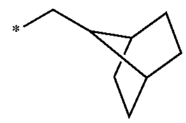

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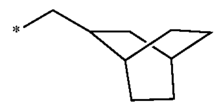

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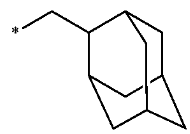

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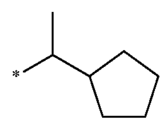

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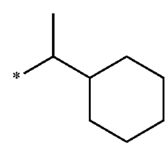

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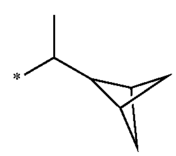

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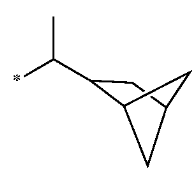

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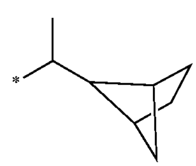

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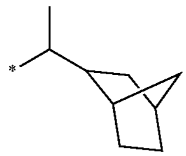

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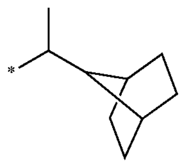

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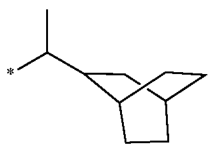

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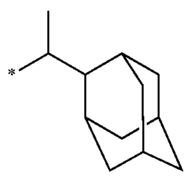

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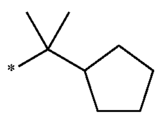

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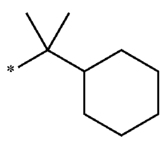

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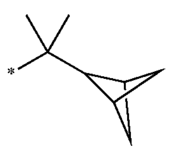

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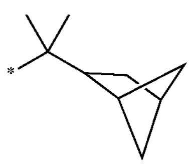

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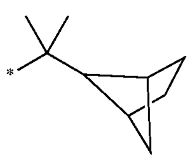

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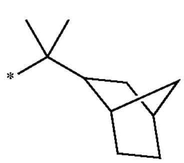

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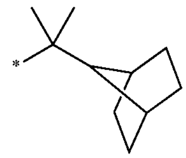

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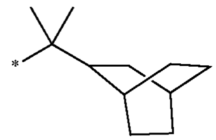

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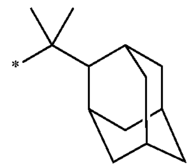

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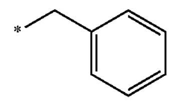

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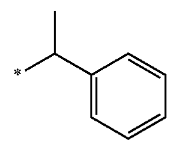

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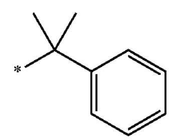

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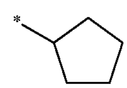

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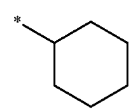

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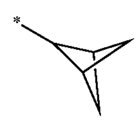

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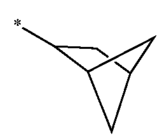

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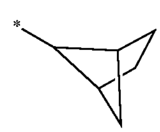

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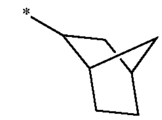

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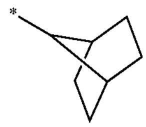

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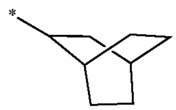

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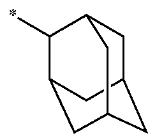

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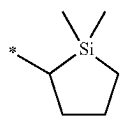

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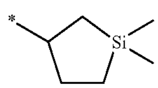

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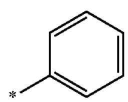

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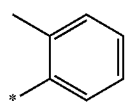

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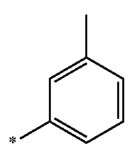

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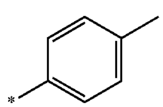

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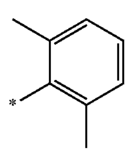

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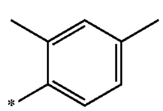

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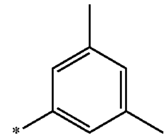

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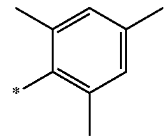

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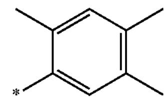

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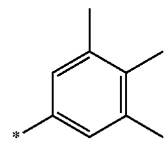

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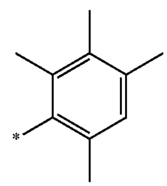

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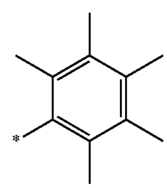

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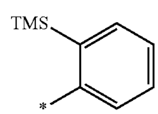

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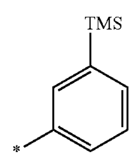

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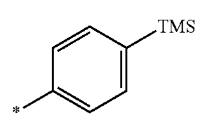

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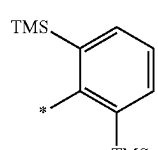

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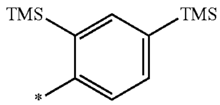

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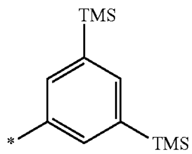

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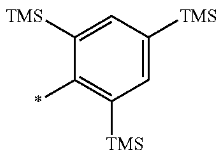

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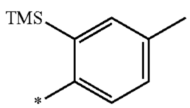

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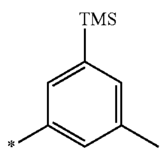

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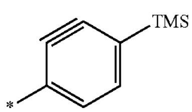

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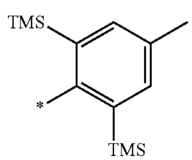

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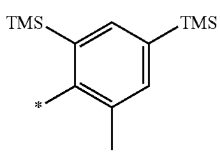

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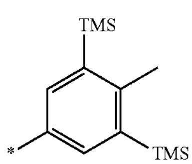

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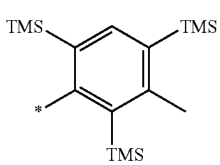

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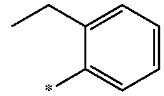

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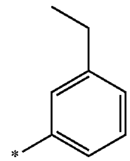

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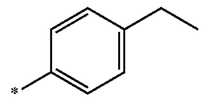

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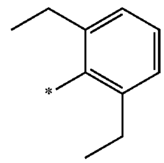

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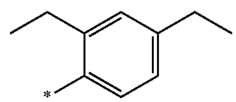

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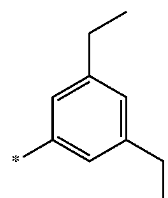

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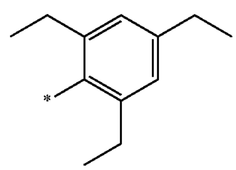

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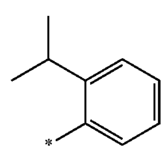

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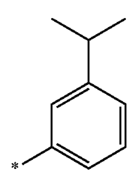

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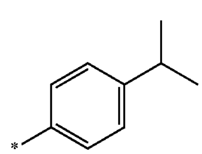

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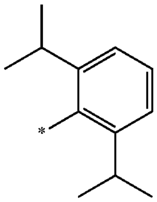

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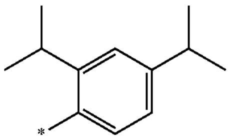

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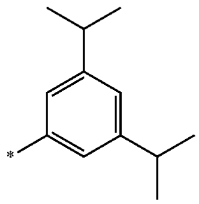

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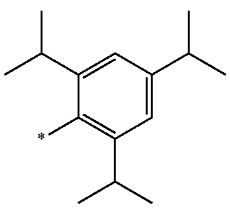

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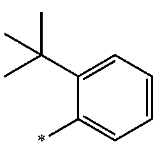

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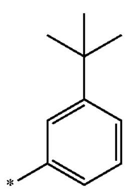

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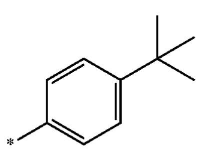

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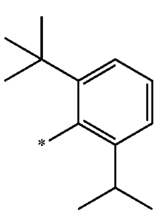

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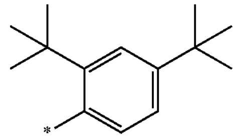

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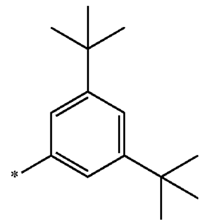

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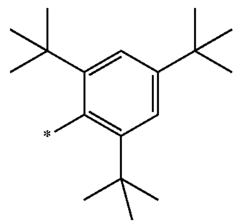

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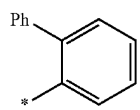

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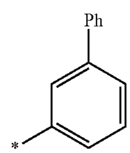

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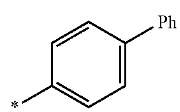

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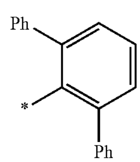

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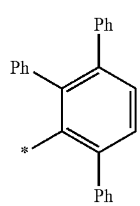

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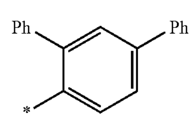

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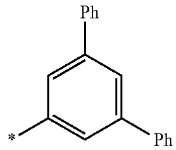

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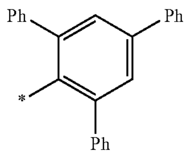

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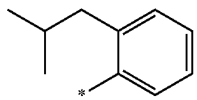

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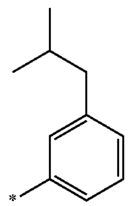

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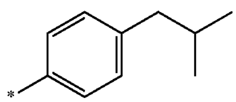

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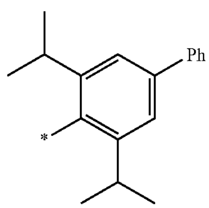

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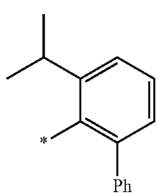

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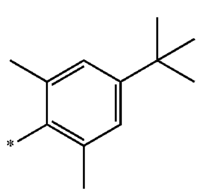

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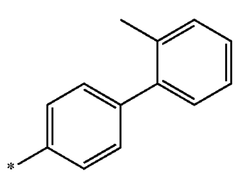

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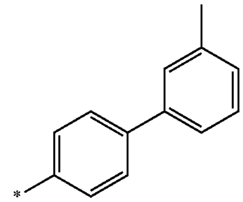

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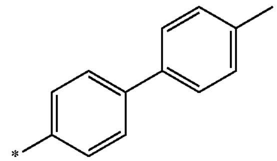

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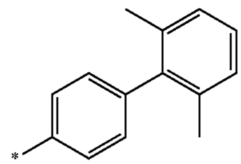

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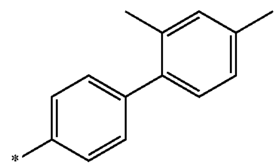

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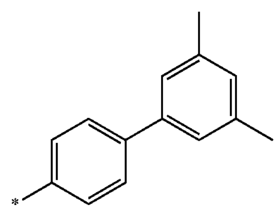

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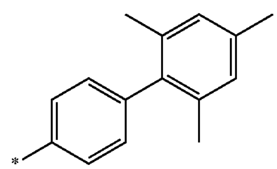

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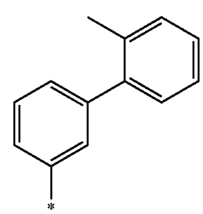

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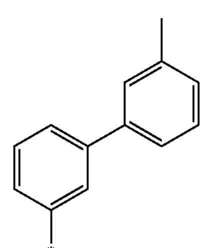

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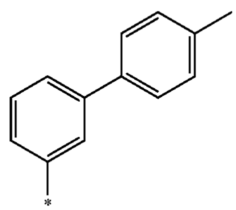

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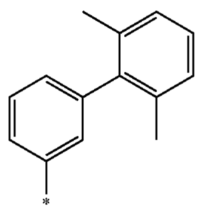

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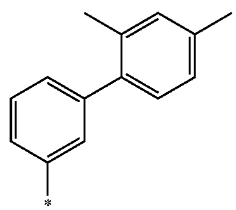

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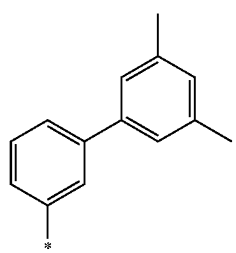

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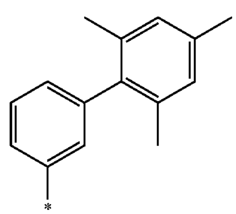

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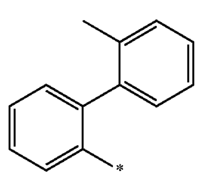

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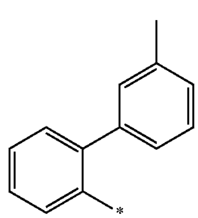

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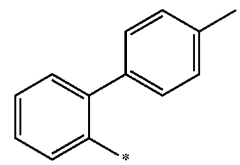

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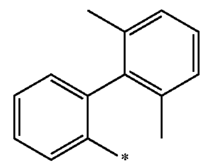

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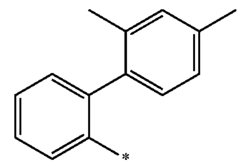

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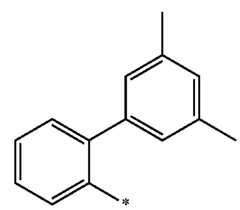

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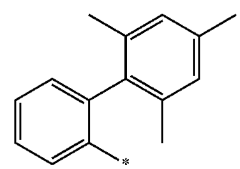

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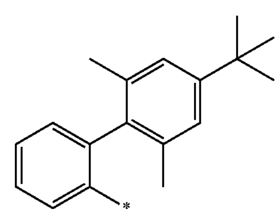

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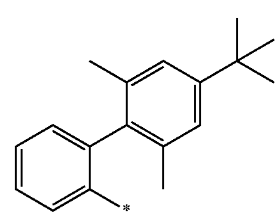

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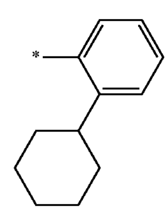

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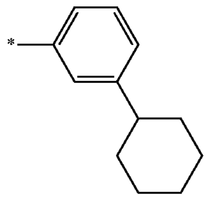

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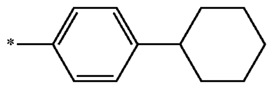

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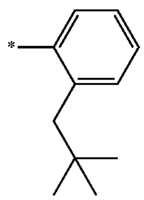

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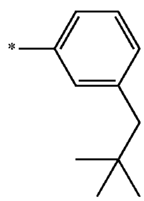

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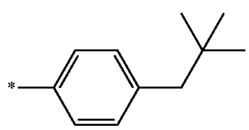

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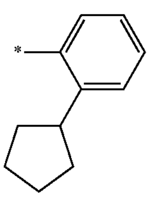

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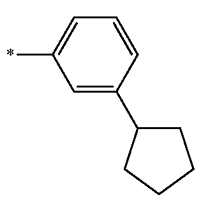

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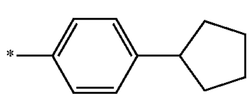

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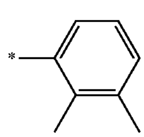

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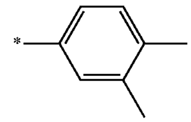

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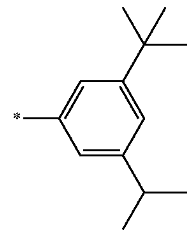

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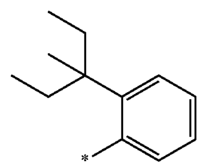

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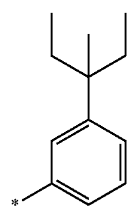

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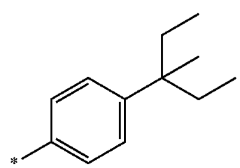

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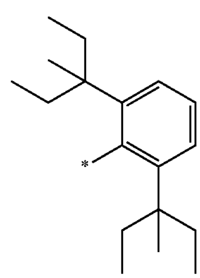

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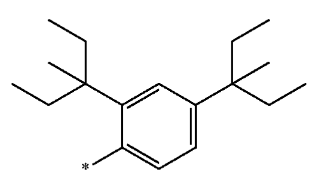

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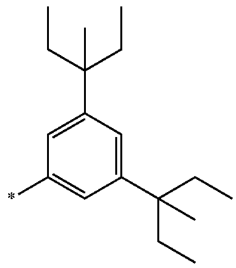

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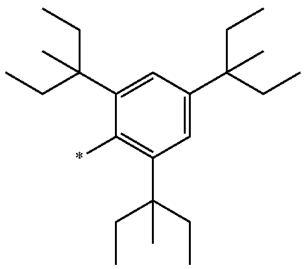

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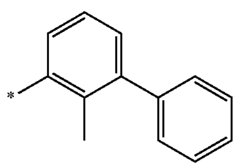

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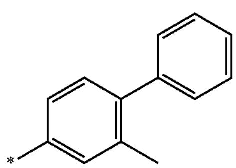

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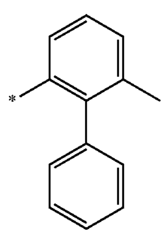

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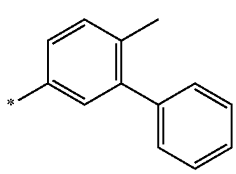

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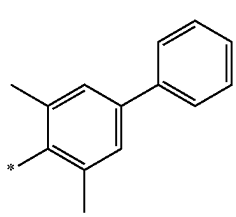

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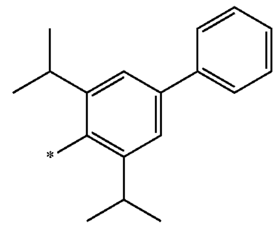

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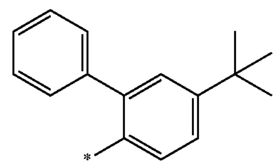

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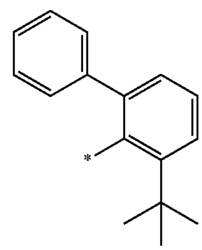

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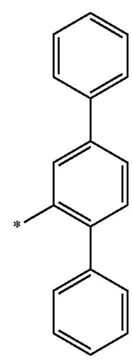

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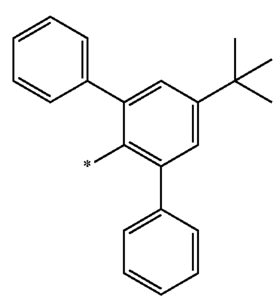

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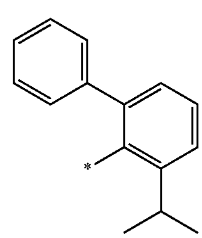

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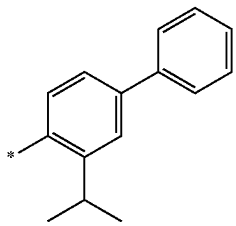

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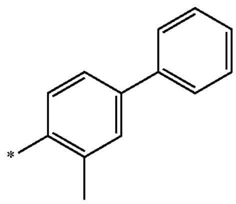

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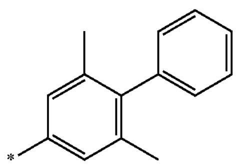

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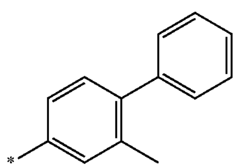

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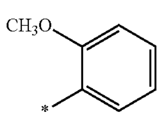

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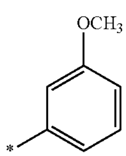

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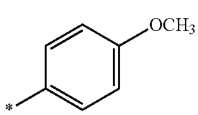

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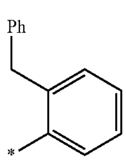

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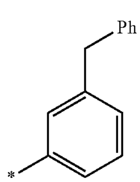

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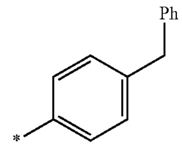

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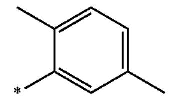

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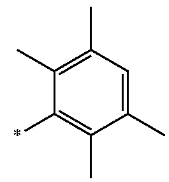

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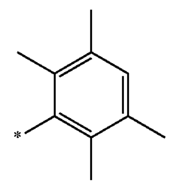

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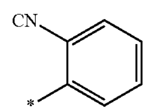

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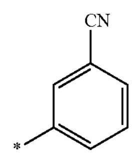

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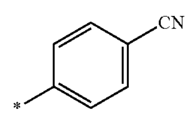

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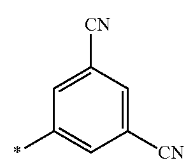

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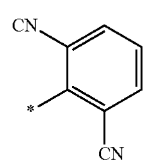

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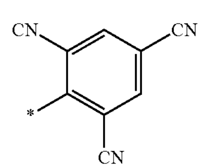

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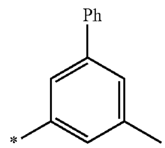

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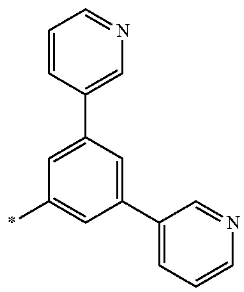

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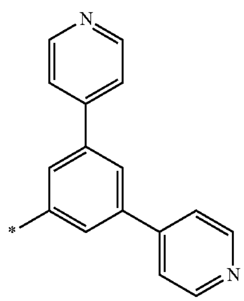

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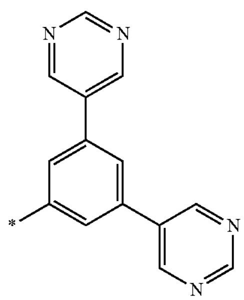

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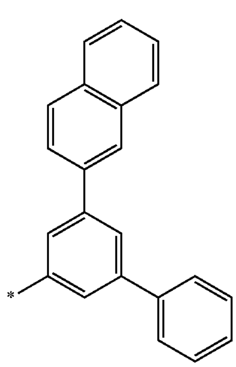

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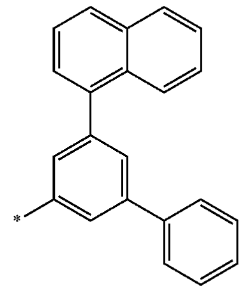

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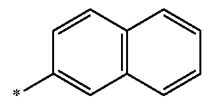

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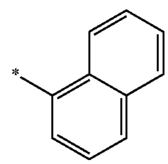

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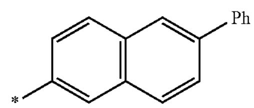

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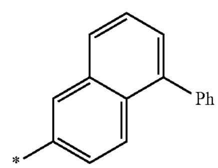

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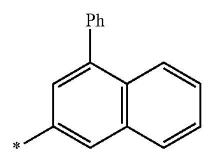

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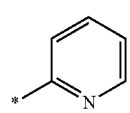

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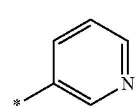

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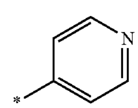

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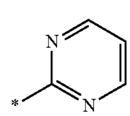

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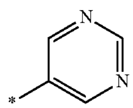

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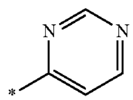

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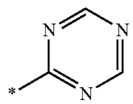

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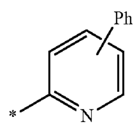

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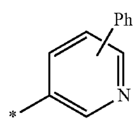

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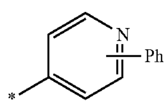

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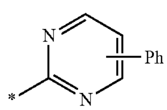

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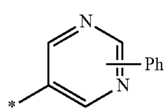

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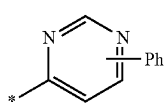

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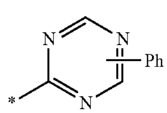

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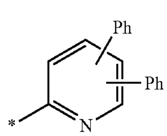

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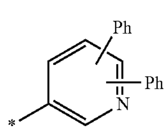

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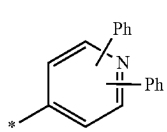

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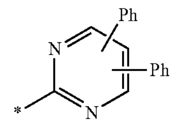

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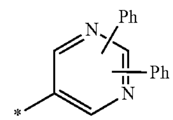

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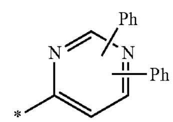

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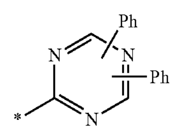

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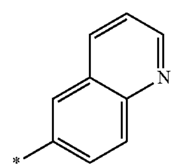

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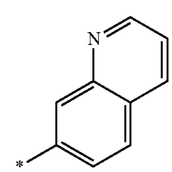

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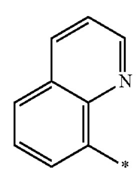

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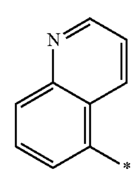

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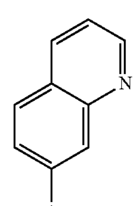

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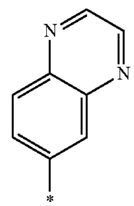

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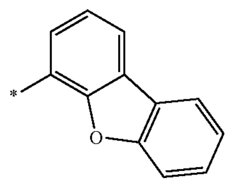

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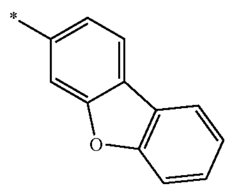

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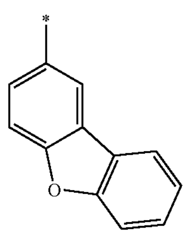

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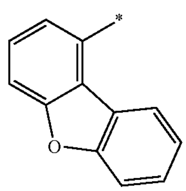

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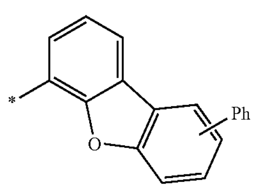

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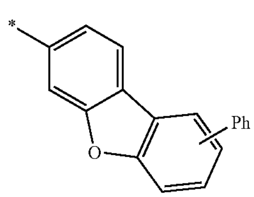

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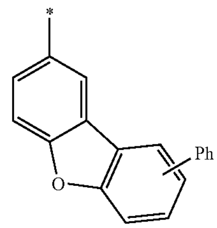

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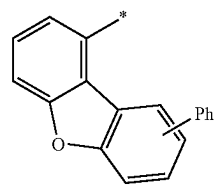

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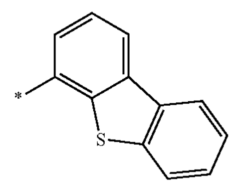

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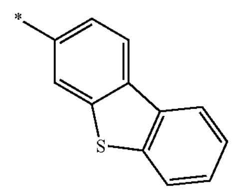

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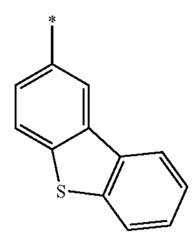

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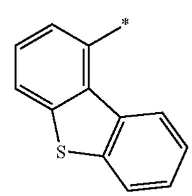

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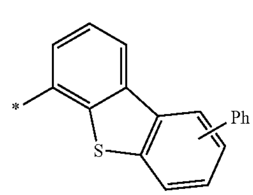

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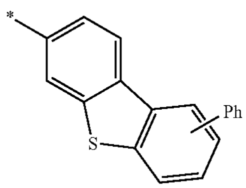

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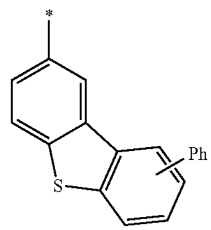

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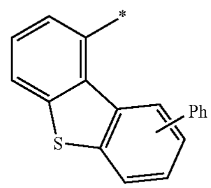

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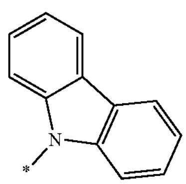

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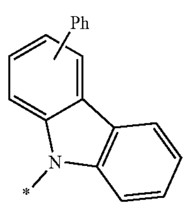

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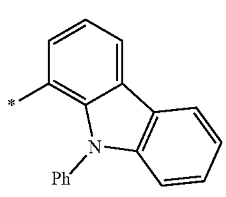

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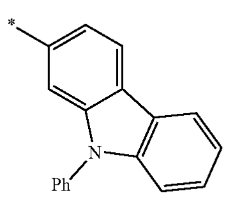

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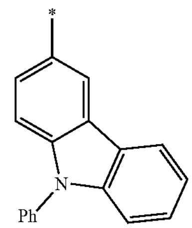

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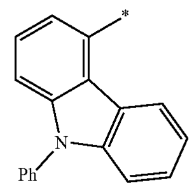

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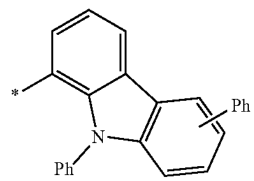

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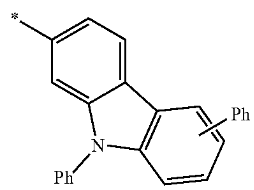

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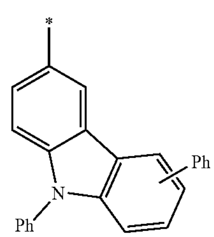

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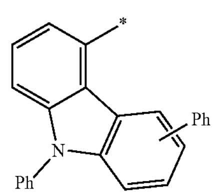

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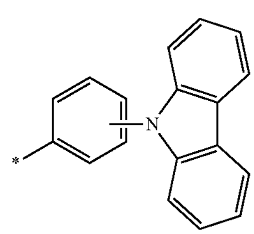

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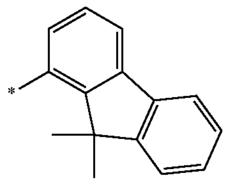

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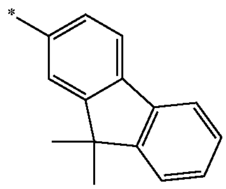

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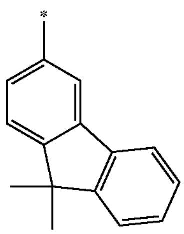

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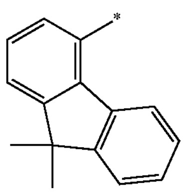

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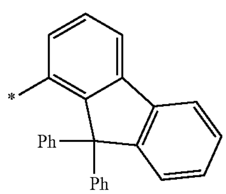

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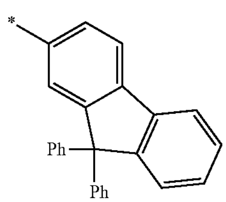

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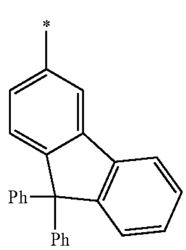

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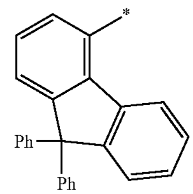

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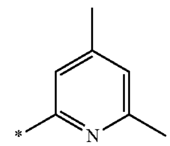

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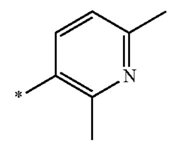

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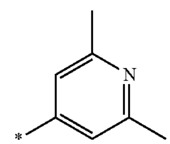

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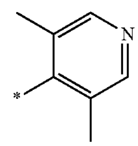

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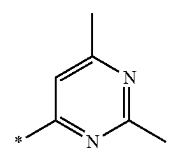

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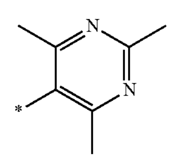

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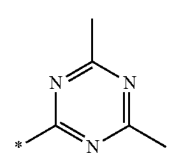

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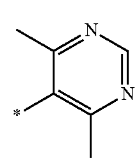

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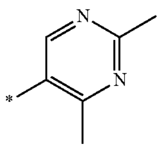

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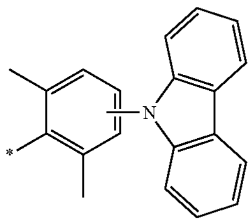

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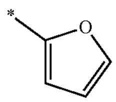

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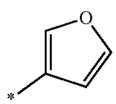

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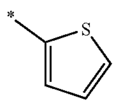

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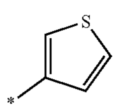

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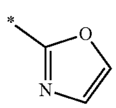

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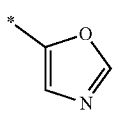

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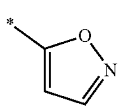

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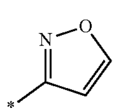

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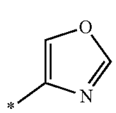

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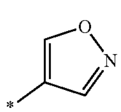

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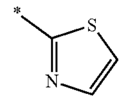

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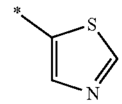

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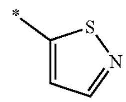

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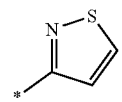

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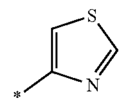

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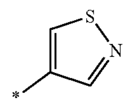

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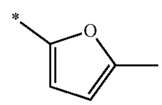

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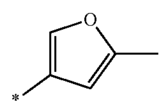

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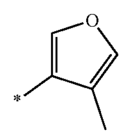

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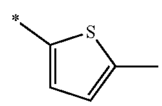

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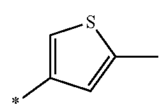

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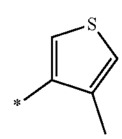

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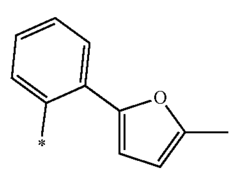

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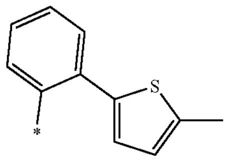

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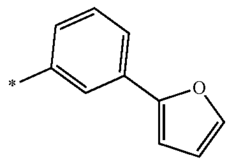

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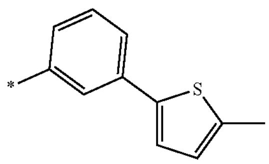

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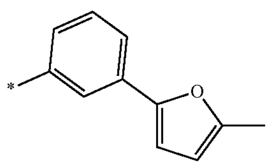

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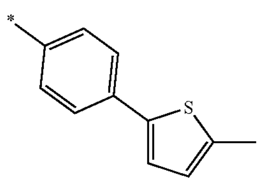

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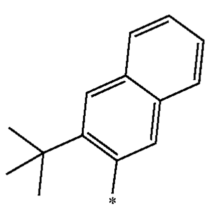

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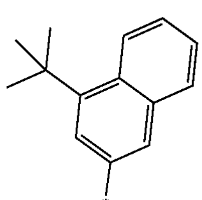

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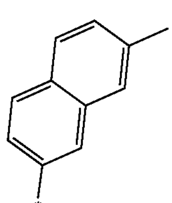

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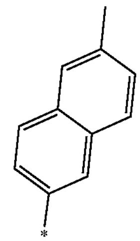

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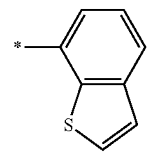

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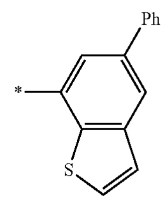

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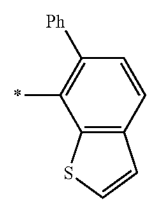

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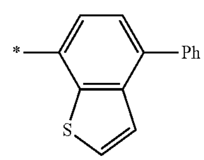

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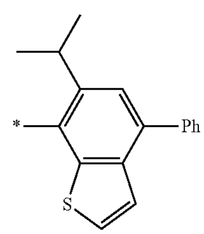

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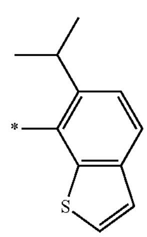

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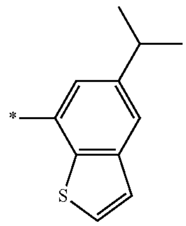

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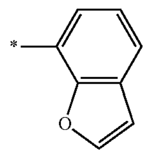

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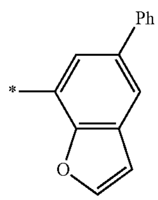

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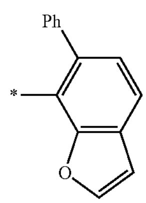

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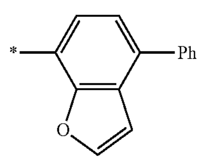

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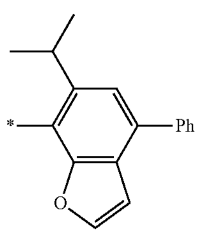

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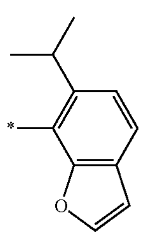

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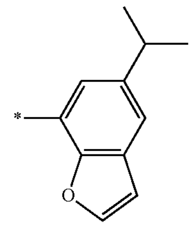

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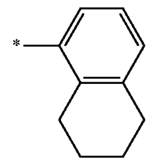

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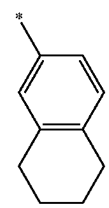

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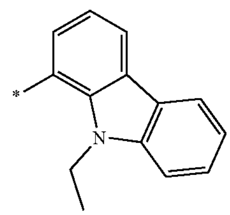

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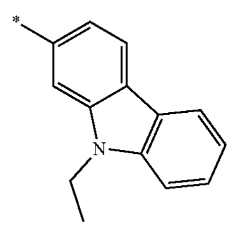

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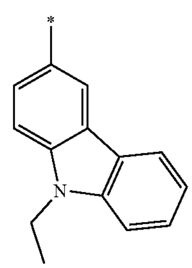

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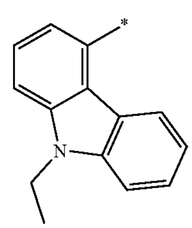

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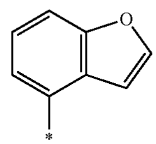

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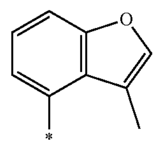

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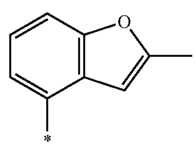

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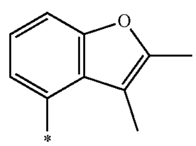

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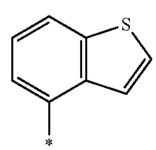

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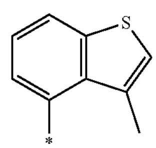

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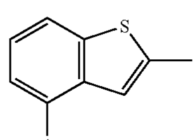

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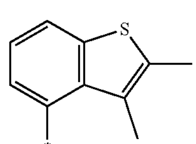

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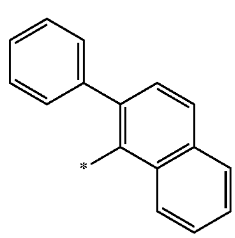

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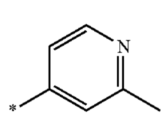

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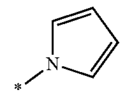

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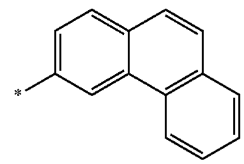

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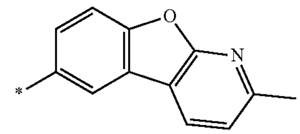

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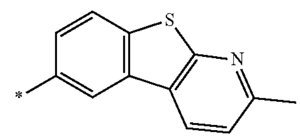

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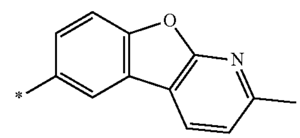

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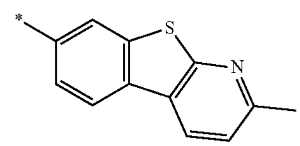

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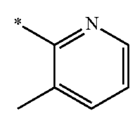

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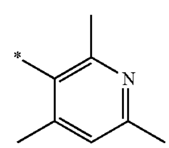

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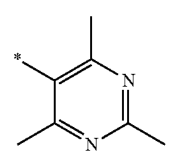

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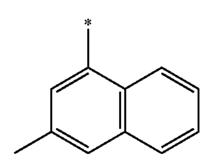
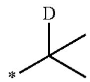
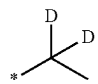


[0132] In Formulae 9-1 to 9-39, 9-201 to 9-230, 10-1 to 10-145 and 10-201 to 10-354, indicates a binding site to a neighboring atom, "Ph" is a phenyl group, "TMS" is a trimethylsilyl group, and "TMG" is a trimethylgermyl group.

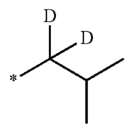
[0133] The "group in which at least one hydrogen of one of Formulae 9-1 to 9-39 is substituted with deuterium" and the "group in which at least one hydrogen of one of Formulae 9-201 to 9-230 is substituted with deuterium" may be, for example, a group represented by one of Formulae 9-501 to 9-514 or 9-601 to 9-637:



9-501



9-502



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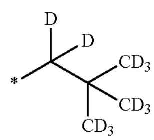

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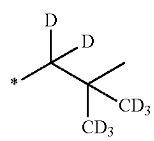

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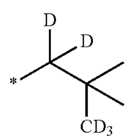

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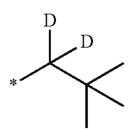

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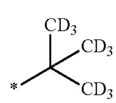

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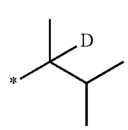

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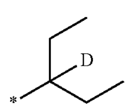

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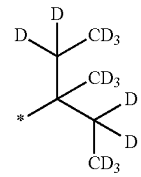

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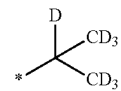

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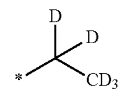

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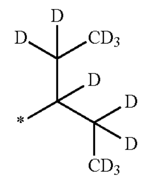

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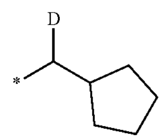

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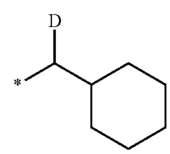

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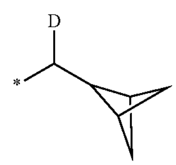

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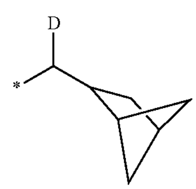

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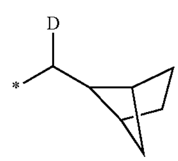

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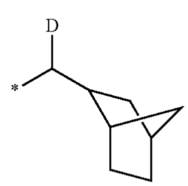

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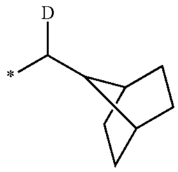
9-607


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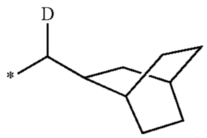

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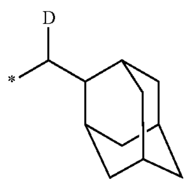

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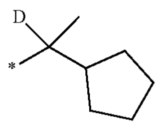

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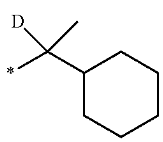

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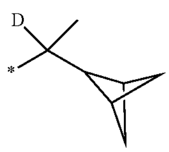

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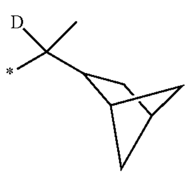

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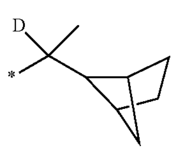

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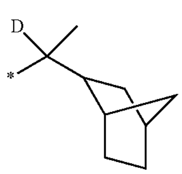

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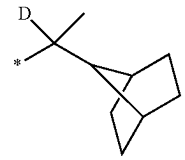

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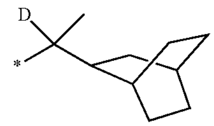

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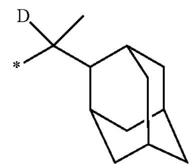

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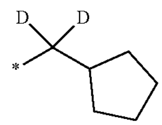

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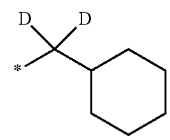
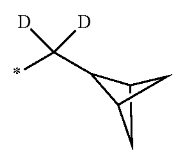
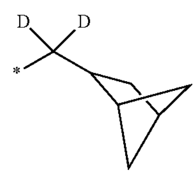


9-621


9-622


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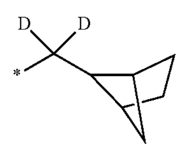

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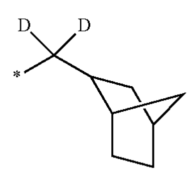

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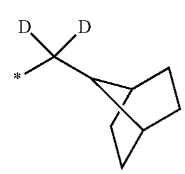

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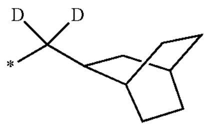
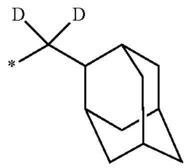
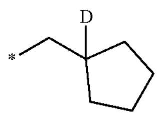
9-626



9-627



9-628



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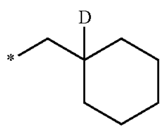
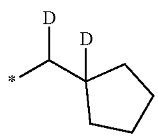
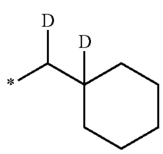


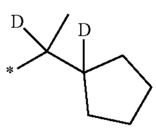
9-630



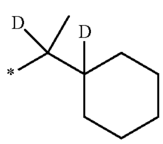
9-631



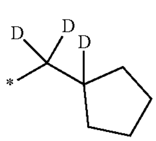
9-632



9-633



9-634



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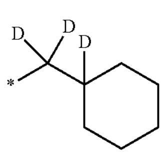

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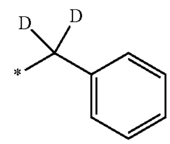
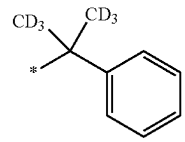
9-636

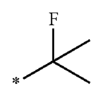


9-637

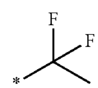


[0134] The "group in which at least one hydrogen of one of Formulae 9-1 to 9-39 is substituted with —F" and the "group in which at least one hydrogen of one of Formulae 9-201 to 9-230 is substituted with —F" may be, for example, a group represented by one of Formulae 9-701 to 9-710:

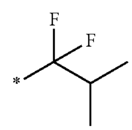
9-701



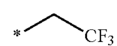
9-702



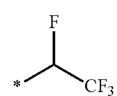
9-703



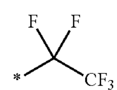
9-704



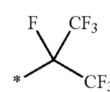
9-705



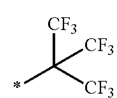
9-706



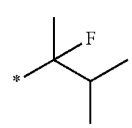
9-707



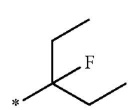
9-708



9-709

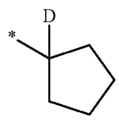


9-710

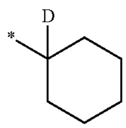


[0135] The "group in which at least one hydrogen of one of Formulae 10-1 to 10-145 is substituted with deuterium" and the "group in which at least one hydrogen of one of Formulae 10-201 to 10-354 is substituted with deuterium" may be, for example, a group represented by one of Formulae 10-501 to 10-553:

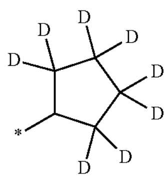
10-501



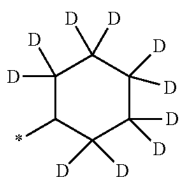
10-502



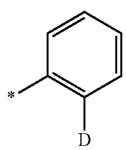
10-503



10-504



10-505



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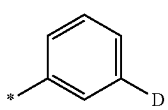

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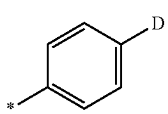
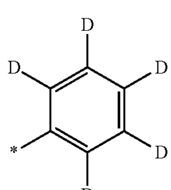
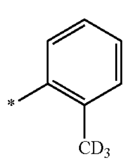


10-508



10-509



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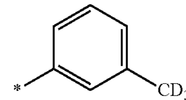
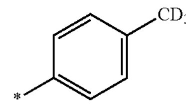
10-510



10-511



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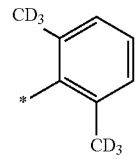

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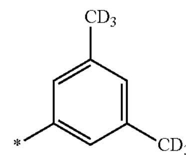

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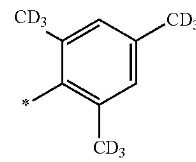

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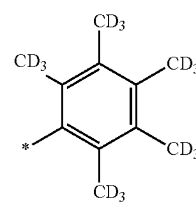

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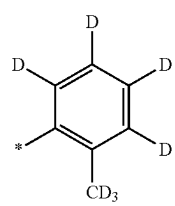
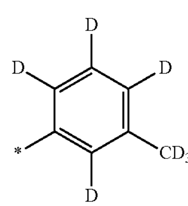
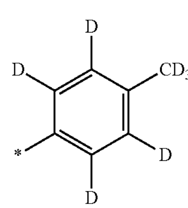


10-517



10-518



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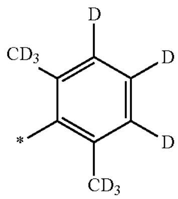
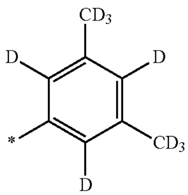
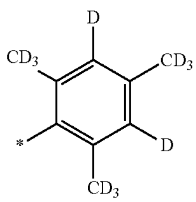
10-519



10-520



10-521



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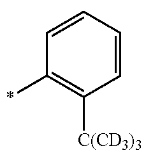

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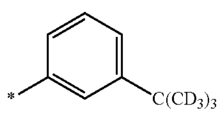
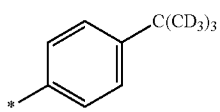
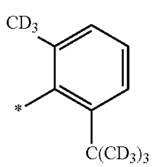


10-524



10-525



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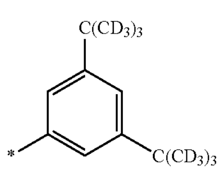

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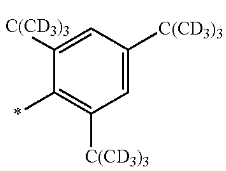

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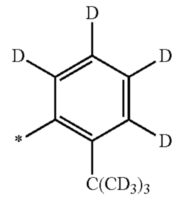
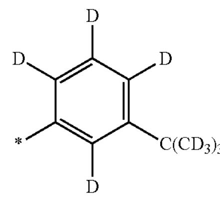
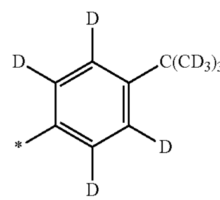
10-528



10-529



10-530



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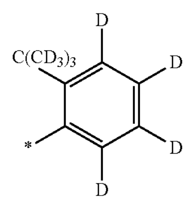

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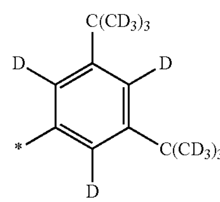
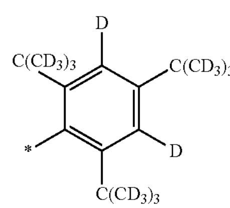


10-533



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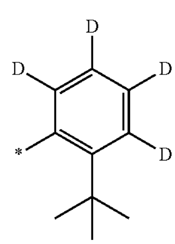

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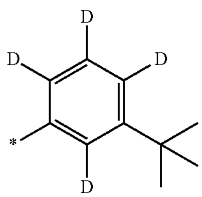
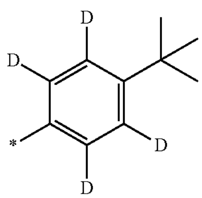
10-535



10-536



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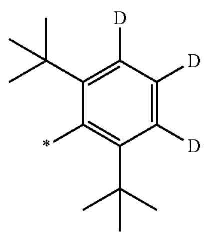
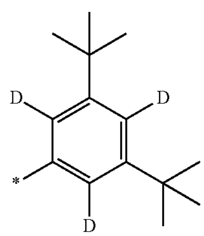


10-538



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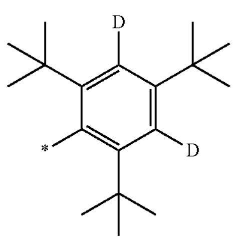
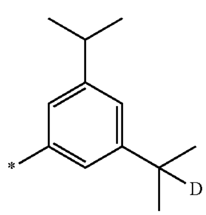
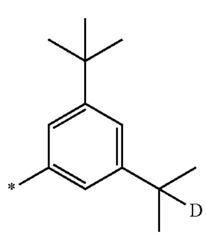


10-541



10-542



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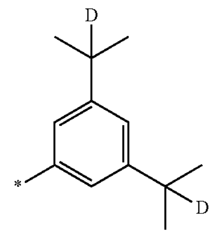
10-543



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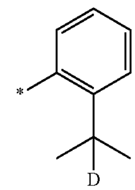
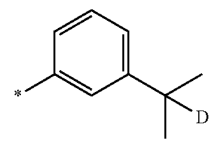
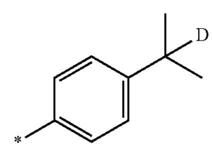


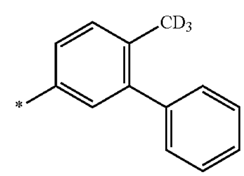
10-545



10-546



10-547



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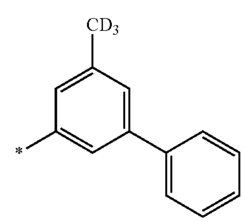

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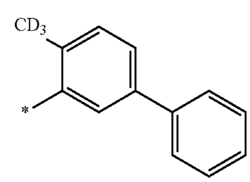

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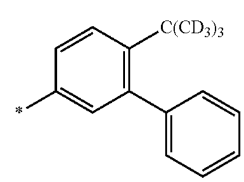

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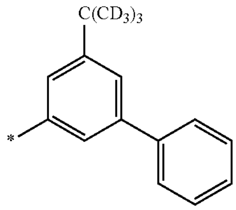

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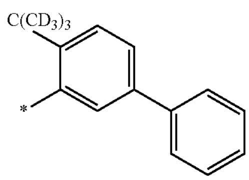

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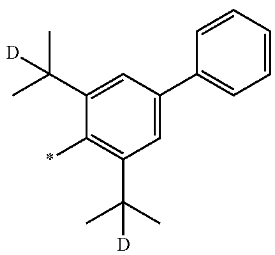
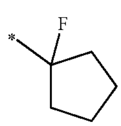


[0136] The "group in which at least one hydrogen of one of Formulae 10-1 to 10-145 is substituted with —F" and the "group in which at least one hydrogen of one of Formulae 10-201 to 10-354 is substituted with —F" may be, for example, a group represented by one of Formulae 10-601 to 10-636:

10-601



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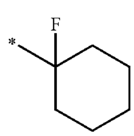

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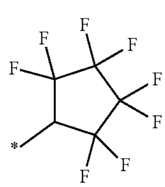

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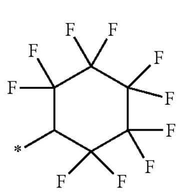

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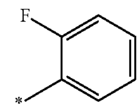

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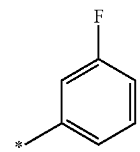

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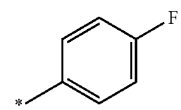

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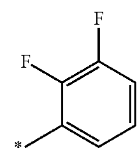

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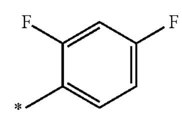

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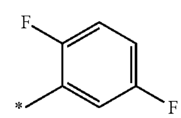

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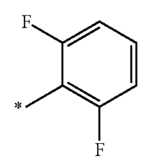

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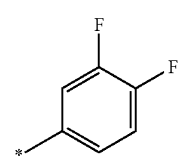

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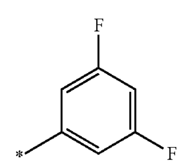

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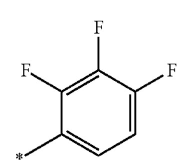

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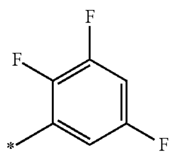

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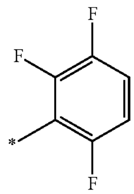

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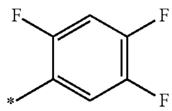

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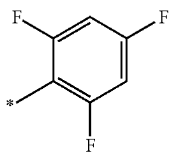

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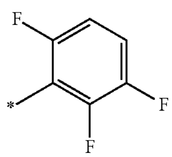

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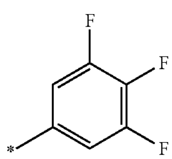

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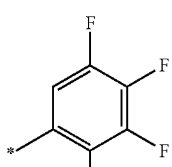

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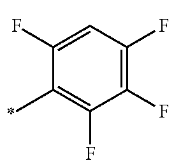

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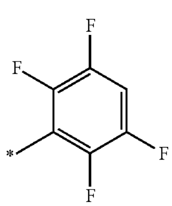

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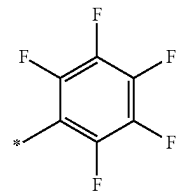

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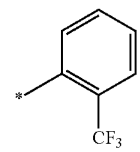

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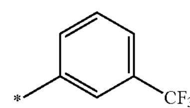

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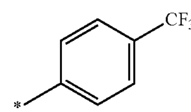

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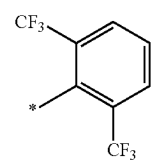

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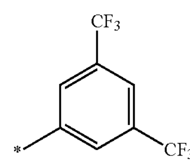

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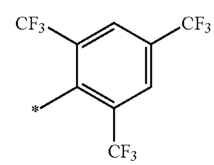

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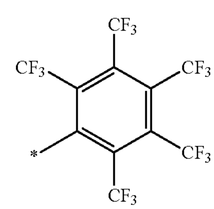

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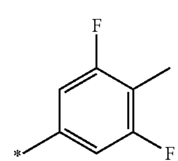

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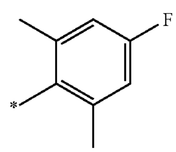

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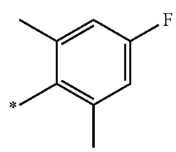

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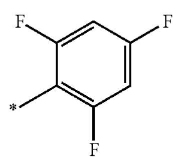

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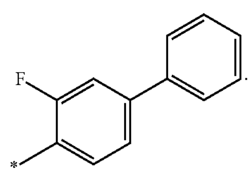


10-636

[0137] According to one or more embodiments, the organometallic compound represented by Formula 1 may include deuterium, a fluoro group (—F), or a combination thereof.

[0138] In one or more embodiments, the organometallic compound represented by Formula 1 may satisfy at least one of Condition 1 to Condition 12:

[0139] Condition 1

[0140] at least one R_1 is not hydrogen, and

[0141] R_1 includes at least one deuterium;

[0142] Condition 2

[0143] at least one R_2 is not hydrogen, and

[0144] R_2 includes at least one deuterium;

[0145] Condition 3

[0146] at least one R_3 is not hydrogen, and

[0147] R_3 includes at least one deuterium;

[0148] Condition 4

[0149] a group represented by $*\text{-}T_{30}\text{-}R_{30}$ includes at least one deuterium;

[0150] Condition 5

[0151] at least one R_4 is not hydrogen, and

[0152] R_4 includes at least one deuterium;

[0153] Condition 6

[0154] at least one W_4 includes deuterium;

[0155] Condition 7

[0156] at least one R_1 is not hydrogen, and

[0157] R_1 includes at least one fluoro group;

[0158] Condition 8

[0159] at least one R_2 is not hydrogen, and

[0160] R_2 includes at least one fluoro group;

[0161] Condition 9

[0162] at least one R_3 is not hydrogen, and

[0163] R_3 includes at least one fluoro group;

[0164] Condition 10

[0165] a group represented by $*\text{-}T_{30}\text{-}R_{30}$ includes at least one fluoro group;

[0166] Condition 11

[0167] at least one R_4 is not hydrogen, and

[0168] R_4 includes at least one fluoro group;

[0169] Condition 12

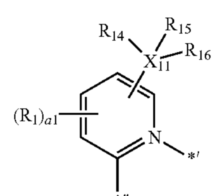
[0170] at least one W_4 includes a fluoro group.

[0171] In Formula 2-1 and Formula 2-2, i) two or more of a plurality of R_1 are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a}, ii) two or more of a plurality of R_2 are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a}, iii) two or more of a plurality of R_3 are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a}, iv) two or more of a plurality of R_4 are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a}, and v) two or more of R_1 to R_4 or R_{30} are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a}.

[0172] R_{10a} is as described in connection with R_2 herein. For example, R_{10a} may be as described in connection with R_2 herein, and may not be hydrogen.

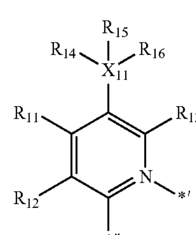
[0173] * and *' in Formula 2-1 and Formula 2-2 each indicates a binding site to M in Formula 1.

[0174] According to one or more embodiments, a group represented by:



in Formula 2-1 may be a group represented by one of Formulae CY1-1 to CY1-3:

CY1-1



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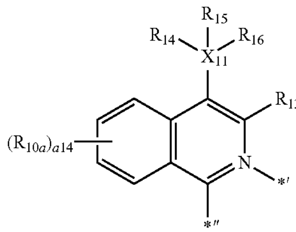
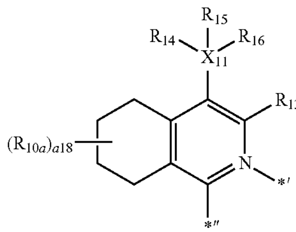
CY1-2



CY1-3



wherein, in Formulae CY1-1 to CY1-3,

[0175] each of X_{11}, R_{14} to R_{16} and R_{10a} is as described herein,

[0176] R_{11} to R_{13} may each be the same as described in connection with R_1,

[0177] a14 may be an integer from 0 to 4,

[0178] a18 may be an integer from 0 to 8,

[0179] *' indicates a binding site to M in Formula 1, and

[0180] *" indicates a binding site to a neighboring atom in Formula 2-1.

[0181] For example, R_{11} in in Formula CY1-1 may not be hydrogen.

[0182] According to one or more embodiments, R_{11} in Formula CY1-1 may not be hydrogen or a methyl group.

[0183] According to one or more embodiments, R_{11} in Formula CY1-1 may be hydrogen or a methyl group.

[0184] According to one or more embodiments, R_{11} in Formula CY1-1 may not be hydrogen, a methyl group, or a cyano group.

[0185] According to one or more embodiments, R_{11} in Formula CY1-1 may not be hydrogen, and R_{12} and R_{13} may each be hydrogen.

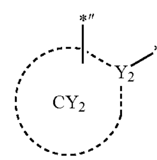
[0186] According to one or more embodiments, R_{11} in Formula CY1-1 may have 2 or more carbon atoms, 3 or more carbon atoms, or 4 or more carbon atoms.

[0187] According to one or more embodiments, R_{11} in Formula CY1-1 may be:

[0188] a methyl group, substituted with at least one of deuterium, —F, a cyano group, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, or a combination thereof; or

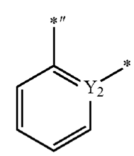
[0189] a C_2-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, or a dibenzothiophenyl group, each unsubstituted or substituted with at least one of deuterium, —F, a cyano group, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, or a combination thereof.

[0190] According to one or more embodiments, a group represented by:

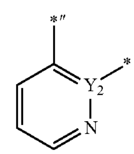


in Formula 2-1 may be a group represented by one of Formulae CY2-1 to CY2-33:

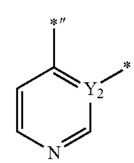
CY2-1



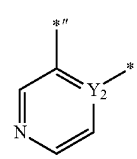
CY2-2



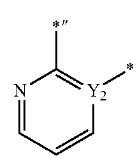
CY2-3



CY2-4



CY2-5



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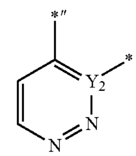
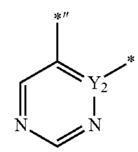
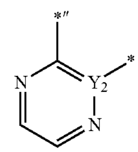
CY2-6



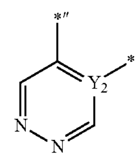
CY2-7



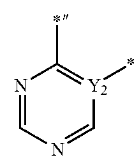
CY2-8



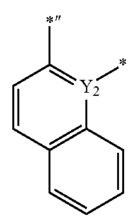
CY2-9



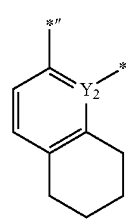
CY2-10



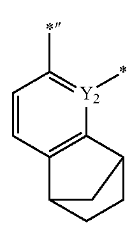
CY2-11



CY2-12



CY2-13



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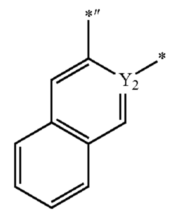
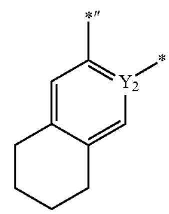
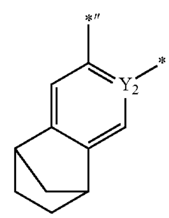
CY2-14



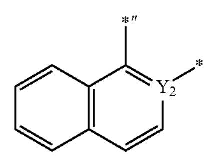
CY2-15



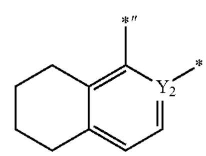
CY2-16



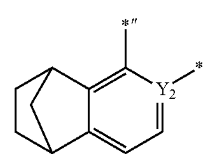
CY2-17



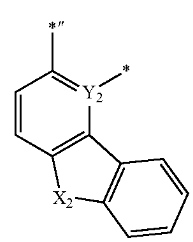
CY2-18



CY2-19



CY2-20



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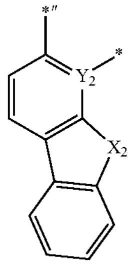
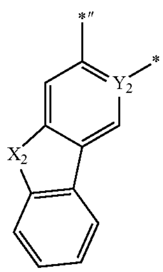
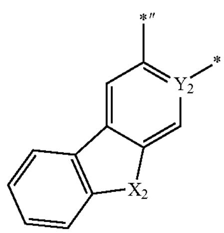
CY2-21



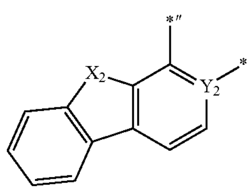
CY2-22



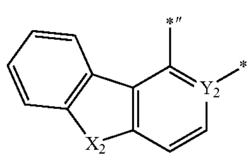
CY2-23



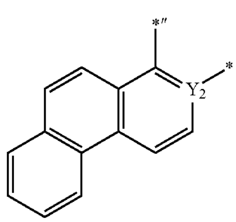
CY2-24



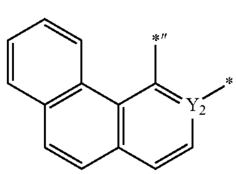
CY2-25



CY2-26



CY2-27



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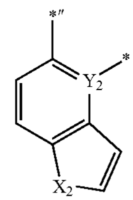
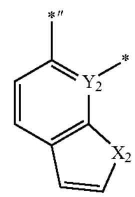
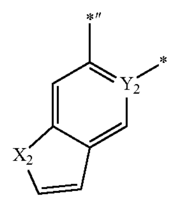
CY2-28



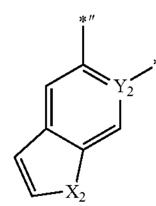
CY2-29



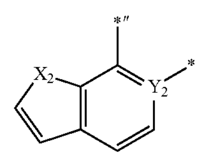
CY2-30



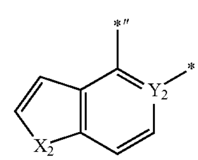
CY2-31



CY2-32



CY2-33



wherein, in Formulae CY2-1 to CY2-33,

[0191] Y_2 may be as described herein,

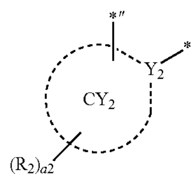
[0192] X_2 may be O, S, Se, $N(R_{28})$, $C(R_{28})(R_{29})$, or $Si(R_{28})(R_{29})$,

[0193] each of R_{28} and R_{29} is as described for R_2 herein,

[0194] *" indicates a binding site to a neighboring atom in Formula 2-1, and

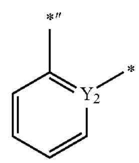
[0195] * indicates a binding site to M in Formula 1.

[0196] In one or more embodiments, a group represented by:

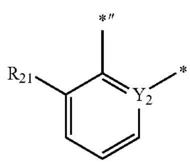


in Formula 2-1 may be a group represented by one of Formulae CY2(1) to CY2(56):

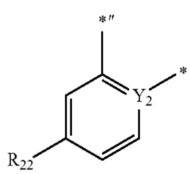
CY2(1)



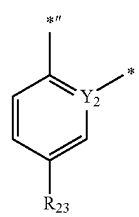
CY2(2)



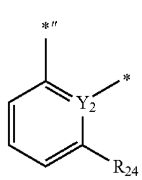
CY2(3)



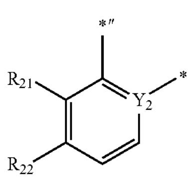
CY2(4)



CY2(5)



CY2(6)



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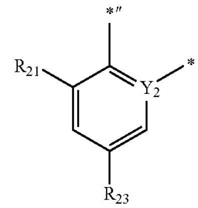
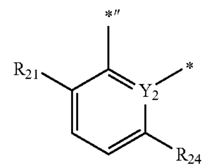
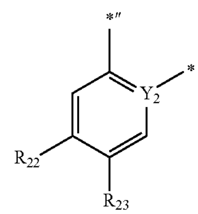
CY2(7)



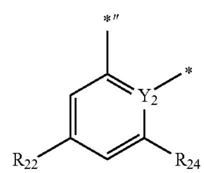
CY2(8)



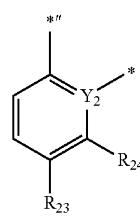
CY2(9)



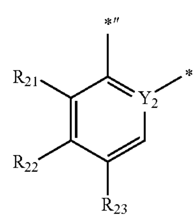
CY2(10)



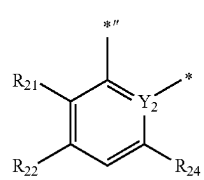
CY2(11)



CY2(12)



CY2(13)



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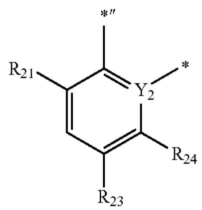
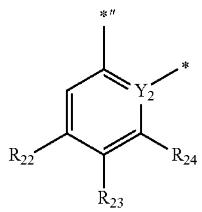
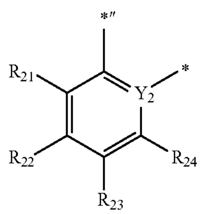
CY2(14)



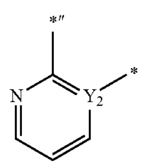
CY2(15)



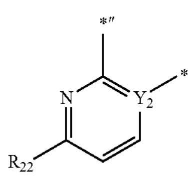
CY2(16)



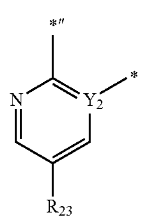
CY2(17)



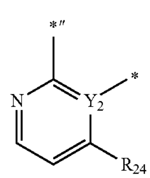
CY2(18)



CY2(19)



CY2(20)



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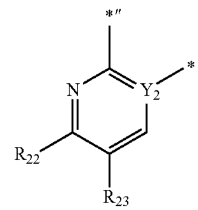
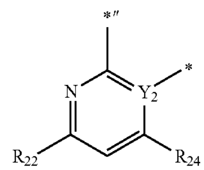
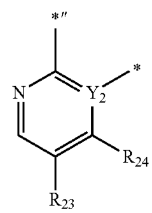
CY2(21)



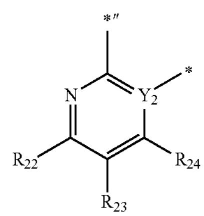
CY2(22)



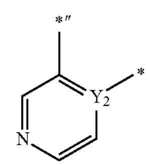
CY2(23)



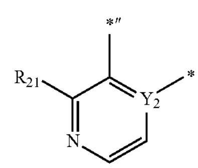
CY2(24)



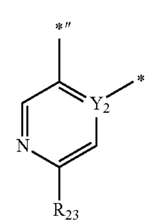
CY2(25)



CY2(26)



CY2(27)



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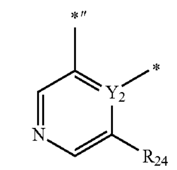
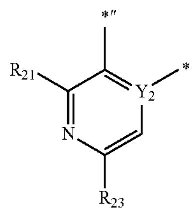
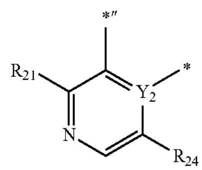
CY2(28)



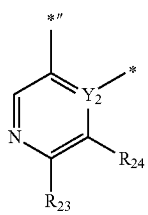
CY2(29)



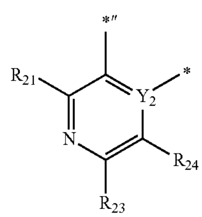
CY2(30)



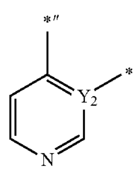
CY2(31)



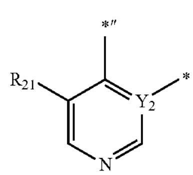
CY2(32)



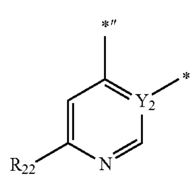
CY2(33)



CY2(34)



CY2(35)



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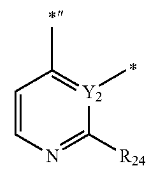
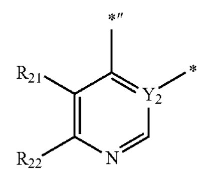
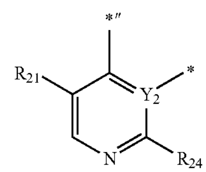
CY2(36)



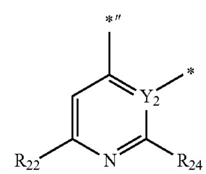
CY2(37)



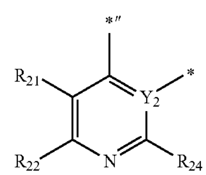
CY2(38)



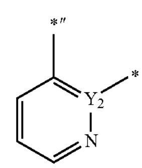
CY2(39)



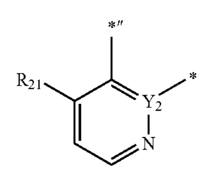
CY2(40)



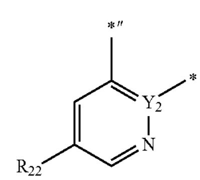
CY2(41)



CY2(42)



CY2(43)



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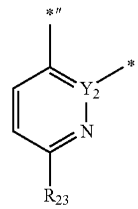
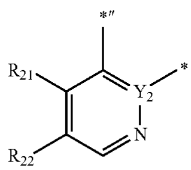
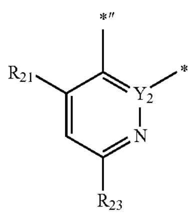
CY2(44)



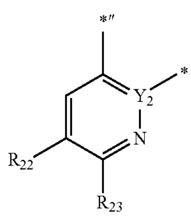
CY2(45)



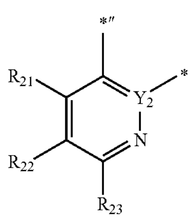
CY2(46)



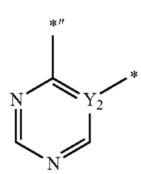
CY2(47)



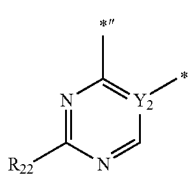
CY2(48)



CY2(49)



CY2(50)



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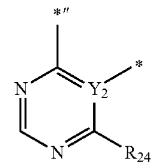
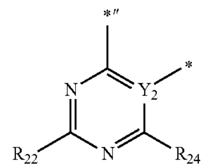
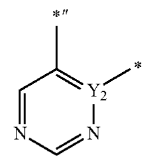
CY2(51)



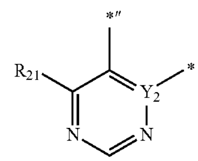
CY2(52)



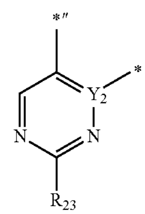
CY2(53)



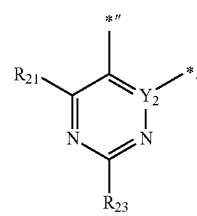
CY2(54)



CY2(55)



CY2(56)



[0197] In Formulae CY2(1) to CY2(56),

[0198] Y_2 may be as described herein,

[0199] R_{21} to R_{24} may each be as described in connection with R_2, wherein each of R_{21} to R_{24} may not be hydrogen,

[0200] *" indicates a binding site to a neighboring atom in Formula 2-1, and

[0201] * indicates a binding site to M in Formula 1.

[0202] According to one or more embodiments, Formula 2-2 may satisfy at least one of Condition A and Condition B:

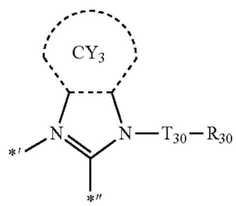
[0203] Condition A

[0204] T_{30} is a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a}, and

[0205] Condition B

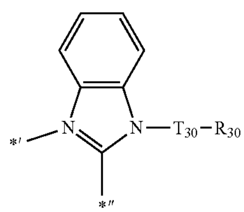
[0206] R_{30} is a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

[0207] In one or more embodiments, a group represented by:

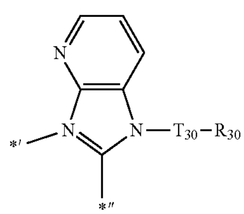


in Formula 2-2 may be a group represented by one of Formulae CY3-1 to CY3-21:

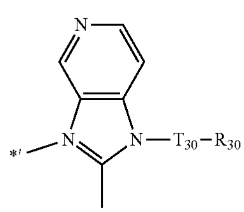
CY3-1



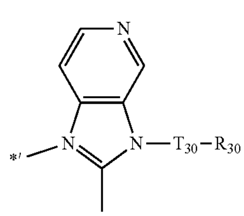
CY3-2



CY3-3



CY3-4



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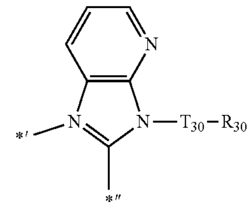
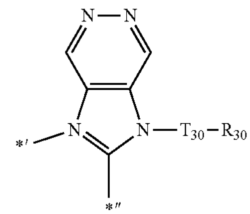
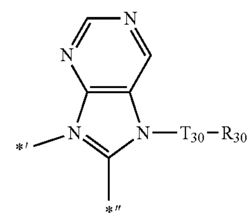
CY3-5



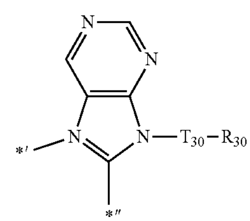
CY3-6



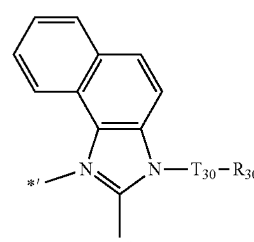
CY3-7



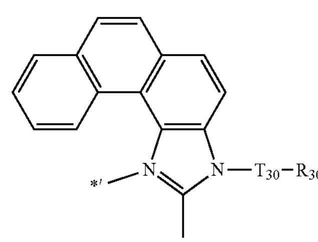
CY3-8



CY3-9



CY3-10



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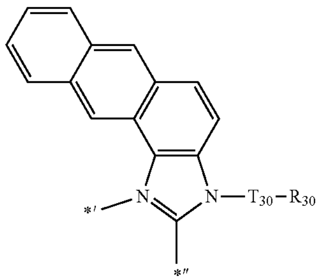
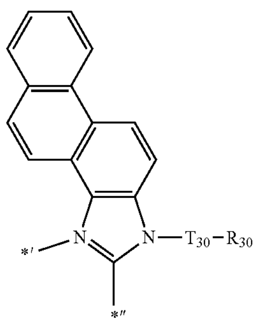
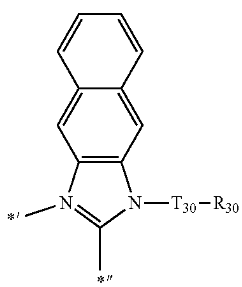
CY3-11



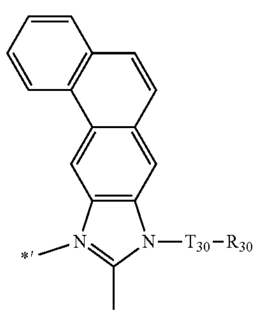
CY3-12



CY3-13



CY3-14



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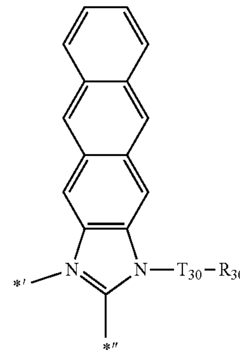
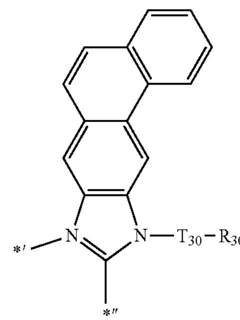
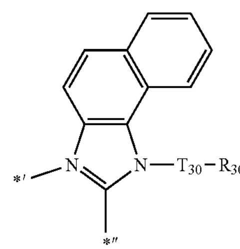
CY3-15



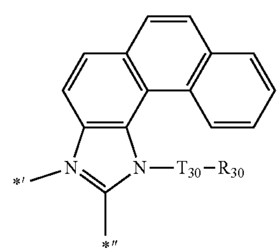
CY3-16



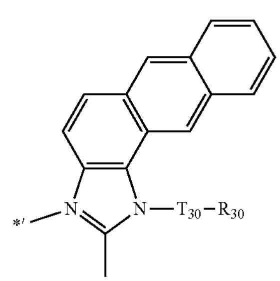
CY3-17



CY3-18



CY3-19



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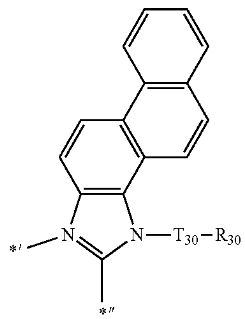
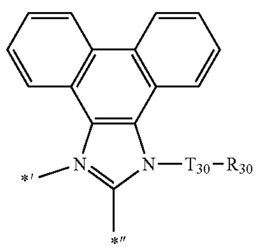
CY3-20



CY3-21



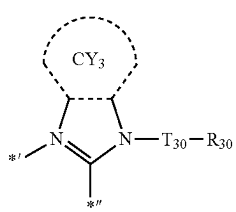
wherein, in Formulae CY3-1 to CY3-21,

[0208] each of T_{30} and R_{30} is as described herein,

[0209] *' indicates a binding site to M in Formula 1, and

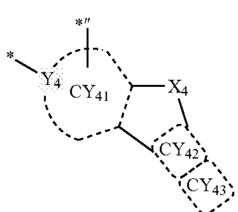
[0210] *" indicates a binding site to a neighboring atom in Formula 2-2.

[0211] According to one or more embodiments, a group represented by:



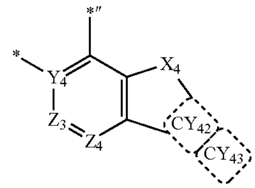
in Formula 2-2 may be a group represented by Formulae CY3-1 or CY3-17.

[0212] According to one or more embodiments, a group represented by:

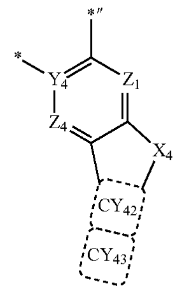


in Formula 2-2 may be a group represented by one of Formulae CY4-1 to CY4-6:

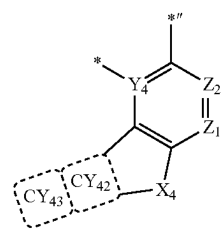
CY4-1



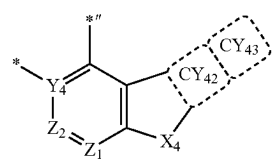
CY4-2



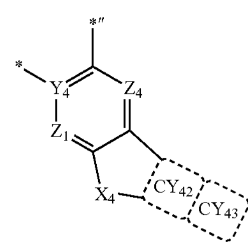
CY4-3



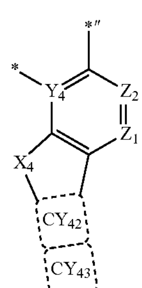
CY4-4



CY4-5



CY4-6



wherein, in Formulae C4-1 to C4-6,

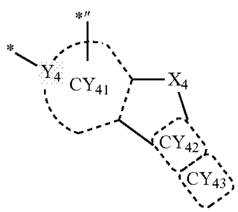
[0213] each of Y_4, X_4, ring CY_{42} and ring CY_{43} is as described herein,

[0214] Z_1 to Z_4 may each independently be N or C,

[0215] * indicates a binding site to M in Formula 1, and

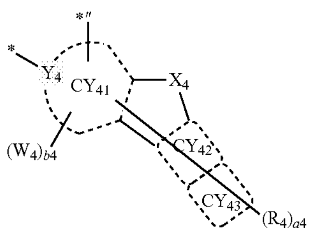
[0216] *" indicates a binding site to a neighboring atom in Formula 2-2.

[0217] According to one or more embodiments, a group represented by:



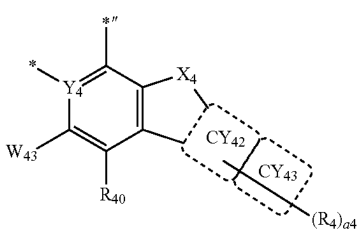
in Formula 2-2 may be a group represented by Formula CY4-1.

[0218] In one or more embodiments, a group represented by:

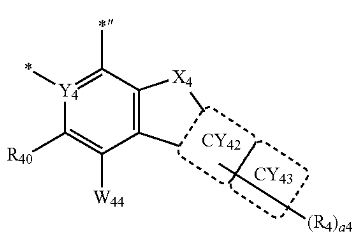


in Formula 2-2 may be a group represented by one of Formulae CY4(1) to CY4(18):

CY4(1)



CY4(2)



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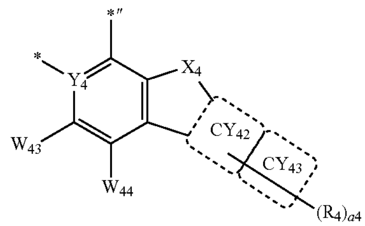
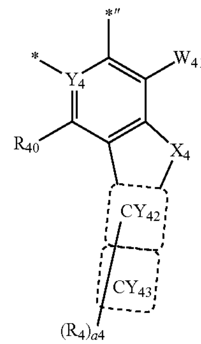
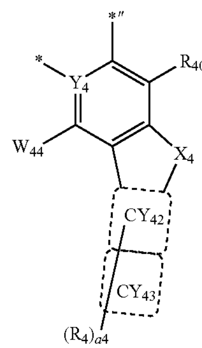
CY4(3)



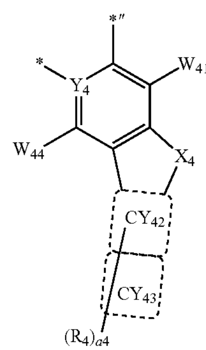
CY4(4)



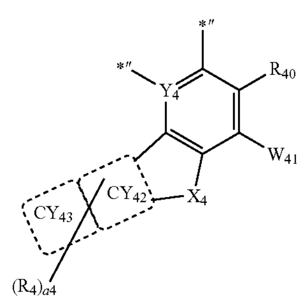
CY4(5)



CY4(6)



CY4(7)



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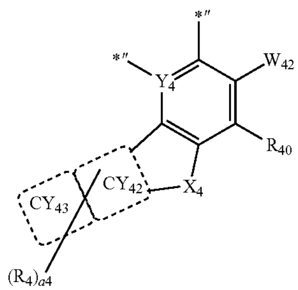
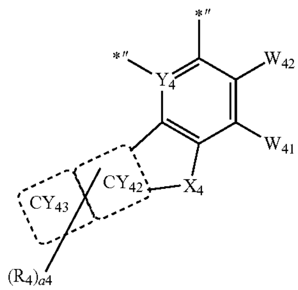
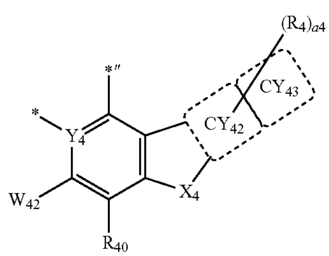
CY4(8)



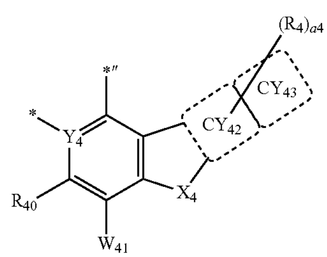
CY4(9)



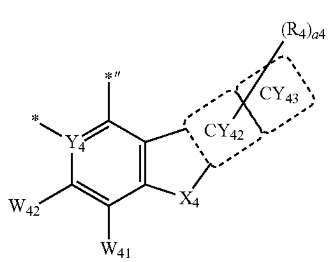
CY4(10)



CY4(11)



CY4(12)



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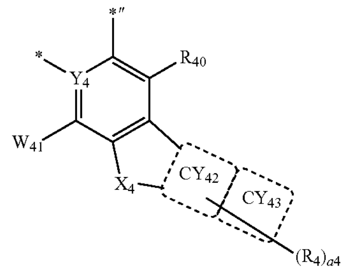
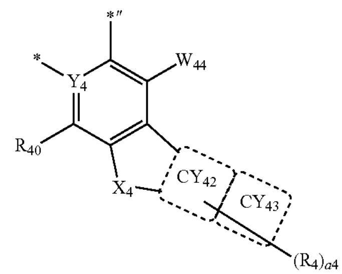
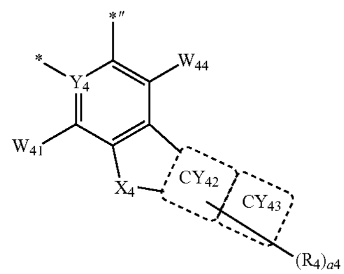
CY4(13)



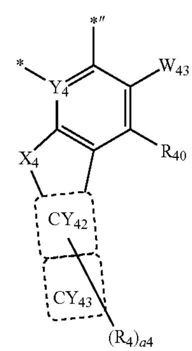
CY4(14)



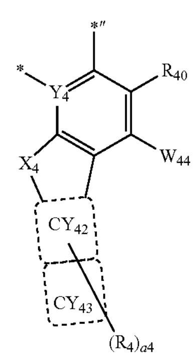
CY4(15)



CY4(16)



CY4(17)



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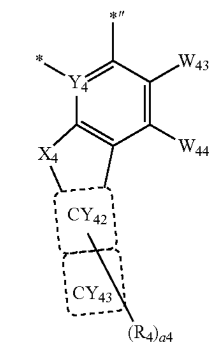
CY4(18)



wherein, in Formulae CY4(1) to CY4(18),

[0219] each of Y_4, ring CY_{42}, ring CY_{43} and R_4 is as described herein,

[0220] each W_{41} to W_{44} is as described in connection with W_4 herein,

[0221] R_{40} is as described in connection with R_4,

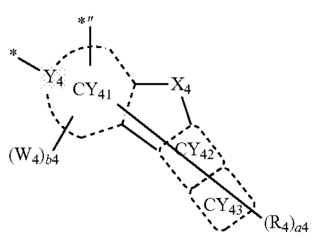
[0222] a4 may be an integer from 0 to 10,

[0223] * indicates a binding site to M in Formula 1, and

[0224] *" indicates a binding site to a neighboring atom in Formula 2-2.

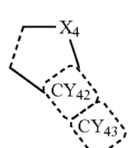
[0225] For example, a4 in Formulae CY4(1) to CY4(18) may be 0, 1, 2, or 3.

[0226] According to one or more embodiments, a group represented by:

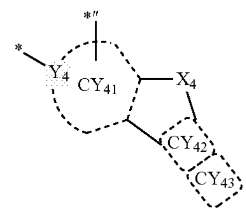


in Formula 2-2 may be a group represented by one of Formulae CY4(1) to CY4(3).

[0227] According to one or more embodiments, a group represented by

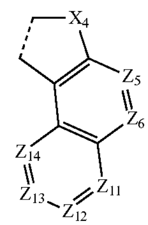


a group represented by

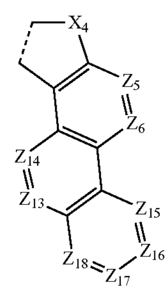


of Formula 2-2 and in Formulae CY4(1) to CY4(18) may be a group represented by one of Formulae CY401 to CY451:

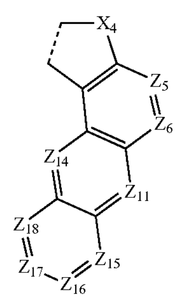
CY401



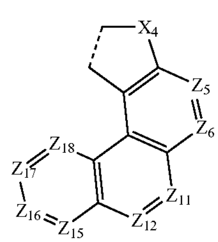
CY402



CY403



CY404



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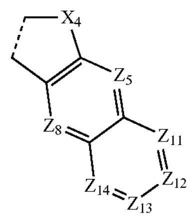
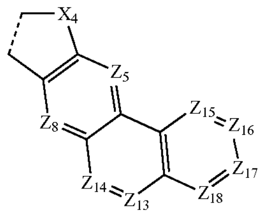
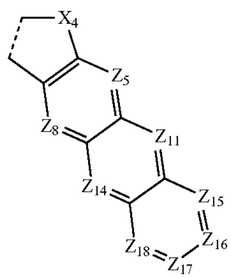
CY405



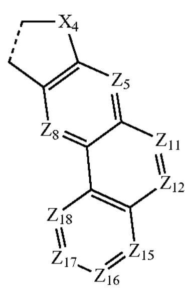
CY406



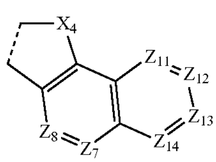
CY407



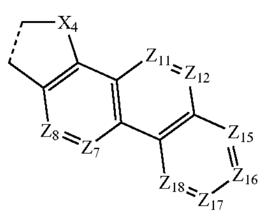
CY408



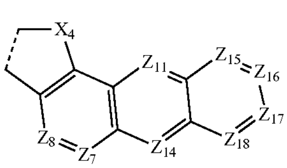
CY409



CY410



CY411



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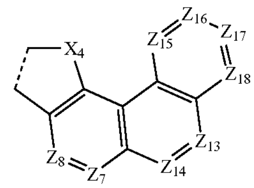
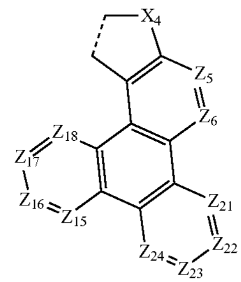
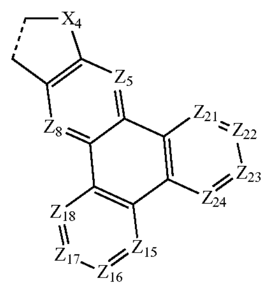
CY412



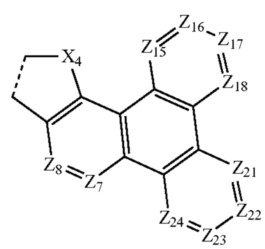
CY413



CY414



CY415



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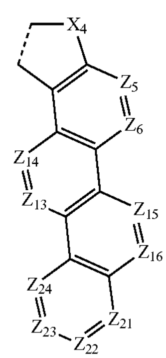

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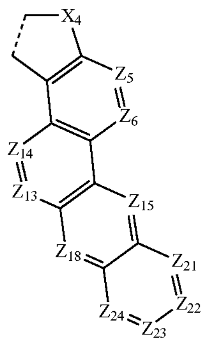
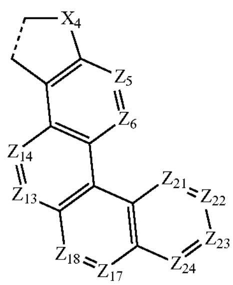
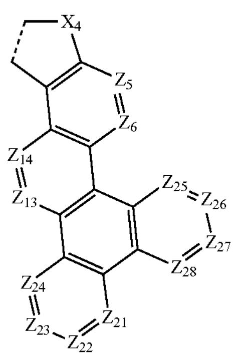
CY417



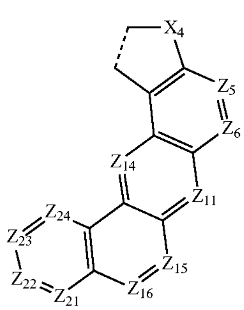
CY418



CY419



CY420



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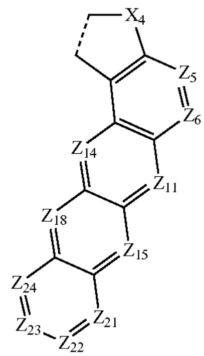
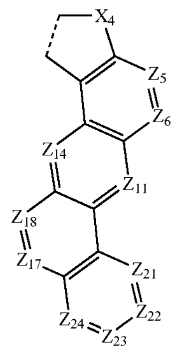
CY421



CY422



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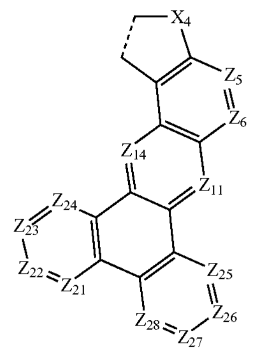
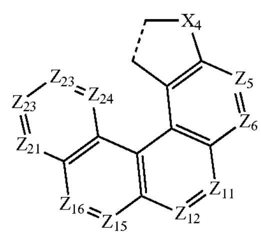
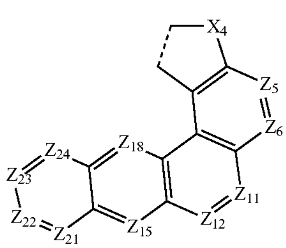


CY424



CY425



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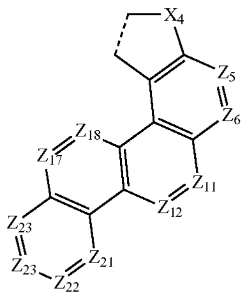
CY426



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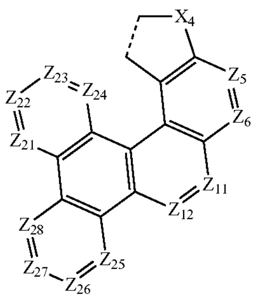

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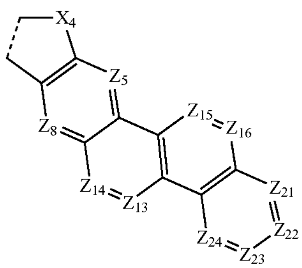

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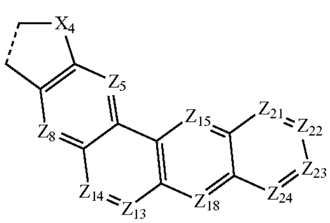

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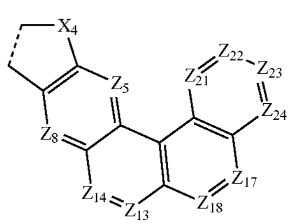

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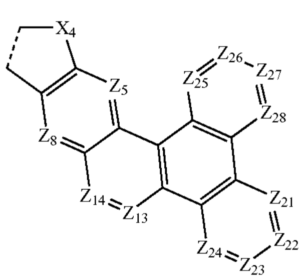

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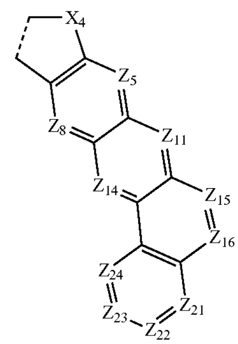

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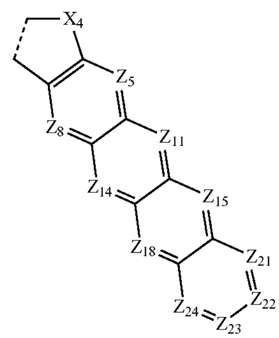

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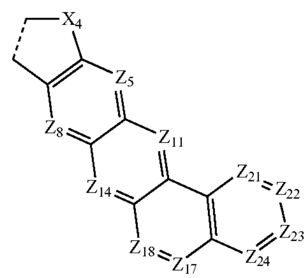

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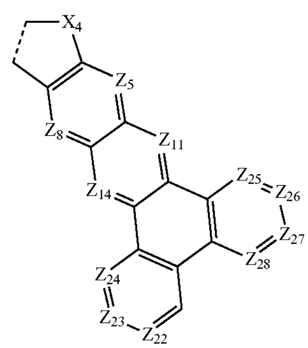

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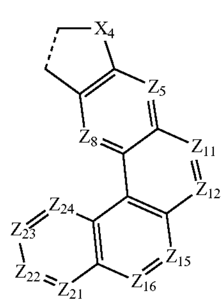

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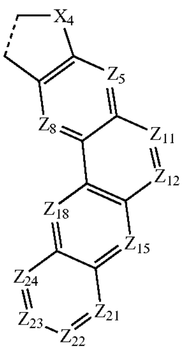
CY437



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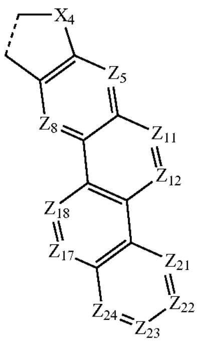

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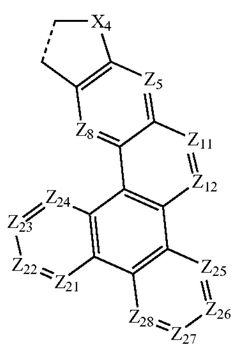

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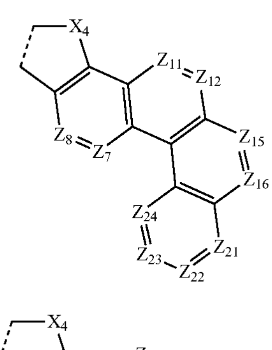
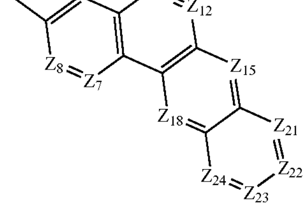


CY441



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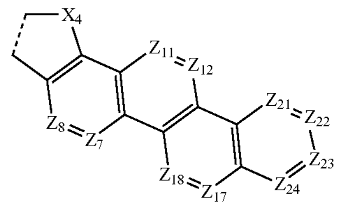
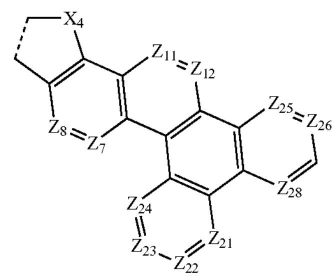
CY442



CY443



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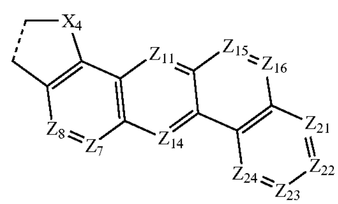

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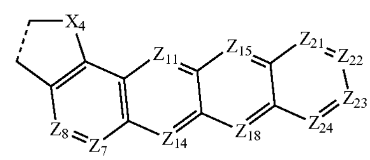
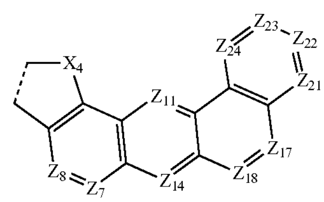
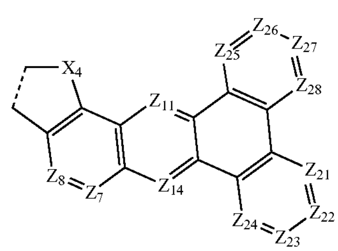


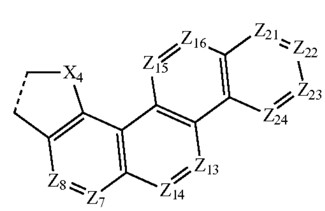
CY446



CY447



CY448



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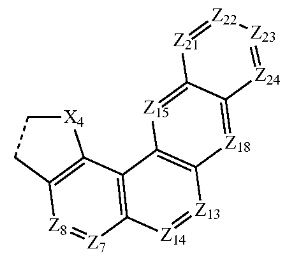
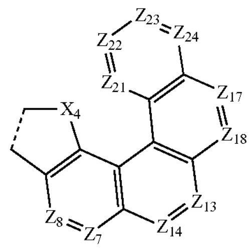
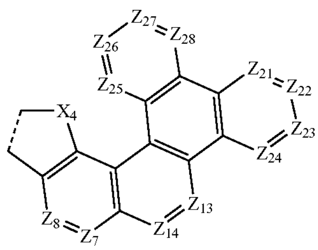
CY449



CY450



CY451



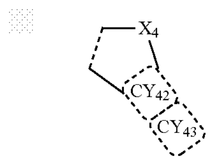
wherein, in Formulae CY401 to CY451,

[0228] X_4 is the same as described herein,

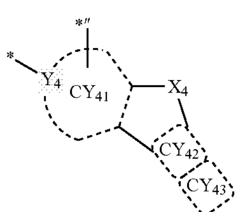
[0229] Z_5 to Z_8, Z_{11} to Z_{18}, and Z_{21} to Z_{28} may each independently be N or C, and

[0230] an X_4-containing five-membered ring may be condensed with ring CY_{41} which is adjacent thereto.

[0231] According to one or more embodiments, a group represented by:

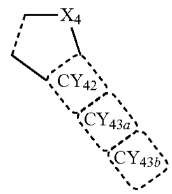


in a group represented by

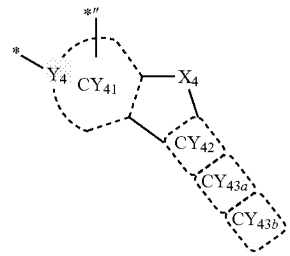


of Formula 2-2 and Formulae CY4(1) to CY4(18) may be a group represented by one of Formulae CY406, CY408, and CY410.

[0232] According to one or more embodiments, a group represented by:



in



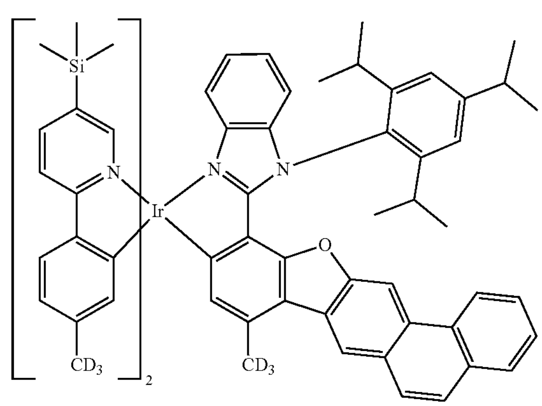
of Formula 2-2A may be a group represented by one of Formulae CY402 to CY404, CY406 to CY408, and CY410 to CY451.

[0233] According to one or more embodiments, the organometallic compound represented by Formula 1 may emit a red light or a green light, for example, a red light or a green light that has the maximum emission wavelength of about 500 nanometers (nm) or greater, or from about 500 nm to about 850 nm. For example, the organometallic compound represented by Formula 1 may emit a green light.

[0234] According to one or more embodiments, the organometallic compound represented by Formula 1 may emit light having the maximum emission wavelength of about 510 nm to about 550 nm, about 510 nm to about 540 nm, about 510 nm to about 529 nm, about 515 nm to about 550 nm, about 515 nm to about 540 nm, about 515 nm to about 529 nm, about 520 nm to about 550 nm, about 520 nm to about 540 nm, about or 520 nm to about 529 nm. For example, the organometallic compound represented by Formula 1 may emit light (for example, a green light) with a maximum emission wavelength of about 510 nm to about 529 nm, or about 520 nm to about 529 nm.

[0235] The organometallic compound represented by Formula 1 may be one of Compounds 1 to 111:

1



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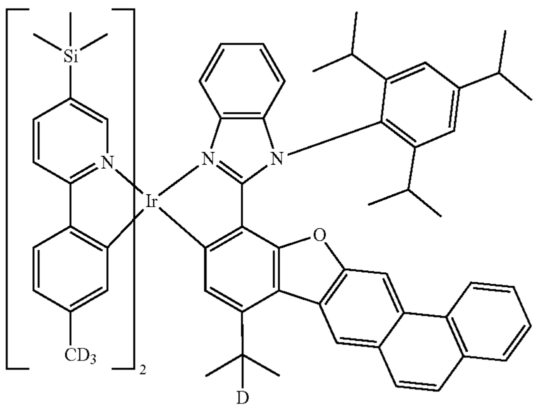
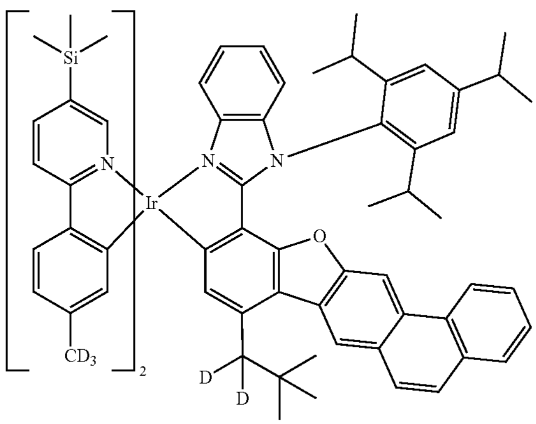
2



3



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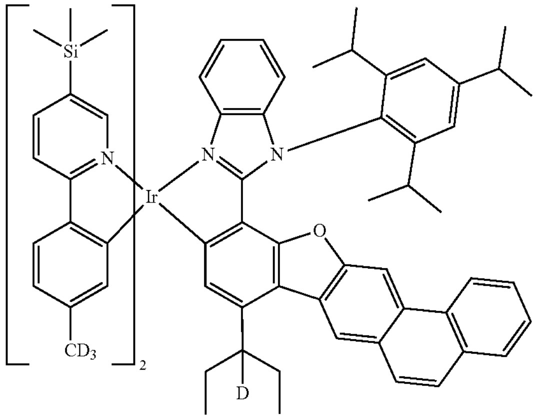

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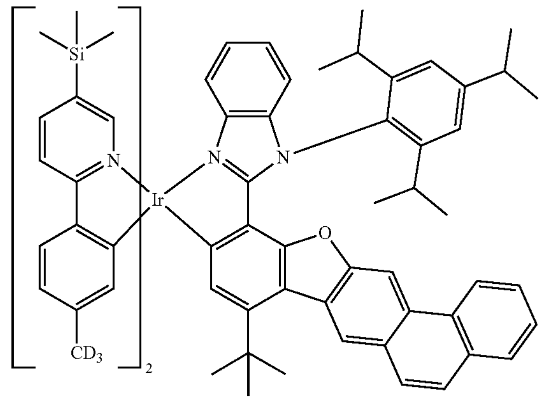

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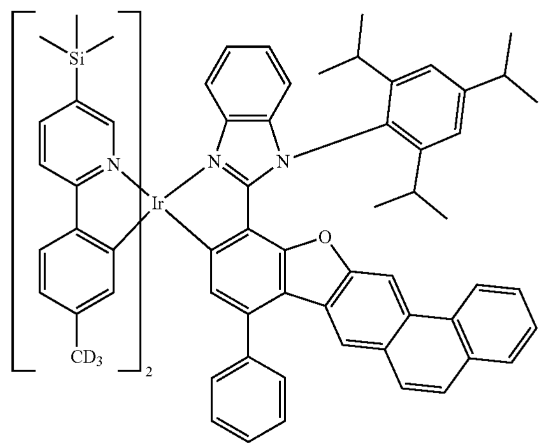

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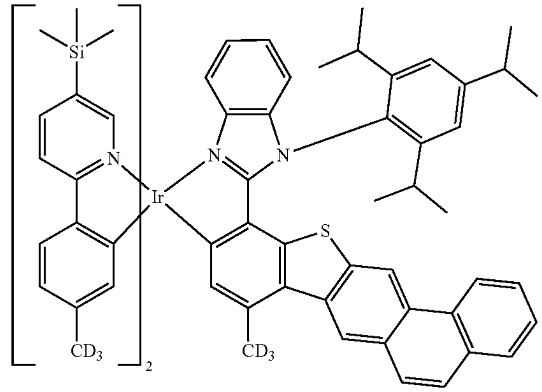

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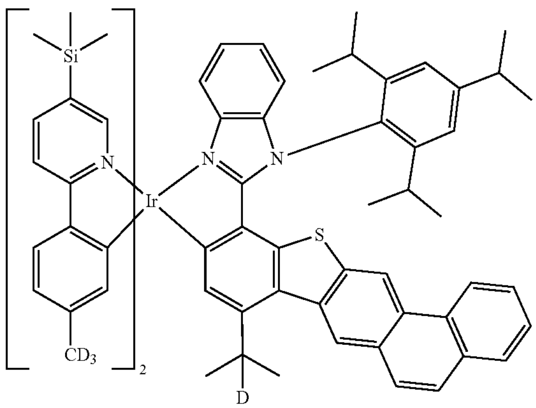
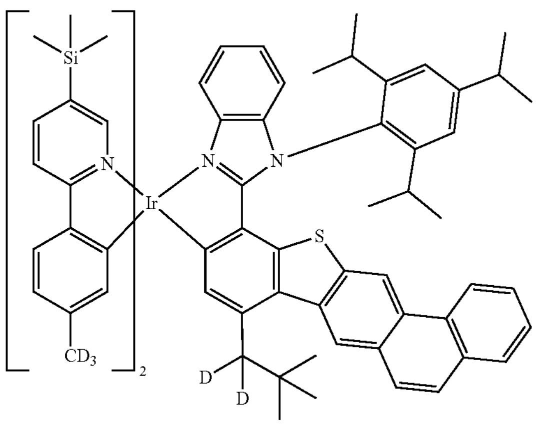
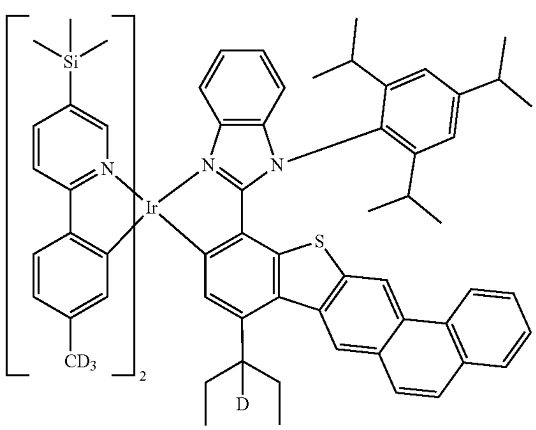
8



9



10



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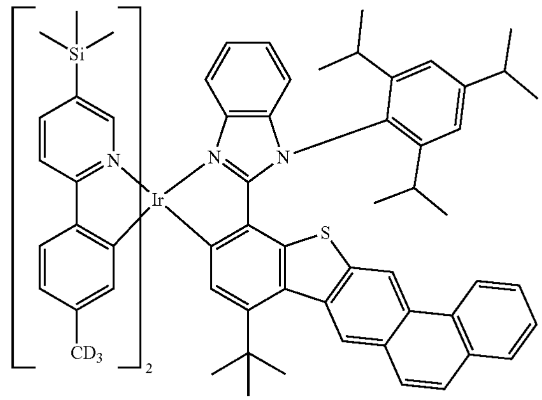
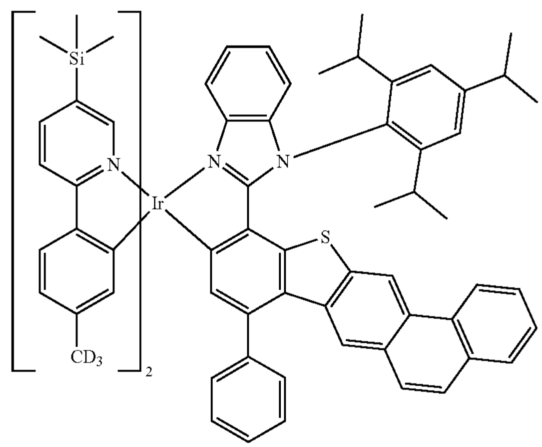
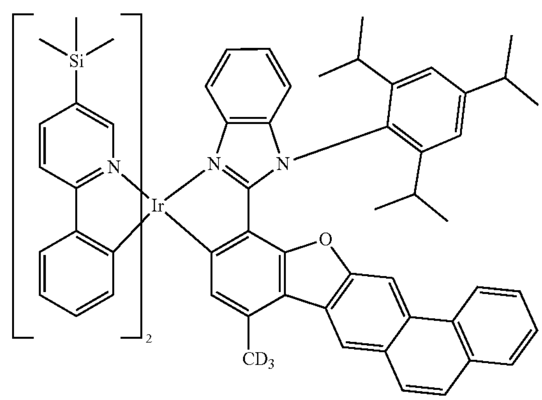
11



12



13



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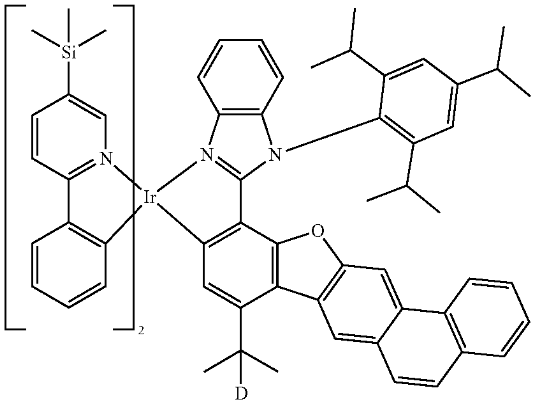
14



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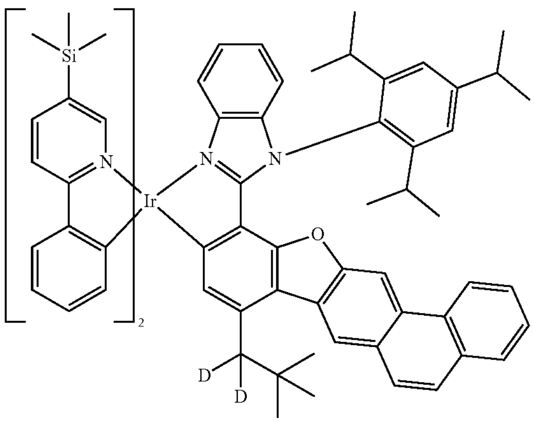

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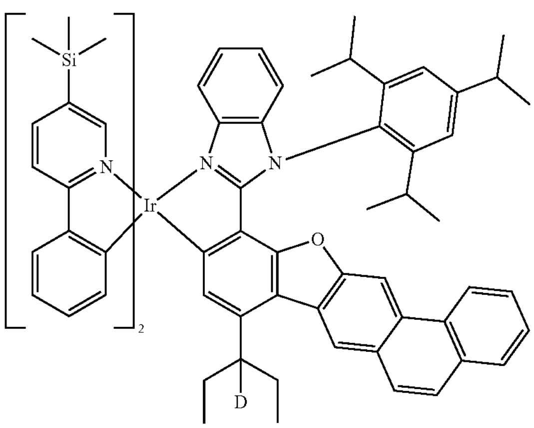

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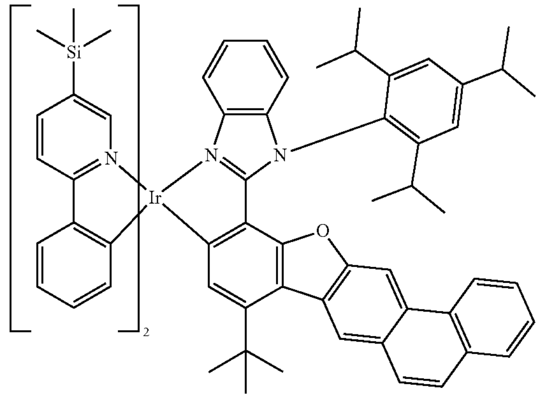

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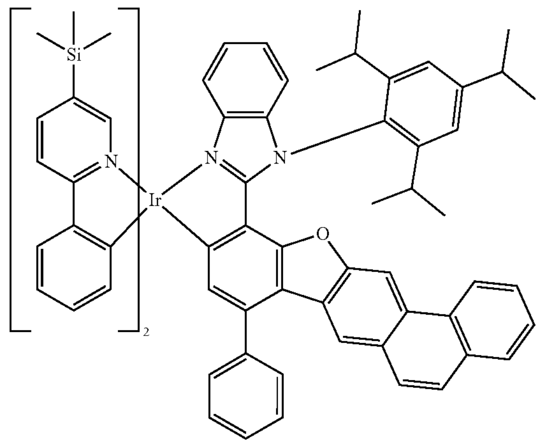

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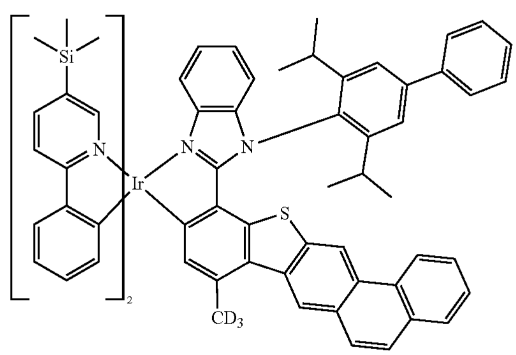

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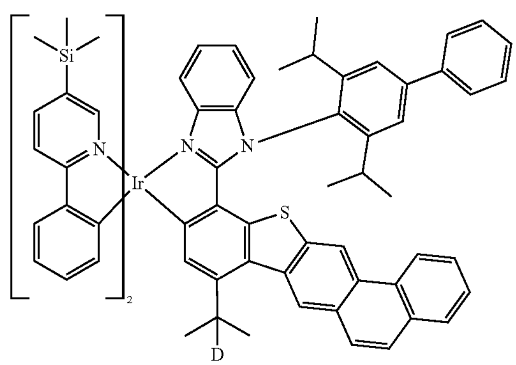

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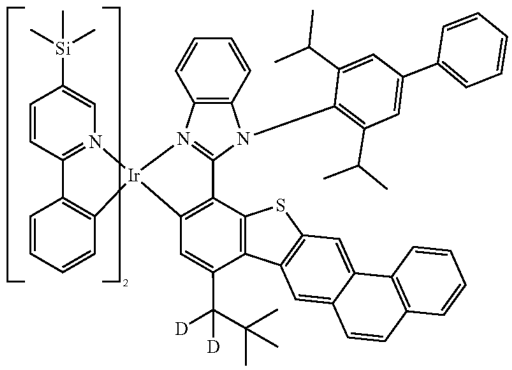

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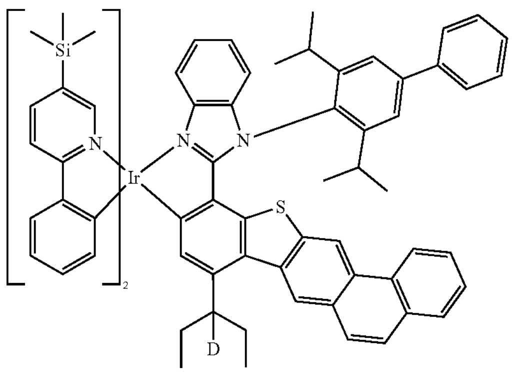

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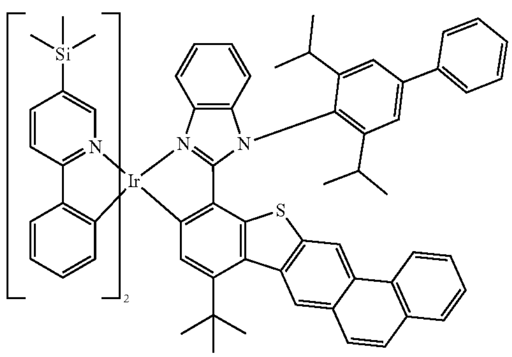

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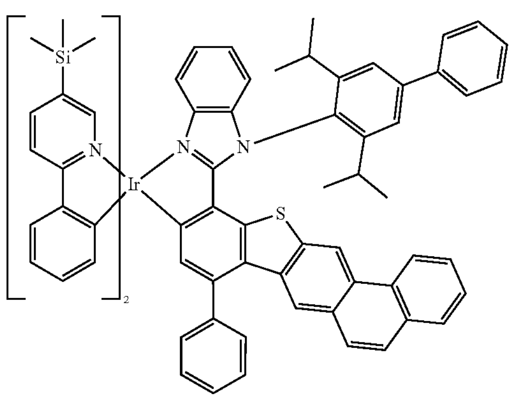

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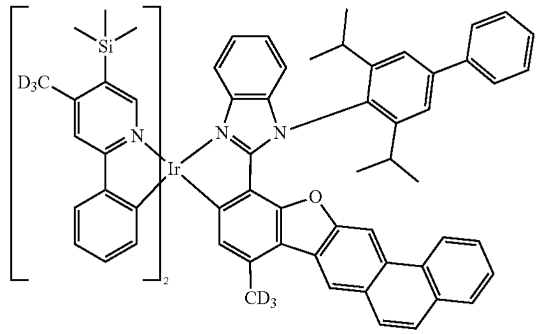

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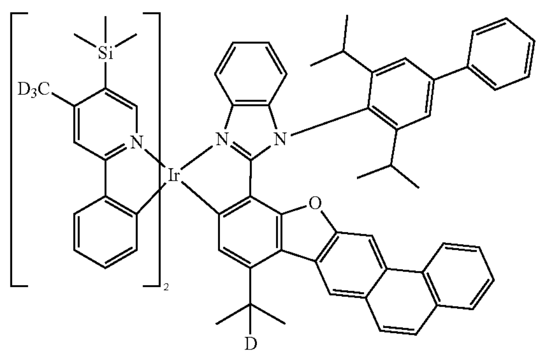

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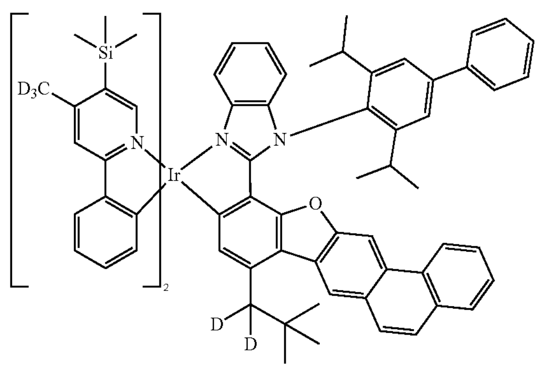

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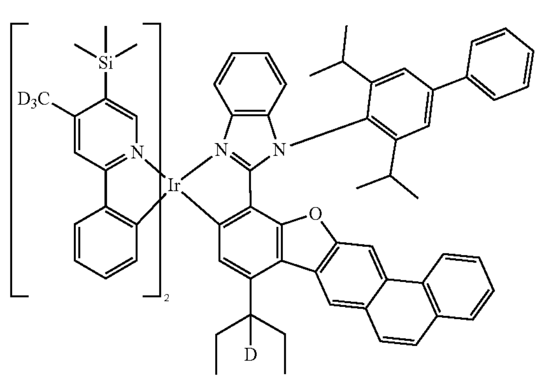

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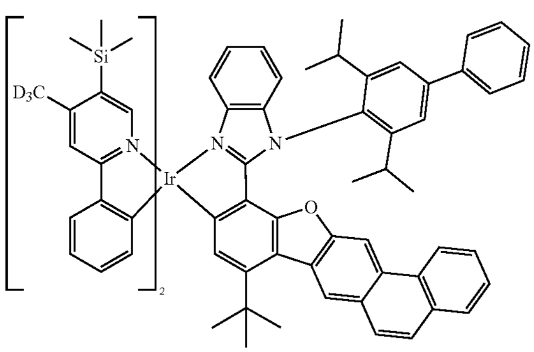

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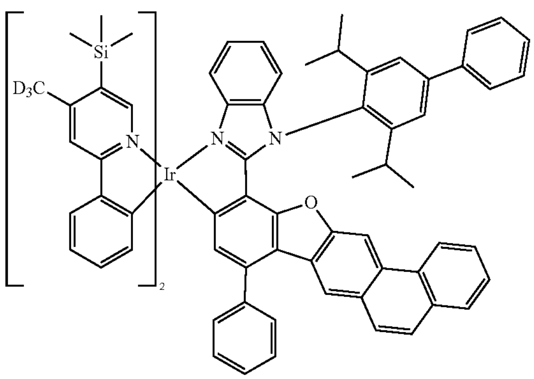

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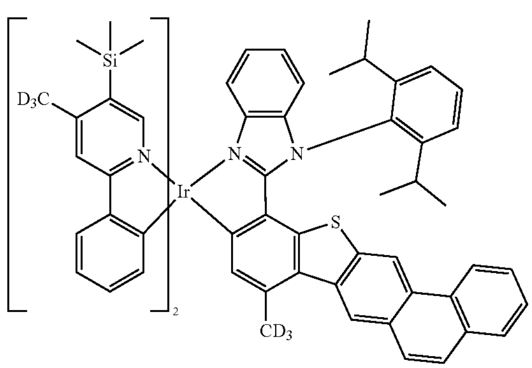

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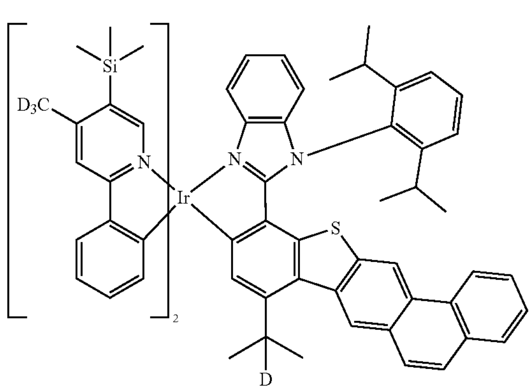

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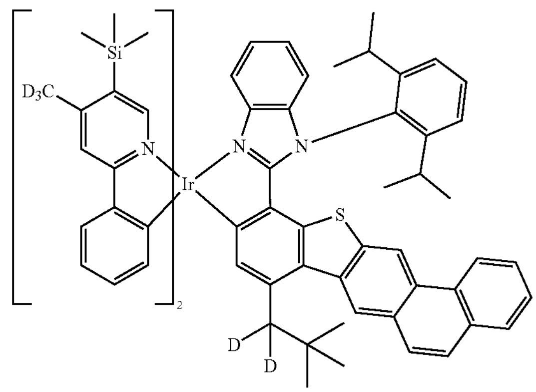

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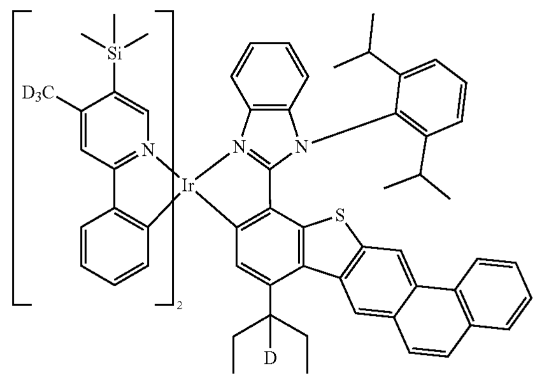

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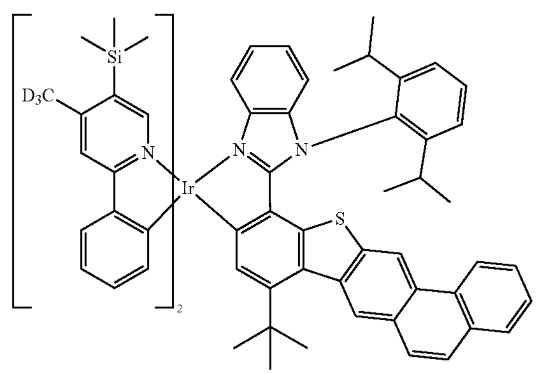

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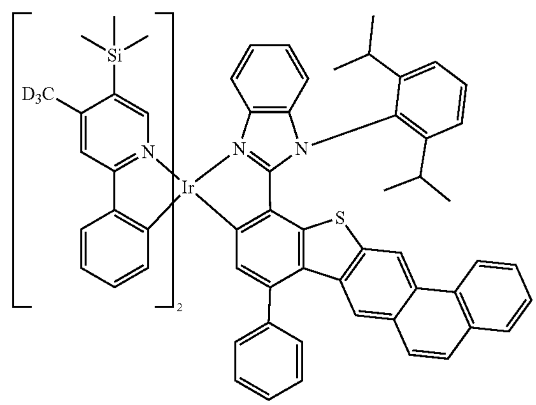

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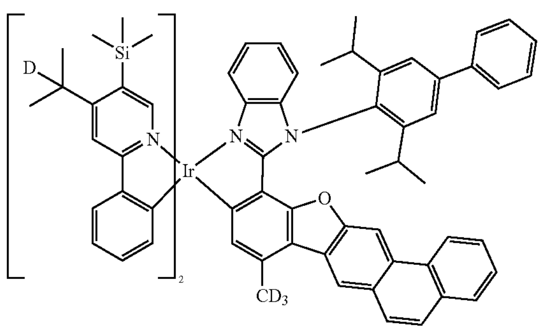

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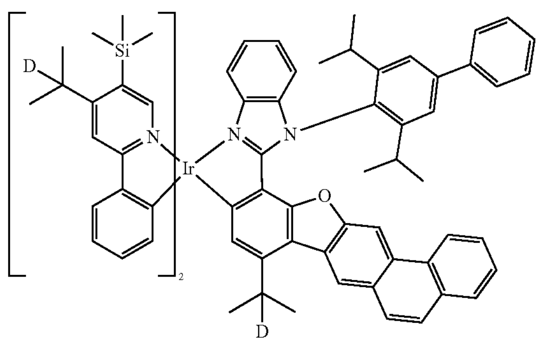

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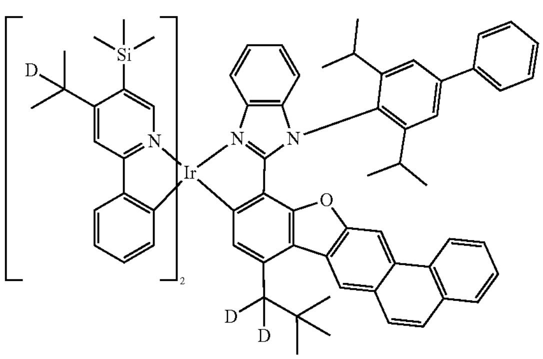

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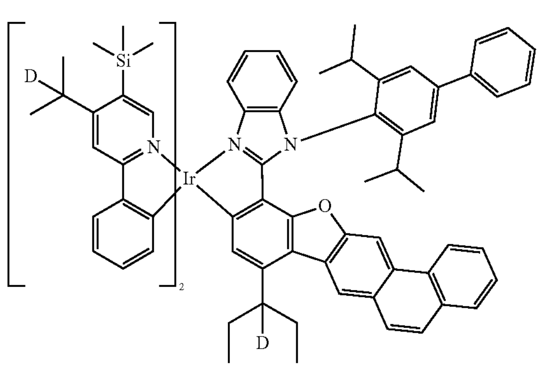

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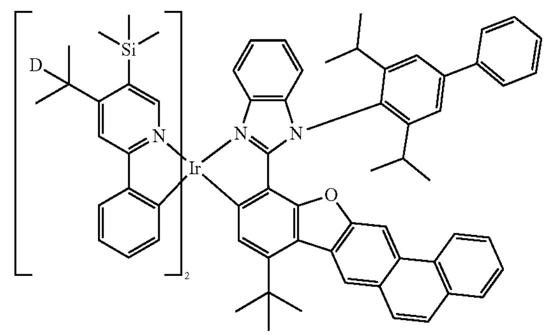

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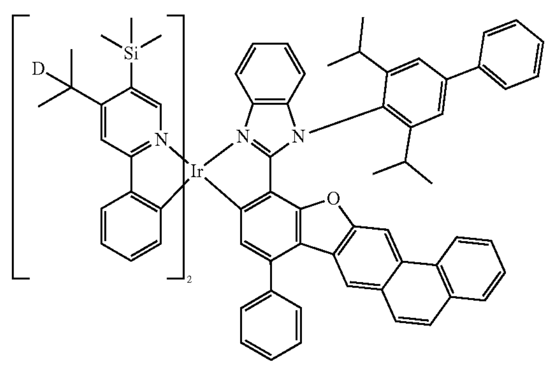

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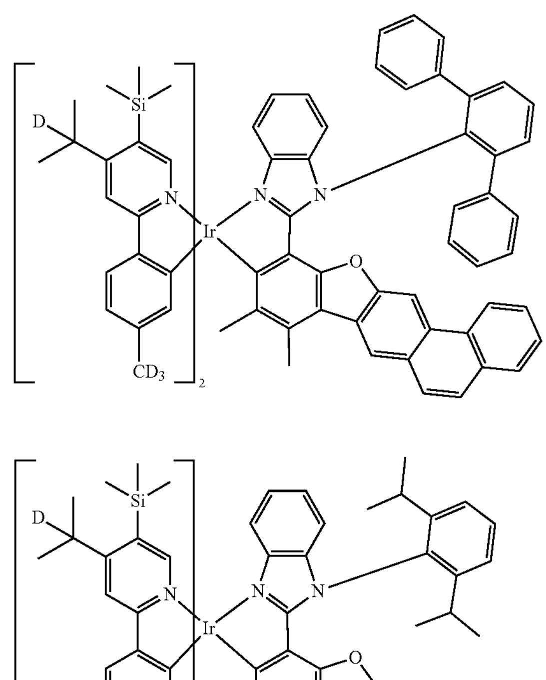

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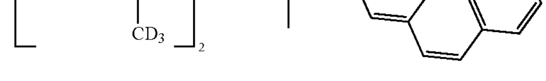

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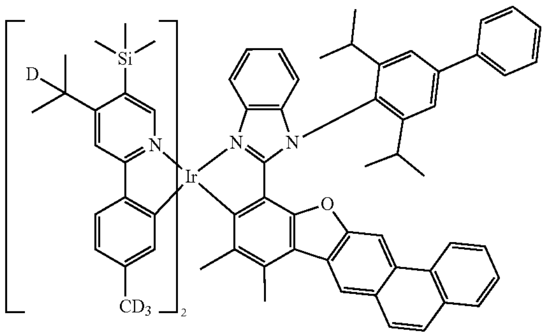

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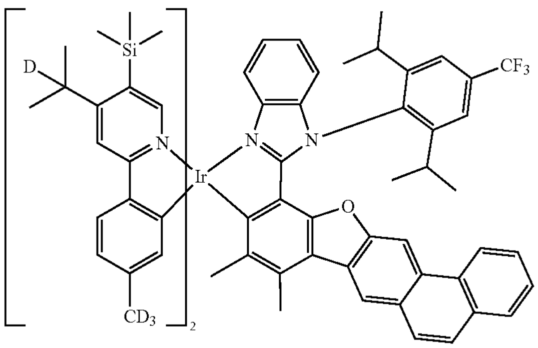

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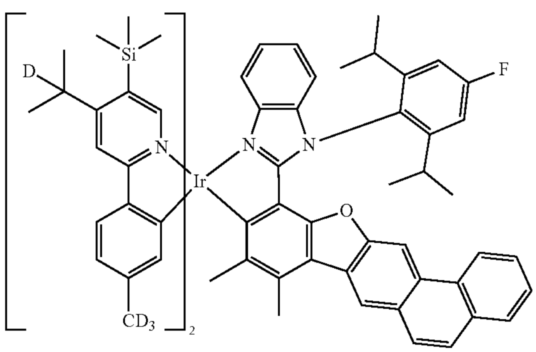

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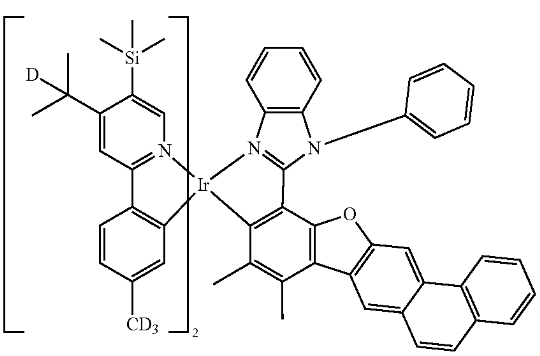

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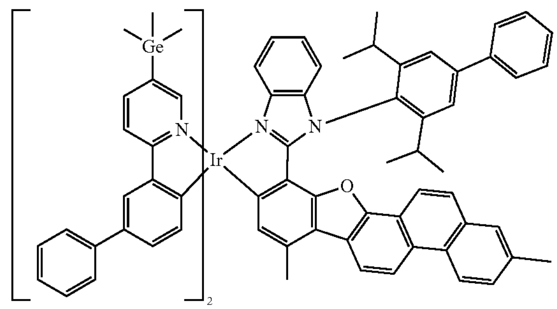

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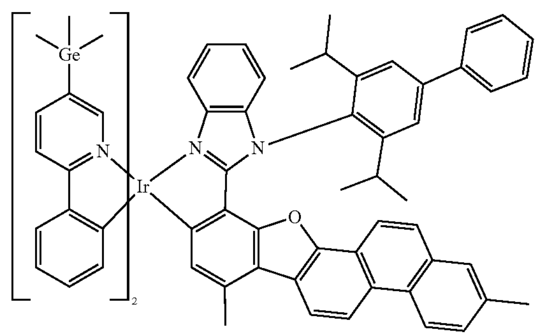

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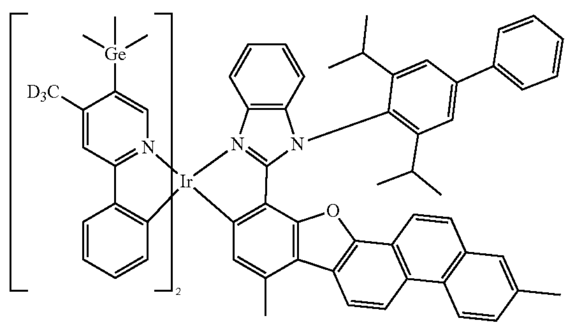

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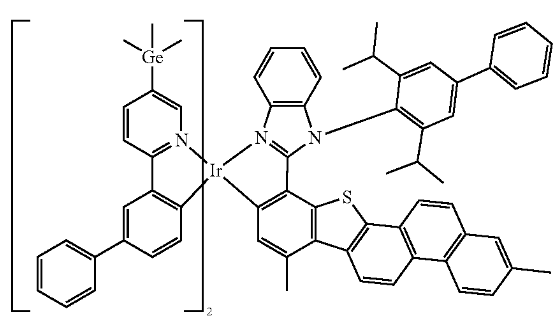

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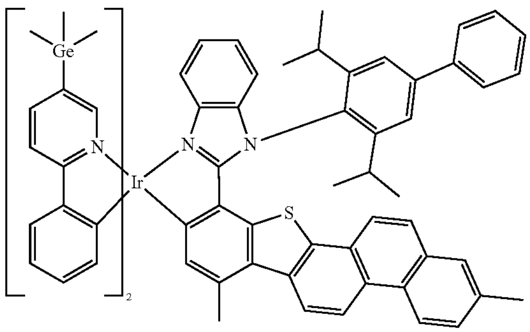

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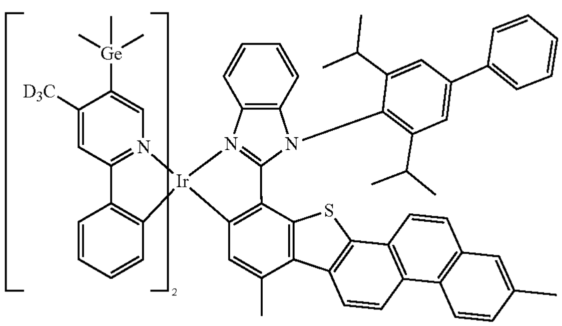

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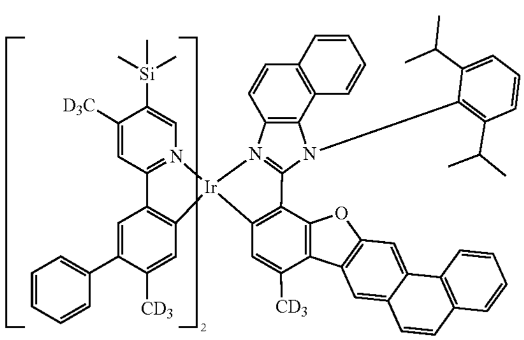

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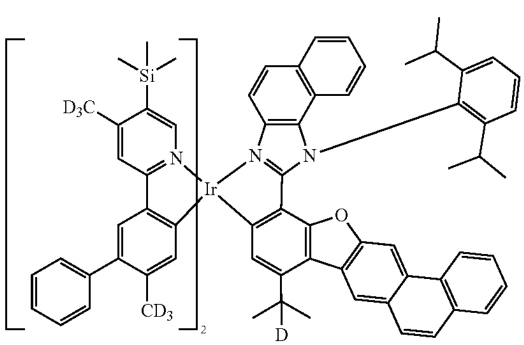

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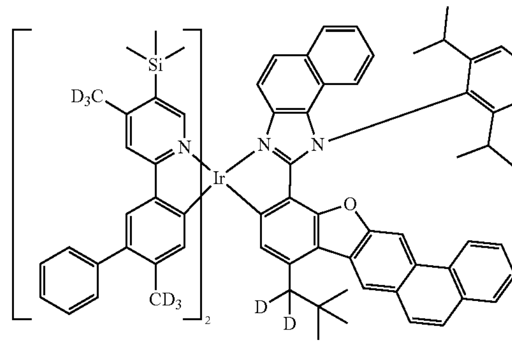

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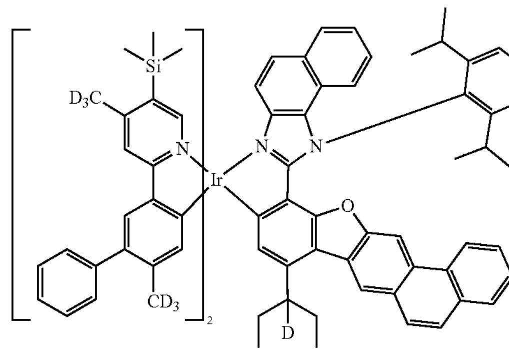

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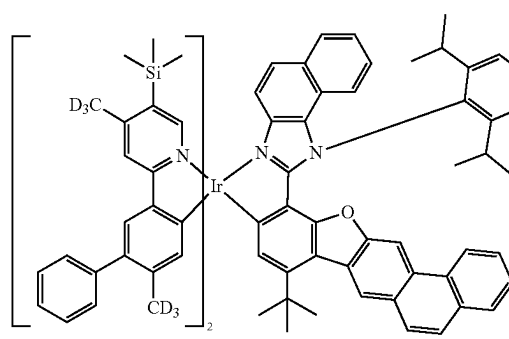

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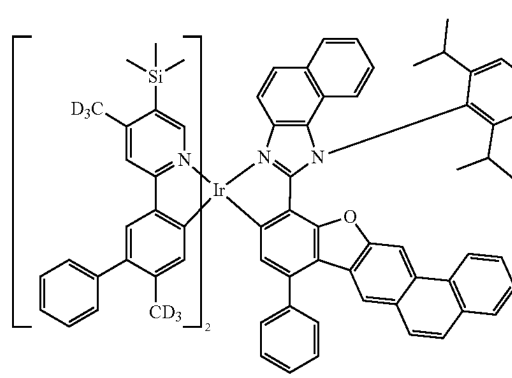

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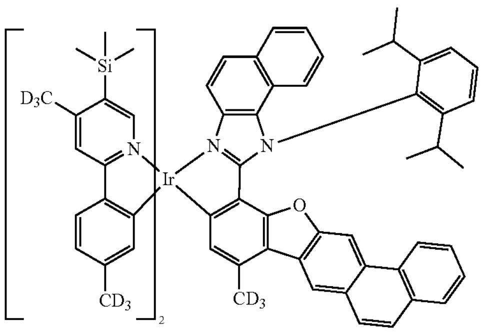

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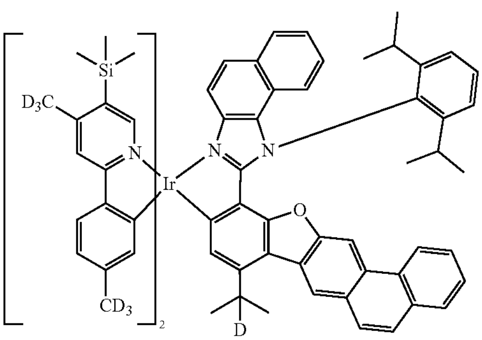

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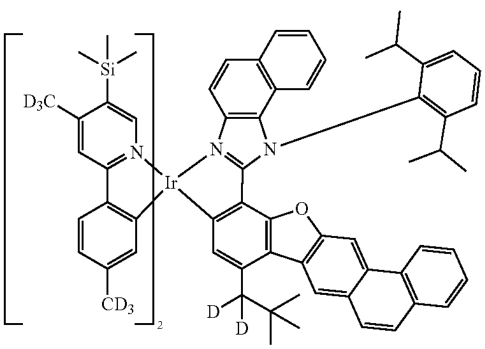

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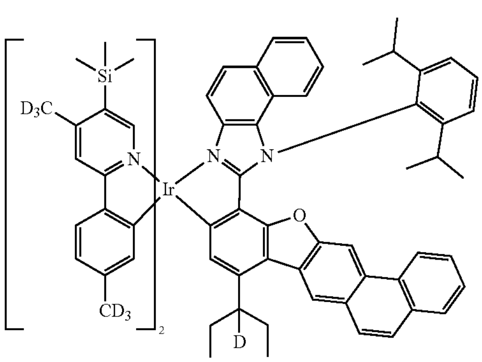

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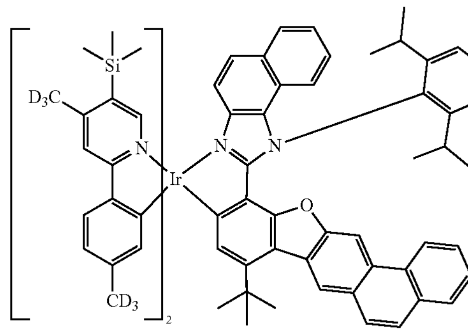

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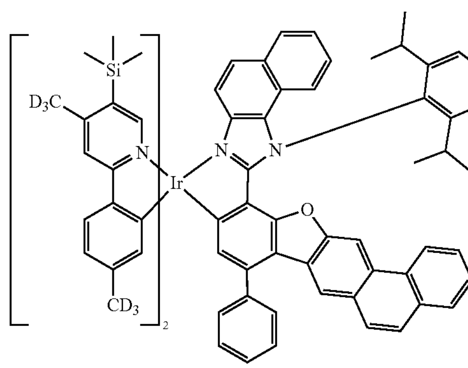

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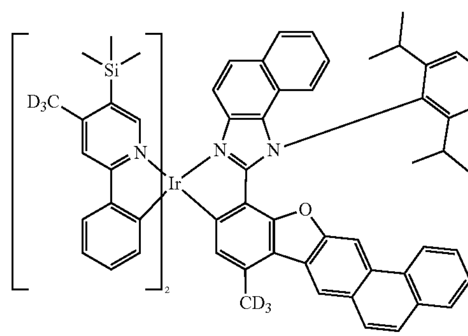

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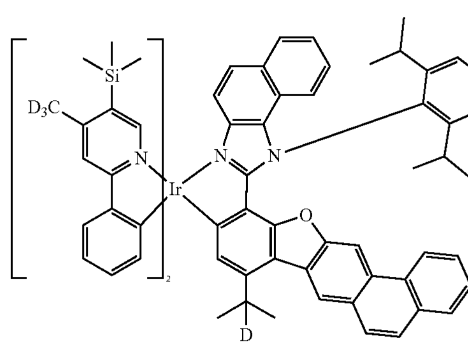

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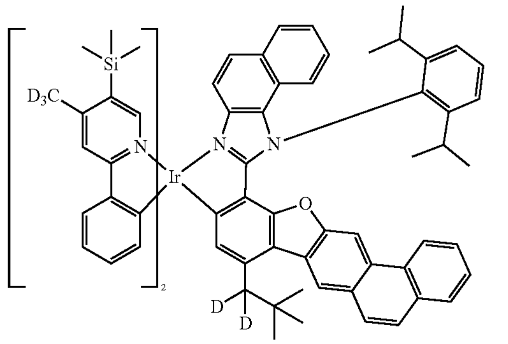

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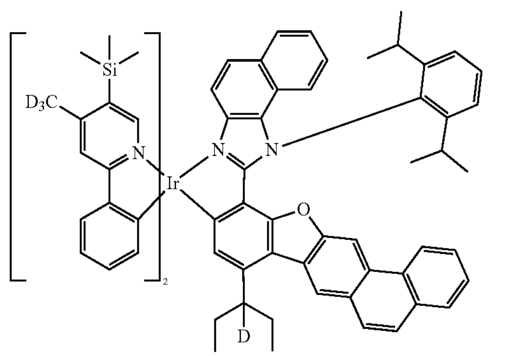

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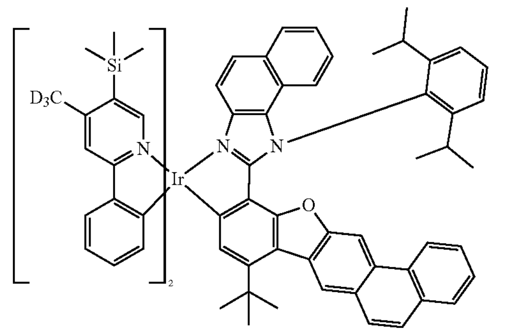

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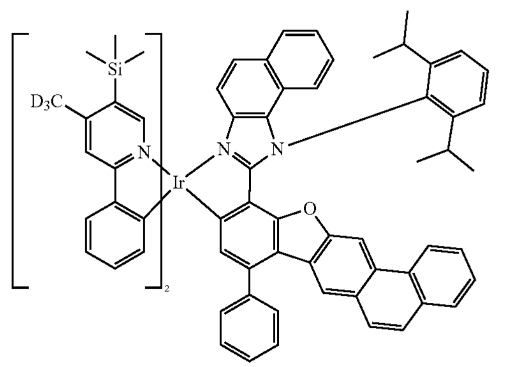

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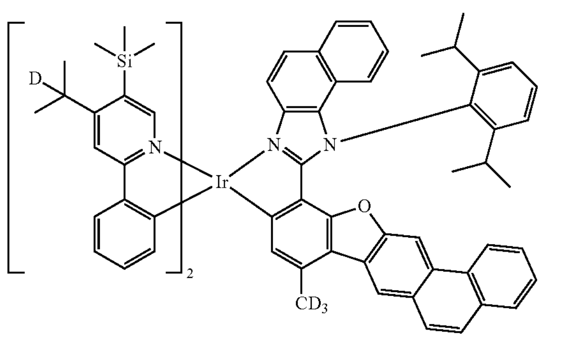

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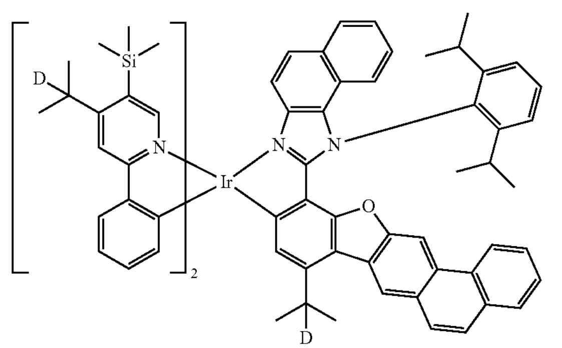

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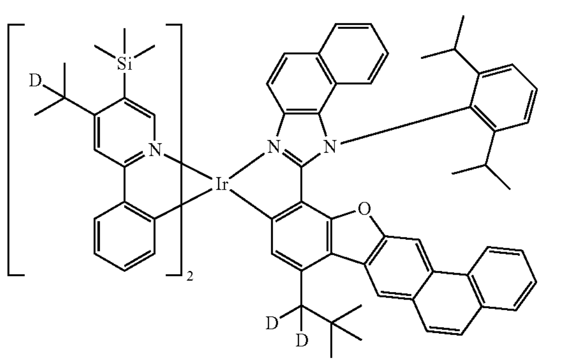

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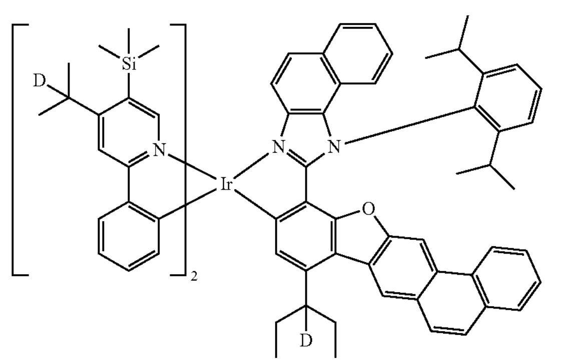

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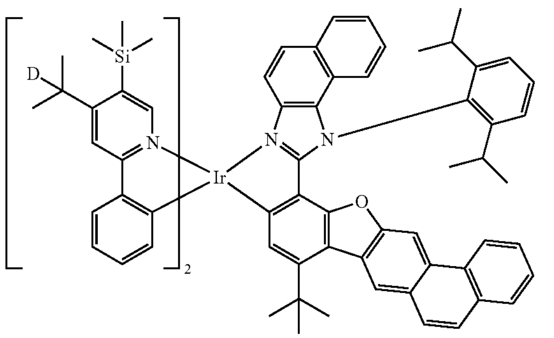

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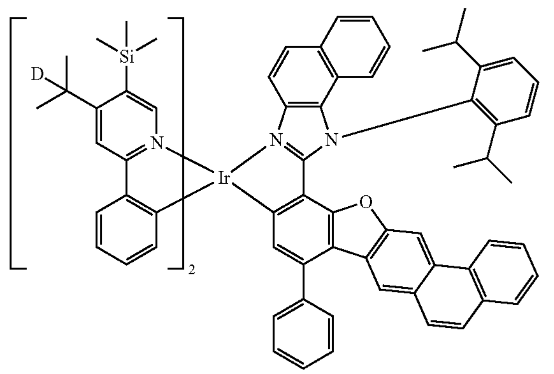

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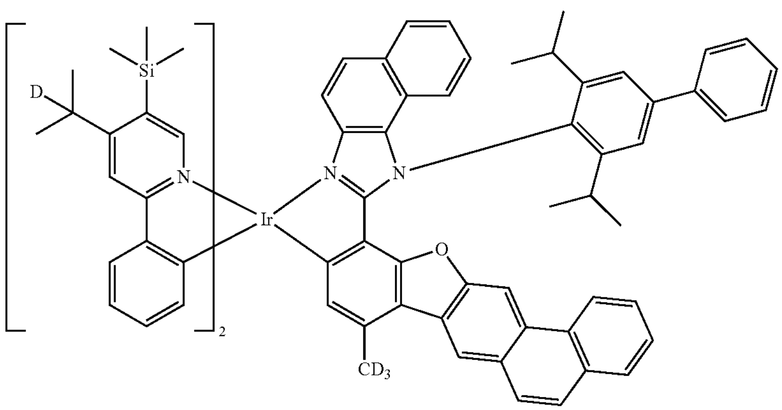

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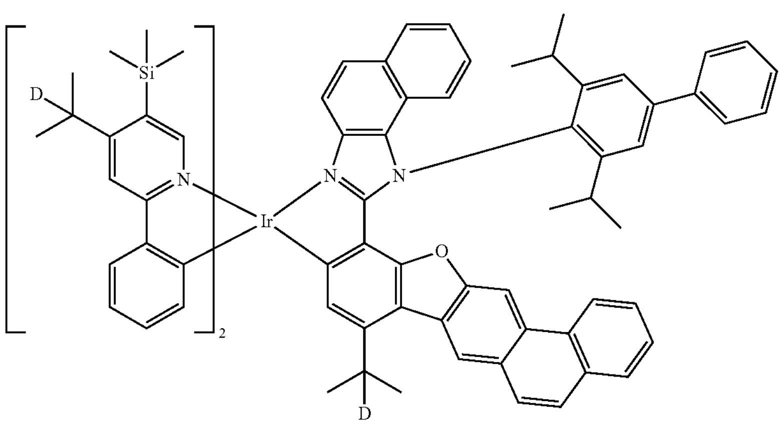

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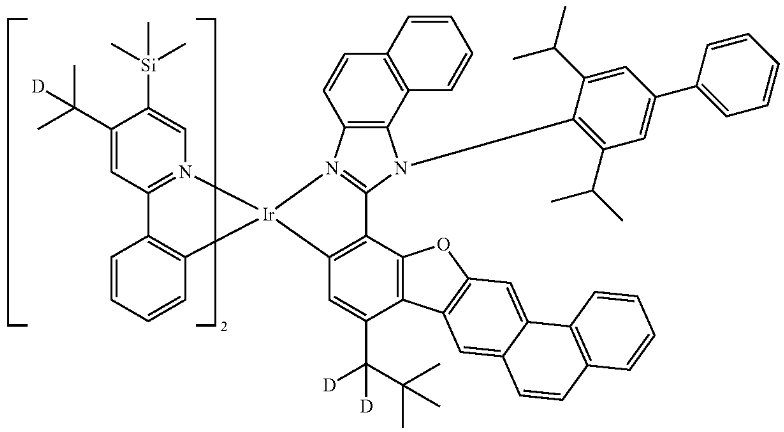

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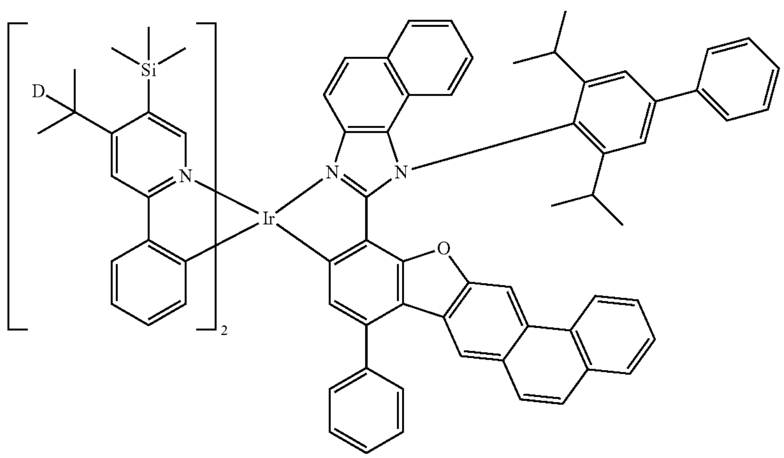

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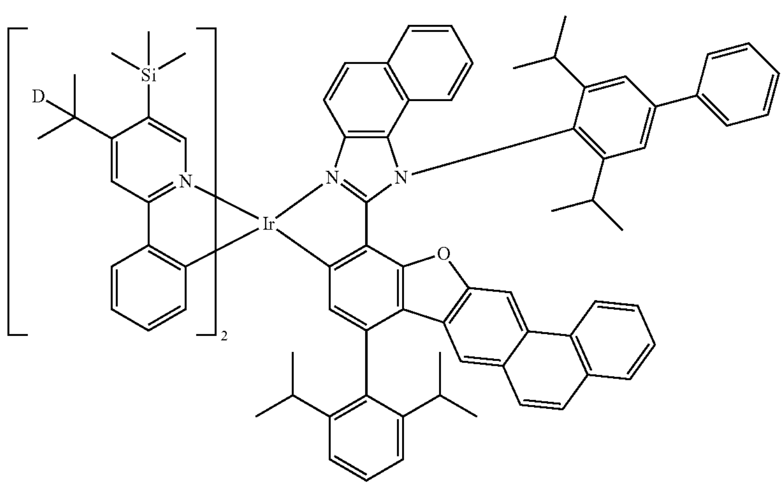

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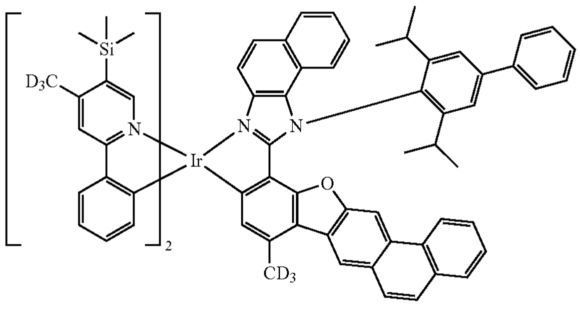

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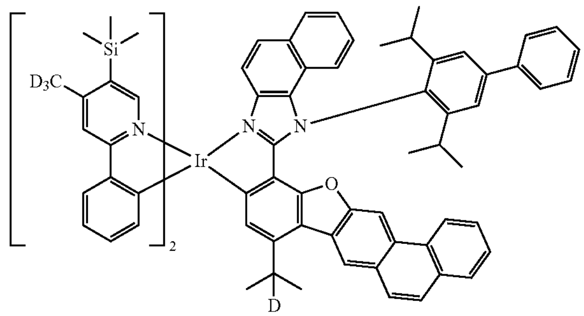

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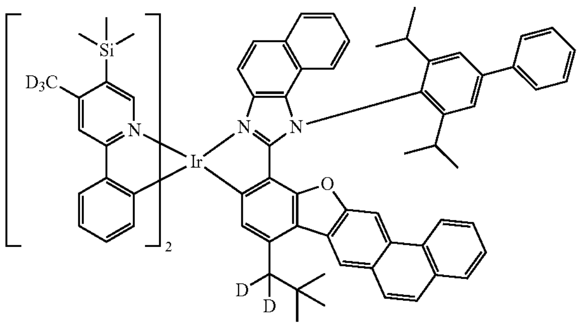

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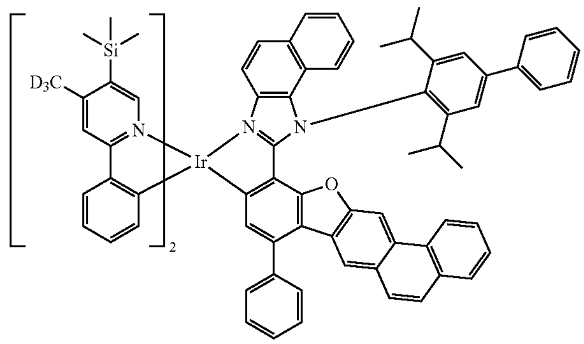

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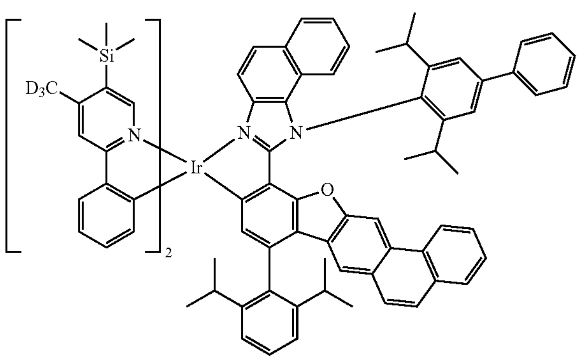

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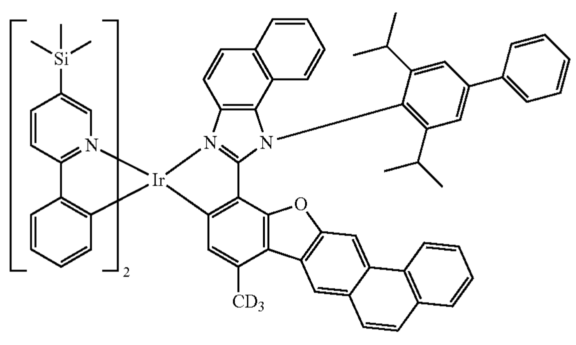

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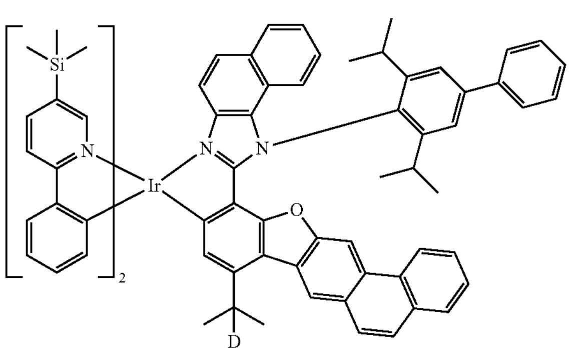

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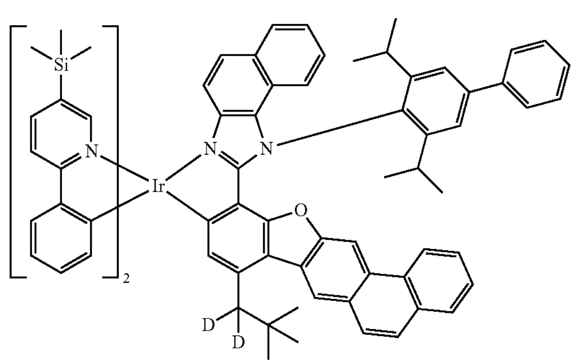

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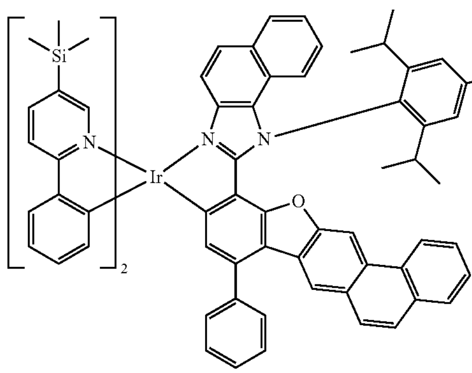

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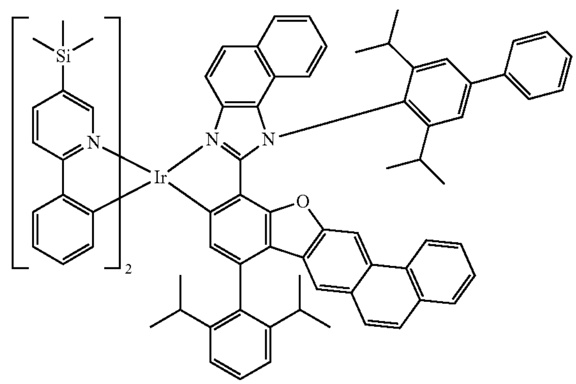

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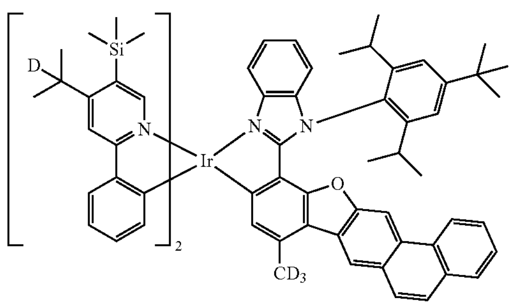

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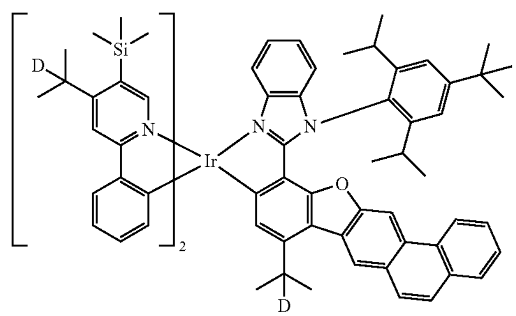

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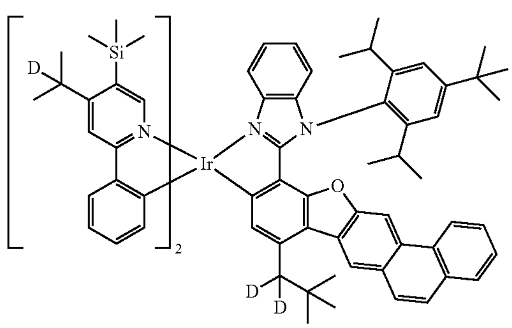

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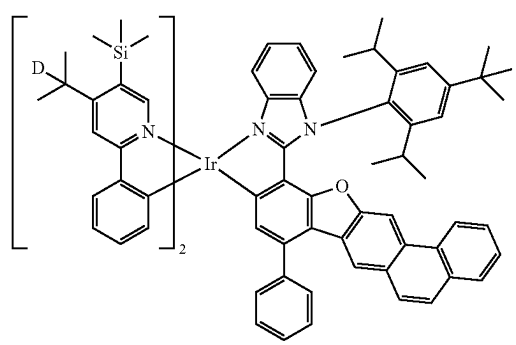

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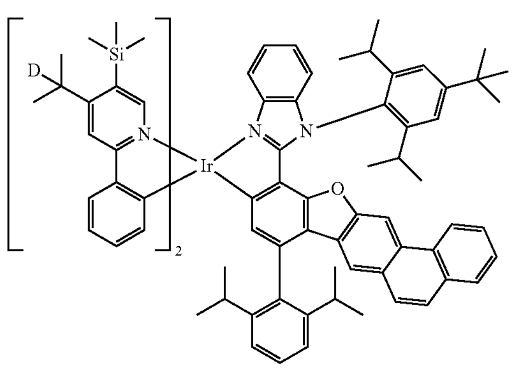

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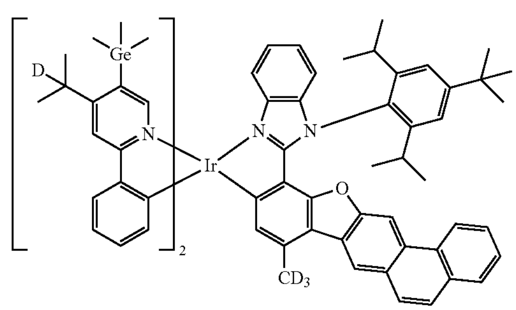

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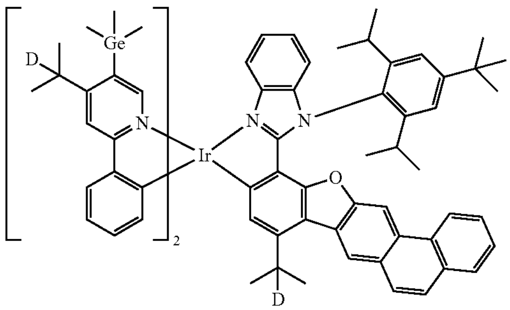

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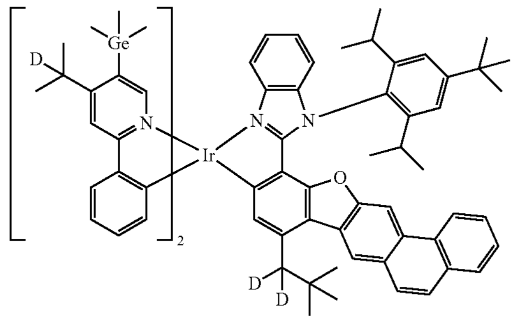

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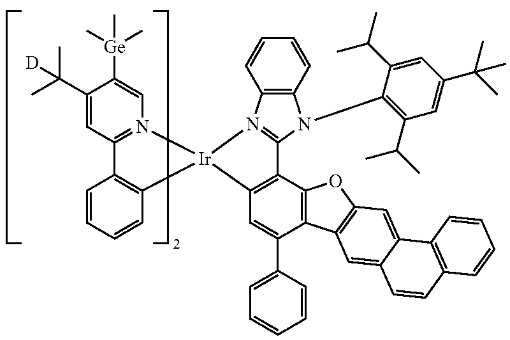

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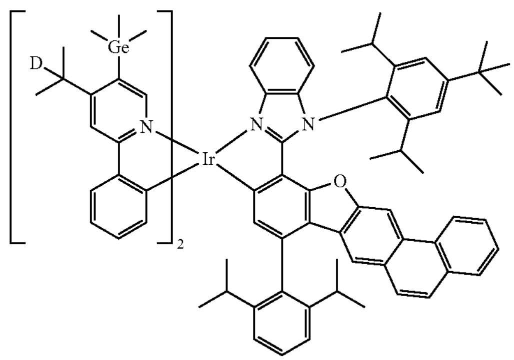

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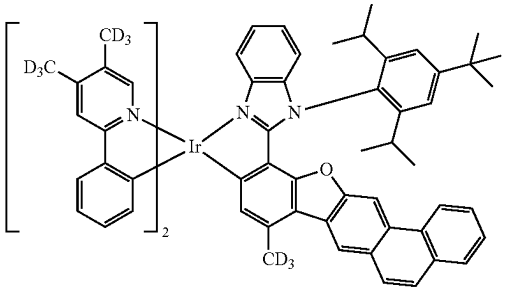

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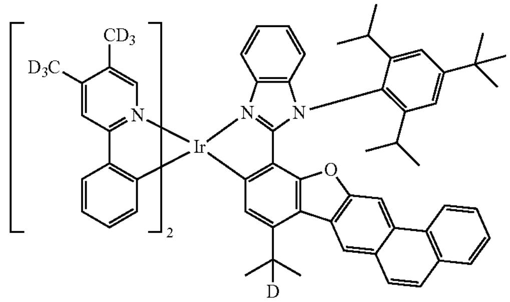

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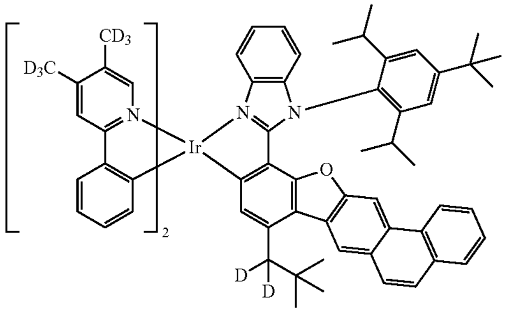

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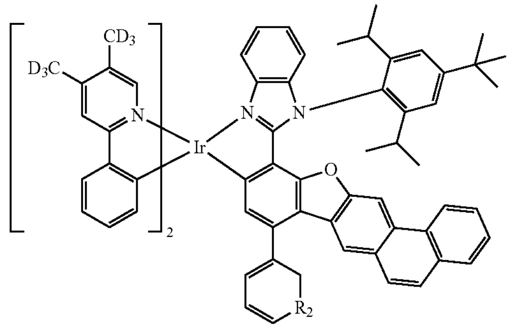

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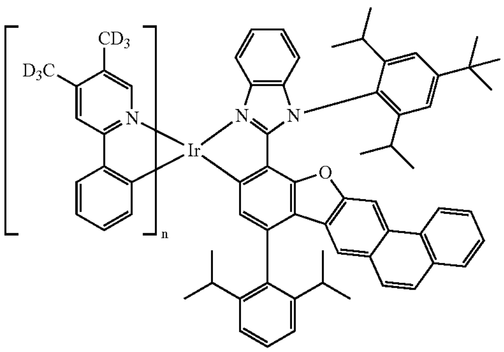

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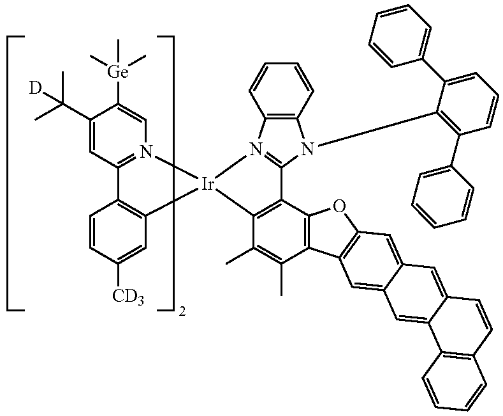

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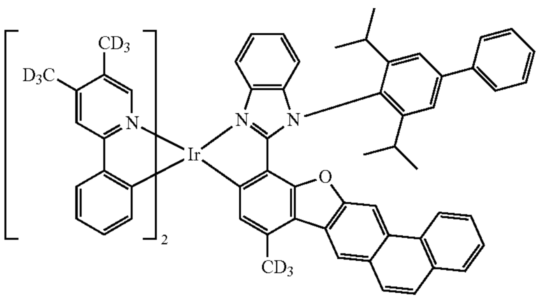

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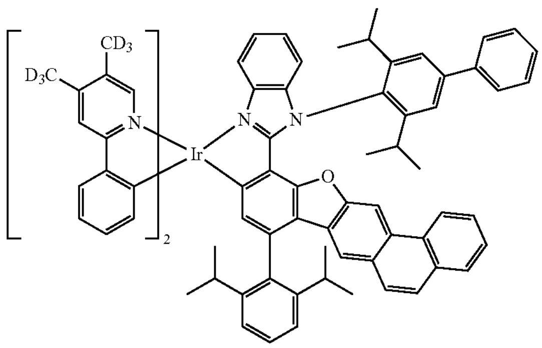


[0236] L_1 and L_2 in the organometallic compound represented by Formula 1 are a ligand represented by Formula 2-1 and a ligand represented by Formula 2-2, respectively, and n1 and n2, which are the number of L_1 and the number of L_2, respectively, may each independently be 1 or 2. In this regard, Formula 2-2 includes ring CY_{43} condensed to ring CY_{42} and includes at least one W_4 as defined herein. As a result, the organometallic compound represented by Formula 1 can emit light having a maximum emission wavelength shifted to a relatively shorter wavelength (for example, a green light having a maximum emission wavelength shifted to a relatively shorter wavelength), and at the same time, has excellent electrical stability and thermal stability. Accordingly, an electronic device, for example, a light-emitting device, including the same, can emit light having a maximum emission wavelength shifted to a relatively shorter wavelength (for example, a green light having a maximum emission wavelength shifted to a relatively shorter wavelength), and at the same time, external quantum efficiency and lifetime thereof can be improved.

[0237] For example, the highest occupied molecular orbital (HOMO) energy level of the organometallic compound represented by Formula 1 may be about −5.0 eV to about −4.0 eV, or about −4.8 eV to about −4.5 eV. In one or more embodiments, the lowest unoccupied molecular orbital (LUMO) energy level of the organometallic compound represented by Formula 1 may be about −1.7 eV to about −1.0 eV, or about −1.3 eV to about −1.05 eV. In one or more embodiments, the Ti energy level of the organometallic compound represented by Formula 1 may be about 2.0 eV to about 3.0 eV, or about 2.4 eV to about 2.5 eV. The HOMO energy level, LUMO energy level, and Ti energy level may be evaluated by using a simulation evaluation method based on density functional theory (DFT), for example, a simulation evaluation method using Gaussian 09 program involving molecular structure optimization by DFT based on B3LYP.

[0238] The HOMO energy level, LUMO energy level, and T_1 energy level of some of the organometallic compounds represented by Formula 1 were evaluated by using the Gaussian 09 program involving molecular structure optimization by DFT based on B3LYP. Results of the evaluation are shown in Table 1.

TABLE 1

Compound No.	HOMO (eV)	LUMO (eV)	T_1 (eV)
94	−4.7013	−1.1652	2.4739
98	−4.7511	−1.2093	2.4551
99	−4.6891	−1.1217	2.4732
103	−4.7386	−1.1796	2.4542
104	−4.6689	−1.1072	2.4665
108	−4.7291	−1.1666	2.4535

[0239] From Table 1, it was confirmed that the organometallic compound represented by Formula 1 has electric characteristics suitable for use as a dopant for an electronic device, for example, a light-emitting device.

[0240] Synthesis methods of the organometallic compound represented by Formula 1 may be recognizable by one of ordinary skill in the art by referring to Synthesis Examples provided below.

[0241] Accordingly, the organometallic compound represented by Formula 1 may be suitable for use as a dopant for an interlayer of a light-emitting device, for example, an emission layer of the interlayer. Thus, according to another aspect, provided is a light-emitting device including a first electrode, a second electrode, and an interlayer that is disposed between the first electrode and the second electrode, wherein the interlayer includes an emission layer, and wherein the interlayer further includes at least one of the organometallic compounds represented by Formula 1.

[0242] Due to the inclusion of the interlayer containing at least one of the organometallic compounds represented by Formula 1 as described herein, the light-emitting device can have improved external quantum efficiency and improved lifetime characteristics.

[0243] The organometallic compound represented by Formula 1 may be used between a pair of electrodes of a light-emitting device. For example, the organometallic compound represented by Formula 1 may be included in the emission layer. In this regard, the organometallic compound may act as a dopant, and the emission layer may further include a host (that is, an amount of the at least one organometallic compound represented by Formula 1 in the emission layer is less than an amount of the host in the emission layer, based on weight).

[0244] According to one or more embodiments, the emission layer (or a light-emitting device including the emission layer) may emit light having the maximum emission wavelength of about 510 nm to about 550 nm, about 510 nm to about 540 nm, about 510 nm to about 529 nm, about 515 nm to about 550 nm, about 515 nm to about 540 nm, about 515 nm to about 529 nm, about 520 nm to about 550 nm, about 520 nm to about 540 nm, about or 520 nm to about 529 nm. For example, the emission layer, or a light-emitting device including the emission layer, may emit light (for example, a green light) having a maximum emission wavelength of about 510 nm to about 529 nm, or about 520 nm to about 529 nm.

[0245] The expression "(an interlayer) includes at least one of organometallic compounds represented by Formula 1" as used herein may include a case in which "(an interlayer) includes identical organometallic compounds represented by Formula 1" and a case in which "(an interlayer) includes two or more different organometallic compounds represented by Formula 1."

[0246] In one or more embodiments, the interlayer may include, as the at least one organometallic compound represented by Formula 1, only Compound 1. In this regard, Compound 1 may be present in the emission layer of the light-emitting device. In one or more embodiments, the interlayer may include, as the at least one organometallic compound represented by Formula 1, Compound 1 and Compound 2. In this regard, Compound 1 and Compound 2 may exist in the same layer (for example, both Compound 1 and Compound 2 may exist in the emission layer).

[0247] The first electrode may be an anode, which is a hole injection electrode, and the second electrode may be a cathode, which is an electron injection electrode, or the first electrode may be a cathode, which is an electron injection electrode, and the second electrode may be an anode, which is a hole injection electrode.

[0248] For example, in the light-emitting device, the first electrode may be an anode, the second electrode may be a cathode, and the interlayer may further include: a hole transport region between the first electrode and the emission layer; and an electron transport region between the emission layer and the second electrode, wherein the hole transport region may include a hole injection layer, a hole transport layer, an electron-blocking layer, a buffer layer, or a combination thereof, and the electron transport region may include a hole-blocking layer, an electron transport layer, an electron injection layer, or a combination thereof.

[0249] The term "interlayer" as used herein refers to a single layer and/or a plurality of layers located between a first electrode and a second electrode of a light-emitting device. The "interlayer" may include, in addition to an organic compound, an organometallic complex including a metal.

[0250] The FIG. **1***s* a schematic cross-sectional view of an organic light-emitting device **10** of a light-emitting device according to one or more embodiments. Hereinafter, the structure and manufacturing method of the organic light-emitting device **10** according to one or more embodiments will be described with reference to the FIGURE. The organic light-emitting device **10** may have a structure in which a first electrode **11**, an interlayer **15**, and a second electrode **19** are sequentially stacked in the stated order.

[0251] A substrate may be further disposed under the first electrode **11** or on the second electrode **19**. The substrate may be a substrate commonly used in organic light-emitting devices, for example, a glass substrate or a transparent plastic substrate, which have excellent mechanical strength, thermal stability, transparency, surface smoothness, ease of handling, and/or water repellency.

[0252] The first electrode **11** may be formed by, for example, depositing or sputtering, onto the substrate, a material for forming the first electrode **11**. The first electrode **11** may be an anode. The material for forming the first electrode **11** may include materials with a high work function to facilitate hole injection. The first electrode **11** may be a reflective electrode, a transflective electrode, or a transmissive electrode. The material for forming the first electrode **11** may be indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO_2), or zinc oxide (ZnO). In one or more embodiments, the material for forming the first electrode **11** may be a metal, such as magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), or magnesium-silver (Mg—Ag).

[0253] The first electrode **11** may have a single-layered structure or a multi-layered structure including two or more layers. For example, the first electrode **11** may have a three-layered structure of ITO/Ag/ITO.

[0254] The interlayer **15** may be arranged on the first electrode **11**.

[0255] The interlayer **15** may include a hole transport region, an emission layer, and an electron transport region.

[0256] The hole transport region may be disposed between the first electrode **11** and the emission layer.

[0257] The hole transport region may include a hole injection layer (HIL), a hole transport layer, an electron-blocking layer, a buffer layer, or a combination thereof.

[0258] The hole transport region may include only either a HIL or a hole transport layer. In one or more embodiments, the hole transport region may have a HIL/hole transport layer structure or a HIL/hole transport layer/electron-blocking layer structure, in which respective layers of each structure are sequentially stacked in the stated order from the first electrode **11**.

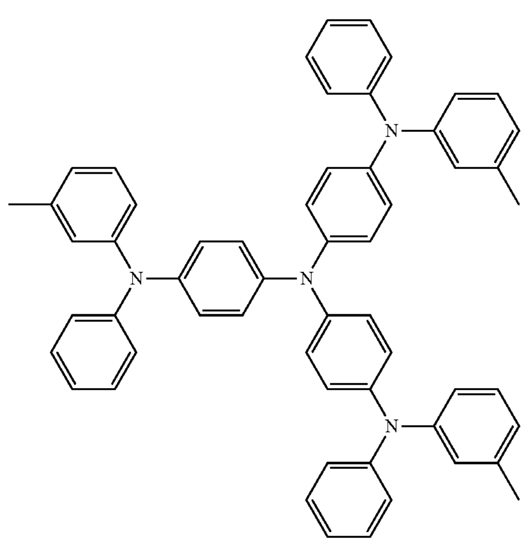
[0259] When the hole transport region includes a HIL, the HIL may be formed on the first electrode **11** by using various methods, for example, vacuum deposition, spin coating, casting, and/or Langmuir-Blodgett (LB) deposition, but embodiments are not limited thereto.

[0260] When a HIL is formed by vacuum deposition, the deposition conditions may vary depending on a material that is used to form the HIL, and the structure and thermal characteristics of the HIL. For example, the deposition conditions may include a deposition temperature of about 100° C. to about 500° C., a vacuum pressure of about 10^{-8} torr to about 10^{-3} torr, and a deposition rate of about 0.01 angstrom per second (Å/sec) to about 100 Å/sec.

[0261] When the HIL is formed by spin coating, the coating conditions may vary depending on a material for forming the HIL, and the structure and thermal characteristics of the HIL. For example, the coating conditions may include a coating speed of about 2,000 revolutions per minute (rpm) to about 5,000 rpm and a heat treatment temperature of about 80° C. to about 200° C. for removing a solvent after coating, but embodiments are not limited thereto.

[0262] The conditions for forming the hole transport layer and the electron-blocking layer may be referred to the description provided for the conditions for forming the HIL.

[0263] The hole transport region may be 4,4',4"-tris(3-methylphenylphenylamino)triphenylamine (m-MTDATA), 4,4',4"-tris(N,N-diphenylamino)triphenylamine (TDATA), 4,4',4"-tris{N-(2-naphthyl)-N-phenylamino}-triphenylamine (2-TNATA), N,N'-di(1-naphthyl)-N,N'-diphenylbenzidine (NPB), p-NPB, N,N'-bis(3-methylphenyl)-N,N'-diphenyl-[1,1-biphenyl]-4,4'-diamine (TPD), Spiro-TPD, Spiro-NPB, methylated NPB, 4,4'-cyclohexylidene bis[N,N-bis(4-methylphenyl)benzenamine](TAPC), 4,4'-bis[N,N'-(3-tolyl)amino]-3,3'-dimethylbiphenyl (HMTPD), 4,4',4"-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), a compound represented by Formula 201, a compound represented by Formula 202, or a combination thereof, but embodiments are not limited thereto:



m-MTDATA

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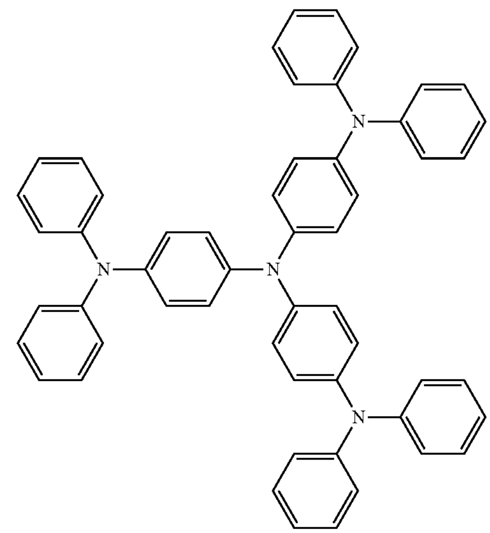
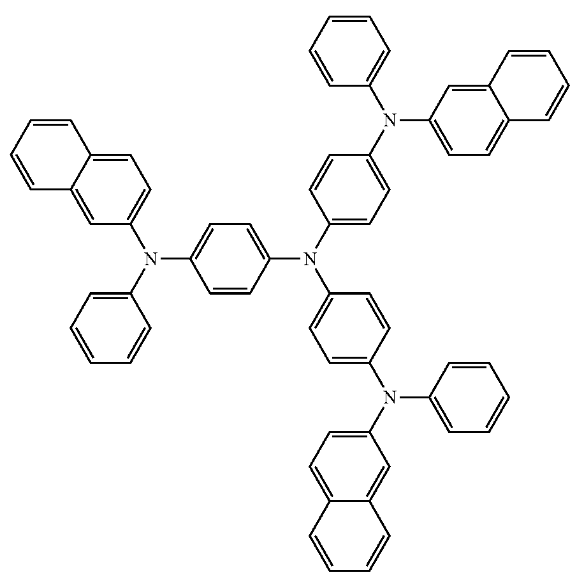
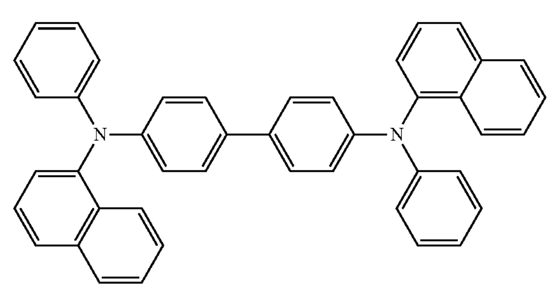


TDATA



2-TNATA



NPB

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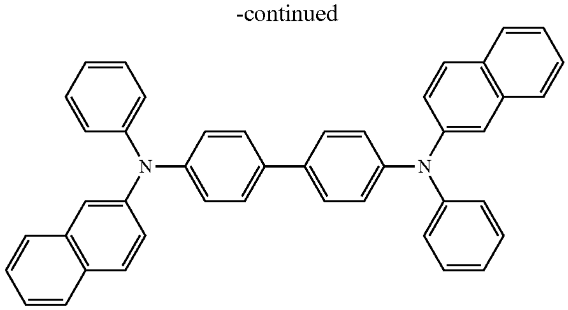
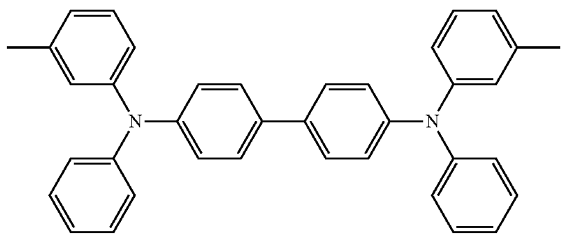
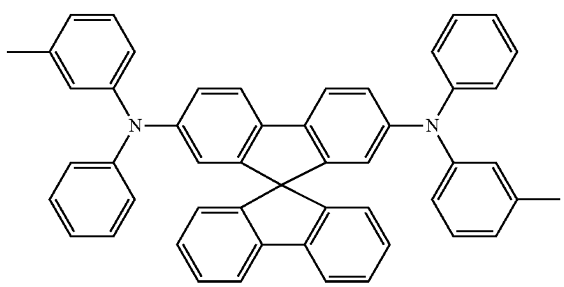


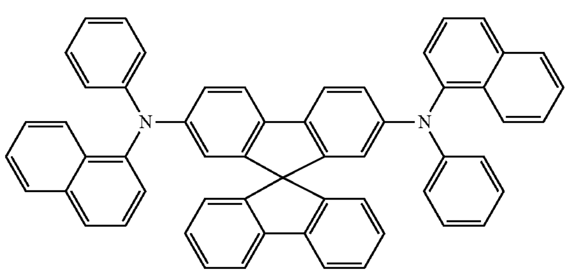
β-NPB



TPD



Spiro-TPD



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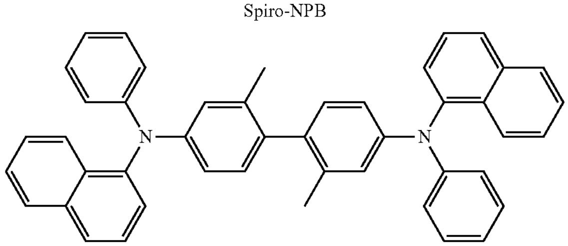

methylated NPB

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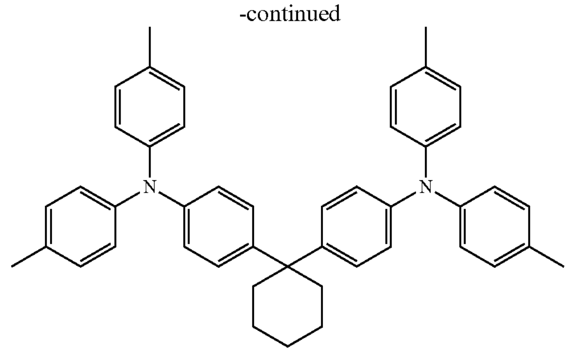
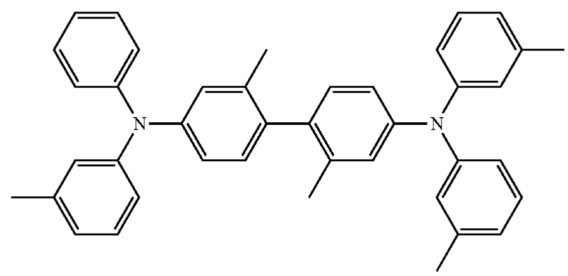
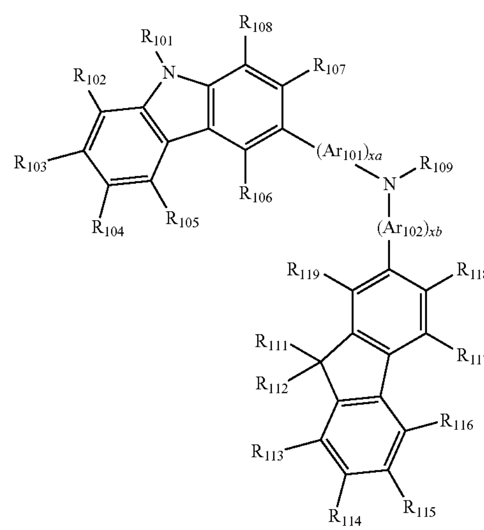


TAPC

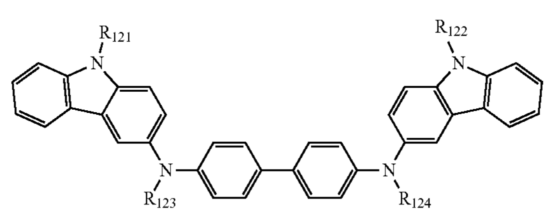


HMTPD

Formula 201



Formula 202



[0264] Ar_{101} and Ar_{102} in Formula 201 may each independently be a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an acenaphthylene group, a fluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, or a pentacenylene group, each unsubstituted or substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy group, a C_1-C_{60} alkylthio group, a C_3-C_{10} cycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{60} heterocycloalkyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_7-C_{60} aryl alkyl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a C_2-C_{60} alkyl heteroaryl group, a C_2-C_{60} heteroaryl alkyl group, a C_1-C_{60} heteroaryloxy group, a C_1-C_{60} heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, or a combination thereof.

[0265] xa and xb in Formula 201 may each independently be an integer from 0 to 5, or 0, 1, or 2. For example, xa may be 1 and xb may be 0.

[0266] R_{101} to R_{108}, R_{111} to R_{119}, and R_{121} to R_{124} in Formulae 201 and 202 may each independently be:

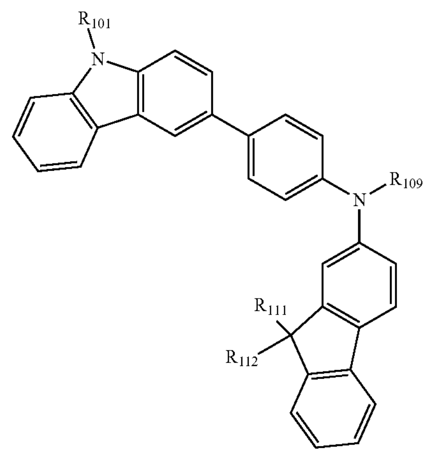
[0267] hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{10} alkyl group (for example, a methyl group, an ethyl group, a propyl group, a butyl group, a pentyl group, a hexyl group, or the like), a C_1-C_{10} alkoxy group (for example, a methoxy group, an ethoxy group, a propoxy group, a butoxy group, a pentoxy group, or the like), or a C_1-C_{10} alkylthio group;

[0268] a C_1-C_{10} alkyl group, a C_1-C_{10} alkoxy group, or a C_1-C_{10} alkylthio group each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, or a combination thereof; or a phenyl group, a naphthyl group, an anthracenyl group, a fluorenyl group, or a pyrenyl group, each unsubstituted or substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amnio group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{10} alkyl group, a C_1-C_{10} alkoxy group, a C_1-C_{10} alkylthio group, or a combination thereof.

[0269] R_{109} in Formula 201 may be a phenyl group, a naphthyl group, an anthracenyl group, or a pyridinyl group, each unsubstituted or substituted deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{20} alkyl group, a C_1-C_{20} alkoxy group, a C_1-C_{20} alkylthio group, a phenyl group, a naphthyl group, an anthracenyl group, a pyridinyl group, or a combination thereof.

[0270] In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A:

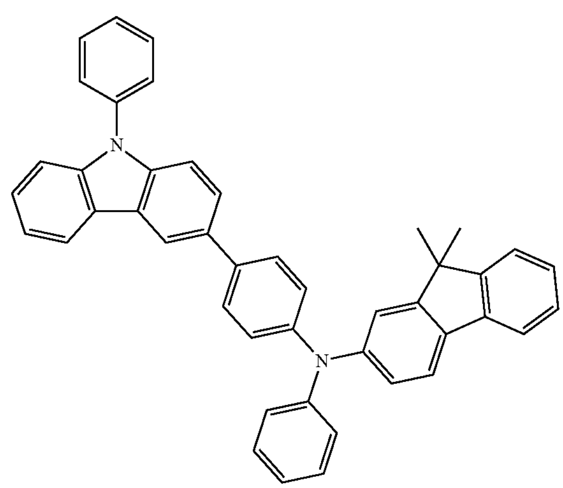
Formula 201A



[0271] R_{101}, R_{111}, R_{112}, and R_{109} in Formula 201A may each be as described herein.

[0272] For example, the compound represented by Formula 201 and the compound represented by Formula 202 may include one or more of Compounds HT1 to HT20:

HT1



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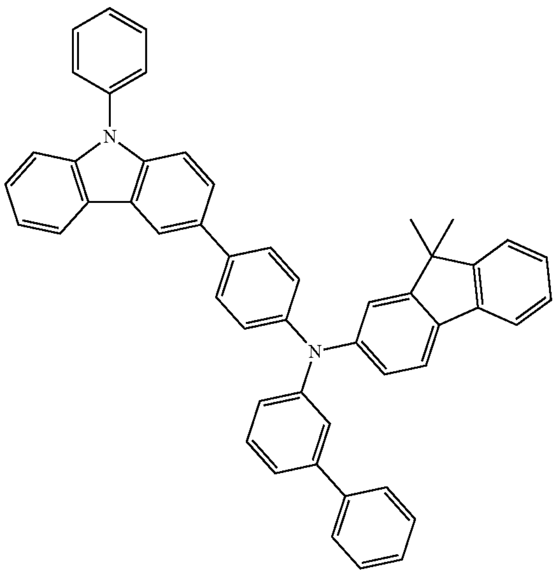
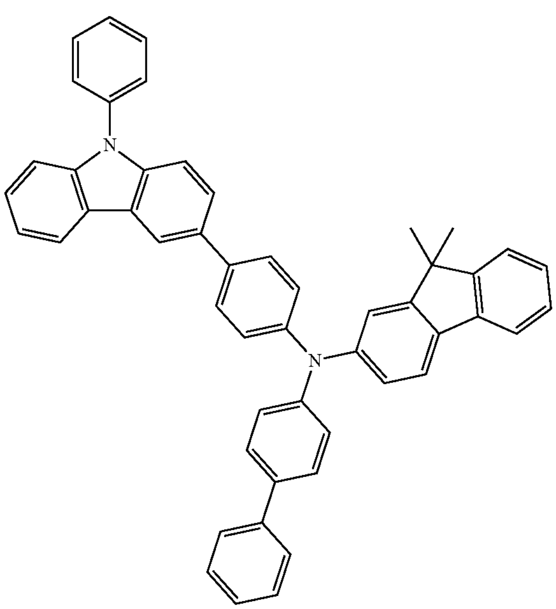
HT2



HT3



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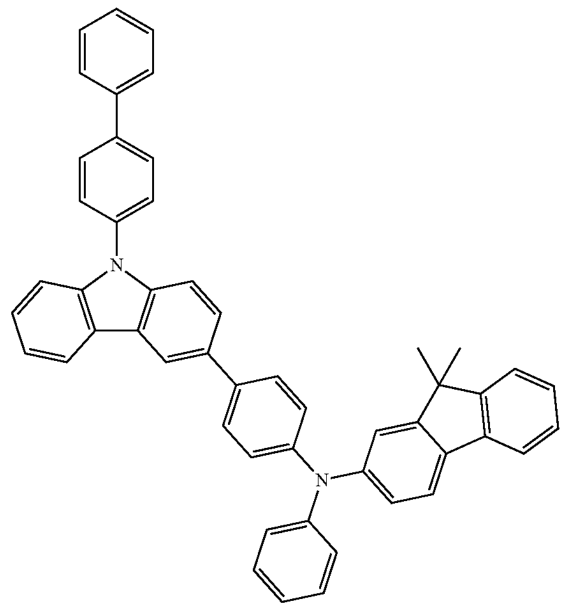
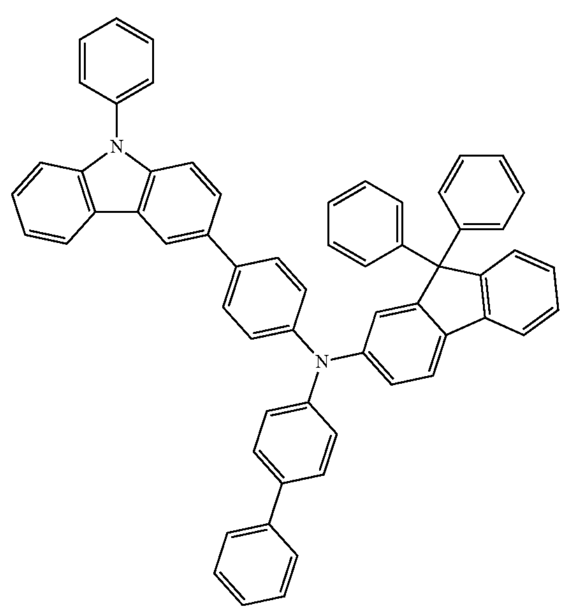
HT4

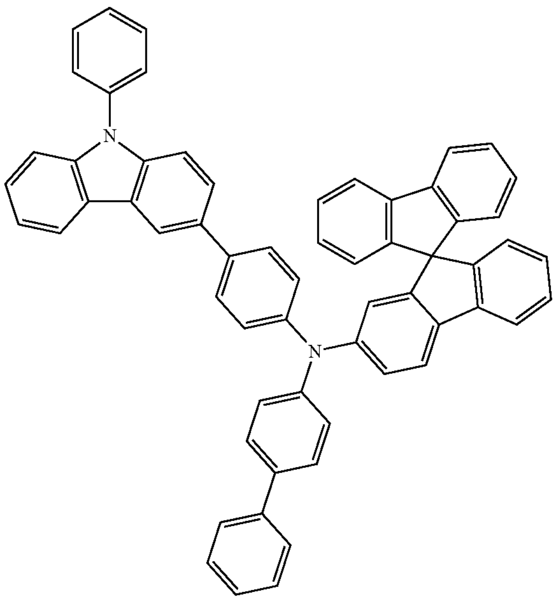


HT5

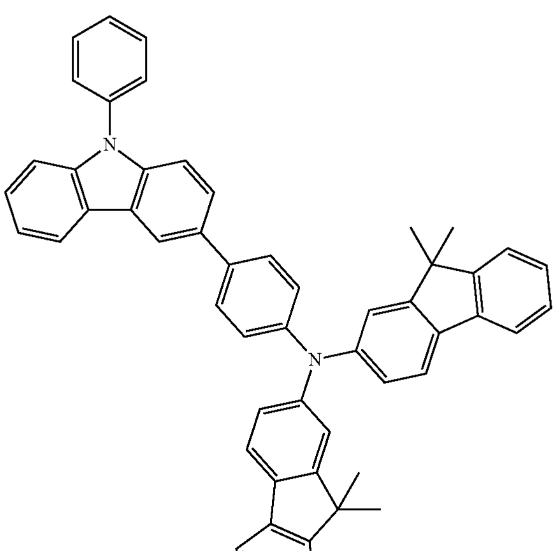


-continued

HT6



HT7



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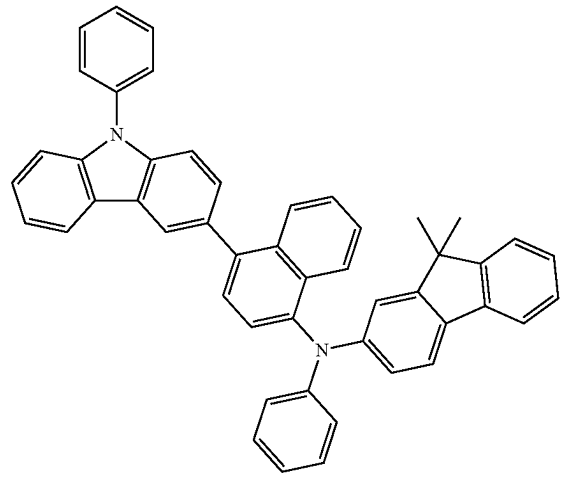
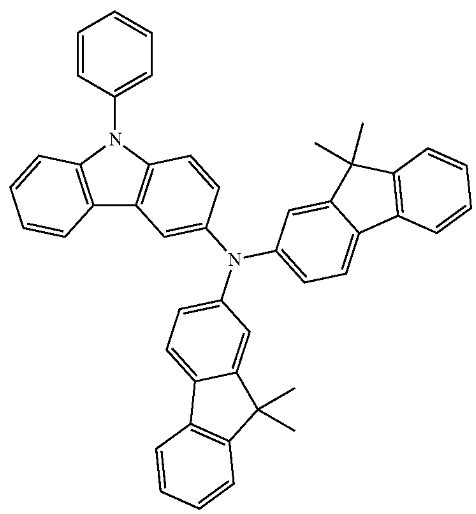
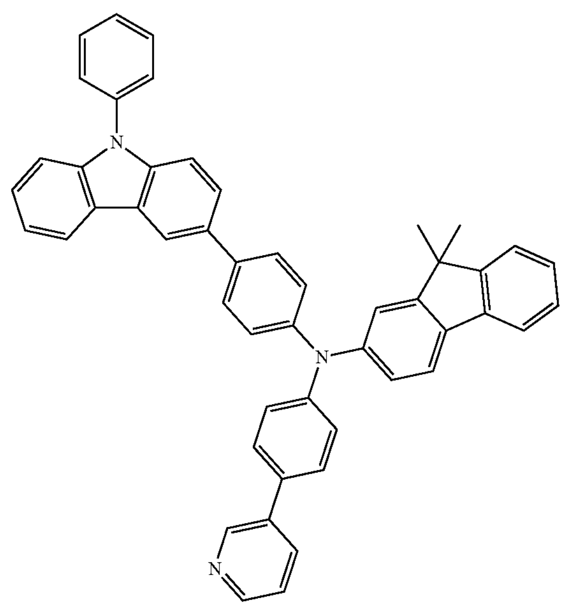
HT8



HT9



HT10



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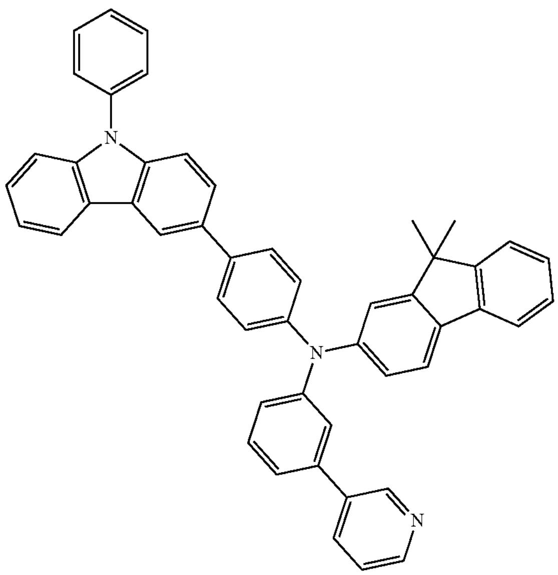
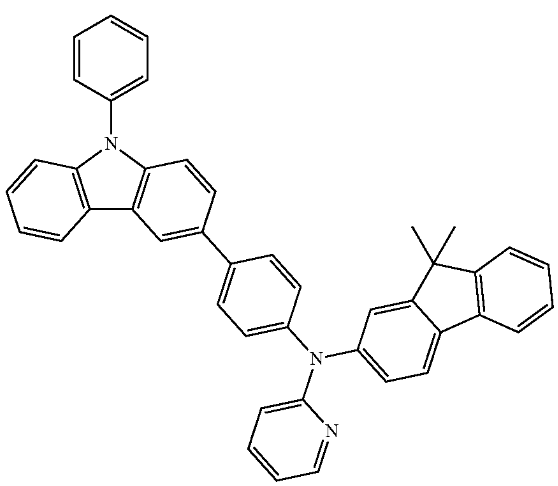
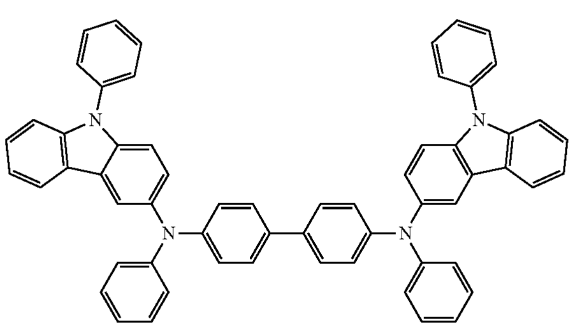
HT11



HT12



HT13



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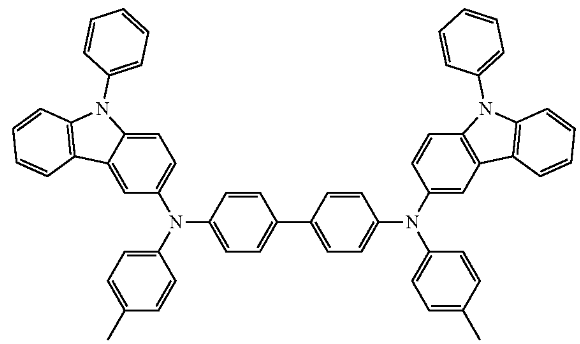
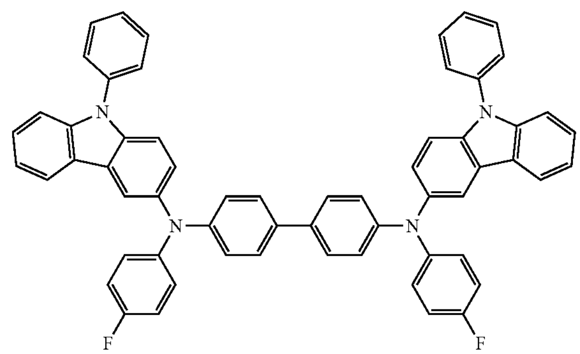
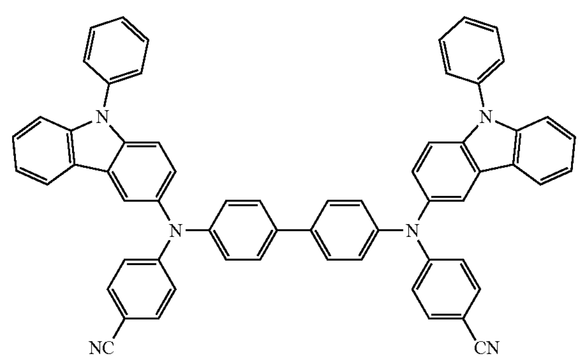
HT14



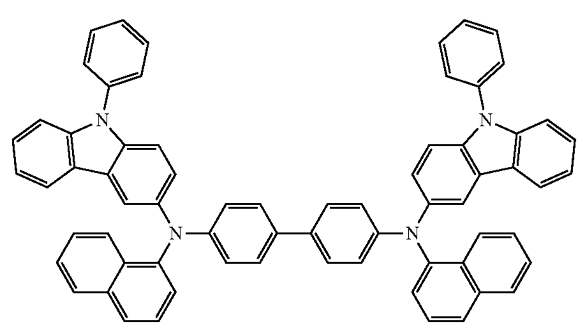
HT15



HT16



HT17



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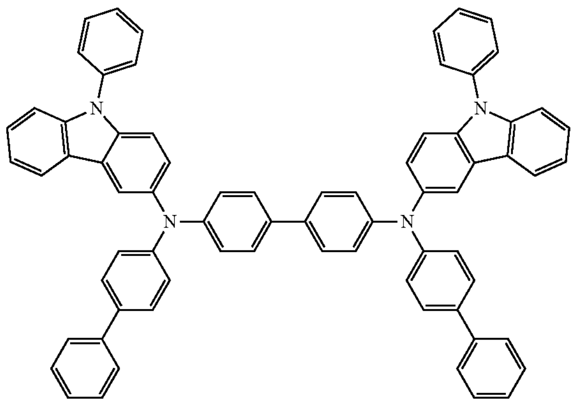
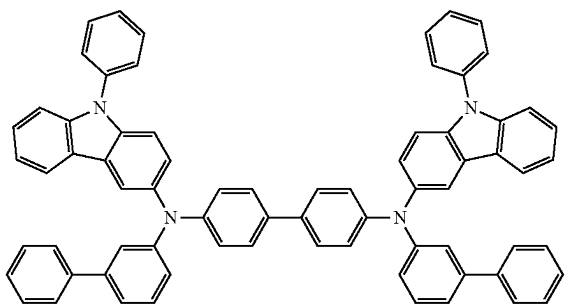
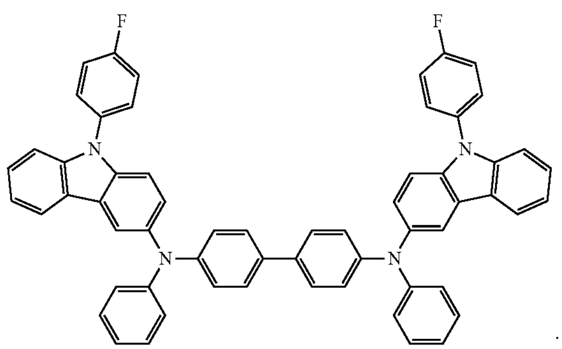
HT18



HT19



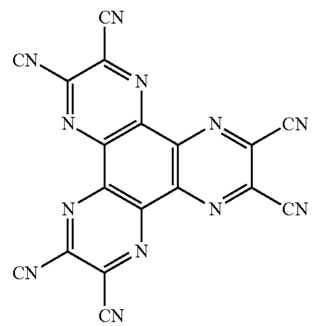
HT20



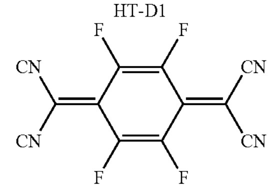
[0273] A thickness of the hole transport region may be about 100 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å. When the hole transport region includes at least one of a HIL and a hole transport layer, the thickness of the HIL may be about 50 Å to about 5000 Å, for example, about 50 Å to about 1000 Å, and the thickness of the hole transport layer may be about 50 Å to about 3000 Å, for example, about 100 Å to about 2000 Å. When the thicknesses of the hole transport region, the HIL, and the hole transport layer are within these ranges, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

[0274] The hole transport region may further include, in addition to the above-described materials, a charge-generation material for improving conductivity. The charge-generation material may be homogeneously or non-homogeneously dispersed in the hole transport region.

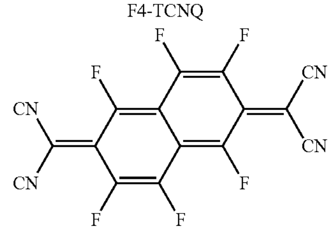
[0275] The charge-generation material may be, for example, a p-dopant. The p-dopant may be one of a quinone derivative, a metal oxide, or a cyano group-containing compound. For example, non-limiting examples of the p-dopant include a quinone derivative, such as tetracyanoquinodimethane (TCNQ), 2,3,5,6-tetrafluoro-tetracyano-1,4-benzoquinonedimethane (F4-TCNQ), 1,3,4,5,7,8-hexafluorotetracyanonaphthoquinodimethane (F6-TCNNQ), HT-D2, or the like; a metal oxide, such as a tungsten oxide, a molybdenum oxide, or the like; or a cyano group-containing compound, such as Compound HT-D1, but embodiments are not limited thereto:



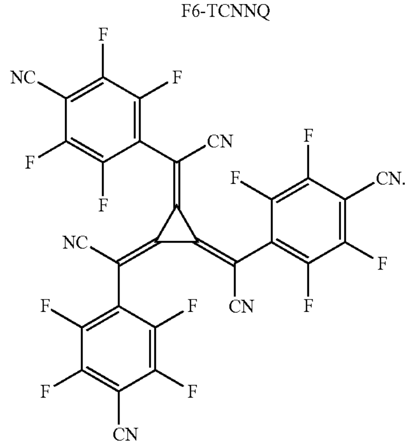
HT-D1



F4-TCNQ



F6-TCNNQ



HT-D2

[0276] The hole transport region may include a buffer layer.

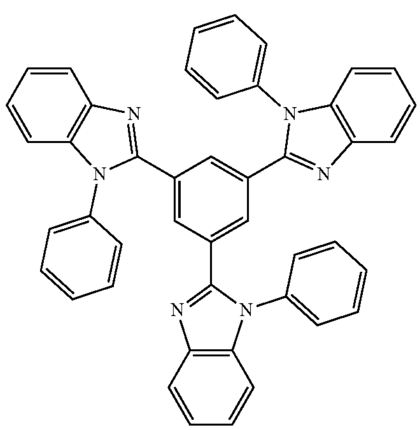
[0277] The buffer layer may compensate for an optical resonance distance according to a wavelength of light emitted from the emission layer, and thus, efficiency of a formed organic light-emitting device may be improved.

[0278] Meanwhile, when the hole transport region includes an electron-blocking layer, a material for forming the electron-blocking layer may include a material that is used in the hole transport region as described herein, a host material described herein, or a combination thereof. For example, when the hole transport region includes an electron-blocking layer, a material for forming the electron-blocking layer may be mCP, H—H1, or the like which are described herein.

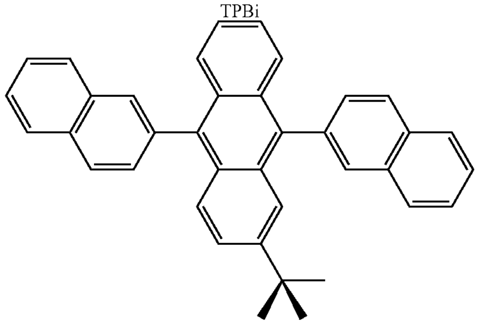
[0279] The emission layer (EML) may be formed on the hole transport region by vacuum deposition, spin coating, casting, LB deposition, or the like. When the emission layer is formed by vacuum deposition or spin coating, the deposition or coating conditions may generally be similar to those applied in forming the HIL although the deposition or coating conditions may vary according to a material that is used to form the emission layer.

[0280] The emission layer may include a host and a dopant, and the dopant may include at least one of the organometallic compounds represented by Formula 1.

[0281] The hosts may include at least one of 1,3,5-tri(1-phenyl-1H-benzo[d]imidazol-2-yl)benzene (TPBi), 3-tert-butyl-9,10-di(naphth-2-yl)anthracene (TBADN), 9,10-di (naphthalene-2-yl)anthracene (ADN) (also referred to as "DNA"), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 4,4'-bis(9-carbazolyl)-2,2'-dimethyl-biphenyl (CDBP), 1,3,5-tris (carbazole-9-yl)benzene (TCP), 1,3-bis(N-carbazolyl)benzene (mCP), Compound H50, Compound H51, Compound H52, Compound H—H1, Compound H-E43, Compound GH3, or a combination thereof, but embodiments are not limited thereto:



TPBi



TBADN

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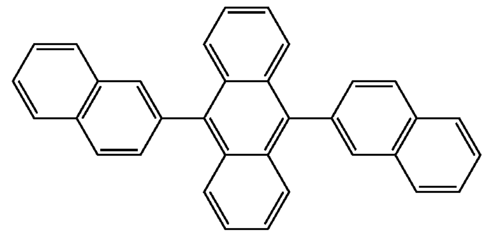
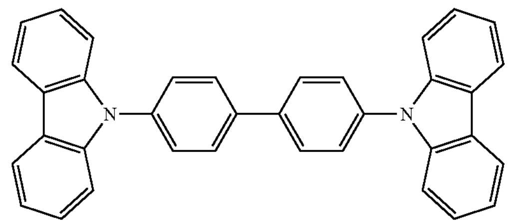
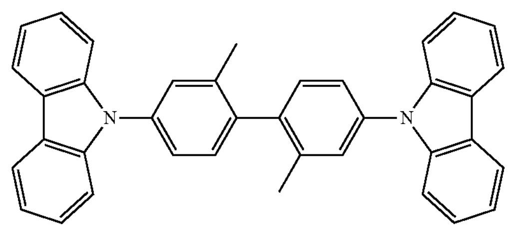


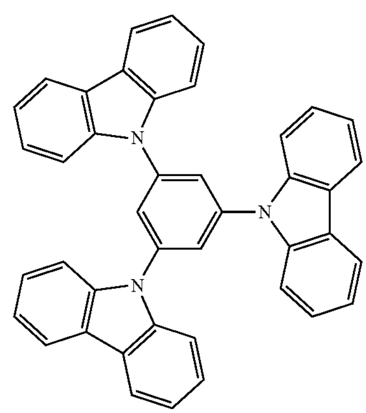
ADN



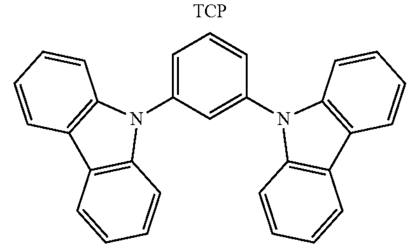
CBP



CDBP



TCP



mCP

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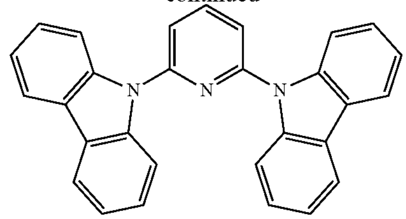
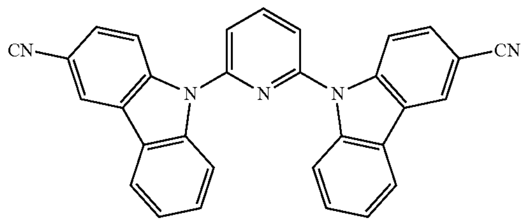
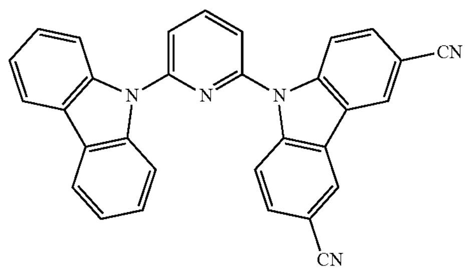


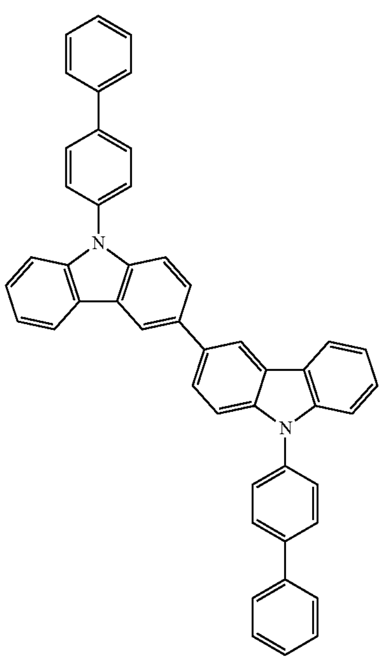
H50



H51



H52



H-H1

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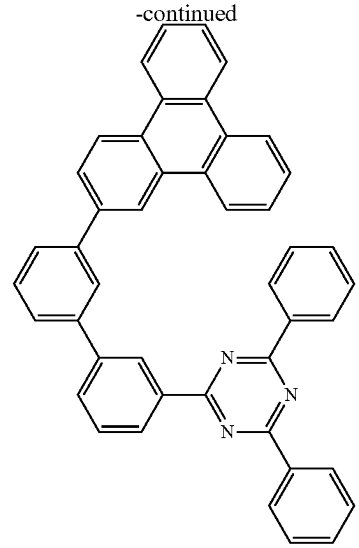
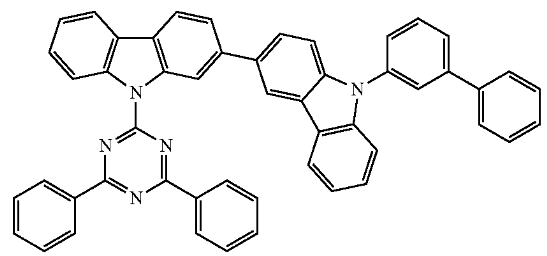


H-E43



GH3

[0282] When the organic light-emitting device is a full-color organic light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, and/or a blue emission layer. In one or more embodiments, the emission layer may have a structure in which a red emission layer, a green emission layer, and/or a blue emission layer are stacked, and thus, various modifications such as emission of white light are possible.

[0283] When the emission layer includes a host and a dopant, the amount (for example, by weight) of the dopant may typically be about 0.01 parts by weight to about 20 parts by weight based on about 100 parts by weight of the emission layer.

[0284] The thickness of the emission layer may be about 100 Å to about 1,000 Å, for example, about 200 Å to about 600 Å. When the thickness of the emission layer is within the range described above, excellent luminescence characteristics may be obtained without a substantial increase in driving voltage.

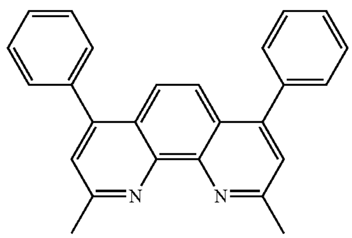
[0285] Next, an electron transport region may be placed on the emission layer.

[0286] The electron transport region may include a hole-blocking layer, an electron transport layer, an EIL, or a combination thereof.

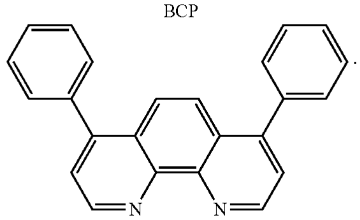
[0287] For example, the electron transport region may have a hole-blocking layer/electron transport layer/electron injection layer structure or an electron transport layer/electron injection layer structure. The electron transport layer may have a single-layered structure or a multi-layered structure including two or more different materials.

[0288] Conditions for forming the hole-blocking layer, the electron transport layer, and the electron injection layer which constitute the electron transport region may be referred to the description provided for the conditions for forming the HIL.

[0289] When the electron transport region includes a hole-blocking layer, the hole-blocking layer may include, for example, at least one of 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), bis(2-methyl-8-quinolinolato-N1,O8)-(1,1'-biphenyl-4-olato)aluminum (BAlq), or a combination thereof, but embodiments are not limited thereto:



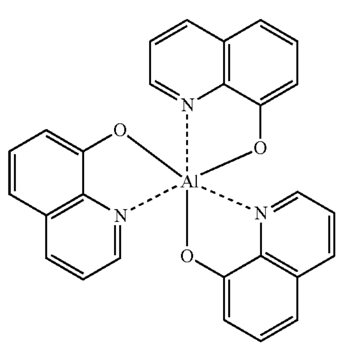
BCP



Bphen

[0290] A thickness of the hole-blocking layer may be about 20 Å to about 1,000 Å, for example, about 30 Å to about 300 Å. When the thickness of the hole-blocking layer is within these ranges, excellent hole-blocking characteristics may be obtained without a substantial increase in driving voltage.

[0291] The electron transport layer may include at least one of 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), tris(8-hydroxy-quinolinato)aluminum (Alq_3), bis(2-methyl-8-quinolinolato-N1,08)-(1,1'-biphenyl-4-olato)aluminum (BAlq), 3-(4-biphenylyl)-4-phenyl-5-tert-butylphenyl-1,2,4-triazole (TAZ), 4-(naphthalen-1-yl)-3,5-diphenyl-4H-1,2,4-triazole (NTAZ), or a combination thereof, but embodiments are not limited thereto:



Alq_3

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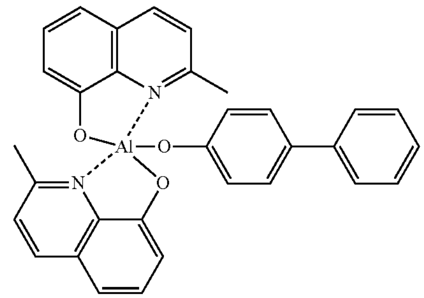
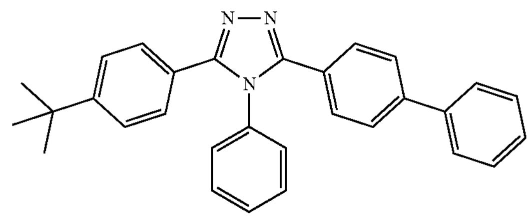
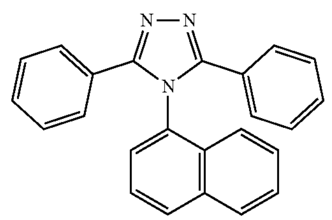


BAlq



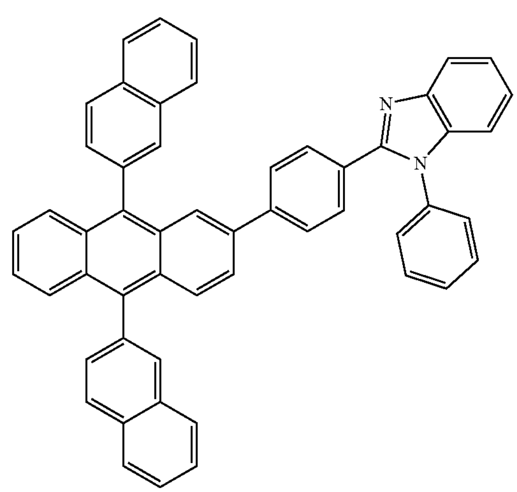
TAZ



NTAZ

[0292] In one or more embodiments, the electron transport layer may include one of Compounds ET1 to ET25, or a combination thereof, but embodiments are not limited thereto:

ET1



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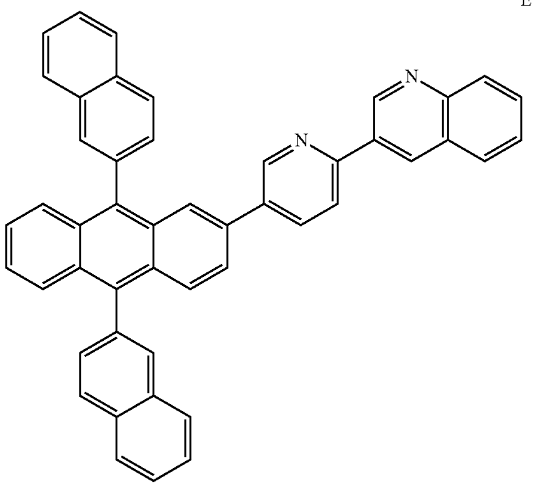
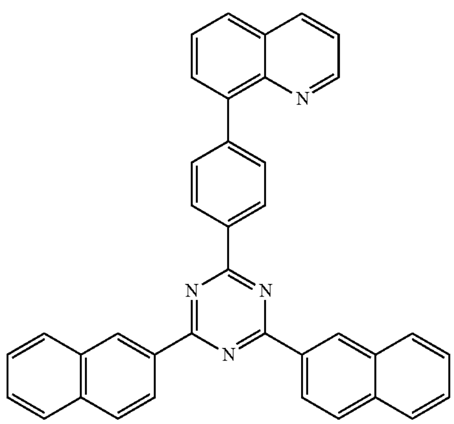
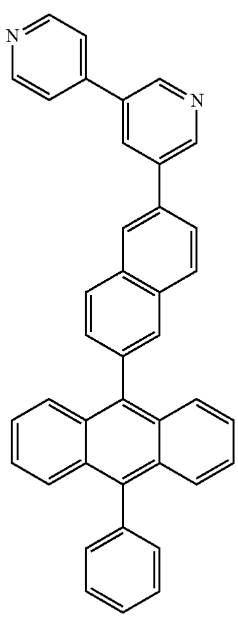
ET2



ET3



ET4



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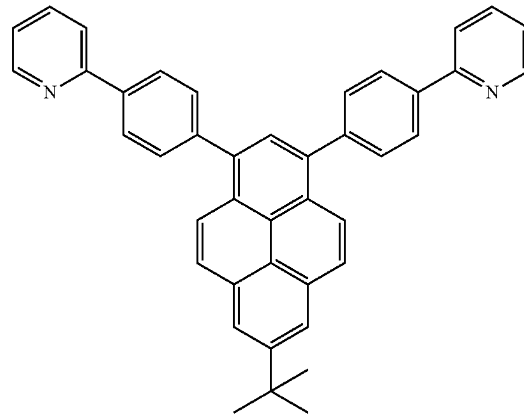
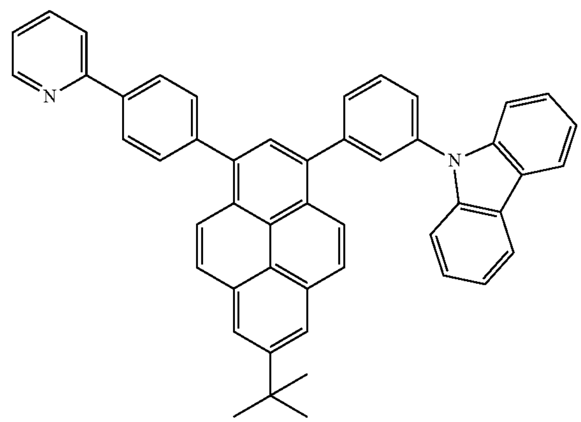
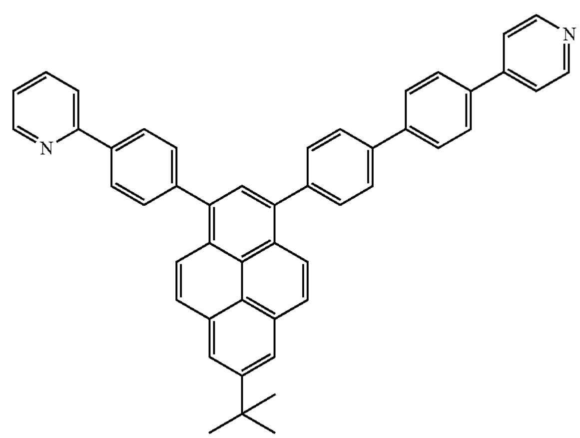
ET5



ET6



ET7



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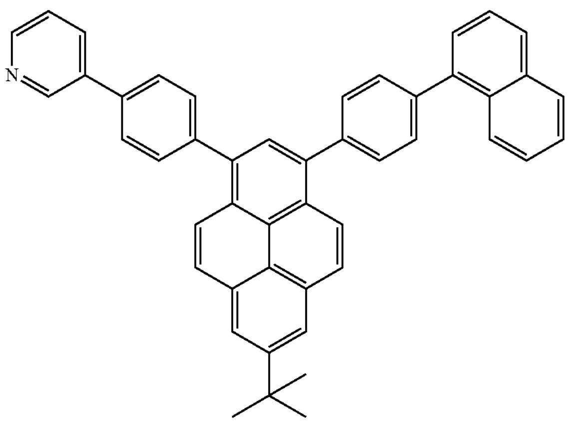
ET8



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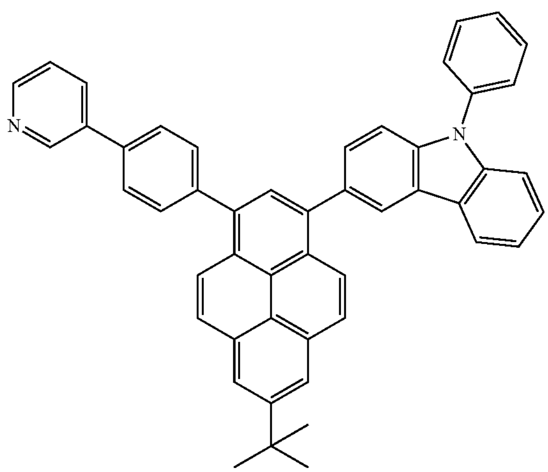

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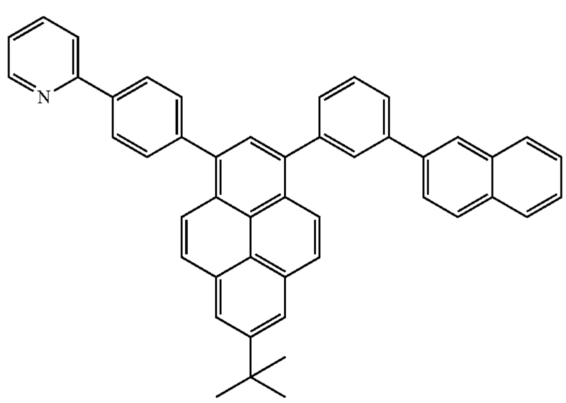

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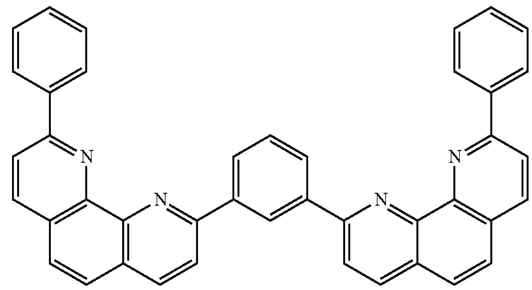

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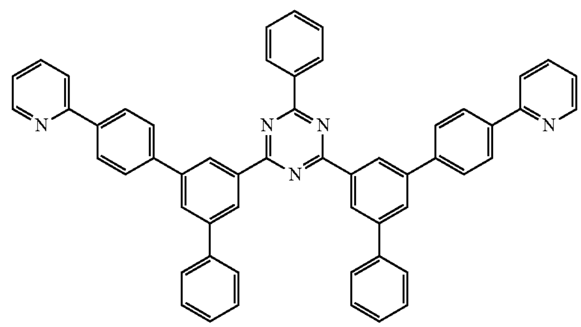

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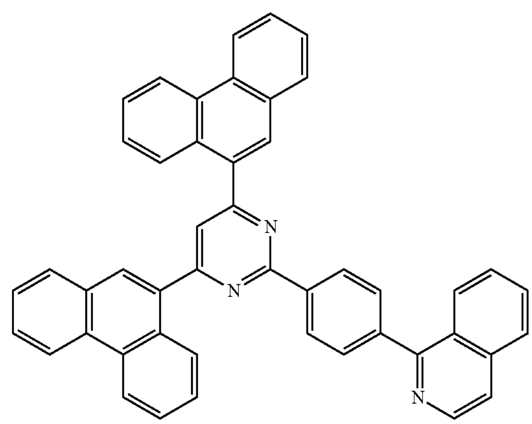

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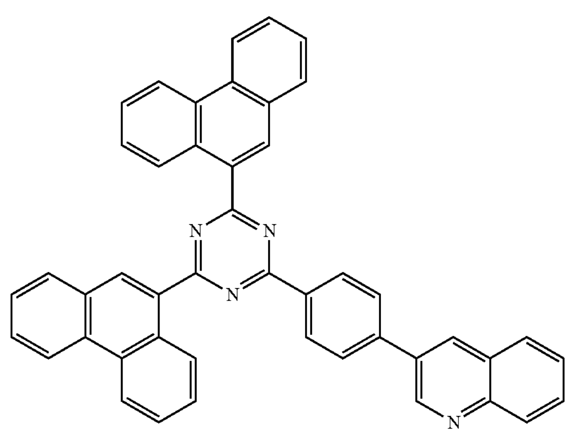

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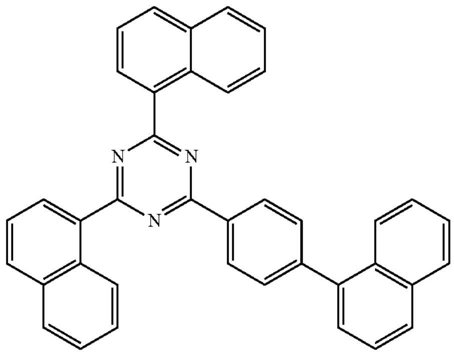

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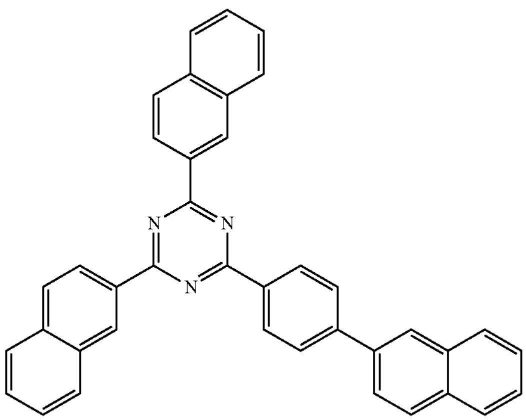

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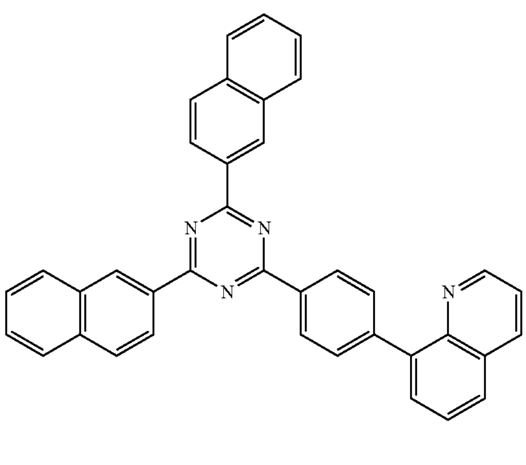

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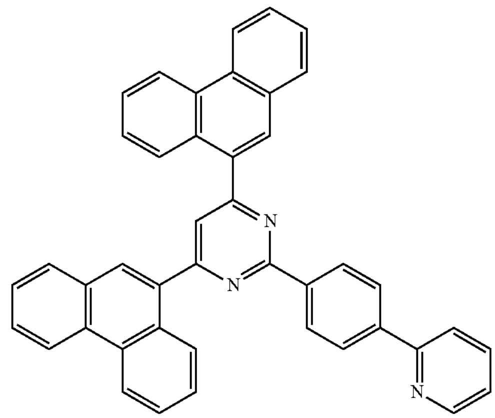

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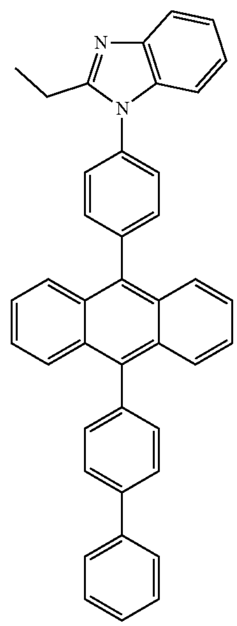

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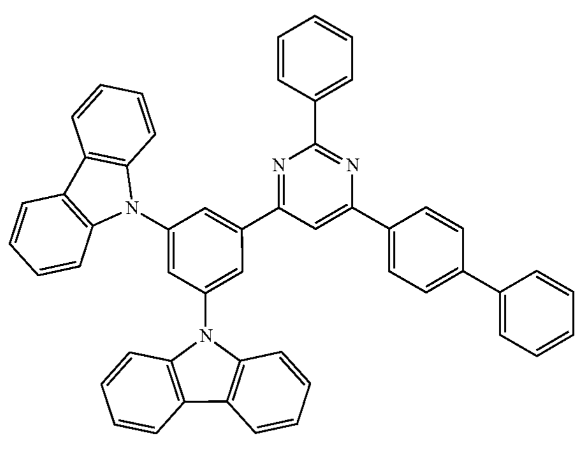

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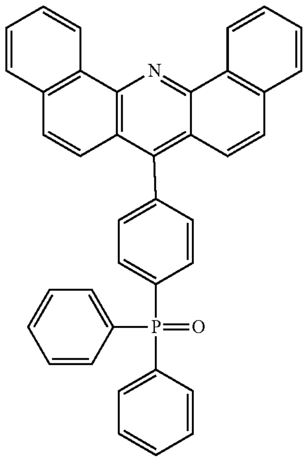

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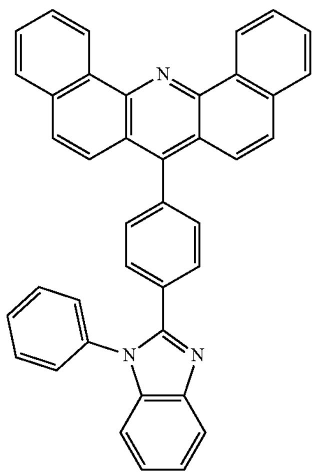

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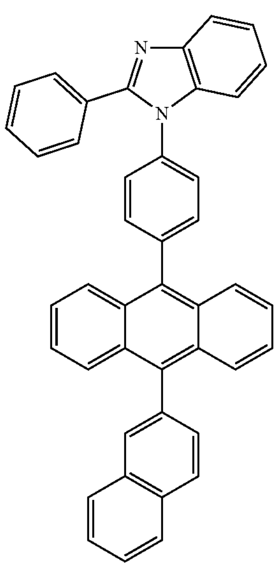

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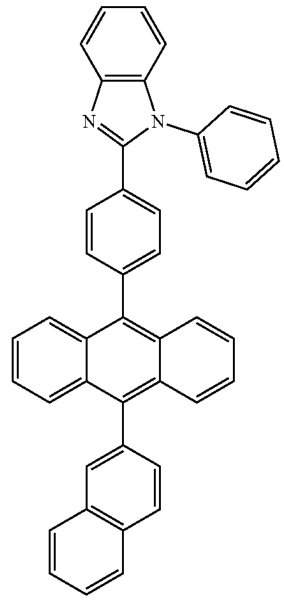
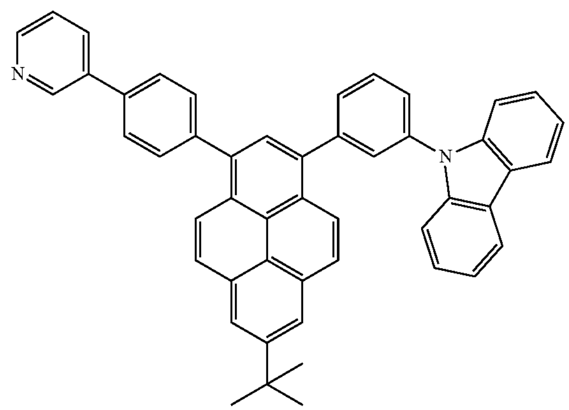
ET24



ET25

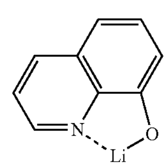


[0293] The thickness of the electron transport layer may be about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thickness of the electron transport layer is within these ranges, satisfactory electron transporting characteristics may be obtained without a substantial increase in driving voltage.

[0294] The electron transport layer may further include a metal-containing material, in addition to the material as described herein.

[0295] The metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (LiQ) or ET-D2, but embodiments are not limited thereto:

ET-D1



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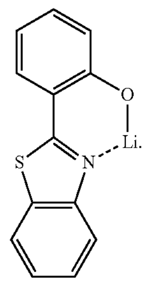
ET-D2



[0296] The electron transport region may include an EIL that facilitates the injection of electrons from the second electrode **19**.

[0297] The electron injection layer may include ET-D1, LiF, NaCl, CsF, Li_2O, BaO, or a combination thereof.

[0298] A thickness of the electron injection layer may be about 1 Å to about 100 Å, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the range as described above, satisfactory electron injection characteristics may be obtained without a substantial increase in driving voltage.

[0299] A second electrode **19** may be provided on the interlayer **15**. The second electrode **19** may be a cathode. A material for the second electrode **19** may be a metal, an alloy, an electrically conductive compound, or a combination thereof, each having a relatively low work function. For example, lithium (Li), silver (Ag), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), or magnesium-silver (Mg—Ag) may be used as the material for forming the second electrode **19**. In one or more embodiments, to manufacture a top-emission type light-emitting device, various modifications, such as formation of a transmissive second electrode using ITO or IZO, is possible.

[0300] Hereinbefore, the light-emitting device **10** according to one or more embodiments has been described in connection with the FIGURE, but embodiments are not limited thereto.

[0301] According to another aspect, the light-emitting device may be included in an electronic apparatus. Thus, an electronic apparatus including the light-emitting device is provided. The electronic apparatus may include, for example, a display, an illumination, a sensor, or the like.

[0302] According to one or more embodiments, a diagnostic composition includes at least one organometallic compound represented by Formula 1.

[0303] The organometallic compound represented by Formula 1 may provide a high luminous efficiency, and thus, a diagnostic composition including at least one of the organometallic compounds represented by Formula 1 may have high diagnostic efficiency.

[0304] The diagnostic composition may be used in various applications including a diagnosis kit, a diagnosis reagent, a biosensor, or a biomarker, but embodiments are not limited thereto.

[0305] The term "C_1-C_{60} alkyl group" as used herein refers to a linear or branched saturated aliphatic hydrocarbon monovalent group having 1 to 60 carbon atoms, and the term "C_1-C_{60} alkylene group" as used herein refers to a divalent group having the same structure as the C_1-C_{60} alkyl group.

[0306] As used herein, the C_1-C_{60} alkyl group may be a C_1-C_{20} alkyl group. As used herein, the C_1-C_{20} alkyl group may be a C_1-C_{10} alkyl group, a C_1-C_6 alkyl group, or a C_1-C_3 alkyl group. The C_1-C_{60} alkyl group, the C_1-C_{20} alkyl group, and the C_1-C_{10} alkyl group may each independently linear or branched, and in the case of the branched alkyl group, the lower limit of the carbon number in each of the alkyl groups becomes 3. Non-limiting examples of the C_1-C_{60} alkyl group, the C_1-C_{20} alkyl group, and/or the C_1-C_{10} alkyl group include a methyl group, an ethyl group, an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, an n-hexyl group, an isohexyl group, a sec-hexyl group, a tert-hexyl group, an n-heptyl group, an isoheptyl group, a sec-heptyl group, a tert-heptyl group, an n-octyl group, an isooctyl group, a sec-octyl group, a tert-octyl group, an n-nonyl group, an isononyl group, a sec-nonyl group, a tert-nonyl group, an n-decyl group, an isodecyl group, a sec-decyl group, a tert-decyl group, or the like, each unsubstituted or substituted with at least one of a methyl group, an ethyl group, an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, an n-hexyl group, an isohexyl group, a sec-hexyl group, a tert-hexyl group, an n-heptyl group, an isoheptyl group, a sec-heptyl group, a tert-heptyl group, an n-octyl group, an isooctyl group, a sec-octyl group, a tert-octyl group, an n-nonyl group, an isononyl group, a sec-nonyl group, a tert-nonyl group, an n-decyl group, an isodecyl group, a sec-decyl group, a tert-decyl group, or the like, or a combination thereof. For example, Formula 9-33 is a branched C_6 alkyl group, for example, a tert-butyl group that is substituted with two methyl groups.

[0307] The term "C_1-C_{60} alkoxy group" as used herein refers to a monovalent group represented by —OA_{101} (wherein A_{101} is the C_1-C_{60} alkyl group), and non-limiting examples thereof include a methoxy group, an ethoxy group, a propoxy group, a butoxy group, a pentoxy group, or the like.

[0308] The term "C_1-C_{60} alkylthio group" as used herein refers to a monovalent group represented by —SA_{101} (wherein A_{101} is the C_1-C_{60} alkyl group).

[0309] The term "C_2-C_{60} alkenyl group" as used herein refers to a structure containing at least one carbon-carbon double bond in the middle or at the end of the C_2-C_{60} alkyl group, and non-limiting examples thereof include an ethenyl group, a propenyl group, a butenyl group, or the like. The term "C_2-C_{60} alkenylene group" as used herein refers to a divalent group having the same structure as the C_2-C_{60} alkenyl group.

[0310] The term "C_2-C_{60} alkynyl group" as used herein refers to a hydrocarbon group formed by substituting at least one carbon-carbon triple bond in the middle or at the terminus of the C_2-C_{60} alkyl group, and non-limiting examples thereof are an ethynyl group, a propynyl group, or the like. The term "C_2-C_{60} alkynylene group" as used herein refers to a divalent group having the same structure as the C_2-C_{60} alkynyl group.

[0311] The term "C_3-C_{10} cycloalkyl group" as used herein refers to a monovalent saturated hydrocarbon ring group having 3 to 10 carbon atoms, and the C_3-Cia cycloalkylene group is a divalent group having the same structure as the C_3-C_{10} cycloalkyl group.

[0312] Non-limiting examples of the C_3-C_{10} cycloalkyl group include a cyclopropyl group, a cyclobutyl group, a cyclopentyl, cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group (or a bicyclo[2.2.1]heptyl group), a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, a bicyclo[2.2.2]octyl group, or the like.

[0313] The term "C_1-C_{10} heterocycloalkyl group" as used herein refers to a monovalent saturated ring group that includes at least one heteroatom selected from N, O, P, Si, S, Se, Ge, and B as a ring-forming atom and 1 to 10 carbon atoms as ring-forming atom(s), and the term "C_1-C_{10} heterocycloalkylene group" as used herein refers to a divalent group having the same structure as the C_1-C_{10} heterocycloalkyl group.

[0314] Non-limiting examples of the C_1-C_{10} heterocycloalkyl group include a silolanyl group, a silinanyl group, tetrahydrofuranyl group, a tetrahydro-2H-pyranyl group, a tetrahydrothiophenyl group, or the like.

[0315] The term "C_3-C_{10} cycloalkenyl group" as used herein refers to a monovalent ring group that has 3 to 10 carbon atoms as ring-forming atoms and has at least one carbon-carbon double bond in the ring thereof and has no aromaticity, and non-limiting examples thereof include a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, or the like. The term "C_3-C_{10} cycloalkenylene group" as used herein refers to a divalent group having the same structure as the C_3-C_{10} cycloalkenyl group.

[0316] The term "C_1-C_{10} heterocycloalkenyl group" as used herein refers to a monovalent ring group that includes at least one heteroatom selected from N, O, P, Si, S, Se, Ge, and B as a ring-forming atom, 1 to 10 carbon atoms as ring-forming atom(s), and at least one double bond in the ring thereof, and has no aromaticity. Non-limiting examples of the C_1-Cia heterocycloalkenyl group include a 2,3-dihydrofuranyl group, a 2,3-dihydrothiophenyl group, or the like. The term "C_1-C_{10} heterocycloalkenylene group" as used herein refers to a divalent group having the same structure as the C_1-C_{10} heterocycloalkenyl group.

[0317] The term "C_6-C_{60} aryl group" as used herein refers to a monovalent group having a carbocyclic aromatic ring system having 6 to 60 carbon atoms as ring-forming atoms, and the term "C_6-C_{60} arylene group" as used herein refers to a divalent group having a carbocyclic aromatic ring system having 6 to 60 carbon atoms as ring-forming atoms. Non-limiting examples of the C_6-C_{60} aryl group include a phenyl group, a naphthyl group, an anthracenyl group, a phenanthrenyl group, a pyrenyl group, a chrysenyl group, or the like. When the C_6-C_{60} aryl group and the C_6-C_{60} arylene group each include two or more rings, the rings may be fused with each other.

[0318] The term "C_7-C_{60} alkyl aryl group" as used herein refers to a C_6-C_{60} aryl group substituted with at least one C_1-C_{60} alkyl group. The term "C_7-C_{60} aryl group alkyl" as used herein refers to a C_1-C_{60} alkyl group substituted with at least one C_6-C_{60} aryl group.

[0319] The term "C_1-C_{60} heteroaryl group" as used herein refers to a monovalent group that includes at least one heteroatom selected from N, O, P, Si, S, Se, Ge, and B as a ring-forming atom and a heteroaromatic ring system having 1 to 60 carbon atoms, and the term "C_1-C_{60} heteroarylene group" as used herein refers to a divalent group that includes at least one heteroatom selected from N, O, P, Si, S, Se, Ge, and B as a ring-forming atom and a heteroaromatic system having 1 to 60 carbon atoms. Non-limiting examples of the C_1-C_{60} heteroaryl group include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, or the like. When the C_1-C_{60} heteroaryl group and the C_1-C_{60} heteroarylene group each include two or more rings, the rings may be fused with each other.

[0320] The term "C_2-C_{60} alkyl heteroaryl group" as used herein refers to a C_1-C_{60} heteroaryl group substituted with at least one C_1-C_{60} alkyl group. The term "C_2-C_{60} heteroaryl alkyl group" as used herein refers to a C_1-C_{60} alkyl group substituted with at least one C_1-C_{60} heteroaryl group.

[0321] The term "C_6-C_{60} aryloxy group" as used herein indicates —OA_{102} (wherein A_{102} indicates the C_6-C_{60} aryl group), the C_6-C_{60} arylthio group indicates —SA_{103} (wherein A_{103} indicates the C_6-C_{60} aryl group).

[0322] The term "C_1-C_{60} heteroaryloxy group" as used herein indicates —OA_{105} (wherein A_{105} is a C_1-C_{60} heteroaryl group), and the term "C_1-C_{60} heteroarylthio group" as used herein indicates —SA_{106} (wherein A_{106} is the C_1-C_{60} heteroaryl group).

[0323] The term "monovalent non-aromatic condensed polycyclic group" as used herein refers to a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed with each other, only carbon atoms as ring-forming atoms, and no aromaticity in its entire molecular structure. Non-limiting examples of the monovalent non-aromatic condensed polycyclic group include a fluorenyl group or the like. The term "divalent non-aromatic condensed polycyclic group" as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed polycyclic group.

[0324] The term "monovalent non-aromatic condensed heteropolycyclic group" as used herein refers to a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings condensed to each other, at least one heteroatom selected from N, O, P, Si, S, Se, Ge, and B, other than carbon atoms, as a ring-forming atom, and no aromaticity in its entire molecular structure. Non-limiting examples of the monovalent non-aromatic condensed heteropolycyclic group include a carbazolyl group or the like. The term "divalent non-aromatic condensed heteropolycyclic group" as used herein refers to a divalent group having the same structure as the monovalent non-aromatic condensed heteropolycyclic group.

[0325] The term "C_5-C_{30} carbocyclic group" as used herein refers to a saturated or unsaturated cyclic group having, as ring-forming atoms, 5 to 30 carbon atoms only. The C_5-C_{30} carbocyclic group may be a monocyclic group or a polycyclic group. Non-limiting examples of the "C_5-C_{30} carbocyclic group (unsubstituted or substituted with at least one R_{10a})" used herein include an adamantane group, a norbornene group, a bicyclo[1.1.1]pentane group, a bicyclo[2.1.1]hexane group, a bicyclo[2.2.1]heptane(norbornane) group, a bicyclo[2.2.2]octane group, a cyclopentane group, a cyclohexane group, a cyclohexene group, a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a 1,2,3,4-tetrahydronaphthalene group, a cyclopentadiene group, a fluorene group, or the like (each unsubstituted or substituted with at least one R_{10a}).

[0326] The term "C_1-C_{30} heterocyclic group" as used herein refers to a saturated or unsaturated ring group having, as a ring-forming atom, at least one heteroatom selected from N, O, P, Si, S, Se, Ge, and B other than 1 to 30 carbon atoms as ring-forming atoms(s). The C_1-C_{30} heterocyclic group may be a monocyclic group or a polycyclic group. The "C_1-C_{30} heterocyclic group (unsubstituted or substituted with at least one R_{10a})" may be, for example, a thiophene group, a furan group, a pyrrole group, a silole group, borole group, a phosphole group, a selenophene group, a germole group, a benzothiophene group, a benzofuran group, an indole group, a benzosilole group, a benzoborole group, a benzophosphole group, a benzoselenophene group, a benzogermole group, a dibenzothiophene group, a dibenzofuran group, a carbazole group, a dibenzosilole group, a dibenzoborole group, a dibenzophosphole group, a dibenzoselenophene group, a dibenzogermole group, a dibenzothiophene 5-oxide group, a 9H-fluoren-9-one group, a dibenzothiophene 5,5-dioxide group, an azabenzothiophene group, an azabenzofuran group, an azaindole group, an azaindene group, an azabenzosilole group, an azabenzoborole group, an azabenzophosphole group, an azabenzoselenophene group, an azabenzogermole group, an azadibenzothiophene group, an azadibenzofuran group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzoborole group, an azadibenzophosphole group, an azadibenzoselenophene group, an azadibenzogermole group, an azadibenzothiophene 5-oxide group, an aza-9H-fluoren-9-one group, an azadibenzothiophene 5,5-dioxide group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isooxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a 5,6,7,8-tetrahydroisoquinoline group, a 5,6,7,8-tetrahydroquinoline group, or the like (each unsubstituted or substituted with at least one R_{10a}).

[0327] Non-limiting examples of the "C_5-C_{30} carbocyclic group" and "C_1-C_{30} heterocyclic group" as used herein include i) a first ring, ii) a second ring, iii) a condensed ring group in which two or more first rings are condensed with each other, iv) a condensed ring group in which two or more second rings are condensed with each other, or v) a condensed ring group in which at least one first ring is condensed with at least one second ring,

[0328] the first ring may be a cyclopentane group, a cyclopentene group, a furan group, a thiophene group, a pyrrole group, a silole group, a borole group, a phosphole group, a germole group, a selenophene group, an oxazole group, an isoxazole group, an oxadiazole group, an oxatriazole group, a thiazole group, an isothiazole group, a thiadiazole group, a thiatriazole group, a pyrazole group, an imidazole group, a triazole group, a tetrazole group, an azasilole group, a diazasilole group, or a triazasilole group, and

[0329] the second ring may be an adamantane group, a norbornane group, a norbornene group, a cyclohexane group, a cyclohexene group, a benzene group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, or a triazine group.

[0330] The terms "fluorinated C_1-C_{60} alkyl group (or a fluorinated C_1-C_{20} alkyl group or the like)", "fluorinated C_3-C_{10} cycloalkyl group", "fluorinated C_1-C_{10} heterocycloalkyl group," and "fluorinated phenyl group" respectively indicate a C_1-C_{60} alkyl group (or a C_1-C_{20} alkyl group or the like), a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, and a phenyl group, each substituted with at least one fluoro group (—F). For example, the term "fluorinated C_1 alkyl group (that is, a fluorinated methyl group)" includes —CF_3, —CF_2H, —CFH_2, or the like. The "fluorinated C_1-C_{60} alkyl group (or, a fluorinated C_1-C_{20} alkyl group, or the like)", "the fluorinated C_3-C_{10} cycloalkyl group", "the fluorinated C_1-C_{10} heterocycloalkyl group", or "the fluorinated phenyl group" may be i) a fully fluorinated C_1-C_{60} alkyl group (or, a fully fluorinated C_1-C_{20} alkyl group, or the like), a fully fluorinated C_3-C_{10} cycloalkyl group, a fully fluorinated C_1-C_{10} heterocycloalkyl group, or a fully fluorinated phenyl group, wherein, in each group, all hydrogen atoms included therein are substituted with a fluoro group, or ii) a partially fluorinated C_1-C_{60} alkyl group (or, a partially fluorinated C_1-C_{20} alkyl group, or the like), a partially fluorinated C_3-C_{10} cycloalkyl group, a partially fluorinated C_1-C_1 heterocycloalkyl group, or partially fluorinated phenyl group, wherein, in each group, all hydrogen atoms included therein are not substituted with a fluoro group.

[0331] The terms "deuterated C_1-C_{60} alkyl group (or a deuterated C_1-C_{20} alkyl group or the like)", "deuterated C_3-C_{10} cycloalkyl group", "deuterated C_1-C_1 heterocycloalkyl group," and "deuterated phenyl group" respectively indicate a C_1-C_{60} alkyl group (or a C_1-C_{20} alkyl group or the like), a C_3-C_{10} cycloalkyl group, a C_1-C_1 heterocycloalkyl group, and a phenyl group, each substituted with at least one deuterium. For example, the "deuterated C_1 alkyl group (that is, the deuterated methyl group)" may include —CD_3, —CD_2H, —CDH_2, or the like, and examples of the "deuterated C_3-C_1 cycloalkyl group" include, for example, Formula 10-501 or the like. The "deuterated C_1-C_{60} alkyl group (or, the deuterated C_1-C_{20} alkyl group or the like)", "the deuterated C_3-C_1 cycloalkyl group", "the deuterated C_1-C_{10} heterocycloalkyl group", or "the deuterated phenyl group" may be i) a fully deuterated C_1-C_{60} alkyl group (or, a fully deuterated C_1-C_{20} alkyl group or the like), a fully deuterated C_3-C_{10} cycloalkyl group, a fully deuterated C_1-C_{10} heterocycloalkyl group, or a fully deuterated phenyl group, in which, in each group, all hydrogen atoms included therein are substituted with deuterium, or ii) a partially deuterated C_1-C_{60} alkyl group (or, a partially deuterated C_1-C_{20} alkyl group or the like), a partially deuterated C_3-C_{10} cycloalkyl group, a partially deuterated C_1-C_{10} heterocycloalkyl group, or a partially deuterated phenyl group, in which, in each group, all hydrogen atoms included therein are not substituted with deuterium.

[0332] The term "(C_1-C_{20} alkyl)'X' group" as used herein refers to a 'X' group that is substituted with at least one C_1-C_{20} alkyl group. For example, the term "(C_1-C_{20} alkyl) C_3-C_{10} cycloalkyl group" as used herein refers to a C_3-C_1 cycloalkyl group substituted with at least one C_1-C_{20} alkyl group, and the term "(C_1-C_{20} alkyl)phenyl group" as used herein refers to a phenyl group substituted with at least one C_1-C_{20} alkyl group. An example of a (C_1 alkyl)phenyl group is a tolyl group.

[0333] As used herein, the terms "an azaindole group, an azabenzoborole group, an azabenzophosphole group, an azaindene group, an azabenzosilole group, an azabenzogermole group, an azabenzothiophene group, an azabenzoselenophene group, an azabenzofuran group, an azacarbazole group, an azadibenzoborole group, an azadibenzophosphole group, an azafluorene group, an azadibenzosilole group, an azadibenzogermole group, an azadibenzothiophene group, an azadibenzoselenophene group, an azadibenzofuran group, an azadibenzothiophene 5-oxide group, an aza-9H-fluoren-9-one group, and an azadibenzothiophene 5,5-dioxide group" respectively refer to heterocyclic groups having the same backbones as "an indole group, a benzoborole group, a benzophosphole group, an indene group, a benzosilole group, a benzogermole group, a benzothiophene group, a benzoselenophene group, a benzofuran group, a carbazole group, a dibenzoborole group, a dibenzophosphole group, a fluorene group, a dibenzosilole group, a dibenzogermole group, a dibenzothiophene group, a dibenzoselenophene group, a dibenzofuran group, a dibenzothiophene 5-oxide group, a 9H-fluoren-9-one group, and a dibenzothiophene 5,5-dioxide group," in which, in each group, at least one carbon selected from ring-forming carbons is substituted with nitrogen.

[0334] A substituent of the substituted C_1-C_{60} alkyl group, the substituted C_2-C_{60} alkenyl group, the substituted C_2-C_{60} alkynyl group, the substituted C_1-C_{60} alkoxy group, the substituted C_1-C_{60} alkylthio group, the substituted C_3-C_{10} cycloalkyl group, the substituted C_1-C_{10} heterocycloalkyl group, the substituted C_3-C_{10} cycloalkenyl group, the substituted C_1-C_{10} heterocycloalkenyl group, the substituted C_6-C_{60} aryl group, the substituted C_7-C_{60} alkyl aryl group, the substituted C_7-C_{60} aryl alkyl group, the substituted C_6-C_{60} aryloxy group, the substituted C_6-C_{60} arylthio group, the substituted C_1-C_{60} heteroaryl group, the substituted C_2-C_{60} alkyl heteroaryl group, the substituted C_2-C_{60} heteroaryl alkyl group, the substituted C_1-C_{60} heteroaryloxy group, the substituted C_1-C_{60} heteroarylthio group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group may be:

[0335] deuterium, —F, —Cl, —Br, —I, —SF_5, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy group, or a C_1-C_{60} alkylthio group;

[0336] a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy group, or a C_1-C_{60} alkylthio group, each substituted with at least one of deuterium, —F, —C_1, —Br, —I, —SF_5, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_3-C_1 cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_1 cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a C_2-C_{60} alkyl heteroaryl group, a C_1-C_{60} heteroaryloxy group, a C_1-C_{60} heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —$Si(Q_{11})(Q_{12})(Q_{13})$, —$Ge(Q_{11})(Q_{12})(Q_{13})$, —$N(Q_{14})(Q_{15})$, —$B(Q_{16})(Q_{17})$, —$P(=O)(Q_{18})(Q_{19})$, —$P(Q_{18})(Q_{19})$, or a combination thereof;

[0337] a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a C_2-C_{60} alkyl heteroaryl group, a C_1-C_{60} heteroaryloxy group, a C_1-C_{60} heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each unsubstituted or substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF_5, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy group, a C_1-C_{60} alkylthio group, a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_1 heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_7-C_{60} aryl alkyl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a C_2-C_{60} alkyl heteroaryl group, a C_2-C_{60} heteroaryl alkyl group, a C_1-C_{60} heteroaryloxy group, a C_1-C_{60} heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —$Si(Q_{21})(Q_{22})(Q_{23})$, —$Ge(Q_{21})(Q_{22})(Q_{23})$, —$N(Q_{24})(Q_{25})$, —$B(Q_{26})(Q_{27})$, —$P(=O)(Q_{28})(Q_{29})$, $P(Q_{28})(Q_{29})$, or a combination thereof;

[0338] —$Si(Q_{31})(Q_{32})(Q_{33})$, —$Ge(Q_{31})(Q_{32})(Q_{33})$, —$N(Q_{34})(Q_{35})$, —$B(Q_{36})(Q_{37})$, —$P(=O)(Q_{38})(Q_{39})$, or —$P(Q_{38})(Q_{39})$; or

[0339] a combination thereof.

[0340] Q_1 to Q_9, Q_{11} to Q_{19}, Q_{21} to Q_{29}, and Q_{31} to Q_{39} may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_7-C_{60} alkyl aryl group, a substituted or unsubstituted C_7-C_{60} aryl alkyl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted C_2-C_{60} alkyl heteroaryl group, a substituted or unsubstituted C_2-C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1-C_{60} heteroaryloxy group, a substituted or unsubstituted C_1-C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

[0341] For example, Q_1 to Q_9, Q_{11} to Q_{19}, Q_{21} to Q_{29}, and Q_{31} to Q_{39} as described herein may each independently be:

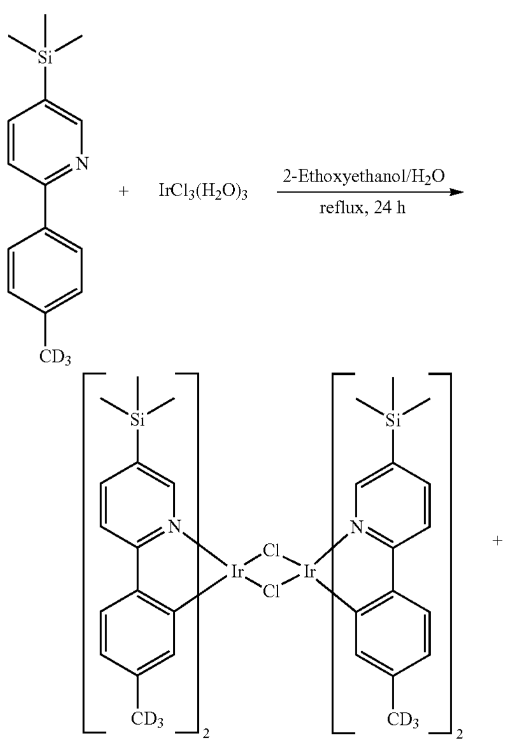
[0342] —CH_3, —CD_3, —CD_2H, —CDH_2, —CH_2CH_3, —CH_2CD_3, —CH_2CD_2H, —CH_2CDH_2, —$CHDCH_3$, —$CHDCD_2H$, —$CHDCDH_2$, —$CHDCD_3$, —CD_2CD_3, —CD_2CD_2H, or —CD_2CDH_2; or

[0343] an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, a phenyl group, a biphenyl group, or a naphthyl group, each unsubstituted or substituted with at least one of deuterium, a C_1-C_{10} alkyl group, a phenyl group, or a combination thereof.

[0344] Hereinafter, organometallic compounds represented by Formula 1 and organic light-emitting devices including the same, according to one or more embodiments, will be described in further detail with reference to Synthesis Example and Examples. However, the following examples are not intended to limit the scope of the disclosure. The wording "B was used instead of A" used in describing Synthesis Examples means that an amount of A used was identical to an amount of B used, in terms of a molar equivalent.

EXAMPLES

Synthesis Example 1 (Compound 1)



1A

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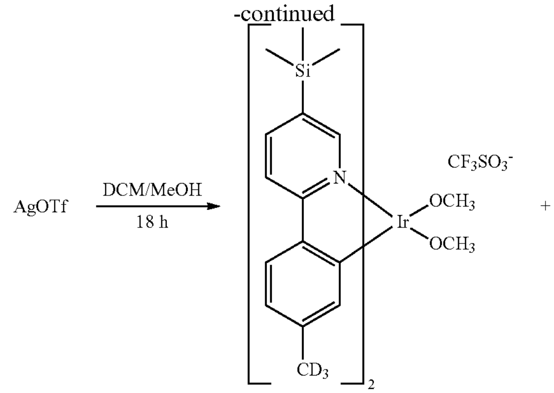
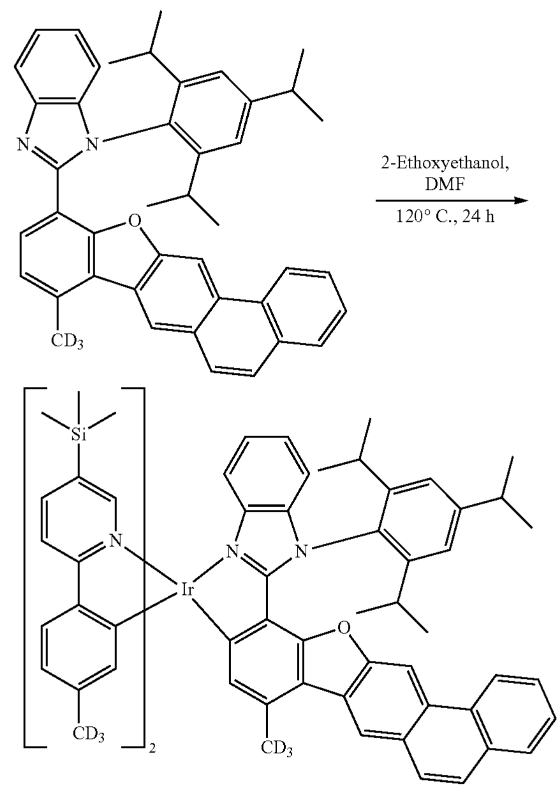


1B



1

Synthesis of Compound 1A

[0345] 2-(4-(methyl-d3)phenyl)-5-(trimethylsilyl)pyridine [3.0 grams (g), 12.27 millimoles (mmol)] and iridium chloride trihydrate (2.06 g, 5.84 mmol) were combined with 90 milliliters (mL) of 2-ethoxyethanol and 30 mL of distilled water. The reaction mixture was stirred under reflux for 24 hours, and then cooled to room temperature. Precipitated solids were separated by filtration, and thoroughly washed with water/methanol/hexane in that order. The obtained solids were dried in a vacuum oven to provide 3.4 g (yield of 81%) of Compound 1A.

Synthesis of Compound 1B

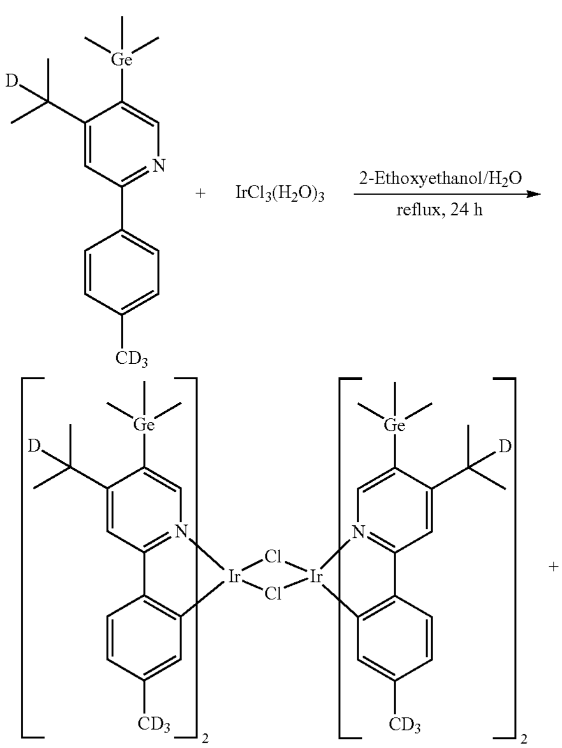
[0346] Compound 1A (3.4 g, 2.38 mmol) was combined with 120 mL of dichloromethane (DCM), and then a solution of silver trifluoromethanesulfonate (AgOTf) (1.28 g, 5 mmol) in 40 mL of methanol (MeOH) was added thereto. Afterwards, the reaction mixture was stirred at room temperature for 18 hours in the dark, then filtered through Celite to remove the precipitated solids, and the filtrate was concentrated to provide a solid residue (Compound 1B), which was then used in the next reaction without further purification.

Synthesis of Compound 1

[0347] Compound 1B (3 g, 3.37 mmol) and 2-(8-(methyl-d3)phenanthro[3,2-b]benzofuran-11-yl)-1-(2,4,6-triisopropylphenyl)-1H-benzo[d]imidazole (2.04 g, 3.37 mmol) were combined with 25 mL of 2-ethoxyethanol and 25 mL of dimethylformamide (DMF). The reaction mixture was stirred at 120° C. for 24 hours, and then cooled. The resulting mixture was concentrated under reduced pressure to provide a solid residue, which was then purified by column chromatography (eluent: hexane and ethyl acetate) to provide 1.03 g (yield of 24%) of Compound 1. The obtained Compound 1 was identified by high resolution mass spectrometry using matrix assisted laser desorption ionization (HRMS (MALDI)) and high-performance liquid chromatography (HPLC) analysis.

[0348] HRMS (MALDI) calculated for $C_{73}H_{66}D_9IrN_4OSi_2$: mass to charge ratio (m/z) 1281.5674, Found: 1281.5678.

Synthesis Example 2 (Compound 109)



109A

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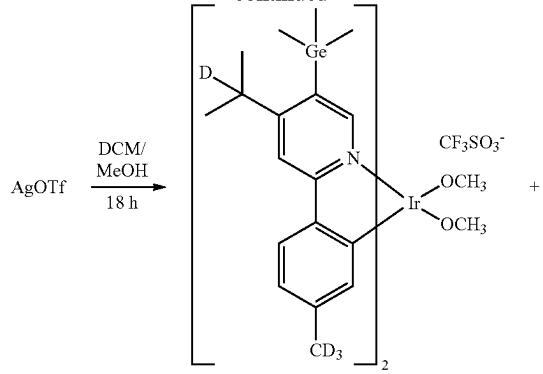
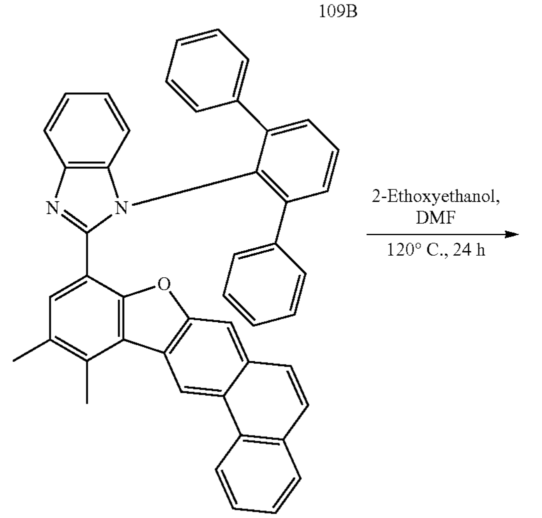
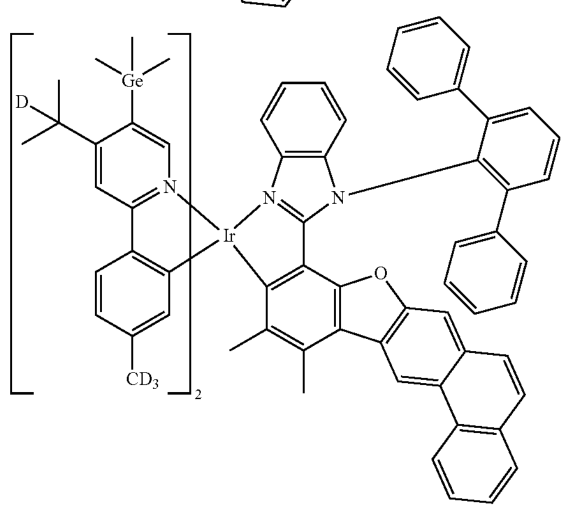


109B





109

Synthesis of Compound 109A

[0349] 2-(4-(methyl-d3)phenyl)-4-(propan-2-yl-2-d)-5-(trimethylgermyl)pyridine (3.0 g, 9.03 mmol) and iridium chloride trihydrate (1.52 g, 4.30 mmol) were combined with 90 mL of 2-ethoxyethanol and 30 mL of distilled water. The reaction mixture was then stirred under reflux for 24 hours, and the resultant reaction mixture was cooled to room temperature. Precipitated solids were separated by filtration, and thoroughly washed with water/methanol/hexane in that order. The obtained solids were dried in a vacuum oven to provide 3.1 g (yield of 81%) of Compound 109A.

Synthesis of Compound 109B

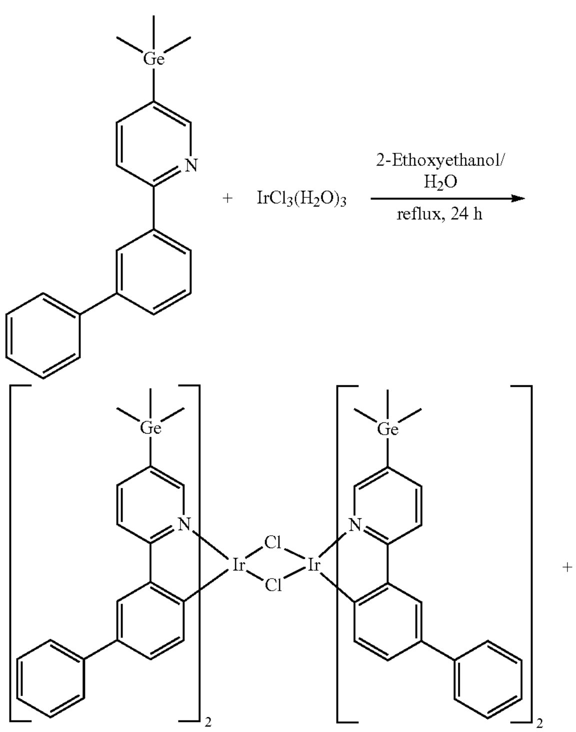
[0350] Compound 109A (3.1 g, 1.74 mmol) was combined with 120 mL of dichloromethane, and then a solution of AgOTf (0.94 g, 3.66 mmol) in 40 mL of methanol was added thereto. Afterwards, the reaction mixture was stirred at room temperature for 18 hours in the dark, then filtered through Celite to remove the precipitated solids, and the filtrate was concentrated to provide a solid residue (Compound 109B), which was used in the next reaction without further purification.

Synthesis of Compound 109

[0351] Compound 109B (2.8 g, 2.63 mmol) and 1-([1,1':3',1"-terphenyl]-2'-yl)-2-(11,12-dimethylphenanthro[2,3-b]benzofuran-9-yl)-1H-benzo[d]imidazole (1.68 g, 2.63 mmol) were combined with 25 mL of 2-ethoxyethanol and 25 mL of dimethylformamide. The reaction mixture was stirred at 120° C. for 24 hours, and then the temperature was lowered. The resulting mixture was concentrated under reduced pressure to provide a solid residue, which was then purified by column chromatography (eluent: hexane and ethyl acetate) to provide 1.01 g (yield of 26%) of Compound 109. The obtained Compound 109 was identified by mass spectrometry and HPLC analysis.

[0352] HRMS(MALDI) calculated for $C_{83}H_{71}D_8Ge_2IrN_4O$: m/z 1496.4809, Found: 1496.4813.

Synthesis Example 3 (Compound 49)



49A

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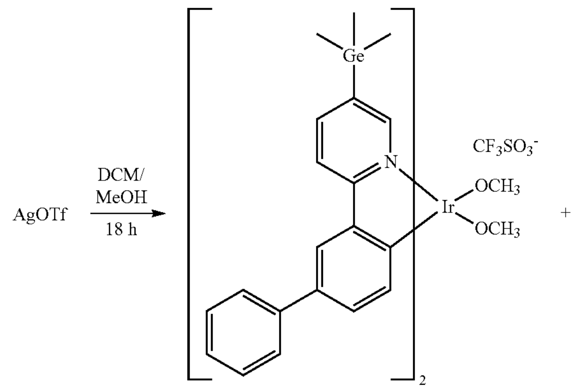
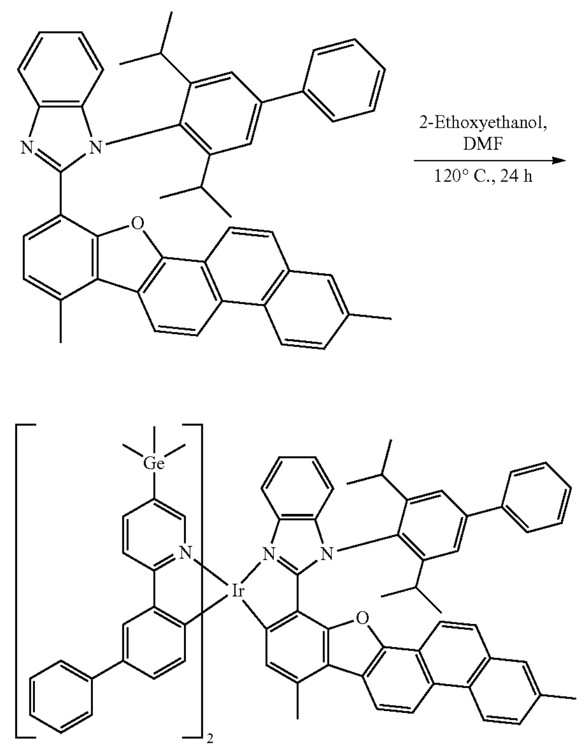


49B



49

Synthesis of Compound 49A

[0353] 2-([1,1'-biphenyl]-3-yl)-5-(trimethylgermyl)pyridine) (3.0 g, 8.62 mmol) and iridium chloride trihydrate (1.45 g, 4.10 mmol) were combined with 90 mL of 2-ethoxyethanol and 30 mL of distilled water. The reaction mixture was then stirred under reflux for 24 hours. Then, the temperature was lowered to room temperature. Precipitated solids were separated by filtration, and thoroughly washed with water/methanol/hexane in that order. The obtained solids were dried in a vacuum oven to provide 3.1 g (yield of 82%) of Compound 49A.

Synthesis of Compound 49B

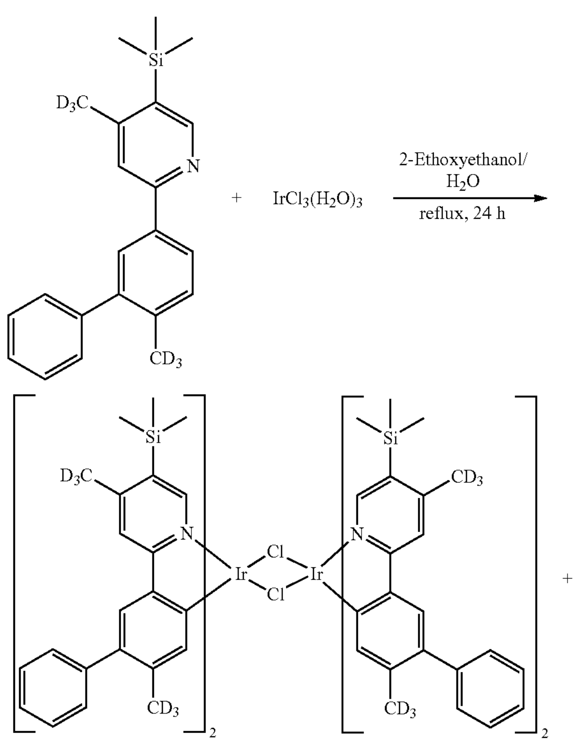
[0354] Compound 49A (3.1 g, 1.68 mmol) was combined with 120 mL of dichloromethane, and then a solution of AgOTf (0.91 g, 3.53 mmol) in 40 mL of methanol was added thereto. Afterwards, the reaction mixture was stirred at room temperature for 18 hours in the dark, then filtered through Celite to remove the precipitated solids, and the filtrate was concentrated to provide a solid residue (Compound 49B), which was used in the next reaction without further purification.

Synthesis of Compound 49

[0355] Compound 49B (2.8 g, 2.55 mmol) and 1-(3,5-diisopropyl-[1,1'-biphenyl]-4-yl)-2-(4,9-dimethylphenanthro[1,2-b]benzofuran-12-yl)-1H-benzo[d]imidazole (1.66 g, 2.55 mmol) were combined with 25 mL of 2-ethoxyethanol and 25 mL of dimethylformamide. The reaction mixture was then stirred at 120° C. for 24 hours, and then the temperature was lowered. The resulting mixture was concentrated under reduced pressure to provide a solid residue, which was then purified by column chromatography (eluent: hexane and ethyl acetate) to provide 0.89 g (yield of 23%) of Compound 49. The obtained compound was identified by mass spectrometry and HPLC analysis.

[0356] HRMS(MALDI) calculated for $C_{87}H_{79}Ge_2IrN_4O$: m/z 1536.4307, Found: 1536.4311.

Synthesis Example 4 (Compound 57)



57A

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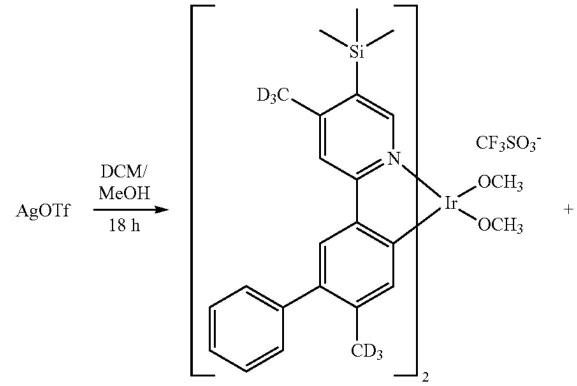
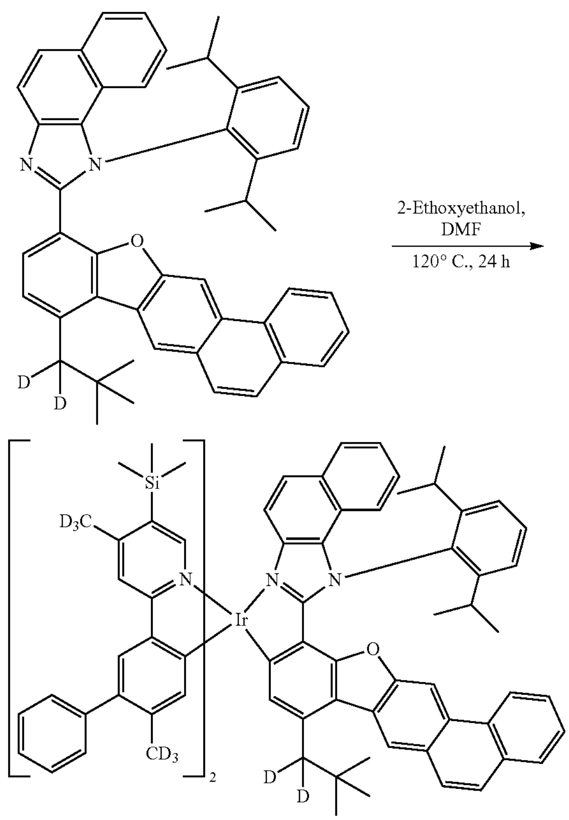


57B



57

Synthesis of Compound 57A

[0357] 4-(methyl-d3)-2-(6-(methyl-d3)-[1,1'-biphenyl]-3-yl)-5-(trimethylsilyl)pyridine (3.0 g, 8.89 mmol) and iridium chloride trihydrate (1.49 g, 4.23 mmol) were combined with 90 mL of 2-ethoxyethanol and 30 mL of distilled water. The reaction mixture was stirred under reflux for 24 hours, and then cooled to room temperature. Precipitated solids were separated by filtration, and thoroughly washed with water/methanol/hexane in that order. The obtained solids were dried in a vacuum oven to provide 3.1 g (yield of 81%) of Compound 57A.

Synthesis of compound 57B

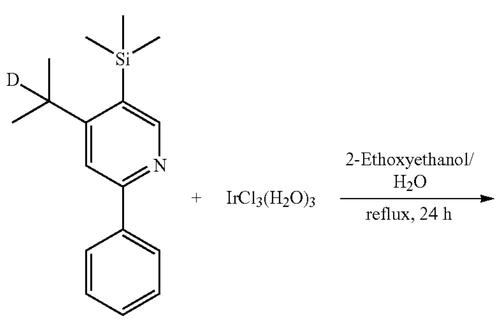
[0358] Compound 57A (3.1 g, 1.68 mmol) was combined with 120 mL of dichloromethane, and then a solution of AgOTf (0.93 g, 3.61 mmol) in 40 mL of methanol was added thereto. Afterwards, the reaction mixture was stirred at room temperature for 18 hours in the dark, then filtered through Celite to remove the precipitated solids, and the filtrate was concentrated to provide a solid residue (Compound 57B), which was used in the next reaction without further purification.

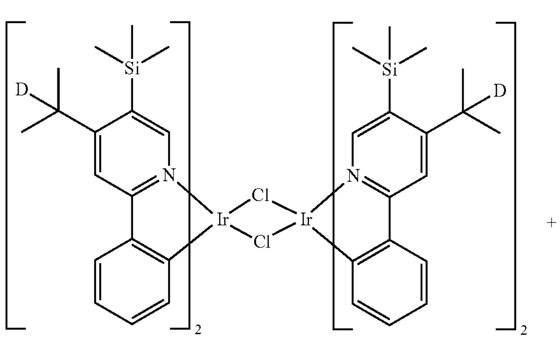
Synthesis of Compound 57

[0359] Compound 57B (2.7 g, 2.51 mmol) and 1-(2,6-diisopropylphenyl)-2-(8-(2,2-dimethylpropyl-1,1-d2)phenanthro[3,2-b]benzofuran-11-yl)-1H-naphtho[1,2-d]imidazole (1.67 g, 2.51 mmol) were combined with 25 mL of 2-ethoxyethanol and 25 mL of dimethylformamide. The reaction mixture was stirred at 120° C. for 24 hours, and then the temperature was lowered. The resulting mixture was concentrated under reduced pressure to provide a solid residue, which was then purified by column chromatography (eluent: hexane and ethyl acetate) to provide 0.92 g (yield of 24%) of Compound 57. The obtained compound was identified by mass spectrometry and HPLC analysis.

[0360] HRMS(MALDI) calculated for $C_{92}H_{77}D_{14}IrN_4OSi_2$: m/z 1530.7239, Found: 1530.7241.

Synthesis Example 5 (Compound 94)





94A

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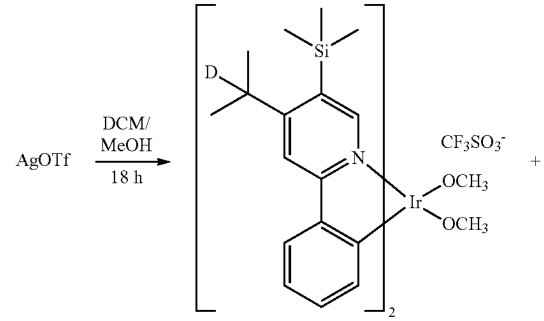
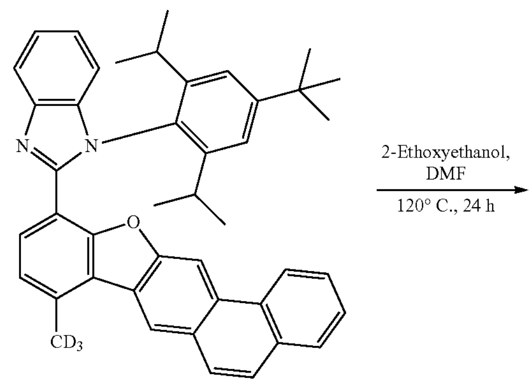
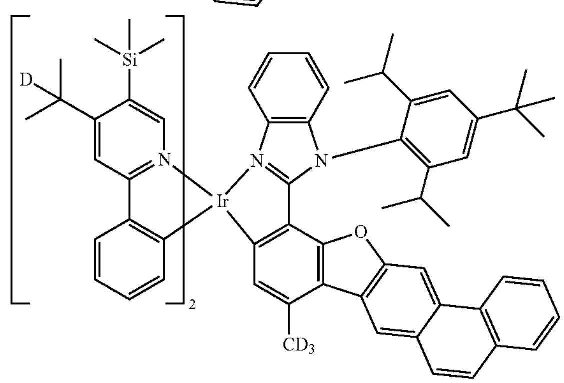


94B





94

Synthesis of Compound 94A

[0361] 2-phenyl-4-(propan-2-yl-2-d)-5-(trimethylsilyl)pyridine (3.0 g, 11.09 mmol) and iridium chloride trihydrate (1.88 g, 5.32 mmol) were combined with 90 mL of 2-ethoxyethanol and 30 mL of distilled water. The reaction mixture was stirred under reflux for 24 hours, and then allowed to cool to room temperature. Precipitated solids were separated by filtration, and thoroughly washed with water/methanol/hexane in that order. The obtained solids were dried in a vacuum oven to provide 3.3 g (yield of 81%) of Compound 94A.

Synthesis of Compound 94B

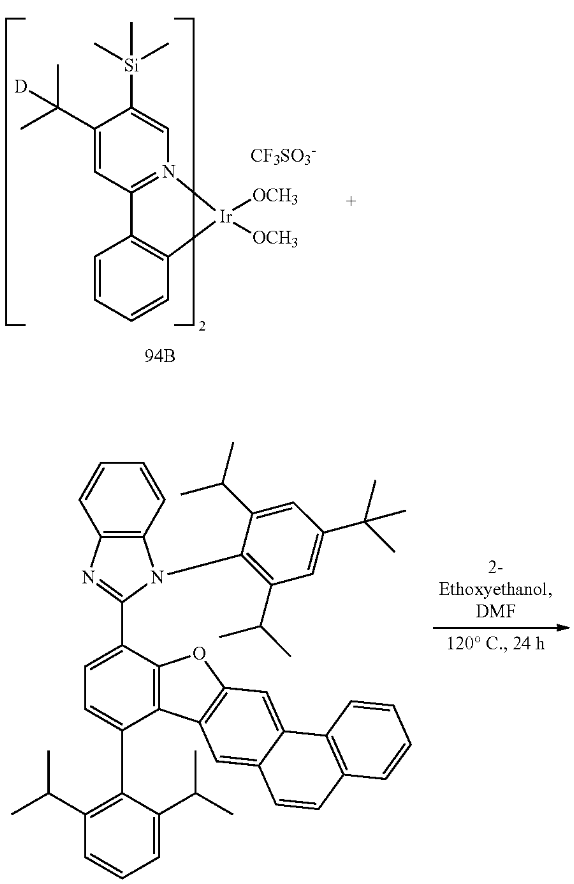
[0362] Compound 94A (3.3 g, 1.68 mmol) was combined with 120 mL of dichloromethane, and then a solution of AgOTf (1.16 g, 4.52 mmol) in 40 mL of methanol was added thereto. Afterwards, the reaction mixture was stirred at room temperature for 18 hours in the dark, then filtered through Celite to remove the precipitated solids, and the filtrate was concentrated to provide a solid residue (Compound 94B), which was used in the next reaction without further purification.

Synthesis of Compound 94

[0363] Compound 94B (3.0 g, 3.18 mmol) and 1-(4-(tert-butyl)-2,6-diisopropylphenyl)-2-(8-(methyl-d3)phenanthro[3,2-b]benzofuran-11-yl)-1H-benzo[d]imidazole(1.77 g, 2.86 mmol) were mixed with 25 mL of 2-ethoxyethanol and 25 mL of dimethylformamide, stirred at 120° C. for 24 hours, and then the temperature was lowered. The resulting mixture was concentrated under reduced pressure to obtain a solid residue, which was then purified by column chromatography (eluent: hexane and ethyl acetate) to provide 0.99 g (yield of 23%) of Compound 94. The obtained compound was identified by mass spectrometry and HPLC analysis.

[0364] HRMS(MALDI) calculated for $C_{78}H_{80}D_5IrN_4OSi_2$: m/z 1347.6205, Found: 1347.6212.

Synthesis Example 6 (Compound 98)



94B

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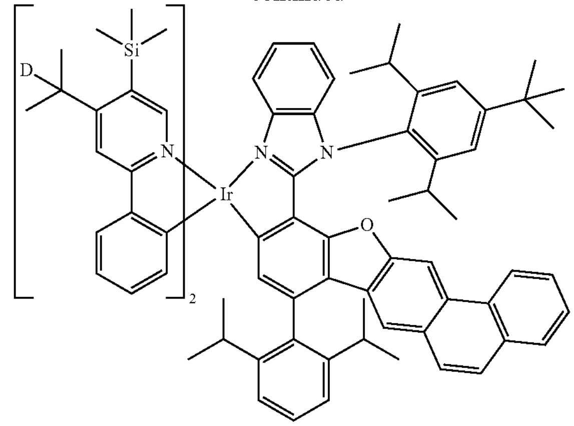
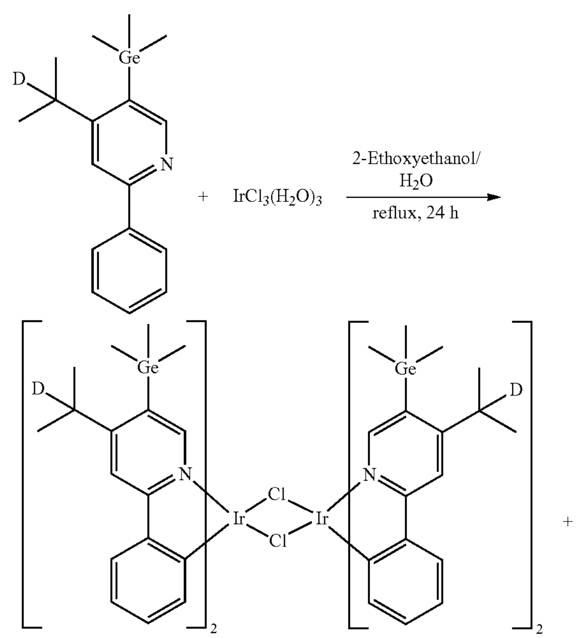


98

[0365] Compound 94B (3.0 g, 3.18 mmol) and 1-(4-(tert-butyl)-2,6-diisopropylphenyl)-2-(8-(2,6-diisopropylphenyl)phenanthro[3,2-b]benzofuran-11-yl)-1H-benzo[d]imidazole (2.18 g, 2.86 mmol) were combined with 25 mL of 2-ethoxyethanol and 25 mL of dimethylformamide. The reaction mixture was stirred at 120° C. for 24 hours, and then the temperature was lowered. The resulting mixture was concentrated under reduced pressure to provide a solid residue, which was then purified by column chromatography (eluent: hexane and ethyl acetate) to provide 0.91 g (yield of 19%) of Compound 98. The obtained compound was identified by mass spectrometry and HPLC analysis.

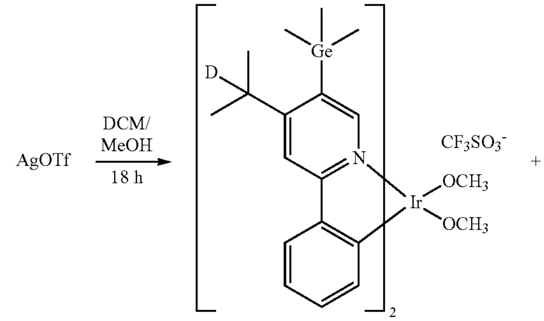
[0366] HRMS(MALDI) calculated for $C_{89}H_{97}D_2IrN_4OSi_2$: m/z 1490.7112, Found: 1490.7116.

Synthesis Example 7 (Compound 99)

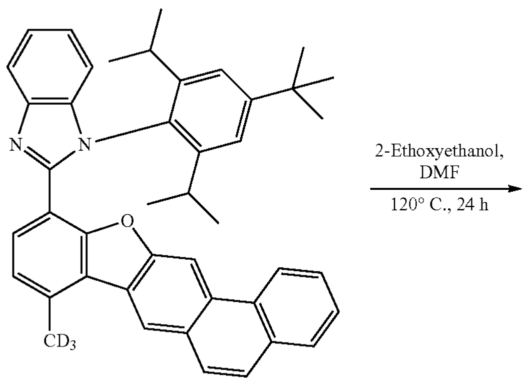


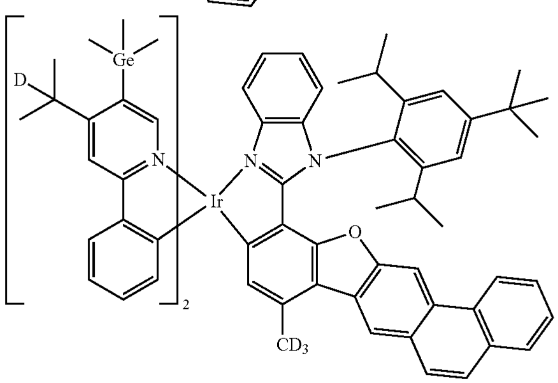
99A

-continued



99B





99

Synthesis of Compound 99A

[0367] 2-phenyl-4-(propan-2-yl-2-d)-**5**-(trimethylgermyl) pyridine (3.0 g, 9.52 mmol) and iridium chloride trihydrate (1.61 g, 4.57 mmol) were combined with 90 mL of 2-ethoxy-ethanol and 30 mL of distilled water. The reaction mixture was stirred under reflux for 24 hours, and then allowed to cool to room temperature. Precipitated solids were separated by filtration, and thoroughly washed with water/methanol/hexane in that order. The obtained solids were dried in a vacuum oven to provide 3.2 g (yield of 82%) of Compound 99A.

Synthesis of Compound 99B

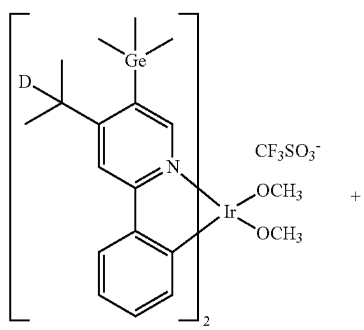
[0368] Compound 99A (3.2 g, 1.87 mmol) was combined with 120 mL of dichloromethane, and then a solution of AgOTf (1.00 g, 3.93 mmol) in 40 mL of methanol was added thereto. Afterwards, the reaction mixture was stirred at room temperature for 18 hours in the dark, then filtered through Celite to remove the resulting precipitated solids, and the filtrate was concentrated to provide a solid residue (Compound 99B), which was used in the next reaction without further purification.

Synthesis of Compound 99

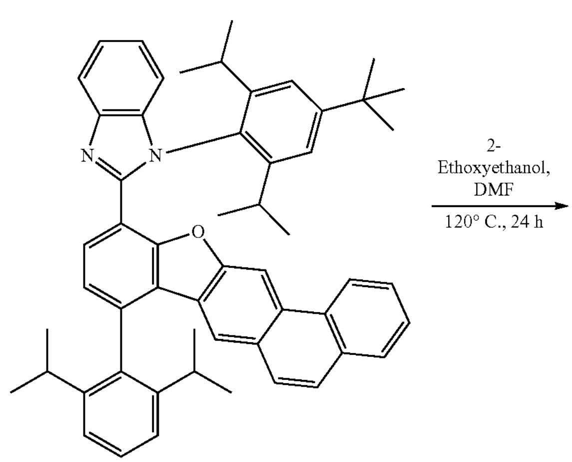
[0369] Compound 99B (3.0 g, 2.90 mmol) and 1-(4-(tert-butyl)-2,6-diisopropylphenyl)-2-(8-(methyl-d3)phenanthro[3,2-b]benzofuran-11-yl)-1H-benzo[d]imidazole(1.61 g, 2.61 mmol) were combined with 25 mL of 2-ethoxyethanol and 25 mL of dimethylformamide. The reaction mixture was stirred at 120° C. for 24 hours, and then the temperature was lowered. The resulting mixture was concentrated under reduced pressure to provide a solid residue, which was then purified by column chromatography (eluent: hexane and ethyl acetate) to provide 0.99 g (yield of 24%) of Compound 99. The obtained compound was identified by mass spectrometry and HPLC analysis.

[0370] HRMS(MALDI) calculated for $C_{78}H_{80}D_5Ge_2IrN_4O$: m/z 1437.5099, Found: 1437.5103.

Synthesis Example 8 (Compound 103)



99B



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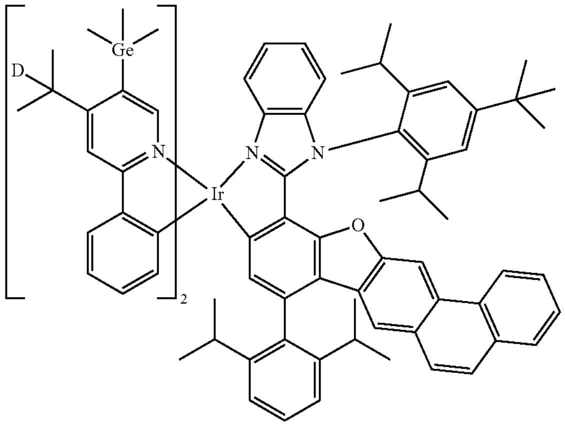
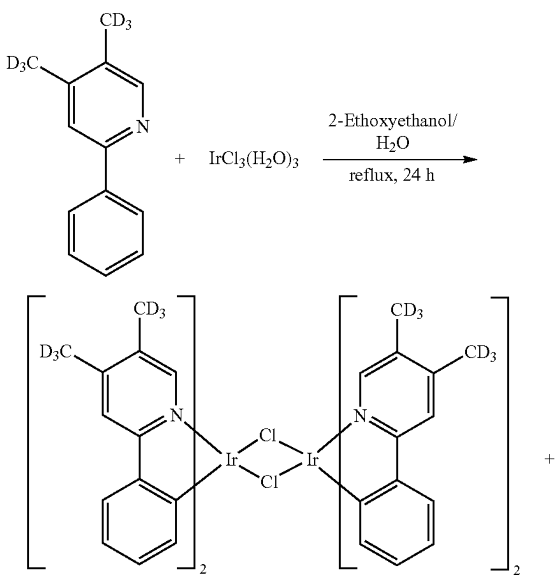


103

[0371] Compound 99B (3.0 g, 2.90 mmol) and 1-(4-(tert-butyl)-2,6-diisopropylphenyl)-2-(8-(2,6-diisopropylphenyl)phenanthro[3,2-b]benzofuran-11-yl)-1H-benzo[d]imidazole (1.99 g, 2.61 mmol) were combined with 25 mL of 2-ethoxy-ethanol and 25 mL of dimethylformamide. The reaction mixture was stirred at 120° C. for 24 hours, and then the temperature was lowered. The resulting mixture was concentrated under reduced pressure to provide a solid residue, which was then purified by column chromatography (eluent: hexane and ethyl acetate) to provide 0.99 g (yield of 22%) of Compound 103. The obtained compound was identified by mass spectrometry and HPLC analysis.

[0372] HRMS(MALDI) calculated for $C_{78}H_{80}D_5Ge_2IrN_4O$: m/z 1580.6006, Found: 1580.6012.

Synthesis Example 9 (Compound 104)



104A

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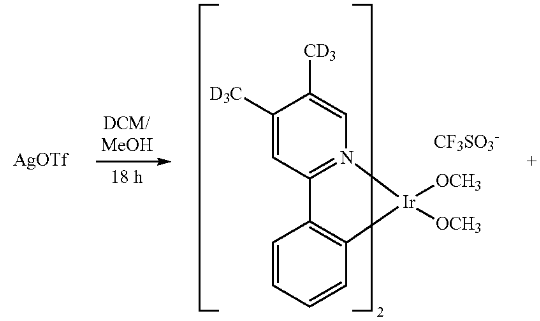
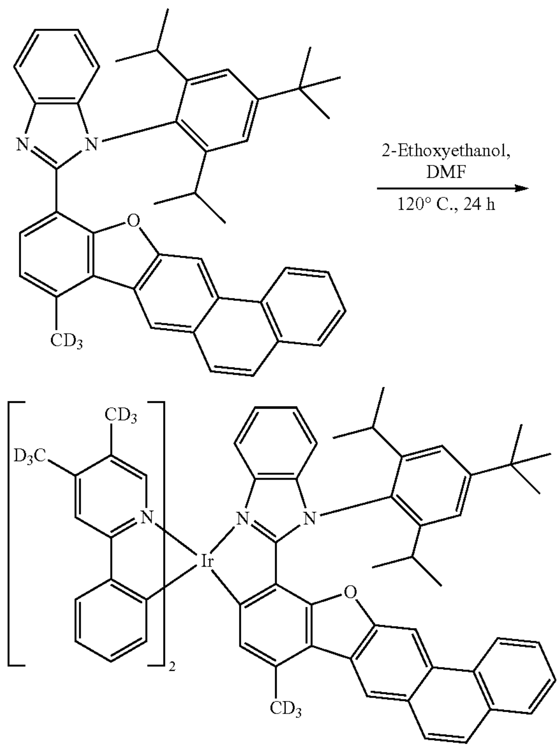


104B



104

Synthesis of Compound 104A

[0373] 4,5-bis(methyl-d3)-2-phenylpyridine (2.0 g, 10.6 mmol) and iridium chloride trihydrate (1.79 g, 5.07 mmol) were combined with 90 mL of 2-ethoxyethanol and 30 mL of distilled water. The reaction mixture was stirred under reflux for 24 hours, and allowed to cool to room temperature. Precipitated solids were separated by filtration, and thoroughly washed with water/methanol/hexane in that order. The obtained solids were dried in a vacuum oven to provide 2.6 g (yield of 85%) of Compound 104A.

Synthesis of Compound 104B

[0374] Compound 104A (2.6 g, 2.15 mmol) was combined with 120 mL of dichloromethane, and then a solution of AgOTf (1.16 g, 4.52 mmol) in 40 mL of methanol was added thereto. Afterwards, the reaction mixture was stirred at room temperature for 18 hours in the dark, then filtered through

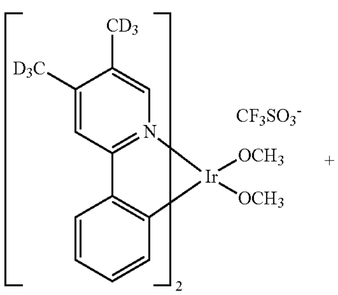
Celite to remove the resulting precipitated solids, and the filtrate was concentrated to provide a solid residue (Compound 104B), which was used in the next reaction without further purification.

Synthesis of Compound 104

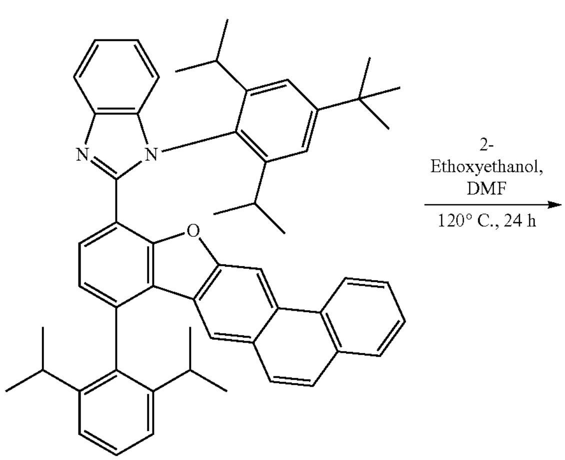
[0375] Compound 104B (2.6 g, 3.33 mmol) and 1-(4-(tert-butyl)-2,6-diisopropylphenyl)-2-(8-(methyl-d3)phenanthro[3,2-b]benzofuran-11-yl)-1H-benzo[d]imidazole (1.85 g, 2.99 mmol) were combined with 25 mL of 2-ethoxyethanol and 25 mL of dimethylformamide. The reaction mixture was stirred at 120° C. for 24 hours, and then the temperature was lowered. The resulting mixture was concentrated under reduced pressure to provide a solid residue, which was then purified by column chromatography (eluent: hexane and ethyl acetate) to provide 0.92 g of Compound 104 (yield of 23%). The obtained compound was identified by mass spectrometry and HPLC analysis.

[0376] HRMS(MALDI) calculated for $C_{70}H_{50}D_{15}IrN_4O$: m/z 1185.5729, Found: 1185.5733.

Synthesis Example 10 (Compound 108)



104B



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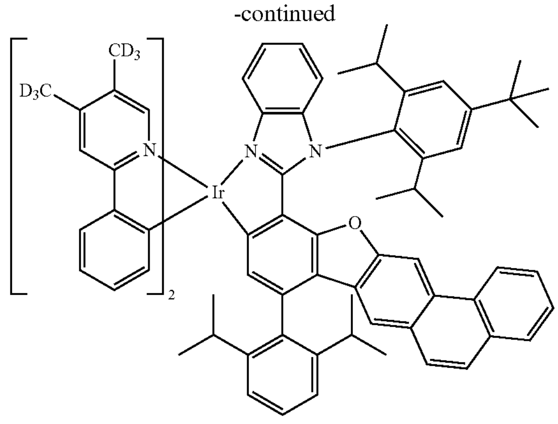
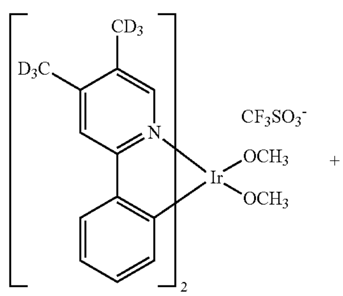


108

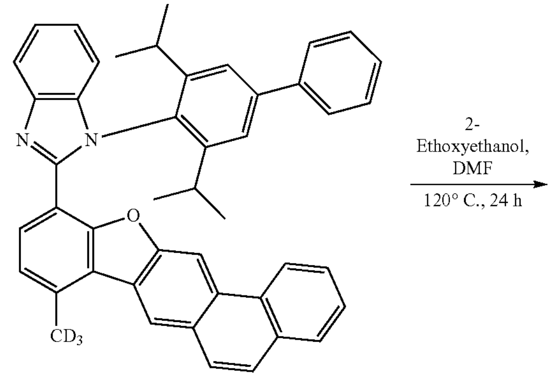
[0377] Compound 104B (2.6 g, 3.33 mmol) and 1-(4-(tert-butyl)-2,6-diisopropylphenyl)-2-(8-(2,6-diisopropylphenyl)phenanthro[3,2-b]benzofuran-11-yl)-1H-benzo[d]imidazole (2.28 g, 2.99 mmol) were combined with 25 mL of 2-ethoxyethanol and 25 mL of dimethylformamide. The reaction mixture was stirred at 120° C. for 24 hours, and then the temperature was lowered. The resulting mixture was concentrated under reduced pressure to provide a solid residue, which was then purified by column chromatography (eluent: hexane and ethyl acetate) to provide 1.0 g (yield of 23%) of Compound 108. The obtained compound was identified by mass spectrometry and HPLC analysis.

[0378] HRMS(MALDI) calculated for $C_{81}H_{67}D_{12}IrN_4O$: m/z 1328.6636, Found: 1328.6640.

Synthesis Example 11 (Compound 110)



104B



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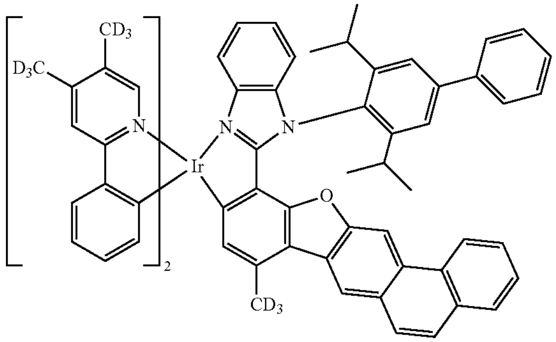
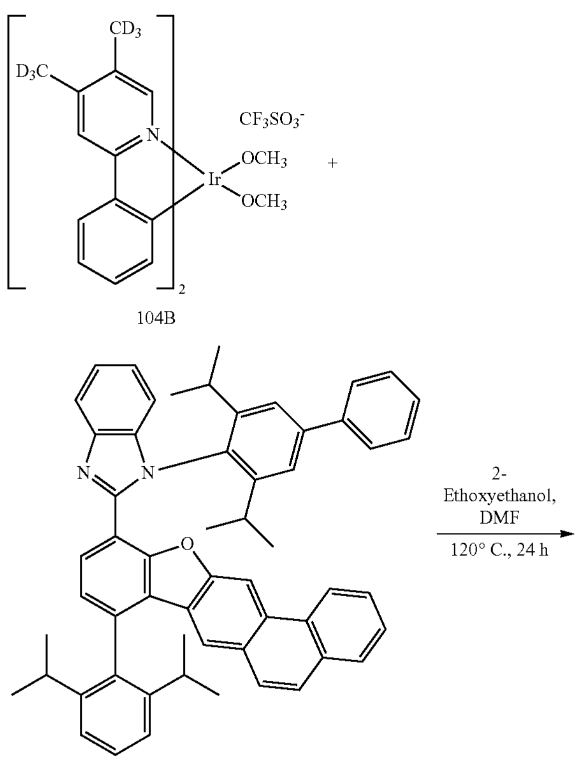


110

[0379] 0.93 g (yield of 23%) of Compound 110 was obtained in the same manner as in Synthesis Example 9, except that 1-(4-(phenyl)-2,6-diisopropylphenyl)-2-(8-(methyl-d3)phenanthro[3,2-b]benzofuran-11-yl)-1H-benzo[d]imidazole was used instead of 1-(4-(tert-butyl)-2,6-diisopropylphenyl)-2-(8-(methyl-d3)phenanthro[3,2-b]benzofuran-11-yl)-1 H-benzo[d]imidazole. The obtained compound was identified by mass spectrometry and HPLC analysis.

[0380] HRMS(MALDI) calculated for $C_{72}H_{46}D_{15}IrN_4O$: m/z 1205.5416, Found: 1205.5420

Synthesis Example 12 (Compound 111)



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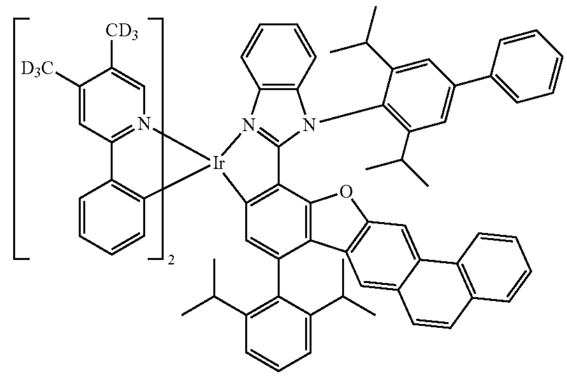


111

[0381] 1.0 g of Compound 111 was obtained in the same manner as in Synthesis Example 9, except that 1-(4-(phenyl)-2,6-diisopropylphenyl)-2-(8-(2,6-diisopropylphenyl)phenanthro[3,2-b]benzofuran-11-yl)-1H-benzo[d]imidazole was used instead of 1-(4-(tert-butyl)-2,6-diisopropylphenyl)-2-(8-(2,6-diisopropylphenyl)phenanthro[3,2-b]benzofuran-11-yl)-1H-benzo[d]imidazole. The obtained compound was identified by mass spectrometry and HPLC analysis.

[0382] HRMS(MALDI) calculated for $C_{83}H_{63}D_{12}IrN_4O$: m/z 1348.6323, Found: 1348.6325.

Example 1

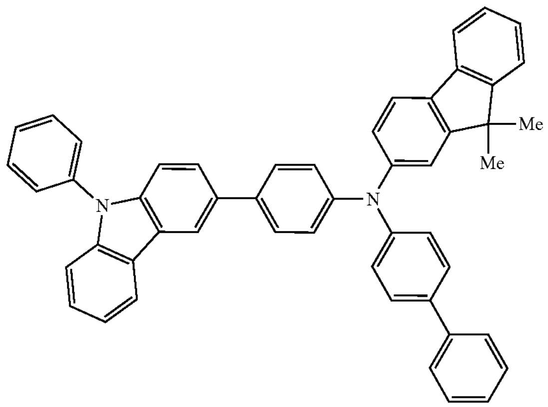
[0383] An ITO(as an anode)-patterned glass substrate was cut to a size of 50 millimeters (mm)×50 mm×0.5 mm, sonicated with isopropyl alcohol and pure water, each for 5 minutes, and then cleaned by radiation with ultraviolet light (UV) and exposure to ozone for 30 minutes. The resulting glass substrate was loaded onto a vacuum deposition apparatus.

[0384] Compounds HT3 and HT-D2 were vacuum-codeposited in a weight ratio of 98:2 on the anode to form a HIL having a thickness of 100 angstroms (Å), and Compound HT3 was vacuum-deposited on the HIL to form a hole transport layer having a thickness of 1650 Å.

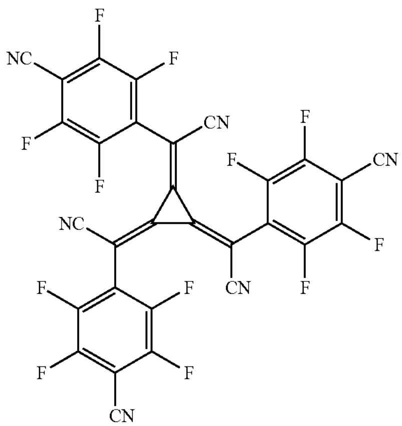
[0385] Next, a host (Compound GH3) and a dopant (Compound 94 listed in Table 2) were vacuum-co-deposited on the hole transport layer at a weight ratio of 92:8 to form an emission layer having a thickness of 400 Å.

[0386] Then, Compound ET3 and ET-D1 were vacuum-co-deposited on the emission layer in a volume ratio of 50:50 to form an electron transport layer having a thickness of 350 Å, ET-D1 was vacuum-deposited on the electron transport layer to form an electron injection layer having a thickness of 10 Å, and Al was vacuum-deposited on the electron injection layer to form a cathode having a thickness of 1,000 Å, thereby completing the manufacture of an organic light-emitting device.

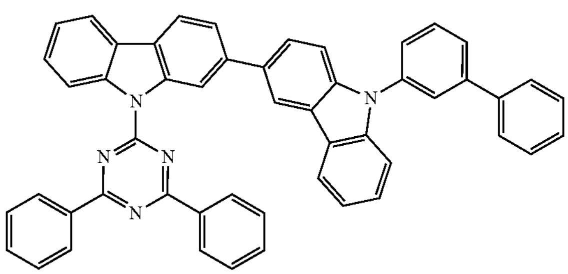
HT3



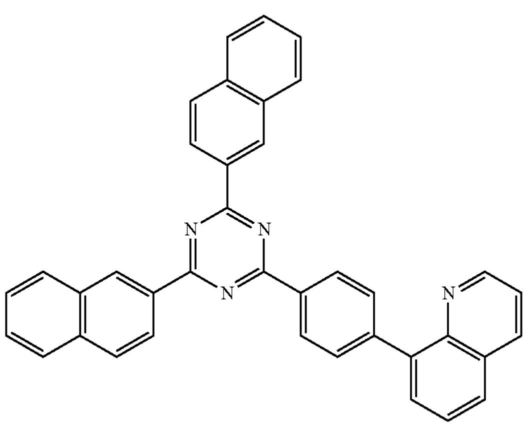
HT-D2



GH3



ET3



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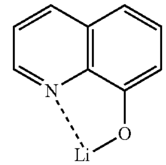
ET-D1



Comparative Example 1R, Example 2 and Comparative Example 2R

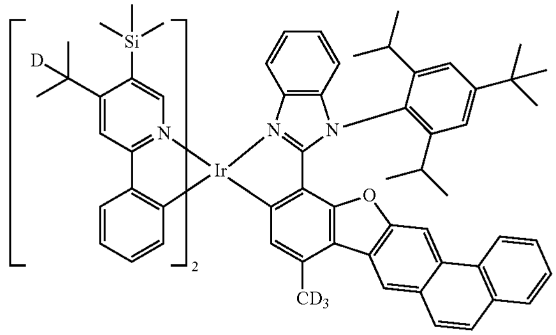
[0387] Organic light-emitting devices were manufactured in the same manner as in Example 1, except that the compounds listed in Table 2 were used as dopants when forming the emission layer.

Evaluation Example 1

[0388] For each of the organic light-emitting devices manufactured in Example 1, Comparative Example 1R, Example 2, and Comparative Example 2R, driving voltage (volts, V), maximum external quantum efficiency (Max EQE), maximum emission wavelength (nm) and lifetime (LT_{97}) were evaluated. Results thereof are shown in Table 2. As evaluation devices, a current-voltage meter (Keithley 2400) and a luminance meter (Minolta Cs-1000A) were used, and for the lifetime (LT_{97}) (at 6,000 nit), the time (hr) required for the luminance to reach 97% of the initial luminance of 100% was evaluated. The maximum external quantum efficiency and lifetime were expressed as relative values (%). The maximum emission wavelength was set/defined as the wavelength corresponding to the top of the emission peak of an emission spectrum measured with a luminance meter.

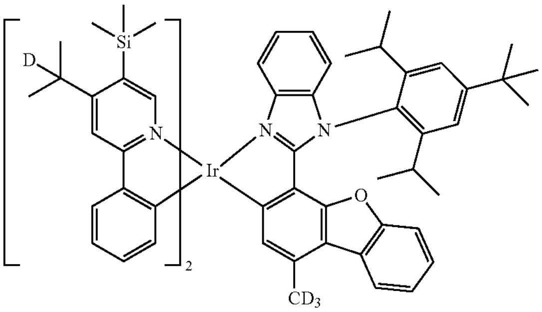
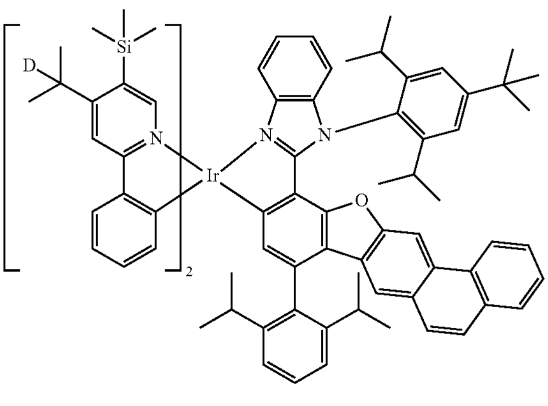
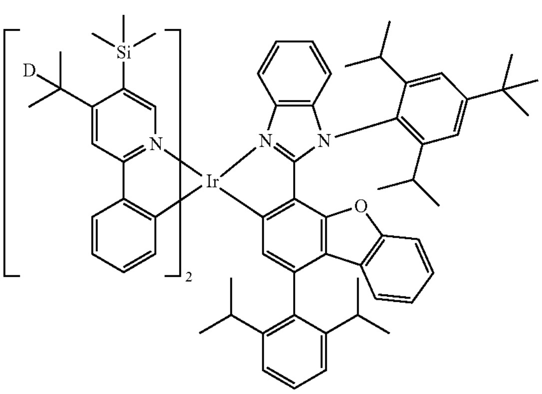
TABLE 2

	Dopant in emission layer	Driving voltage (V)	Max EQE (relative value, %)	Maximum emission wavelength (nm)	LT97 (relative value, %) (at 6,000 nit)
Example 1	94	4.5	108	522	140
Comparative Example 1R	94R	4.6	100	521	100
Example 2	98	4.4	105	526	120
Comparative Example 2R	98R	4.5	98	524	85



94

TABLE 2-continued

Dopant in emission layer	Driving voltage (V)	Max EQE (relative value, %)	Maximum emission wavelength (nm)	LT97 (relative value, %) (at 6,000 nit)
 94R				
 98				
 98R				

[0389] As shown in Table 2, the organic light-emitting device of Example 1 has an improved driving voltage, improved external quantum efficiency, and improved lifetime compared to the organic light-emitting device of Comparative Example 1R. Similarly, the organic light-emitting device of Example 2 has an improved driving voltage, improved external quantum efficiency, and improved lifetime compared to the organic light-emitting device of Comparative Example 2R.

Comparative Example 3R

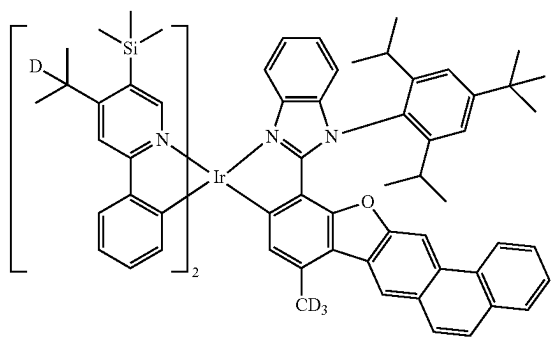
[0390] An organic light-emitting device was manufactured in the same manner as in Example 1, except that the compound listed in Table 3 was used as a dopant when forming the emission layer.

Evaluation Example 2

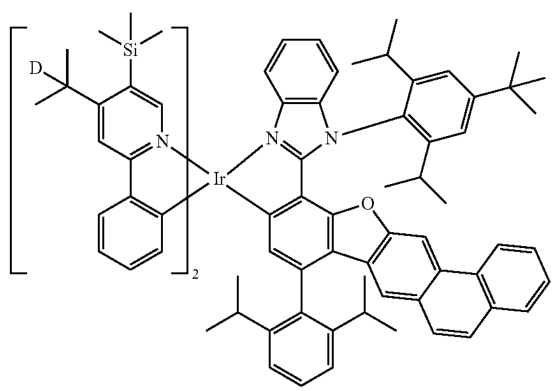
[0391] The maximum emission wavelength (nm) of the organic light-emitting device manufactured in Comparative Example 3 was evaluated by using the same method as described in Evaluation Example 1, and the result is shown in Table 3 together with the maximum emission wavelengths of the organic light-emitting devices of Examples 1 and 2.

TABLE 3

	Dopant in emission layer	Maximum emission wavelength (nm)
Example 1	94	522
Example 2	98	526
Comparative Example 3R	Ref1	530



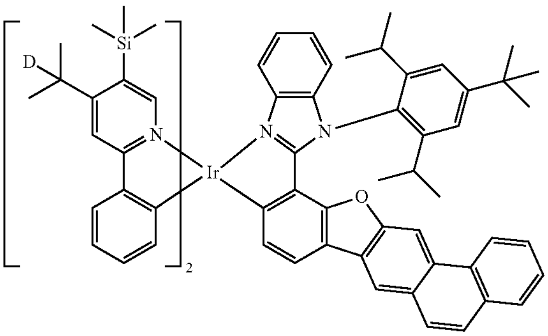
94



98

TABLE 3-continued

	Dopant in emission layer	Maximum emission wavelength (nm)



Ref1

[0392] As shown in Table 3, the organic light-emitting devices of Examples 1 and 2 have a smaller maximum emission wavelength, for example, in the range of 510 nm to 529 nm, than the organic light-emitting device of Comparative Example 3R.

[0393] Example 11, Comparative Example 11R, Example 12, and Comparative Example 12R

[0394] Organic light-emitting devices were manufactured in the same manner as in Example 1, except that the compounds listed in Table 4 were used as dopants when forming the emission layer.

Evaluation Example 3

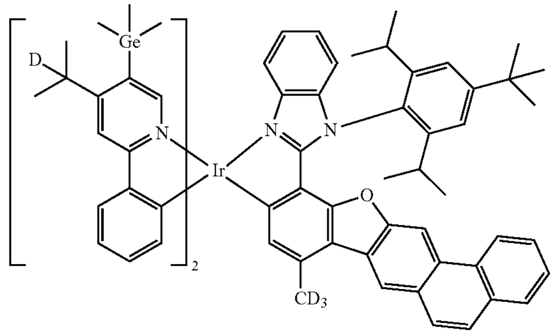
[0395] For each of the organic light-emitting devices manufactured in Example 11, Comparative Example 11R, Example 12, and Comparative Example 12R, driving voltage (V), maximum external quantum efficiency (Max EQE), maximum emission wavelength (nm) and lifetime (LT_{97}) were evaluated in the same manner as used in Evaluation Example 1. Results thereof are shown in Table 4. The maximum external quantum efficiency and lifetime were expressed as relative values (%).

TABLE 4

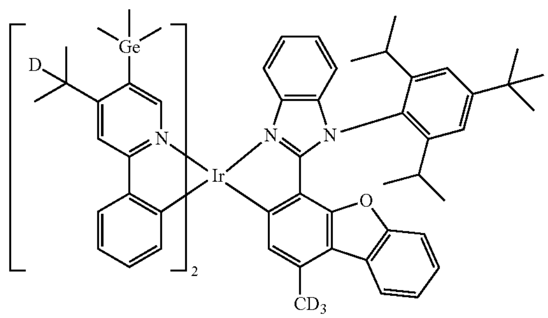
	Dopant in emission layer	Driving voltage (V)	Max EQE (relative value, %)	Maximum emission wavelength (nm)	LT97 (relative value, %) (at 6,000 nit)
Example 11	99	4.3	109	522	135
Comparative Example 11R	99R	4.5	100	521	100
Example 12	103	4.3	106	525	120
Comparative Example 12R	103R	4.3	96	524	90

TABLE 4-continued

	Dopant in emission layer	Driving voltage (V)	Max EQE (relative value, %)	Maximum emission wavelength (nm)	LT97 (relative value, %) (at 6,000 nit)



99



99R

[0396] As shown in Table 4, the organic light-emitting device of Example 11 has an improved driving voltage, improved external quantum efficiency, and improved lifetime compared to the organic light-emitting device of Comparative Example 11R. Similarly, the organic light-emitting device of Example 12 has an improved driving voltage, equivalent external quantum efficiency, and improved lifetime compared to the organic light-emitting device of Comparative Example 12R.

Comparative Example 13R

[0397] An organic light-emitting device was manufactured in the same manner as in Example 1, except that the compound listed in Table 5 was used as a dopant when forming the emission layer.

Evaluation Example 4

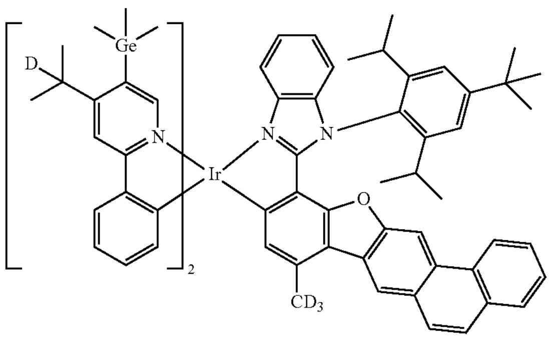
[0398] The maximum emission wavelength (nm) of the organic light-emitting device manufactured in Comparative Example 13R was evaluated by using the same method as described in Evaluation Example 1, and the result is shown in Table 5 together with the maximum emission wavelengths of the organic light-emitting devices of Examples 11 and 12.

TABLE 5

	Dopant in emission layer	Maximum emission wavelength (nm)
Example 11	99	522
Example 12	103	525

TABLE 5-continued

	Dopant in emission layer	Maximum emission wavelength (nm)
Comparative Example 13R	Ref2	531



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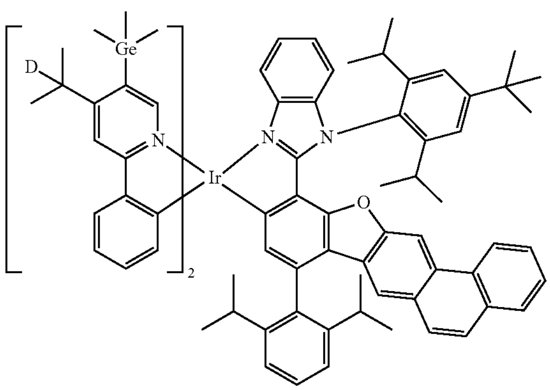
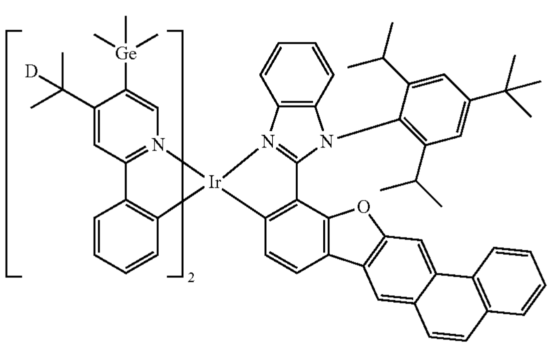


103



Ref2

[0399] As shown in Table 5, the organic light-emitting devices of Examples 11 and 12 have a smaller maximum emission wavelength, for example, in the range of 510 nm to 529 nm, than the organic light-emitting device of Comparative Example 13R.

Example 21, Comparative Example 21R, Example 22, and Comparative Example 22R

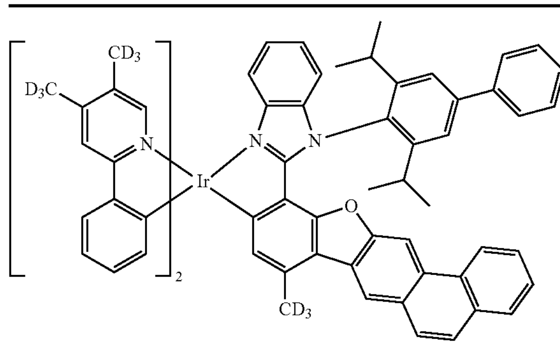
[0400] Organic light-emitting devices were manufactured in the same manner as in Example 1, except that the compounds listed in Table 6 were used as dopants when forming the emission layer.

Evaluation Example 5

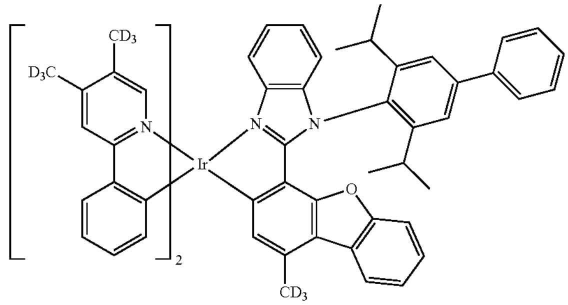
[0401] For each of the organic light-emitting devices manufactured in Example 21, Comparative Example 21R, Example 22, and Comparative Example 22R, driving voltage (V), maximum external quantum efficiency (Max EQE), maximum emission wavelength (nm) and lifetime (LT_{97}) were evaluated in the same manner as used in Evaluation Example 1. Results thereof are shown in Table 6. The maximum external quantum efficiency and lifetime were expressed as relative values (%).

TABLE 6

	Dopant in emission layer	Driving voltage (V)	Max EQE (relative value, %)	Maximum emission wavelength (nm)	LT_{97} (relative value, %) (at 6,000 nit)
Example 21	110	4.1	107	522	135
Comparative Example 21R	110R	4.2	100	522	100
Example 22	111	4.1	104	526	125
Comparative Example	111R	4.2	94	525	90



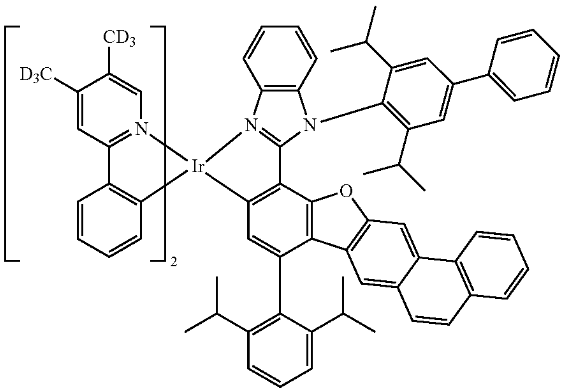
110



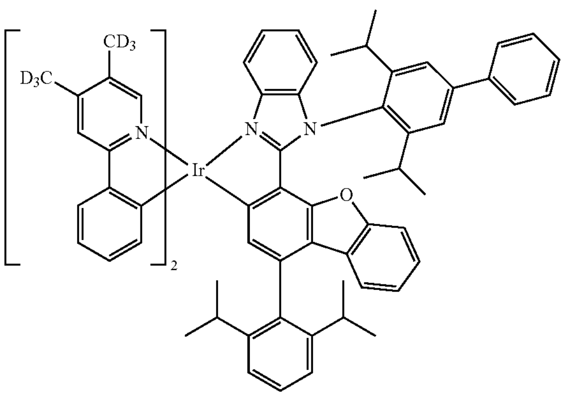
110R

TABLE 6-continued

Dopant in emission layer	Driving voltage (V)	Max EQE (relative value, %)	Maximum emission wavelength (nm)	LT_{97} (relative value, %) (at 6,000 nit)



111



111R

[0402] As shown in Table 6, the organic light-emitting device of Example 21 has an improved driving voltage, improved external quantum efficiency, and improved lifetime compared to the organic light-emitting device of Comparative Example 21R. Similarly, the organic light-emitting device of Example 22 has an improved driving voltage, improved external quantum efficiency, and improved lifetime compared to the organic light-emitting device of Comparative Example 22R.

Comparative Example 23R

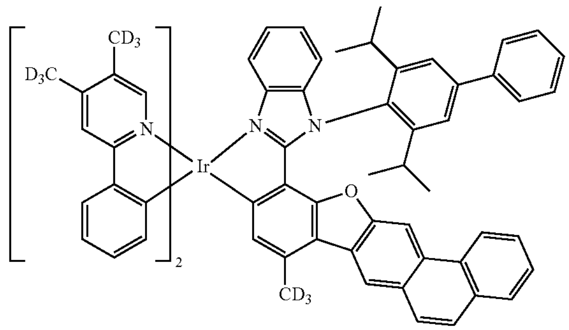
[0403] An organic light-emitting device was manufactured in the same manner as in Example 1, except that the compound listed in Table 7 was used as a dopant when forming the emission layer.

Evaluation Example 6

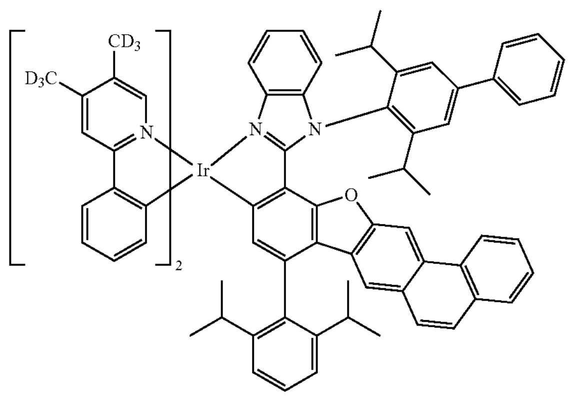
[0404] The maximum emission wavelength (nm) of the organic light-emitting device manufactured in Comparative Example 23R was evaluated by using the same method as described in Evaluation Example 1, and the result is shown in Table 7 together with the maximum emission wavelengths of the organic light-emitting devices of Examples 21 and 22.

TABLE 7

	Dopant in emission layer	Maximum emission wavelength (nm)
Example 21	110	522
Example 22	111	526
Comparative Example 23R	Ref3	532



110



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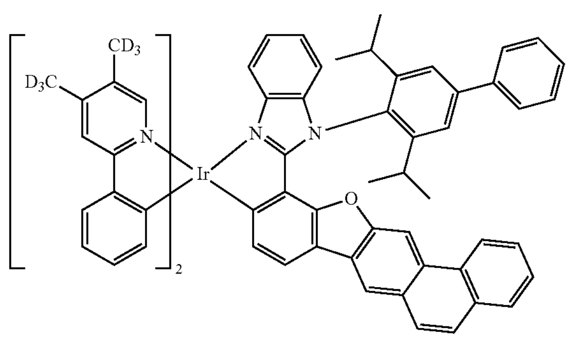


Ref3

[0405] As shown in Table 7, the organic light-emitting devices of Examples 21 and 22 have a smaller maximum emission wavelength, for example, in the range of 510 nm to 529 nm, than the organic light-emitting device of Comparative Example 23R.

Examples 31 to 34 and Comparative Example 31R

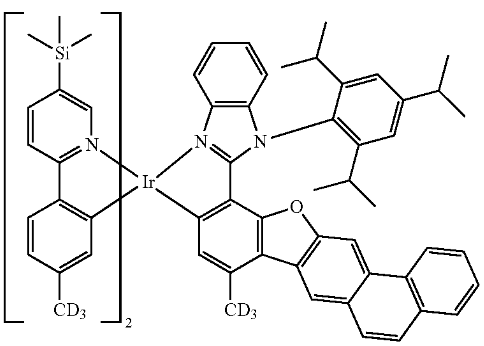
[0406] Organic light-emitting devices were manufactured in the same manner as in Example 1, except that the compounds listed in Table 8 were used as dopants when forming the emission layer.

Evaluation Example 7

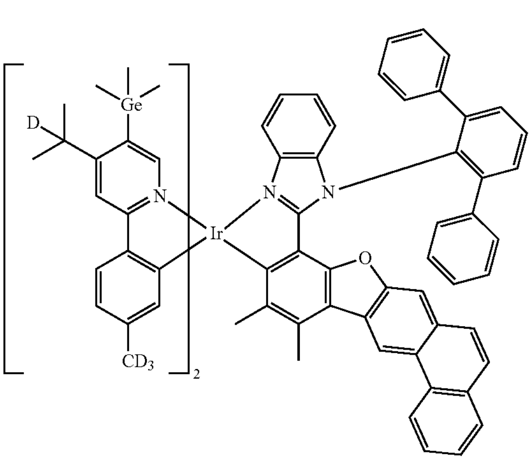
[0407] For each of the organic light-emitting devices manufactured in Examples 31 to 34 and Comparative Example 31R, driving voltage (V), maximum external quantum efficiency (Max EQE), maximum emission wavelength (nm), and lifetime (LT_{97}) were evaluated in the same manner as used in Evaluation Example 1. Results thereof are shown in Table 8. The maximum external quantum efficiency and lifetime were expressed as relative values (%).

TABLE 8

	Dopant in emission layer	Driving voltage (V)	Max EQE (relative value, %)	Maximum emission wavelength (nm)	LT_{97} (relative value, %) (at 6,000 nit)
Example 31	1	4.5	26	523	130
Example 32	109	4.5	26	527	130
Example 33	49	4.6	27	525	140
Example 34	57	4.6	27	524	130
Comparative Example 31R	Ref4	4.5	26	523	100



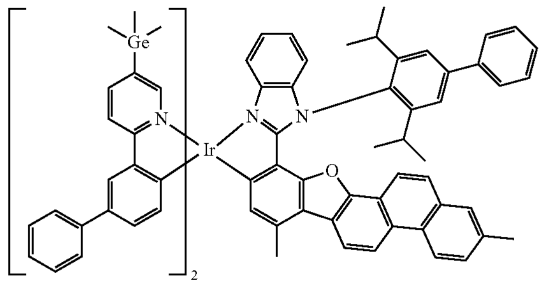
1



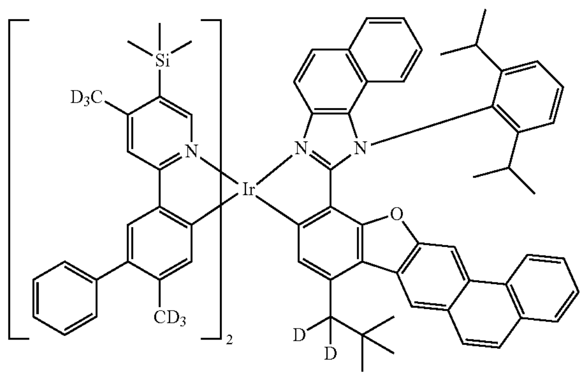
109

TABLE 8-continued

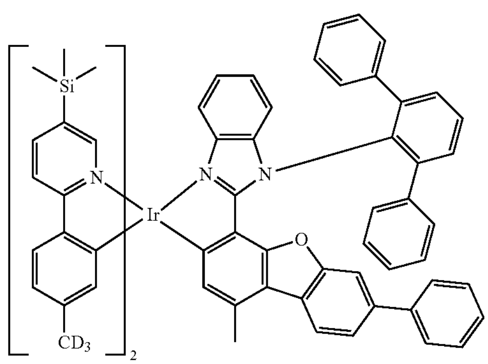
Dopant in emission layer	Driving voltage (V)	Max EQE (relative value, %)	Maximum emission wavelength (nm)	LT_{97} (relative value, %) (at 6,000 nit)



49



57



Ref4

[0408] As shown in Table 8, it can be seen that the organic light-emitting devices of Examples 31 to 34 emit green light while having equivalent or improved external quantum luminescence efficiency and improved lifetime characteristics compared to the organic light-emitting device of Comparative Example 31R.

[0409] The organometallic compound according to the disclosure has excellent electrical properties, and accordingly, an electronic device, for example, a light-emitting device, using the organometallic compound, may have improved driving voltage, improved external quantum efficiency, and improved lifetime characteristics. The light-emitting device enables manufacture of high-quality electronic apparatuses.

[0410] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

What is claimed is:

1. An organometallic compound represented by Formula 1:

$$M(L_1)_{n1}(L_2)_{n2} \qquad \text{Formula 1}$$

wherein, in Formula 1,

M is a transition metal,

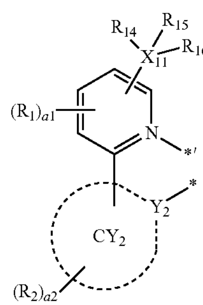
L_1 is a ligand represented by Formula 2-1,

L_2 is a ligand represented by Formula 2-2,

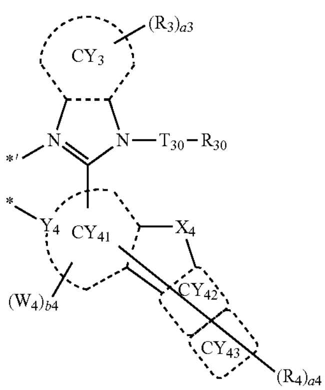
n1 and n2 are each independently 1 or 2, wherein, when n1 is 2, two of L_1 are identical to or different from each other, and when n2 is 2, two of L_2 are identical to or different from each other, and

L_1 and L_2 are different from each other,

Formula 2-1



Formula 2-2



wherein, in Formulae 2-1 and 2-2,

Y_2 and Y_4 are each independently C or N,

ring CY_2, ring CY_3, and ring CY_{41} to ring CY_{43} are each independently a C_5-C_{30} carbocyclic group or a C_1-C_{30} heterocyclic group,

X_{11} is C, Si, or Ge,

T_{30} is a single bond, a C_1-C_{20} alkylene group unsubstituted or substituted with at least one R_{10a}, a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a},

X_4 is O, S, Se, $N(R_{48})$, $C(R_{48})(R_{49})$, or $Si(R_{48})(R_{49})$,

R_1 to R_4, R_{14} to R_{16}, R_{30}, R_{48}, and R_{49} are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted C_1-C_{60} alkoxy group, a substituted or unsubstituted C_1-C_{60} alkylthio group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_7-C_{60} alkyl aryl group, a substituted or unsubstituted C_7-C_{60} aryl alkyl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted C_2-C_{60} alkyl heteroaryl group, a substituted or unsubstituted C_2-C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1-C_{60} heteroaryloxy group, a substituted or unsubstituted C_1-C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —$Si(Q_1)(Q_2)(Q_3)$, —$Ge(Q_1)(Q_2)(Q_3)$, —$N(Q_4)(Q_5)$, —$B(Q_6)(Q_7)$, —$P(=O)(Q_8)(Q_9)$, or —$P(Q_8)(Q_9)$,

a1 is an integer from 0 to 3,

a2 to a4 are each independently an integer from 0 to 20,

W_4 is a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted C_1-C_{60} alkoxy group, a substituted or unsubstituted C_1-C_{60} alkylthio group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_7-C_{60} alkyl aryl group, a substituted or unsubstituted C_7-C_{60} aryl alkyl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted C_2-C_{60} alkyl heteroaryl group, a substituted or unsubstituted C_2-C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1-C_{60} heteroaryloxy group, a substituted or unsubstituted C_1-C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group,

b4 is an integer from 1 to 10,

two or more of a plurality of R_1 are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a},

two or more of a plurality of R_2 are optionally linked to each other to form a C_5-C_{30} carbocyclic group that is unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group that is unsubstituted or substituted with at least one R_{10a},

two or more of a plurality of R_3 are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a},

two or more of a plurality of R_4 are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a},

two or more of R_1 to R_4 or R_{30} are optionally linked to each other to form a C_5-C_{30} carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C_1-C_{30} heterocyclic group unsubstituted or substituted with at least one R_{10a},

R_{10a} is as described in connection with R_2,

* and *' each indicate a binding site to M in Formula 1,

a substituent of the substituted C_1-C_{60} alkyl group, the substituted C_2-C_{60} alkenyl group, the substituted C_2-C_{60} alkynyl group, the substituted C_1-C_{60} alkoxy group, the substituted C_1-C_6o alkylthio group, the substituted C_3-C_{10} cycloalkyl group, the substituted C_1-C_{10} heterocycloalkyl group, the substituted C_3-C_{10} cycloalkenyl group, the substituted C_1-C_1 heterocycloalkenyl group, the substituted C_6-C_{60} aryl group, the substituted C_7-C_{60} alkyl aryl group, the substituted C_7-C_6o aryl alkyl group, the substituted C_6-C_6o aryloxy group, the substituted C_6-C_6o arylthio group, the substituted C_1-C_6o heteroaryl group, the substituted C_2-C_{60} alkyl heteroaryl group, the substituted C_2-C_{60} heteroaryl alkyl group, the substituted C_1-C_6o heteroaryloxy group, the substituted C_1-C_6o heteroarylthio group, the substituted monovalent non-aromatic condensed polycyclic group, and the substituted monovalent non-aromatic condensed heteropolycyclic group is:

deuterium, —F, —Cl, —Br, —I, —SF_5, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_6o alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_6o alkoxy group, or a C_1-C_{60} alkylthio group;

a C_1-C_6o alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_0a alkoxy group, or a C_1-C_6o alkylthio group, each substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF_5, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_3-C_1 cycloalkyl group, a C_1-C_1a heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_1 heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_7-C_{60} aryl alkyl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_6o heteroaryl group, a C_2-C_{60} alkyl heteroaryl group, a C_2-C_{60} heteroaryl alkyl group, a C_1-C_6o heteroaryloxy group, a C_1-C_6o heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —$Si(Q_{11})(Q_{12})(Q_{13})$, —$Ge(Q_{11})(Q_{12})(Q_{13})$, —$N(Q_{14})(Q_{15})$, —$B(Q_{16})(Q_{17})$, —$P(=O)(Q_{18})(Q_{19})$, —$P(Q_{18})(Q_{19})$, or a combination thereof;

a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_{10} heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a C_2-C_{60} alkyl heteroaryl group, a C_1-C_{60} heteroaryloxy group, a C_1-C_{60} heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, or a monovalent non-aromatic condensed heteropolycyclic group, each unsubstituted or substituted with at least one of deuterium, —F, —Cl, —Br, —I, —SF_5, —CD_3, —CD_2H, —CDH_2, —CF_3, —CF_2H, —CFH_2, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a C_1-C_{60} alkyl group, a C_2-C_{60} alkenyl group, a C_2-C_{60} alkynyl group, a C_1-C_{60} alkoxy group, a C_1-C_{60} alkylthio group, a C_3-C_{10} cycloalkyl group, a C_1-C_{10} heterocycloalkyl group, a C_3-C_{10} cycloalkenyl group, a C_1-C_1 heterocycloalkenyl group, a C_6-C_{60} aryl group, a C_7-C_{60} alkyl aryl group, a C_7-C_{60} aryl alkyl group, a C_6-C_{60} aryloxy group, a C_6-C_{60} arylthio group, a C_1-C_{60} heteroaryl group, a C_2-C_{60} alkyl heteroaryl group, a C_2-C_{60} heteroaryl alkyl group, a C_1-C_{60} heteroaryloxy group, a C_1-C_{60} heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, —$Si(Q_{21})(Q_{22})(Q_{23})$, —$Ge(Q_{21})(Q_{22})(Q_{23})$, —$N(Q_{24})(Q_{25})$, —$B(Q_{26})(Q_{27})$, —$P(=O)(Q_{28})(Q_{29})$, $P(Q_{28})(Q_{29})$, or a combination thereof;

—$Si(Q_{31})(Q_{32})(Q_{33})$, —$Ge(Q_{31})(Q_{32})(Q_{33})$, —$N(Q_{34})(Q_{35})$, —$B(Q_{36})(Q_{37})$, —$P(=O)(Q_{38})(Q_{39})$, or —$P(Q_{38})(Q_{39})$; or

a combination thereof, and

Q_1 to Q_9, Q_{11} to Q_{19}, Q_{21} to Q_{29} and Q_{31} to Q_{39} are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, —SF_5, a hydroxyl group, a cyano group, a nitro group, an amino group, an amidino group, a hydrazine group, a hydrazone group, a carboxylic acid group or a salt thereof, a sulfonic acid group or a salt thereof, a phosphoric acid group or a salt thereof, a substituted or unsubstituted C_1-C_{60} alkyl group, a substituted or unsubstituted C_2-C_{60} alkenyl group, a substituted or unsubstituted C_2-C_{60} alkynyl group, a substituted or unsubstituted C_3-C_{10} cycloalkyl group, a substituted or unsubstituted C_1-C_1 heterocycloalkyl group, a substituted or unsubstituted C_3-C_{10} cycloalkenyl group, a substituted or unsubstituted C_1-C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6-C_{60} aryl group, a substituted or unsubstituted C_7-C_{60} alkyl aryl group, a substituted or unsubstituted C_7-C_{60} aryl alkyl group, a substituted or unsubstituted C_6-C_{60} aryloxy group, a substituted or unsubstituted C_6-C_{60} arylthio group, a substituted or unsubstituted C_1-C_{60} heteroaryl group, a substituted or unsubstituted C_2-C_{60} alkyl heteroaryl group, a substituted or unsubstituted C_2-C_{60} heteroaryl alkyl group, a substituted or unsubstituted C_1-C_{60} heteroaryloxy group, a substituted or unsubstituted C_1-C_{60} heteroarylthio group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, or a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

2. The organometallic compound of claim **1**, wherein

ring CY_2 is a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a 1,2,3,4-tetrahydronaphthalene group, a carbazole group, a fluorene group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, a dibenzoselenophene group, a pyridine group, a benzoxazole group, a benzothiazole group, or a benzene group that is condensed with a norbornane group.

3. The organometallic compound of claim **1**, wherein ring CY_3 and ring CY_{41} to ring CY_{43} are each independently:

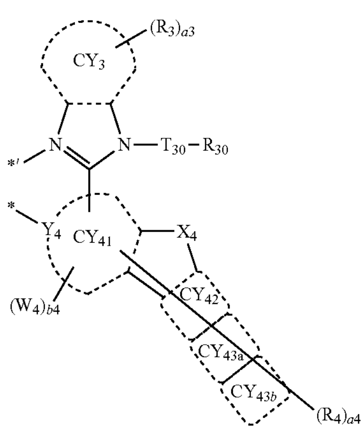
a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, a triazine group, a naphthalene group, a quinoline group, an isoquinoline group, a quinazoline group, a quinoxaline group, a phenanthrene group, a benzoquinoline group, a benzoisoquinoline group, an anthracene group, or a chrysene group; or

a benzene group, a pyridine group, a pyrimidine group, a pyridazine group, a pyrazine group, a triazine group, a naphthalene group, a quinoline group, an isoquinoline group, a quinazoline group, a quinoxaline group, a phenanthrene group, a benzoquinoline group, a benzoisoquinoline group, an anthracene group, or a chrysene group, each condensed with a cyclohexane group, a norbornane group, an adamantane group, or a combination thereof.

4. The organometallic compound of claim **1**, wherein

L_2 is a ligand represented by Formula 2-2A:

Formula 2-2A



wherein, in Formula 2-2A,

Y_4, ring CY_3, ring CY_{41}, ring CY_{42}, T_{30}, X_4, R_3, R_4, R_{30}, a3, a4, W_4, and b4 are as described in claim **1**, and

ring CY_{43a} and ring CY_{43b} are each independently a C_5-C_{15} carbocyclic group or a C_1-C_{15} heterocyclic group.

5. The organometallic compound of claim **1**, wherein T_{30} is:

a single bond; or

a C_1-C_{20} alkylene group, a benzene group, a naphthalene group, a dibenzofuran group, or a dibenzothiophene group, each unsubstituted or substituted with at least one of deuterium, —F, a cyano group, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, or a combination thereof.

6. The organometallic compound of claim **1**, wherein R_1 to R_4, R_{30}, R_{48} and R_{49} are each independently:

hydrogen, deuterium, —F, or a cyano group; or

a C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, or a dibenzothiophenyl group, each unsubstituted or substituted with at least one of deuterium, —F, a cyano group, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, —Si(Q_{31})(Q_{32})(Q_{33}), —Ge(Q_{31})(Q_{32})(Q_{33}), or a combination thereof,

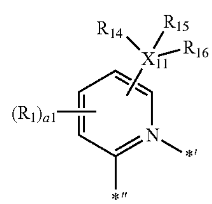
R_{14} to R_{16} are each independently a C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, or a dibenzothiophenyl group, each unsubstituted or substituted with at least one of deuterium, —F, a cyano group, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, or a combination thereof.

7. The organometallic compound of claim **1**, wherein W_4 is a C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, or a dibenzothiophenyl group,

each unsubstituted or substituted with at least one of:

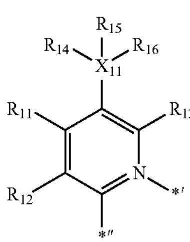
deuterium, —F, a cyano group, a C_1-C_{20} alkyl group, a deuterated C_1-C_{20} alkyl group, a fluorinated C_1-C_{20} alkyl group, a C_3-C_{10} cycloalkyl group, a deuterated C_3-C_{10} cycloalkyl group, a fluorinated C_3-C_{10} cycloalkyl group, a (C_1-C_{20} alkyl)C_3-C_{10} cycloalkyl group, a phenyl group, a deuterated phenyl group, a fluorinated phenyl group, a (C_1-C_{20} alkyl)phenyl group, a naphthyl group, a pyridinyl group, a furanyl group, a thiophenyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, —Si(Q_{31})(Q_{32})(Q_{33}), —Ge(Q_{31})(Q_{32})(Q_{33}), or a combination thereof.

8. The organometallic compound of claim **1**, wherein a group represented by:

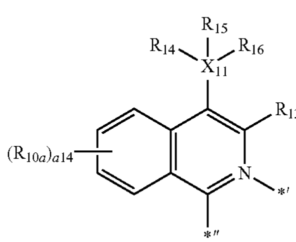


in Formula 2-1 is a group represented by one of Formulae CY1-1 to CY1-3:

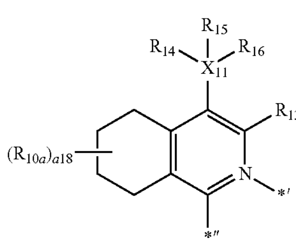
CY1-1



CY1-2



CY1-3



wherein, in Formulae CY1-1 to CY1-3,

X_{11}, R_{14} to R_{16}, and R_{10a} are as described in claim **1**,

each of R_{11} to R_{13} is as described in connection with R_1 in claim **1**,

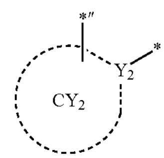
a14 is an integer from 0 to 4,

a18 is an integer from 0 to 8,

*' indicates a binding site to M in Formula 1, and

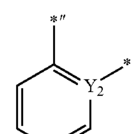
*" indicates a binding site to a neighboring atom in Formula 2-1.

9. The organometallic compound of claim **1**, wherein a group represented by:

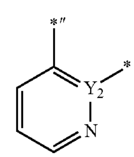


in Formula 2-1 is a group represented by one of Formulae CY2-1 to CY2-33:

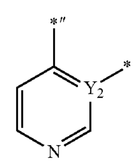
CY2-1



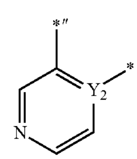
CY2-2



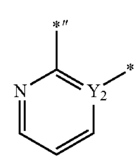
CY2-3



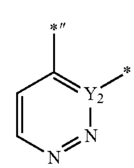
CY2-4



CY2-5



CY2-6



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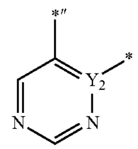
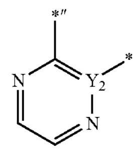
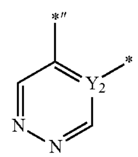
CY2-7



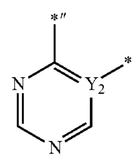
CY2-8



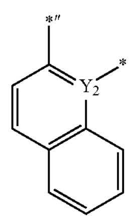
CY2-9



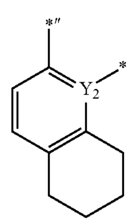
CY2-10



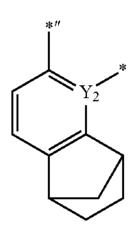
CY2-11



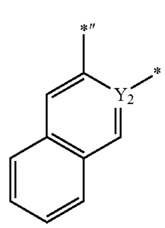
CY2-12



CY2-13



CY2-14



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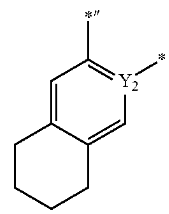
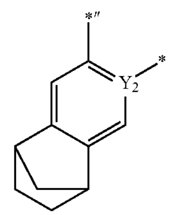
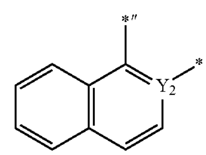
CY2-15



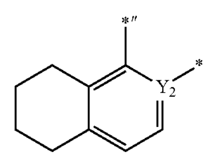
CY2-16



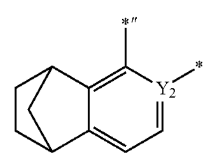
CY2-17



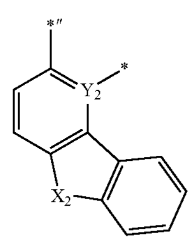
CY2-18



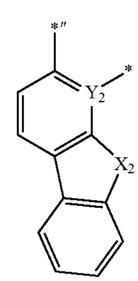
CY2-19



CY2-20



CY2-21



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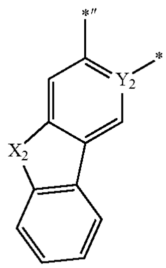
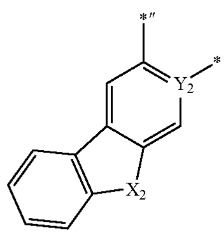
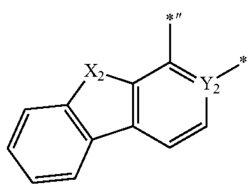
CY2-22



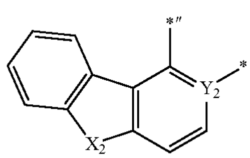
CY2-23



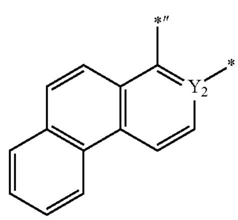
CY2-24



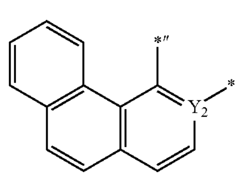
CY2-25



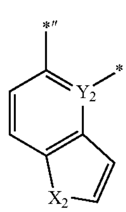
CY2-26



CY2-27



CY2-28



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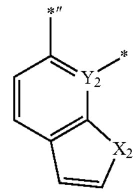
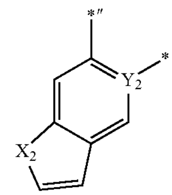
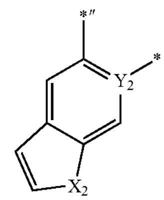
CY2-29



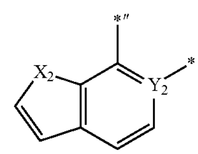
CY2-30



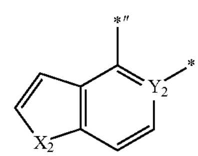
CY2-31



CY2-32



CY2-33



wherein, in Formulae CY2-1 to CY2-33,

Y_2 is as described in claim **1**,

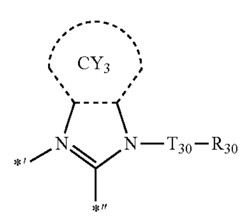
X_2 is O, S, Se, $N(R_{28})$, $C(R_{28})(R_{29})$, or $Si(R_{28})(R_{29})$,

each of R_{28} and R_{29} is as described in connection with R_2 in claim **1**,

*" indicates a binding site to a neighboring atom in Formula 2-1, and

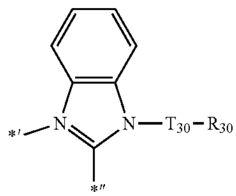
* indicates a binding site to M in Formula 1.

10. The organometallic compound of claim **1**, wherein a group represented by:

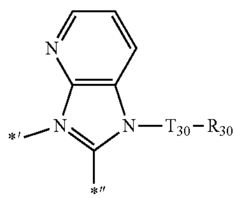


in Formula 2-2 is a group represented by one of Formulae CY3-1 to CY3-21:

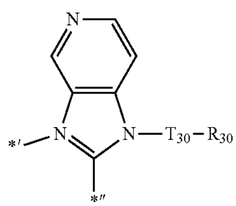
CY3-1



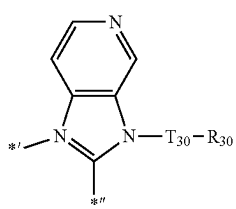
CY3-2



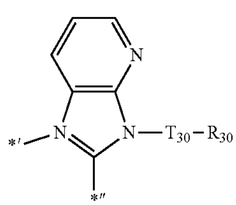
CY3-3



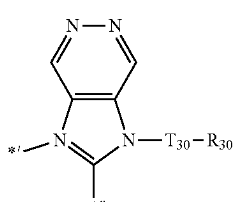
CY3-4



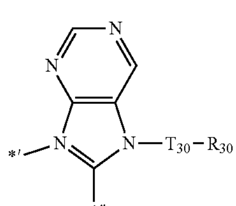
CY3-5



CY3-6



CY3-7



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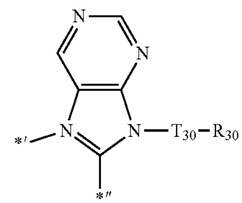
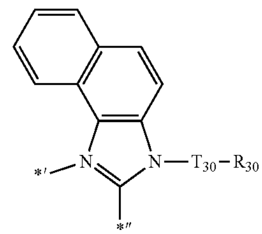
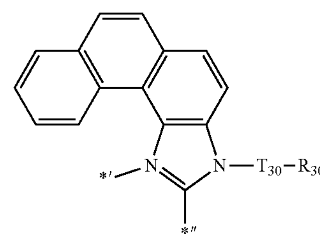
CY3-8



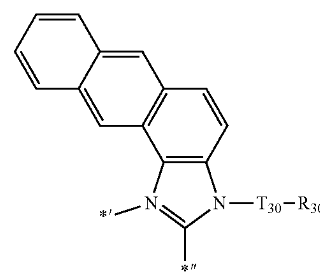
CY3-9



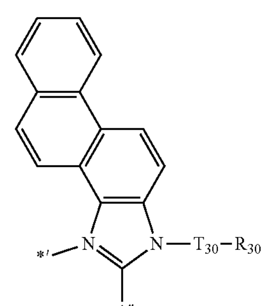
CY3-10



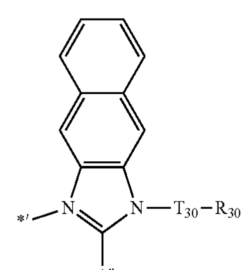
CY3-11



CY3-12



CY3-13



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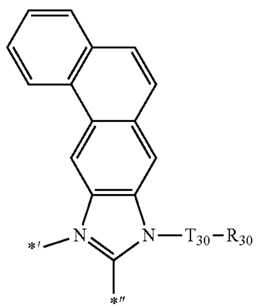
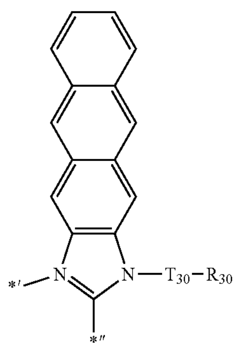
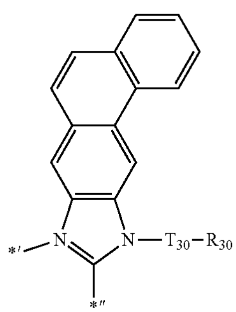
CY3-14



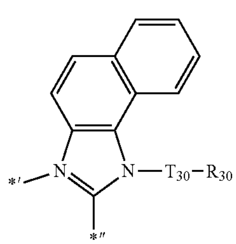
CY3-15



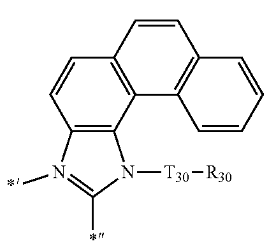
CY3-16



CY3-17



CY3-18



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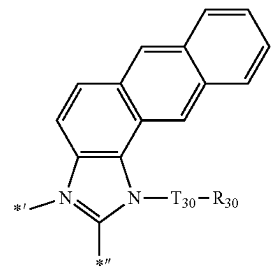
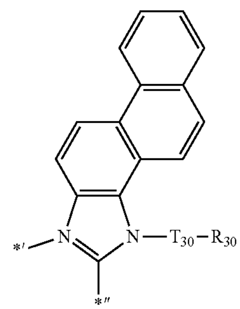
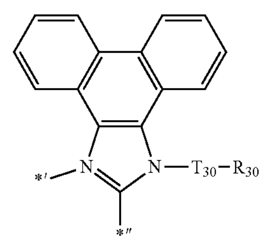
CY3-19



CY3-20



CY3-21



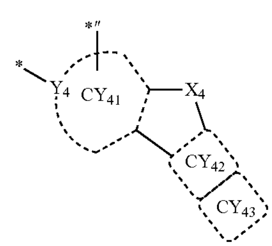
wherein, in Formulae CY3-1 to CY3-21,

T_{30} and R_{30} are as described in claim **1**,

*' indicates a binding site to M in Formula 1, and

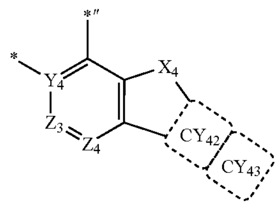
*" indicates a binding site to a neighboring atom in Formula 2-2.

11. The organometallic compound of claim **1**, wherein a group represented by:

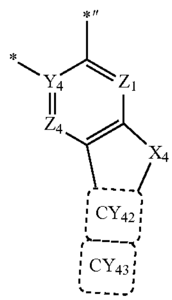


in Formula 2-2 is represented by one of Formulae CY4-1 to CY4-6:

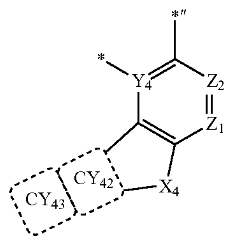
CY4-1



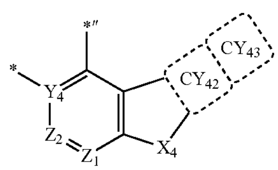
CY4-2



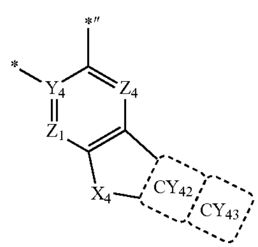
CY4-3



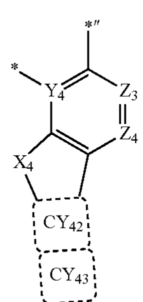
CY4-4



CY4-5



CY4-6



wherein, in Formulae C_4-1 to C_4-6,

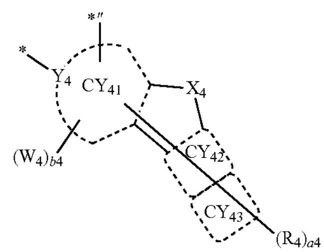
Y_4, X_4, ring CY_{42} and ring CY_{43} are each as described in claim **1**,

Z_1 to Z_4 are each independently N or C,

* indicates a binding site to M in Formula 1, and

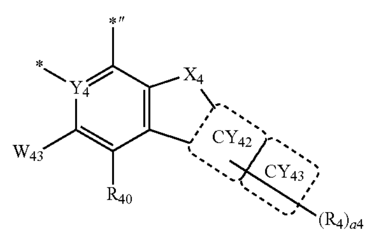
*" indicates a binding site to a neighboring atom in Formula 2-2.

12. The organometallic compound of claim **1**, wherein a group represented by:

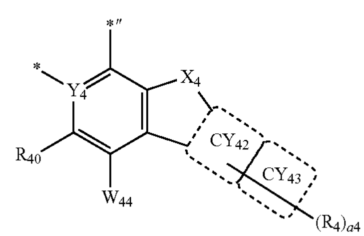


in Formula 2-2 is represented by one of Formulae CY4(1) to CY4(18):

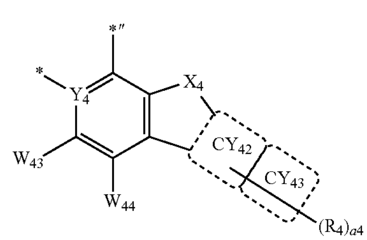
CY4(1)



CY4(2)



CY4(3)



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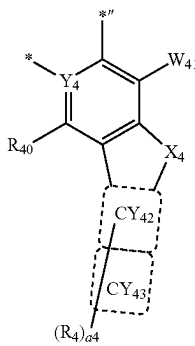
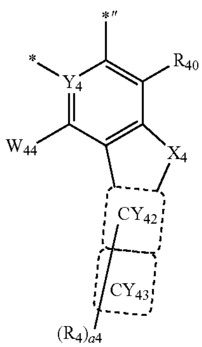
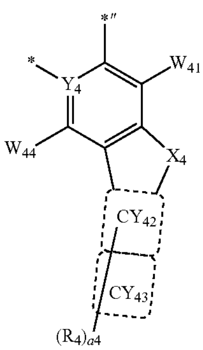
CY4(4)



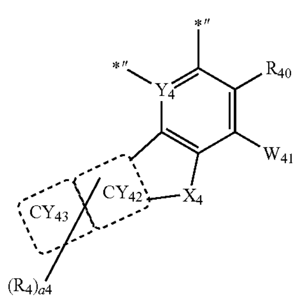
CY4(5)



CY4(6)



CY4(7)



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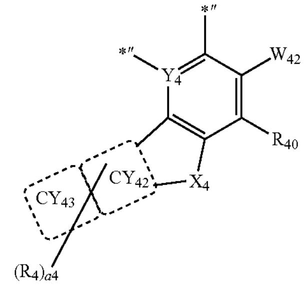
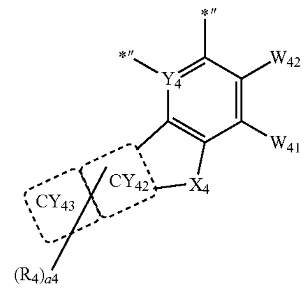
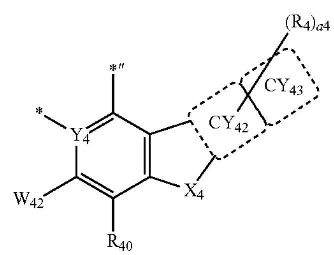
CY4(8)



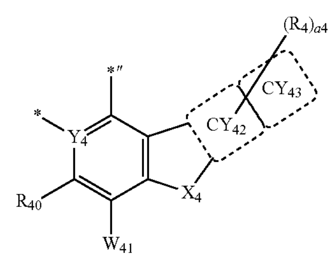
CY4(9)



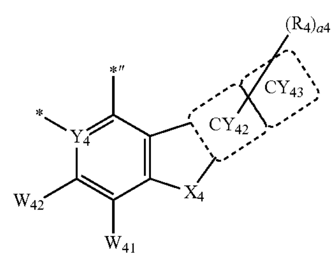
CY4(10)



CY4(11)



CY4(12)



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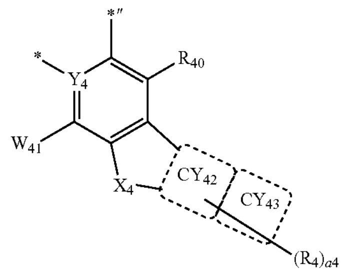
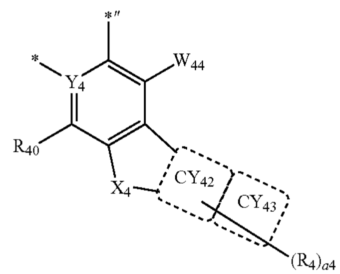
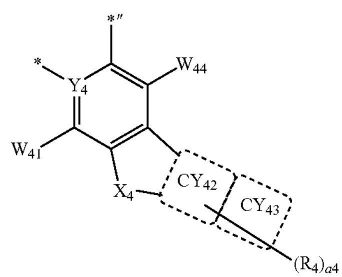
CY4(13)



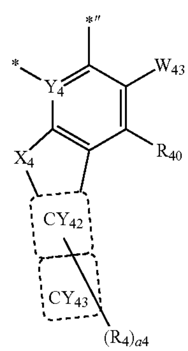
CY4(14)



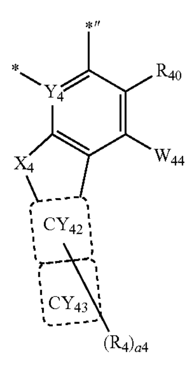
CY4(15)



CY4(16)



CY4(17)



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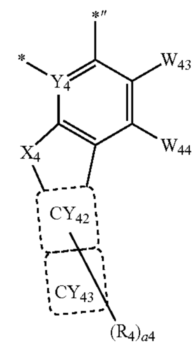
CY4(18)



wherein, in Formulae CY4(1) to CY4(18),

Y_4, ring CY_{42}, ring CY_{43}, and R_4 are as described in claim **1**,

each of W_{41} to W_{44} is as described in connection with W_4 in claim **1**,

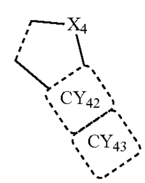
R_{40} is as described in connection with R_4 in claim **1**,

a4 is an integer from 0 to 10,

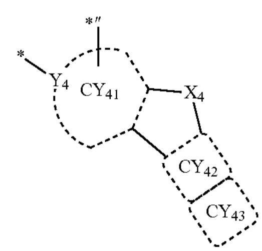
* indicates a binding site to M in Formula 1, and

*" indicates a binding site to a neighboring atom in Formula 2-2.

13. The organometallic compound of claim **1**, wherein a group represented by:

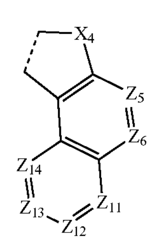


in



of Formula 2-2 is represented by one of Formulae CY401 to CY451:

CY401



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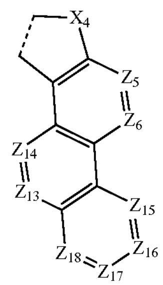
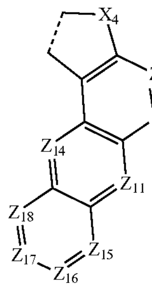
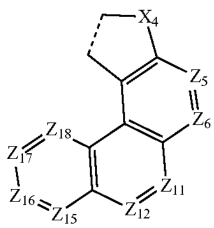
CY402



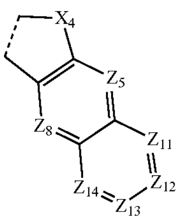
CY403



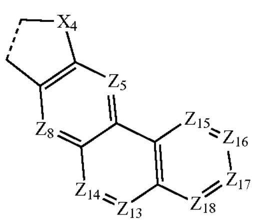
CY404



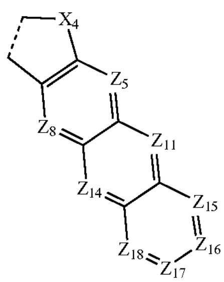
CY405



CY406



CY407



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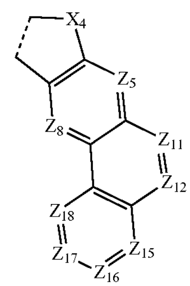
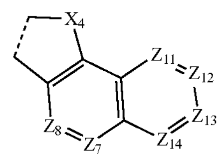
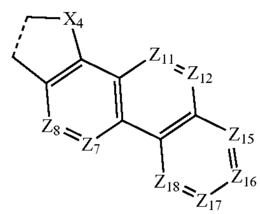
CY408



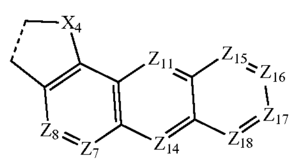
CY409



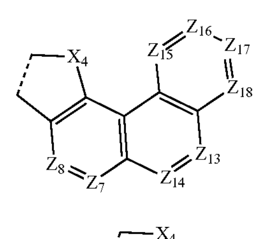
CY410



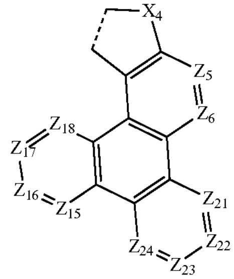
CY411



CY412



CY413



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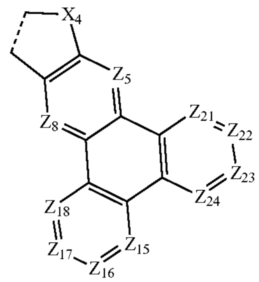
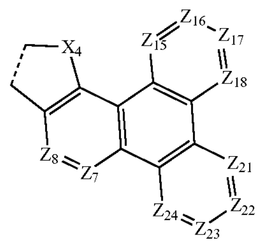
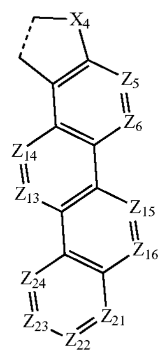
CY414



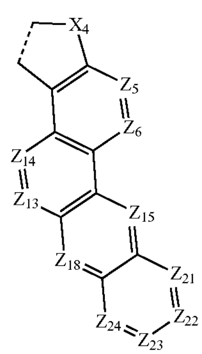
CY415



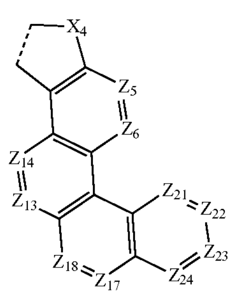
CY416



CY417



CY418



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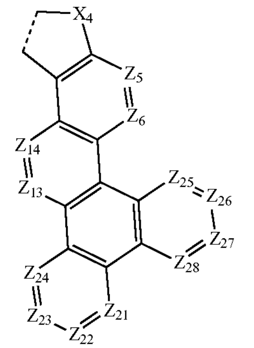
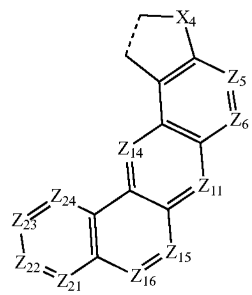
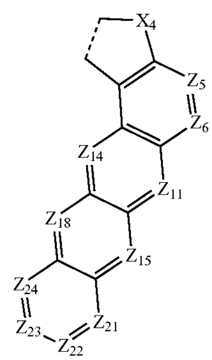
CY419



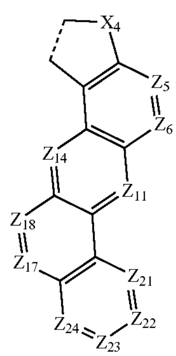
CY420



CY421



CY422



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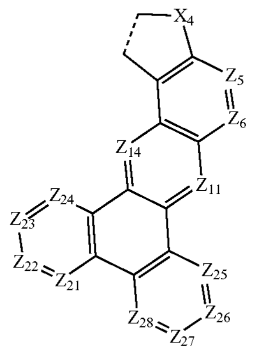
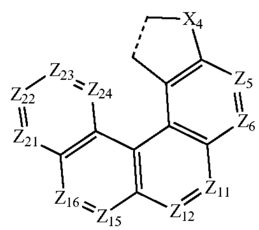
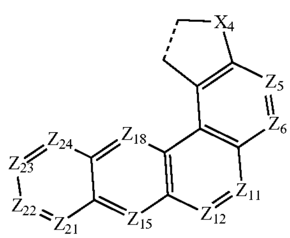
CY423



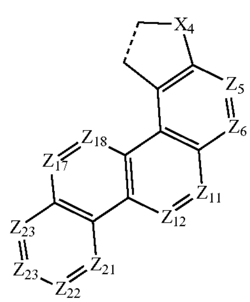
CY424



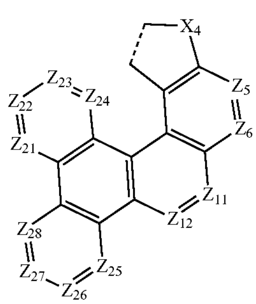
CY425



CY426



C7427



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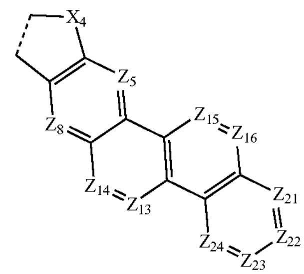
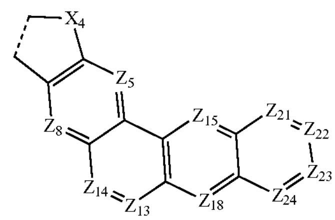
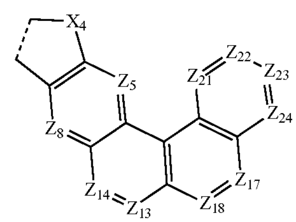
CY428



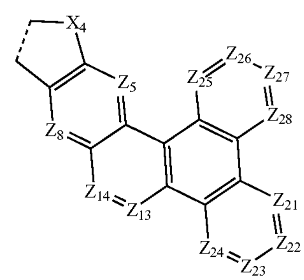
CY429



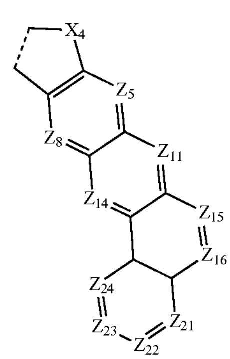
CY430



CY431



CY432



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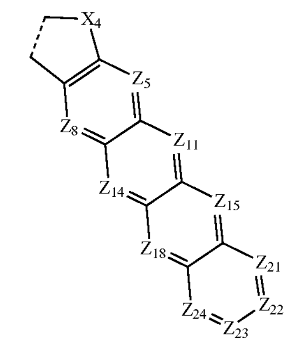
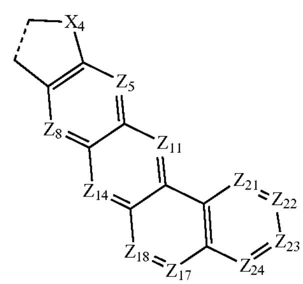
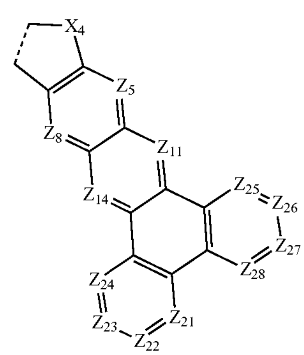
CY433



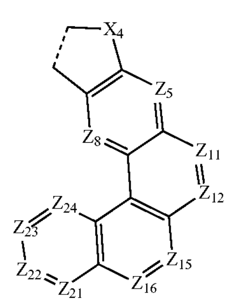
CY434



CY435



CY436



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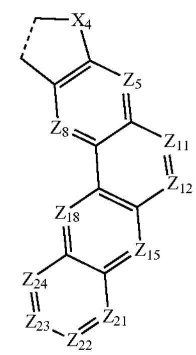
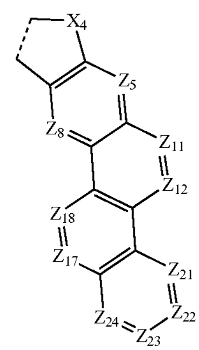
CY437



CY438



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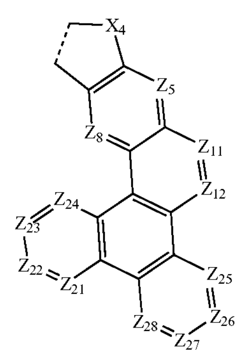

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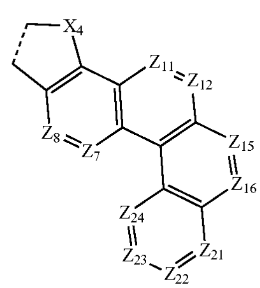

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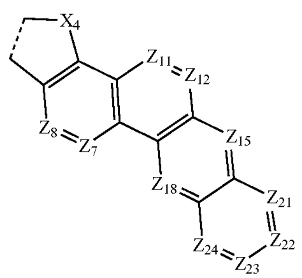

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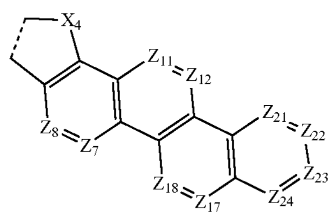

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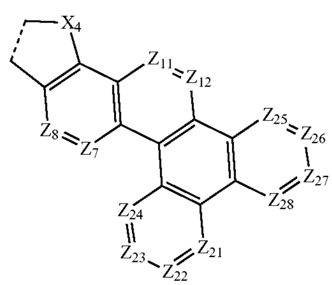

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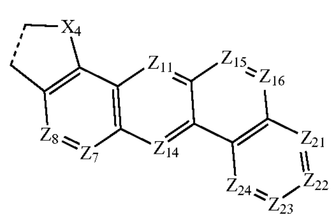

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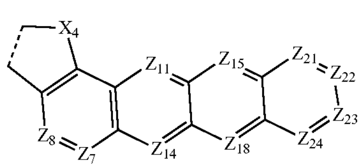

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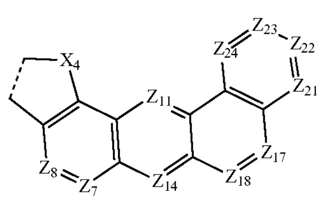

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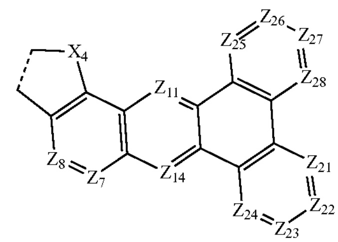

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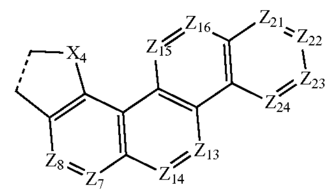

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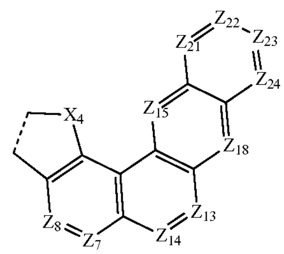

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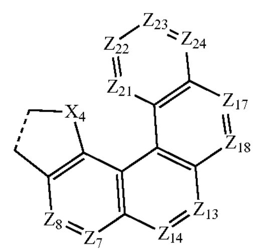

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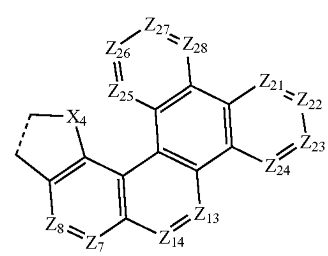

wherein, in Formulae CY401 to CY451,

X_4 is as described in claim **1**,

Z_5 to Z_8, Z_{11} to Z_{18}, and Z_{21} to Z_{28} are each independently N or C, and

an X_4-containing five-membered ring is condensed with ring CY_{41} which is adjacent thereto.

14. The organometallic compound of claim **1**, wherein the organometallic compound emits light having a maximum emission wavelength of about 510 nanometers to about 550 nanometers.

15. A light-emitting device, comprising:

a first electrode;

a second electrode; and

an interlayer arranged between the first electrode and the second electrode,

wherein the interlayer comprises an emission layer, and

wherein the interlayer further comprises at least one of the organometallic compound of claim **1**.

16. The light-emitting device of claim **15**, wherein

the first electrode is an anode,

the second electrode is a cathode,

the interlayer further comprises a hole transport region arranged between the first electrode and the emission layer, and an electron transport region arranged between the emission layer and the second electrode,

the hole transport region comprises a hole injection layer, a hole transport layer, an electron-blocking layer, a buffer layer, or a combination thereof, and

the electron transport region comprises a hole-blocking layer, an electron transport layer, an electron injection layer, or a combination thereof.

17. The light-emitting device of claim **15**, wherein the emission layer comprises at least one of the organometallic compound.

18. The light-emitting device of claim **17**, wherein the emission layer emits a green light.

19. The light-emitting device of claim **17**, wherein the emission layer further comprises a host, and an amount of the host in the emission layer is greater than an amount of the at least one of the organometallic compound in the emission layer, based on weight.

20. An electronic apparatus, comprising the light-emitting device of claim **15**.

* * * * *